



NANAJI DESHMUKH KRUSHI SANJIVANI PRAKALP PHASE II

PROJECT ON CLIMATE RESILIENT AGRICULTURE PHASE II

PROJECT IMPLEMENTATION PLAN



Project Management Unit (PMU), Mumbai,
Department of Agriculture

Nanaji Deshmukh Krishi Sanjivani Prakalp (NDKSP)
Project on Climate Resilient Agriculture (PoCRA) – Phase II

Project Implementation Plan



Project Management Unit (PMU)

Prepared by



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Abbreviations

AA	: Agriculture Assistant	CLFs	: Cluster Level Federations
AE	: Agronomic efficiency	CNB	: Cement Nala Bund
	Aadhaar-enabled Payment	CO	: Carbon Monoxide
AePS	: System	CO ₂	: carbon dioxide
	Agriculture, Forestry, and	CRA	: Climate Resilient Agriculture
AFOLU	: Other Land Use		Climate Resilient Agriculture
	Animal Husbandry	CRATs	: Technologies
AHD	: Department	CRFs	: Controlled-Release Fertilizers
AI	: Artificial Intelligence		Central Research Institute for
AWC	: Available Water Capacity	CRIDA	: Dryland Agriculture
AWD	: alternate wetting and drying	CSOs	: Civil Society Organisations
BAU	: business-as-usual	CVI	: climate vulnerability index
BBF	: Broad Bed Furrow	DBT	: Direct Benefit Transfer
BCM	: bn cubic metres	DBTs	: Direct Beneficiary Transfers
	Bankers Institute of Rural	D-DSR	: dry-direct seeded rice
BIRD	: Development		Department of Economics
BIS	: Bureau of Indian Standard	DEA	: Affairs
BREB	: Bowen Ratio Energy Balance		Department of Financial
BRLF	: Bharati Rural Livelihood Funds	DFS	: Services
CA	: Conservation Agriculture		Directorate General of
	Controller of Aid, Accounts		Information and Public
CAAA	: and Audit	DGIPR	: Relations
	Central Agroforestry Research	DoA	: Department of Agriculture
CAFRI	: Institute		Department of Public
	Controller and Auditor	DPE	: Enterprises
CAG	: General		District Project
CAO	: Circle Agriculture Officer	DPIU	: Implementation Unit
CAs	: Cluster Assistants		District Project Monitoring
CBOs	: Civil Body Organizations	DPMU	: Unit
CCC	: Carbon Credit Certificates	DPR	: Detailed Project Report
	conditions, continuous	DR	: data recovery
CCT	: contour trenches		District Superintending
CCTS	: Carbon Credit Trading Scheme	DSAO	: Agriculture Officer
CEC	: cation exchange capacity	DSR	: Direct Seeded Rice
	Capacity Enhancement Needs	DSS	: Decision Support System
CENA	: Assessment	EAP	: Externally Aided Programmes
	Centra Food Technological		environmental impacts, the
CFTRI	: Research Institute		project applying
	Central Institute of Cotton		Environmental Codes of
CICR	: Research	ECOPs	: Practice
CIG	: Commodity Interest Group		

	Entrepreneurship	FUE	: Fertilizer Use Efficiency
EDP	: Development Program	GAP	: Gender Action Plan
	ESMF The Environmental	GBV	: Gender-Based Violence
EMF	: Management Framework	GDP	: Gross Domestic Product
	Environmental Management	GeM	: Government e-marketplace
EMP	: Plan	GHG	: Greenhouse gases
eNAM	: National Agriculture Market		Geographic Information
ENB	: Earthen Nala Bunds	GIS	: System
Eoi	: Expression of Interest	GKVS	: Gram Krishi Vikas Samiti
	environmental impacts, the	GOI	: Government of India
	project will implement	GoM	: Government of Maharashtra
	Environmental Codes of	GPs	: Gram Panchayats
ESCOPs	: Practice	GPS	: Global Positioning System
	Environmental and Social	GR	: grievance-redress
ESF	: Framework		Groundwater Surveys and
	Environmental and Social	GSDA	: Development Agency
ESMF	: Management Framework	GVKS	: Gram Vikas Krishi Samiti
ESs	: Environmental Standards	GW	: Groundwater
ET	: Evapotranspiration	HDBP	: High-Density Block Plantations
ETL	: economic threshold level	HDPS	: High-Density Polythene Sheet
EX-ACT	: EX-Ante Carbon Balance Tool	HTC	: Horticulture Training Centre
F2FE	: Farmer-to-Farmer Extension		Indian Agricultural Statistics
	Food and Agriculture	IASRI	: Research Institute
FAO	: Organization		Indian Council of Agriculture
FAs	: Field Assistants	ICAR	: Research
	Field Activities Supervision and	ICM	: Indian Carbon Market
FAST	: Tracking System		International Crops Research
	Forest Development		Institute for the Semi-Arid
FDCM	: Corporation of Maharashtra	ICRISAT	: Tropics
FFS	: Farmer Field Schools		Information and
FGDs	: Focus Group Discussions	ICT	: Communications Technology
FIG	: Farmer's Interest Group	IEA	: International Energy Agency
	Finance Information		Information Education
FIMS	: Management System	IEC	: Communication
Fis	: Financial Institutions	IFS	: Integrated Farming Systems
FMB	: Farm Machinery Banks		Integrated Fertilizer and
FMCGs	: Fast Moving Consumer Goods	IFWM	: Water Management
FPCs	: Farmer Producer Companies		Indian Institute of
FPGs	: Farmer Producer Groups	IIM	: Management
FPO	: Farmer Producer Organisation	IISS	: Indian Institute of Soil Science
	Food Safety and Standard		Indian Institute of Social
FSSAI	: Authority of India	IISS	: Science

IIT : Indian Institute of Technology
IIMN Foundation for Entrepreneurship
InFED : Development
Integrated Nutrient
INM : Management
INR : Indian Rupees
IoT : Internet of Things
Intergovernmental Panel on
IPCC : Climate Change
Integrated Pest & Disease
IPDM : Management
IPF : Investment Project Financing
IPM : Integrated Pest Management
Integrated Pest and Nutrient
IPNM : Management
Integrated Pest and Nutrient
IPNMP : Management Plan
International Rice Research
IRRI : Institute
integrated soil fertility
ISFM : management
integrate Information
IT : Technology
Integrated Weed
IWM : Management
Integrated Watershed
IWMP : Management Programme
JDA : Joint Director, Agriculture
JJAS : June-September
KPI : Key Performance Indicators
KVKs : Krishi Vigyan Kendras
LBS : Loose boulder structures
LLP : Limited Liabe Properties
Labor Management
LMP : Procedures
LOC : Lines of Credit
LUP : Land Use Pattern
Maharashtra Agricultural
MACP : Competitiveness Project
Maharashtra Agribusiness
MAGNET : Network Project

MANAGE : Management
Mahila Arthik Vikas
MAVIM : Mahamandal
MB : Measurement books
Maharashtra Cooperative
MCDC : Development Cooperation
Maharashtra Centre for
Entrepreneurship
MCED : Development
Multilateral Development
MDB : Bank
Maharashtra Energy
MEDA : Development Authority
Mahatma Gandhi National
Rural Employment Guarantee
MGNREGS : Scheme
Maharashtra Industrial
MIDC : Development Corporation
Mission for Integrated
MIDH : Development of Horticulture
Management Information
MIS : System
MLP : Micro Level Planning
Million metric ton carbon
MMtCO₂eq : equivalent
Ministry of New and
MNRE : Renewable Energy
Ministry of Environment,
MoEFCC : Forest and Climate Change
Memorandum of
MOUs : Understanding
Maharashtra Remote Sensing
MRSAC : Application Centre
Measurement, reporting, and
MRV : verification
Maharashtra State
MSAMB : Agricultural Marketing Board
Maharashtra State Action Plan
MSAPCC : on Climate Change
Micro Small Medium
MSME : Enterprises
The Maharashtra State Rural
MSRLM : Livelihood Mission

MSWC	Maharashtra state : warehousing corporations		
MT	: Metric Tons	PESA	: The Provision of the Panchayat (Extension to the Schedule Areas) Act
NAIP	National Agricultural : Innovation Project	PMFBY	: Pradhan Mantri Fasal Bima Yojna
NAPCC	National Action Plan on : Climate Change		Pradhan Mantri Formalization of Micro Food Processing
NBM	: National Bamboo Mission	PMFME	: Entreprises
NBSS	: National Bureau of Soil Science	PMI	: performance monitoring index
NC	: National Communication, Nanaji Deshmukh Krishi	PMKSY	: Pradhan Mantri Krishi Sinchai Yojana
NDKSP	: Sanjivani Prakalp		Pradhan Mantri Kisan Urja Suraksha Evam Utthaan
NFE	: Non-formal Education	PM-Kusum	: Mahabhiyan
NHM	: National Horticulture Mission		Pradhan Mantri Kisan Urja Suraksha evam Utthaan
NICRA	National Innovations in : Climate Resilient Agriculture	PM-KUSUM	: Mahabhiyan
NIPHM	The National Institute of Plant : Health Management	PMU	: Project Management Unit
NIPHT	National Institute of Post : Harvest Technology	PoC	: Proof-of-Concept
NMNF	National Mission on Natural : Farming	PoCRA	: Project on Climate Resilient Agriculture
NMOOP	National Mission on Oilseed : and Oil Palm	PoP	: package of practices personal protective
NMSA	National Mission for : Sustainable Agriculture	PPE	: equipment
NPSA	National Programme for : Sustainable Agriculture"	PPSD	: Project Procurement Strategy Development
NREGA	National Rural Employment : Guarantee Act, 2005	PV	: Photovoltaic
NRM	Natural Resource : Management	PWD	: Persons with Disabilities
NRRI	National Rice Research : Institute		Regional Agriculture Extension Management Training
OC	: Organic Carbon	RAMETI	: Institutes
PAO	: Pay and Account Office	RCP	: Representative Concentration Pathways
PDKV	Panjabrao Deshmukh Krishi : Vidyapeeth	REDP	: Residential Entrepreneurship Development Program
PDO	Project Development : Objective	RFID	: Result Framework Indicators
		RIDF	: Rural Infrastructure Development Fund
		RKVY	: Rashtriya Krishi Vikas Yojana
		ROC	: Registration of Companies
		RTI	: Right-To-Information

	Seed Authentication, Traceability and Holistic	ToT	: Training of Trainers
SATHI	: Inventory	TPPF	: Tribal Peoples Planning Framework
SAU	: State Agriculture Universities	TSF	: Taluka Seed Farms
	Science Based Targets	UNDP	: United Nations Development Programs
SBTi	: initiative		United Nations Framework
SC	: Scheduled Castes	UNFCCC	: Convention on Climate Change
SCA	: Seed Certification Agency		United Nations General
	Sub-Divisional Agriculture	UNGA	: Assembly
SDAO	: Officer	US	: United States
	Sustainable Development	USD	: US Dollar
SDGs	: Goals	UTs	: Union Territories
SEA	: Sexual Exploitation and Abuse		Vaikunth Mehta National
	Socio Economic and Caste		Institute of Cooperative
SECC	: Census	VAMNICOM	: Management
	small farmer agri-business		Vasantrao Naik State
SFAC	: consortium		Agriculture Extension
SH	: Sexual Harassment		Management Training
SHG	: Self-Help Group	VANAMATI	: Institute
	Small Industrial Development	VAP	: Village Adaptation Plan
SIDBI	: Bank of India		Village Climate Resilient
	State of Maharashtra's	VCRMCMC	: Management Committee
	Agribusiness and Rural	VDP	: Village Development Plan
SMART	: Transformation project		Village Fertility Index the
SOC	: Soil Organic Carbon	VFI	: Village Fertility Index
SOE	: Statement of Expenditure		Vaikunth Mehta Cooperative
SOM	: Soil Organic Matter	VMNICOM	: Management
	Standard Operating	VPD	: vapour pressure deficit
SOPs	: Procedures		Virtual Personal Deposit
	State Project Management	VPDA	: Account
SPMU	: Unit		Water, and Land Management
	Special Report on Climate	WALMI	: Institute
SRCLL	: Change and Land	WOTR	: Watershed Organization Trust
	Systemic Intensification of		Watershed Potential
SRI	: Rice	WPT	: Treatment
ST	: Scheduled Tribes	WUE	: water use efficiency
TAO	: Taluka Agriculture Officer		Yashwantrao Chavan Academy
TERI	: The Energy Research Institute		of Development
	The total tree covers outside	YASHADA	: Administration
TOF	: the forest		
ToR	: Term Of Reference		

Executive Summary

India, with the fifth-largest economy by GDP and third by purchasing power parity, is experiencing a steady decline in the contribution of agriculture to its economy. Despite averaging a 7% growth rate, the sector continues to lose cultivators—15 million fewer since 1991—largely due to climate variability, low productivity, and lack of adaptive infrastructure. Most Indian farmers, particularly in regions like Maharashtra, struggle to cope due to inadequate irrigation infrastructure, limited access to markets, and poor climate awareness. Climate change is significantly impacting India's food systems, energy consumption, and water management.

Maharashtra's analysis from 2012–2022 revealed that 55% of tehsils saw over a 10% increase in southwest monsoon rainfall compared to historical baselines, while 11% experienced significant decreases. Short-duration heavy rainfall events have become more frequent, leading to flash floods and crop damage. Post-monsoon rainfall (October–December) has also risen, often linked to increased cyclonic activity. In the last decade, 87% of talukas experienced increased monsoon rainfall. The changing monsoon patterns and increasing incidence of delayed withdrawal have severely affected agricultural planning. Meanwhile, rising temperatures—especially heatwaves and warm nights from March to May—are likely to become more frequent, posing additional stress on crops. This will affect plant physiology, soil moisture retention, and increase evapotranspiration, leading to further reductions in yield.

The Maharashtra State Action Plan on Climate Change (MSAPCC) provides a blueprint for integrating adaptation and mitigation strategies across sectors such as agriculture, water resources, forestry, health, and urban development. Key interventions include promoting water conservation, afforestation, climate-resilient crops, and efficient irrigation systems. The revised SAPCC incorporates newer climate models and data to provide better sector-specific recommendations. It highlights how rainfed crops like sorghum, pulses, and oilseeds—covering 60% of the sown area—are vulnerable to both droughts and floods. Key cash crops such as cotton, soybean, and rice are already experiencing productivity changes due to temperature increases, erratic rainfall, and humidity-related diseases. Climate change is also negatively impacting soil and water resources. Excessive use of groundwater and flooding have led to salinity problems in western Maharashtra, particularly in sugarcane-growing regions. Soil fertility is declining due to erosion caused by high-intensity rainfall.

To combat these multifaceted challenges, the MSAPCC and associated programs like Project on Climate Resilient Agriculture (PoCRA) also known as Nanaji Deshmukh Krishi Sanjeevani Prakalp (NDKSP) recommend strengthening institutional capacity, expanding crop and livestock insurance, and mainstreaming climate-resilient agricultural and animal husbandry practices. The

concept of Climate Resilient Agriculture (CRA) aims to build resilience through adaptive and risk-absorbing practices at the farm and community levels. CRA promotes technologies and institutional frameworks that respond effectively to climate-induced shocks.

The first phase of the PoCRA (2018–2024), supported by the World Bank with a \$600 million investment, covered 16 drought-prone districts and over 5,000 villages. Through a combination of digital platforms, community mobilization, and microplanning, the project improved yields, increased incomes, and expanded the adoption of climate-resilient technologies. Nearly 1.3 million farmers benefited from Farmer Field Schools, soil health interventions, protected cultivation techniques, and integrated farming systems. Water productivity improved through widespread adoption of drip and sprinkler irrigation and construction of farm ponds. The success of Phase I has set a strong foundation for scaling up in Phase II, focusing on institutional strengthening, convergence with other schemes, and promoting renewable energy solutions.

The **Project on Climate Resilient Agriculture (PoCRA) Phase II**, officially known as the **Nanaji Deshmukh Krishi Sanjeevani Prakalp (NDKSP)**, is a strategic initiative by the Government of Maharashtra (GoM) aimed at improving the resilience and profitability of smallholder farmers. This project aligns with the GoM's Vision 2030 and the Maharashtra State Action Plan on Climate Change (MSAPCC), striving to transform the agricultural sector into a climate-resilient, resource-efficient, and emission-competitive system. The project builds upon the strong foundation of PoCRA Phase I and proposes to scale up its impact by expanding coverage to nearly 20% of Maharashtra's agricultural landscape across 21 vulnerable districts, including eleven districts from Vidarbha, eight districts from Marathwada, and two from North Maharashtra. A scientific village selection process was followed, adopting the IPCC AR5 framework and using a mini-watershed, cluster-based approach. The project selected 7196 villages based on a calculated vulnerability index that integrates historical hazard data, exposure indicators, and socio-ecological vulnerability parameters. Indicators include rainfall variability, soil water-holding capacity, literacy, gender gap, access to irrigation, and proximity to markets. This structured method ensures that the most climate-sensitive areas are prioritized for intervention.

The **Project Development Objective (PDO)** is to improve resilience and profitability of smallholder farmers across selected districts in Maharashtra.

Project is organized into four major components:

- (A) Resource Efficient Productivity Enhancement,
- (B) Promoting Precision Agriculture for GHG Emissions Management,
- (C) Improving and Building Income Resilience, and
- (D) Project Management and Strengthening Institutions.

The estimated cost of the project is ₹6000 crore (USD 700 million), with the World Bank tentatively funding USD 490 million. The implementation period is six years, starting from the date of loan approval. The project will be regularly monitored using a robust system of Key Performance Indicators (KPIs), including the number of beneficiaries with enhanced climate resilience, percentage increase in household farm income, improvement in water-use efficiency, and adoption of digital advisory services.

The project promotes soil health management, biofertilizer production, micro-irrigation, catchment treatment, and conservation agriculture. Simultaneously, it seeks to reduce emissions through sustainable paddy cultivation, solar energy promotion, and crop diversification, including agroforestry, horticulture, and millet production. To strengthen income resilience, the project supports post-harvest infrastructure, value chain development, and agri-entrepreneurship for landless families through activities like goat rearing, backyard poultry, and fisheries.

To ensure success, the implementation strategy involves six major operational pillars: scientific village selection, microplanning, capacity building, climate-adaptive infrastructure, technology deployment, and robust monitoring and evaluation.

The proposed project approach is conceptualized around building resilience in the farming systems vulnerable to the climate variabilities. Salient features of the project are mentioned below.

- The project targets small and marginal farmers (those owning less than 5 hectares), landless farmers, Self-Help Groups (SHGs), Farmer Producer Organizations (FPOs), and public institutions engaged in agriculture. Priority is given to marginalized communities, such as ST/SC farmers, women, and differently-abled individuals.
- Participatory planning is central to the project strategy, with Village Adaptation Plans (VAPs) being developed through community engagement. These plans guide the implementation of customized interventions tailored to each village's agro-climatic and socio-economic conditions.
- The project focuses on resource conservation through natural resource management systems and development of decision support systems for the community by using advanced technologies.
- The project creates an enabling system for the farmers to adopt the climate resilient technologies for production of the commodities suited to the agro-climatic zones. The real-time farm level advisories will be generated in consortium mode, considering the local area weather forecast, cropping pattern, crop condition, water balance, and soil health prevailing at the level of the revenue circles, and or villages

- The resilience package in the project would include diversified cropping systems, agroforestry, horticulture, afforestation, small livestock systems, sustainable agri and allied enterprises.
- Resilience will be enhanced by promotion of micro-irrigation, precision farming, protected cultivation, use of non-conventional energy, and optimum farm mechanization through Farm Machinery Banks which can run on custom hiring basis.
- Improved productivity levels will be supported by efficient post-harvest infrastructure managed by Self Help Groups, and the Farmer Producer Companies, to increase the share of farmers in the final price of agriculture commodities.
- The project will make conscious efforts to bring in the advanced knowledge through enhanced consortia approach involving the research and academic organisations. The project will also have non-financial MOUs with civil society organizations which have common goals of climate resilient agriculture and people's participation.
- The project is aligned with the Government norms for agriculture and allied sector activities.

The project is constructed with comprehensive and interconnected components, with the overarching goal of enhancing climate resilience, reducing input costs, and increasing farm profitability through resource efficiency and precision agriculture

Component A: Resource Efficient Productivity Enhancement

This component focuses on Resource-Efficient Productivity Enhancement, which lays the foundation for building a climate-resilient production system at the village level. This includes improving soil and water use efficiency while promoting soil health restoration. Depletion of soil organic carbon, a key issue in many areas, is addressed through digital soil mapping and the development of village-level soil fertility indices. In partnership with ICAR-NBSS&LUP, these maps guide targeted interventions, and soil health advisories are disseminated via digital platforms.

Sub-component A1 promotes sustainable production practices such as no-till farming, use of biofertilizers, and organic fertilizers. Conservation agriculture is central to the strategy, integrating practices like permanent soil cover and crop rotation to improve adaptation and mitigate climate risks. Farmers are trained in precision fertilizer application using technologies like fertigation and slow-release fertilizers, thereby increasing fertilizer-use efficiency.

Simultaneously, Sub-component A2 enhances water-use efficiency through a combination of infrastructure development and agronomic strategies. Micro-irrigation systems, groundwater recharge structures such as farm ponds and recharge shafts, and watershed treatment measures aim to secure water availability at the farm level. Real-time evapotranspiration monitoring using

satellite imagery and ground-based sensors ensures precision irrigation. Additionally, structural measures like compartment bunding, graded bunding, and gabion structures help prevent soil erosion and conserve water.

The component also includes initiatives for long-term soil and water productivity, including organic farming and integrated nutrient management. Local production and marketing of organic inputs are encouraged through SHGs and FPOs, creating rural entrepreneurship opportunities. In Sub-component A3, technology transformation is pursued through the integration of scientific innovations, ICT-based tools, and climate-smart agricultural technologies that boost resilience and productivity while minimizing greenhouse gas emissions.

Component B: Promoting Precision Agriculture for GHG Emissions Management

This component aims to reduce greenhouse gas emissions and promote carbon sequestration through precision farming and climate-smart practices. By accounting for the carbon and water footprints of various crops, the project introduces climate-adaptive interventions such as agroforestry, moisture conservation, and reduced methane emissions in rice fields.

Sub-component B1 addresses the issue of methane emissions from flooded paddy cultivation. Techniques such as Direct Seeded Rice (DSR), Alternate Wetting and Drying (AWD), and System of Rice Intensification (SRI) are promoted to reduce emissions while conserving water and improving soil health. These efforts are supported by the EX-ACT carbon balance tool developed by the FAO and technical input from ICAR-CRRI. The project also explores the potential of linking farmers to voluntary carbon markets through robust emission monitoring systems.

In Sub-component B2, the project emphasizes reducing fossil fuel dependency in agriculture by supporting renewable energy-based technologies. The use of crop residues for bioenergy—through briquettes, pellets, and biochar—enhances circularity and adds value to agricultural waste.

Agroforestry, bamboo cultivation, and orchard plantations are also being promoted for carbon sequestration and climate resilience. Planting of Bamboo along bunds and wastelands, while orchards of mango, citrus, and guava are supported to diversify income and stabilize farm ecosystems. Protected cultivation methods such as polyhouses and shade net houses are introduced to boost productivity and reduce resource use.

Sub-component B3 focuses on crop diversification by expanding the cultivation of millets. Millets like jowar, bajra, and Ragi are suitable for semi-arid zones and have high nutritional value. To promote millet-based entrepreneurship, the project supports seed systems, processing units,

branding, and FPO-led market development, overcoming constraints related to demand, processing, and value addition.

Component C: Improving and Building Income Resilience

This component focuses on improving and building Income Resilience by supporting climate-smart agribusiness and value-added agriculture. Recognizing the vulnerability of farmers to climate shocks such as droughts and floods, this component aims to stabilize incomes through the promotion of FPOs, SHGs, and other farmer collectives.

Through matching grants and strategic investments, the project empowers local institutions to implement sustainable practices adapted to local conditions. These include support for organic farming, allied activities such as dairy, fisheries, and apiculture, and value chains for crops like pulses, cotton, and forest produce. Maharashtra's growing network of women-led FPOs and SHGs is central to this component, enhancing input efficiency and facilitating the transition to sustainable livelihoods.

Sub-component C1 aims to strengthen agribusiness value chains by helping FPOs and SHGs identify viable business models, create detailed project reports, and implement income-generating activities. These groups receive training and handholding support to build their managerial and marketing capacities. Women are given special attention, with targeted programs facilitated by MAVIM and MSRLM to integrate them into value chains and post-harvest operations.

Sub-component C2 establishes Agribusiness Growth Centers, which serve as hubs for input supply, post-harvest storage, and marketing. Infrastructure under this component includes biofuel units, and water harvesting systems, contributing to climate resilience and GHG reduction.

Sub-component C3 Promotion of Agri-entrepreneurship - Apart from the focus on groups, the project will also support interventions to foster development of individual agri-entrepreneurship by supporting allied activities such as small ruminants, fisheries, sericulture etc. which provide regular cashflow to the farmers and importantly, reduce their dependence on a single source of income.

Digital innovation plays a crucial role in value chain development. E-learning platforms, AI-based advisory tools, and mobile-based market information systems are developed to improve planning, decision-making, and transparency. The project also promotes registration with

platforms like NCDEX and certification agencies to ensure market access and traceability for farm products.

Component D: Project Management and Strengthening Institutions

This component of the project ensures effective project management, institutional strengthening, and technological integration. The strategy includes coordinated implementation by the Department of Agriculture, technical support from institutions like ICAR, and SAUs, and collaboration with agri-tech startups and civil society organizations.

Sub-component D1 creates a unique consortium of scientific institutions to support the project to achieve the goal of climate resilience and profitability, by partnering with the project both in financial and non-financial modes. The partnerships are intended in the areas of precision agriculture, climate resilience planning, and soil-water productivity improvements etc.

Sub-component D2 contains the digital initiatives by the project, which can be mainstreamed in the Department of Agriculture by the State Government. An Innovation Lab develops adaptive methodologies, emphasizes the role of AI, IoT, GIS, and blockchain in advancing real-time, climate-resilient farming. The project's digital ecosystem supports services such as personalized farmer advisories, smart irrigation planning, crop health monitoring, and market forecasting. Legacy platforms developed in Phase I, including DBT portals and GIS dashboards, are being upgraded. The Agriculture Data Universe aggregates real-time data from multiple sources to power a state-level Decision Support System for policymakers and farmers alike.

Sub-component D3 gives an overall idea about the robust Monitoring and Evaluation framework, which ensures accountability and learning through MIS systems, GIS dashboards, third-party evaluations, and social audits. A grievance redressal system enables transparent resolution of complaints, including those related to gender-based violence or exclusion.

Sub-component D4 focuses on capacity building across project levels. Training programs, both online and offline, are delivered through institutions like YASHADA, MANAGE, and VANAMATI. Participatory methods like Farmer Field Schools and seasonal campaigns are employed, with a strong focus on women's leadership and inclusion. Gender Action Plans aim to reduce systemic inequalities and increase women's participation in planning and implementation.

Sub-component D5 outlines the staffing and governance framework, with a Project Management Unit (PMU) supported by the implementing units at Division, District and Subdivision levels. Staff from the Agriculture Department and domain experts oversee implementation at divisional,

district, and sub-district levels. A Project Steering Committee chaired by the Chief Secretary ensures high-level coordination and problem-solving.

Financially, the project operates through a decentralized structure where funds flow from the World Bank and Government of Maharashtra to implementing units. The Direct Benefit Transfer system ensures that subsidies reach farmers' Aadhaar-linked bank accounts efficiently. Audits are conducted both internally and by the Comptroller and Auditor General (CAG), ensuring compliance with World Bank and national standards.

Procurement follows World Bank guidelines and emphasizes community-led, transparent practices through GKVS and FPOs. Participatory Village Adaptation Plans guide procurement decisions, and systems like the Maha Tender portal and STEP are used for contract management. Manuals are available in local languages to ensure accessibility and compliance.

The comprehensive approach is adopted by the project to ensure environmental and social sustainability. Central to this effort is the Environmental and Social Management Framework (ESMF), developed to identify, assess, and mitigate potential environmental and social risks and impacts associated with project activities. The framework provides a clear structure through ten standards covering issues such as labour, land, vulnerable communities, and gender. The ESMF encompasses several key tools and plans, including the Labor Management Procedures (LMP), the Tribal Peoples Planning Framework (TPPF), and a Gender-Based Violence (GBV), Sexual Exploitation and Abuse (SEA), and Sexual Harassment (SH) Mitigation Plan. To operationalize its commitment to inclusivity and equity, the project integrates a Social Management Framework (SMF) and Stakeholder Engagement Plan (SEP). The project prioritizes the inclusion of vulnerable groups such as Scheduled Castes (SC), Scheduled Tribes (ST), women, and persons with disabilities, ensuring they have equal access to information, decision-making processes, and project benefits. The Gender Action Plan (GAP) is particularly crucial in recognizing and addressing the significant role that women play in agriculture while also tackling gender disparities in access to land, technology, extension services, and income. The project also incorporates a Grievance Redressal Mechanism (GRM) to handle complaints, including those related to SEA/SH, and provides channels for timely and confidential resolution. Social audits will be conducted annually by the GKVS to ensure transparency and accountability, evaluating how benefits and assets are distributed, especially to vulnerable groups.

The EMF methodology involves environmental and legal screenings, adherence to state and national environmental laws, and implementation of the Integrated Pest and Nutrient Management Plan (IPNMP). This plan supports the safe and sustainable use of pesticides and

fertilizers, ensuring that farmers receive proper training in the application, storage, and disposal of agrochemicals.

Capacity building is a core pillar of NDKSP Phase II. Extensive training programs, exposure visits, and digital learning platforms are planned for farmers, SHGs, project staff, and other stakeholders. Collaborations with research institutions, agricultural universities, and civil society organizations will strengthen scientific and technical support. Non-financial Memorandums of Understanding (MoUs) will ensure knowledge exchange and help embed climate resilience practices into local institutions.

In conclusion, PoCRA (NDKSP Phase II) presents a comprehensive and forward-looking model for climate-resilient agriculture in Maharashtra. By integrating environmental sustainability, technological innovation, institutional capacity, and social equity, it aims to build a future-ready agricultural sector capable of thriving in the face of climate change.

Section 1

Introduction



Section 1: Introduction

1.1 Climate Change and Sectoral Context

Country Context

India's economy is growing rapidly, but the agricultural sector's share in GDP is declining. Farmers are facing numerous challenges in agriculture due to high climate variability and inadequate adaptive capacity, leading to low productivity and income. Monsoon patterns play a crucial role in the agricultural success, but inconsistent rainfall leads to significant economic insecurities. Water availability for irrigation is decreasing due to population growth, urbanization, and industrial demands.

Maharashtra faces similar challenges, with high climate variability causing reduced productivity and increased risks for farmers, particularly in regions like Vidarbha and Marathwada. The state has seen an increase in small and marginal landholdings, making it harder for farmers to cope with climate shocks.

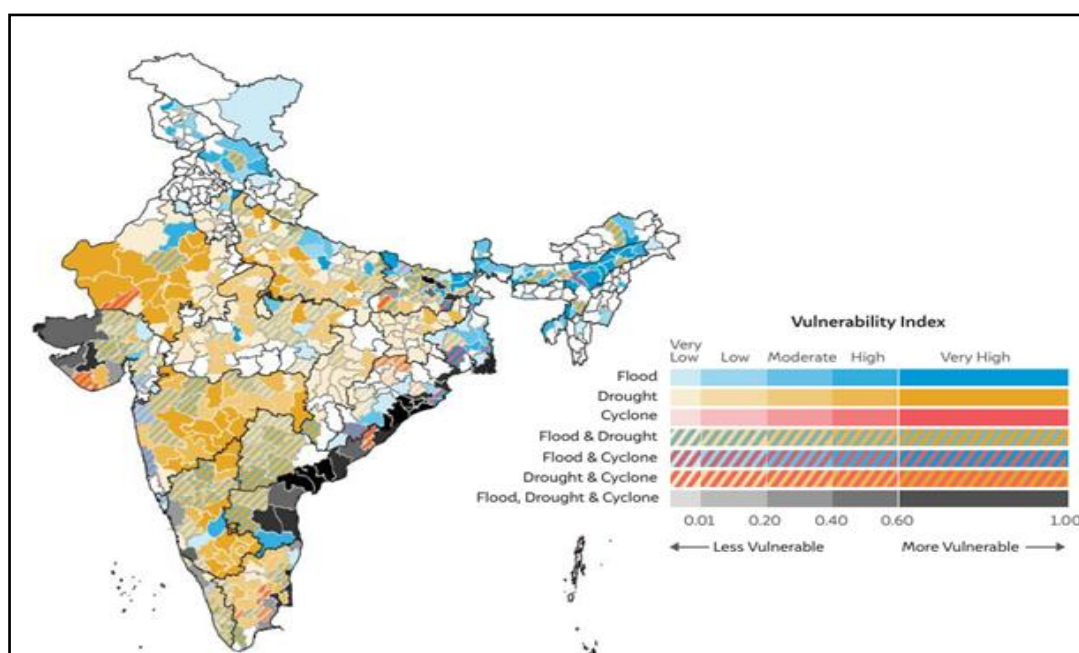
Climate change

India's climate profoundly impacts energy, water management, and food production. With over 50% of the net sown area dependent on rainfall, which contributes about 40% of total agricultural output, the stability of monsoon patterns is crucial. Variations and extreme weather events tied to the Indian monsoons often lead to floods, droughts, and cyclones, significantly affecting agriculture. India's economy is growing rapidly, but the agricultural sector's contribution to GDP is declining due to high climate variability and inadequate adaptive capacity. Farmers face significant challenges from inconsistent monsoon patterns, reduced water availability for irrigation, and lack of infrastructure.

In Maharashtra, similar issues lead to decreased productivity, increased risks for small and marginal farmers, and a high rate of farmer suicides. Climate change impacts, such as extreme weather events, exacerbate these challenges, necessitating shifts toward improved input-use efficiency and climate-resilient practices in agriculture.

A statistically significant JJAS rainfall increase was observed in the traditionally drier tehsils of Rajasthan, Gujarat, central Maharashtra, and parts of Tamil Nadu, as shown in Figure 1.

Figure 1 : Climate vulnerability index for India



Maharashtra Context

Maharashtra's agriculture sector contributes 12 percent to the state GDP (FY23), with a further 7 percent coming from the food processing industry. Growth in the agriculture sector has remained between 3.5 percent and 4.4 percent from 2004 to 2021. Nearly 50 percent of the state's total population depends on agriculture, and the sector is dominated by small and marginal farmers, with 15.28 million operational holdings with an average size of 1.34 ha. The share of rural household dependent on agriculture and agricultural casual labour in Maharashtra has decreased from 63.9 percent in 2019-20 to 55 percent in 2023-24. Similarly, the share of the agricultural sector in terms of the percentage distribution of usually working persons in rural areas of the state decreased from 75 percent to 67.5 percent during this period¹. This reduction is largely attributed to increased sectoral challenges primarily associated with the erratic climatic conditions in the state. One-third of the state's area falls within a rain-shadow region, where rainfall is low and erratic. The state is characterized by high spatial and temporal variation in rainfall, along with the predominance of rainfed agriculture. With nearly 81.8 percent of the crop area being rainfed, the monsoon is considered as a backbone of the agriculture sector. Monsoon pattern analysis revealed that over the last 40 years (1982-2022), 21 percent of talukas in the state experienced both high number of deficient as well as excessive rainfall years. Analysis of extreme rainfall events

¹ Periodic Labour Force Survey, 2019-20 and 2023-24

found that nearly 69 percent of talukas observed a rise in heavy rainfall days in the last decade (2012-2022)².

Around 77 percent of the state's cropped area is highly vulnerable to climate change impacts. According to the Maharashtra State Adaptation Action Plan on Climate Change (MSAAPCC, 2023), climate modelling for 2030, 2050, and 2070 predicts that a hotter climate and more erratic rainfall will impact agriculture, leading to lower yields and higher vulnerability to pest and fungal attacks. This will exacerbate existing challenges, including the increased frequency of extreme climate events such as droughts, hailstorms, and floods. In 2023, 66 percent of the state experienced drought. Simultaneously, some drought-prone areas faced unexpected floods due to high rainfall. These unpredictable weather shifts disproportionately affect small and marginal farmers, leading to increased risks, farm distress, and high rates of farmer suicides.

Agriculture across the State must shift to improved input-use efficiency in order to deliver productivity gains. Agriculture uses nearly 80 per cent of all groundwater withdrawals in the country and overall Water Use Efficiency (WUE) in India is only 3 US\$/m³, much lower than the global average of 21 US\$/m in 2021³. Despite the state's climatic vulnerability, area under paddy cultivation has increased, from 1.51 million hectares in 2010-11 to 1.68 million hectares in 2022-23, while the area under climate-resilient crops like millet has decreased from 5 million hectares to 2 million hectares during the same period⁴. Data from 2014-15 to 2021-22 show that the production of cereals, pulses, and oilseeds has increased in the state. However, productivity levels for most crops are still significantly below the national average. While the Maharashtra Project on Climate Resilient Agriculture (PoCRA) showcased the package of practices to build agriculture resilience, it didn't directly target paddy cultivation or crop diversification. Under PoCRA-II the focus will shift towards resource-use efficiency that by extension will target paddy cultivation and promote climate-resilient crop varieties. WUE in agriculture will need concurrent interventions in agronomic practices (tillage, mulching, planting density), a shift to micro-irrigation, and investments in mechanical measures such as contour farming, terrace farming etc. Recent research⁵ indicates that nitrogen application per unit area (ha) in India has increased from 1.56 kg in 1961 to 104.2 kg in 2021, but nitrogen use efficiency has remained low at about 33.5 per cent. Increments in Fertilizer Use Efficiency will need strategies to improve soil health and soil management, use of innovative fertilizer application techniques and adoption of improved varieties of crops.

Greenhouse Gas (GHG) emissions from the agriculture sector have been on a rise in India. India is the world's third-largest emitter of greenhouse gases (GHG). In 2020, the agriculture

² Department of Agriculture, Government of Maharashtra

³ Progress on Water Use efficiency (SDG Target 6.4), United Nations.

⁴ Economic Survey of Maharashtra 2023-24

⁵ Nitrogen Use Efficiency in Crop Production in India: Trends, Issues, and Challenges, Springer Nature, Vol 12 (2023), 32-44.

sector contributed 406 MtCO₂e of GHG emissions, accounting for 16.65 percent of the country's total emissions (including LULUCF⁶). GHG emissions from the sector have risen by 14 percent since 2000. Within agriculture, emissions are distributed as follows: 23.26 percent from agricultural soils (18.41 percent from direct N₂O and 4.85 percent from indirect N₂O), 16.68 percent from rice cultivation, 3.22 percent from manure management, and 2 percent from the management of agricultural residues in 2020⁷. In light of this, with nearly 7 percent of the Gross Cropped Area under paddy⁸, rice cultivation in Maharashtra has become a key test case to explore whether technical efficiency in crop production, through improved input use, can help reduce the emissions intensity of agriculture. A detailed climate profile for Maharashtra is annexed in Annexure 1.

1.2 Context for mainstreaming Climate Resilient Agriculture

Climate Resilient Agriculture (CRA) helps agriculture adapt to and recover from climate disturbances, reducing risks while promoting growth, food security, and Sustainable Development Goals (SDGs). It provides context-specific solutions for climate adaptation and mitigation.

1.3 Project on Climate Resilient Agriculture Phase- I

Project Overview:

In response to climate vulnerabilities, the Project on Climate Resilient Agriculture (PoCRA) was implemented in Maharashtra from 2018 to 2024. The aim of PoCRA was to develop a drought-proof and climate-resilient strategy for the agriculture sector to address the likely impacts of higher climate variability and climate change. The approach under PoCRA was to design, pilot, and roll out innovative Climate Resilient Agriculture (CRA) practices in the project areas, which constituted approximately 10 percent of the state's cropped area. These included: (i) comprehensive data-driven microplanning exercises led by the communities to develop village plans aimed at the optimal utilization of natural resources, the adoption of appropriate cropping patterns and the latest technologies, and improved access to markets; and (ii) establishing a cutting-edge digital agriculture platform for the delivery of services, including the efficient transfer of funds directly to farmers through Direct Benefit Transfers (DBT) program. The project covered approximately 10% of the state's cropped area, focusing on 16 districts, including 670 vulnerable mini-watershed clusters and 5220 villages, with special attention to 932 salinity-affected villages in the Purna river basin. The details of PoCRA-I are attached in Annexure-2.

⁶ Land Use, Land-Use Change, and Forestry.

⁷ MoEFCC (2024). India: Fourth Biennial Update Report to the United Nations Framework Convention on Climate Change. Ministry of Environment, Forest and Climate Change, Government of India.

⁸ Economic Survey of Maharashtra, 2023-24.

1.4 Project on Climate Resilient Agriculture – Phase II

The Project on Climate Resilient Agriculture (PoCRA) is also known as Nanaji Deshmukh Krishi Sanjeevani Prakash (NDKSP). PoCRA-II aims to position agriculture as a profitable vocation for smallholder farmers while reducing its variable environmental impacts in Maharashtra. Demand for adoption of climate-resilient technologies and practices helped achieve stable and increased crop yields in the previous phase of the project. To transform agriculture into a profitable and resilient productive system for smallholder farmers, there is a need to enhance the efficiency of both natural and external resources for strengthening ecosystem resilience, while ensuring an emission-competitive and profitable production system. Emissions competitiveness refers to the net reduction in GHG emissions due to improved practices in pilot projects, compared to similar cropping systems under conventional practices. The design of interventions will be guided by the creation of a data-driven decision support system.

1.5 Sustainable Development Goals and the NDKSP-II

The Sustainable Development Goals (SDGs), established by the United Nations in 2015, outline a framework for promoting sustainable and inclusive growth worldwide by 2030. Agriculture is a key driver for several SDGs and the NDKSP Phase II is designed to support sustainable and inclusive agricultural development by aligning with key Sustainable Development Goals (SDGs). The key SDGs that relate to the NDKSP are:

	NDKSP Phase II aims to eradicate poverty by enhancing farmer incomes, improving market access, and promoting sustainable agricultural practices.		NDKSP Phase II focuses on reducing inequalities by ensuring fair access to resources, markets, and support systems for marginalized farmers, including smallholders and women
	Through sustainable farming, helping farmers grow more resilient crops to secure their livelihoods and nutritious food.		The project advocates eco-friendly farming through organic methods, soil health management, and reduced chemical use, fostering sustainable and responsible agriculture.
	NDKSP promotes women's empowerment in agriculture through training, leadership, and		NDKSP supports SDG 13 by helping farmers adapt to climate change, reducing the impacts of droughts, unpredictable rainfall,

financial inclusion, ensuring equal opportunities.	and rising temperatures on their crops and livelihoods.
 SDG 6: CLEAN WATER AND SANITATION NDKSP focuses on managing water requirements through water structures, better irrigation methods, and rainwater harvesting.	 SDG 15: LIFE ON LAND NDKSP helps restore ecosystems and prevent land degradation by promoting soil conservation, agroforestry, and sustainable land management practices.
 SDG 8: DECENT WORK AND ECONOMIC GROWTH NDKSP empowers rural farmers by enhancing sustainable farming practices, supporting livelihoods, and promoting entrepreneurship through FPOs and improved market linkages.	 SDG 17: PARTNERSHIPS FOR THE GOALS NDKSP is a collaborative initiative by the Government of Maharashtra, the World Bank, local communities, FPOs, and experts, driving sustainable development through collective efforts.

Figure 2: SDGs align with the NDKSP's integrated approach to climate-resilient agriculture

1.6 Project Development Objective, Components and Results Chain

1.7 Project Development Objective (PDO):

The Project Development Objective (PDO) is ***to improve resilience and profitability of smallholder farmers across selected districts in Maharashtra.***

Innovations for resilience: In the context of the project development objective (PDO), the NDKSP Phase II proposes to position Maharashtra in a leadership role to address the emerging climate context and climate change impact. The Phase II of the project aims to make Maharashtra a leader in climate-resilient agriculture. It focuses on improving smallholder farmers' resilience and profitability by enhancing resource efficiency and reducing greenhouse gas (GHG) emissions through data-driven decision-making. The project will build on the successes of Phase I, expanding its reach to 20% of the state and leveraging a digital agriculture platform to provide tailored on-farm technical services. These services will help reduce input costs and integrate climate resilience into everyday farming practices, ultimately boosting farmers' incomes and establishing sustainable, emission-competitive production systems.

1.8 Theory of Change

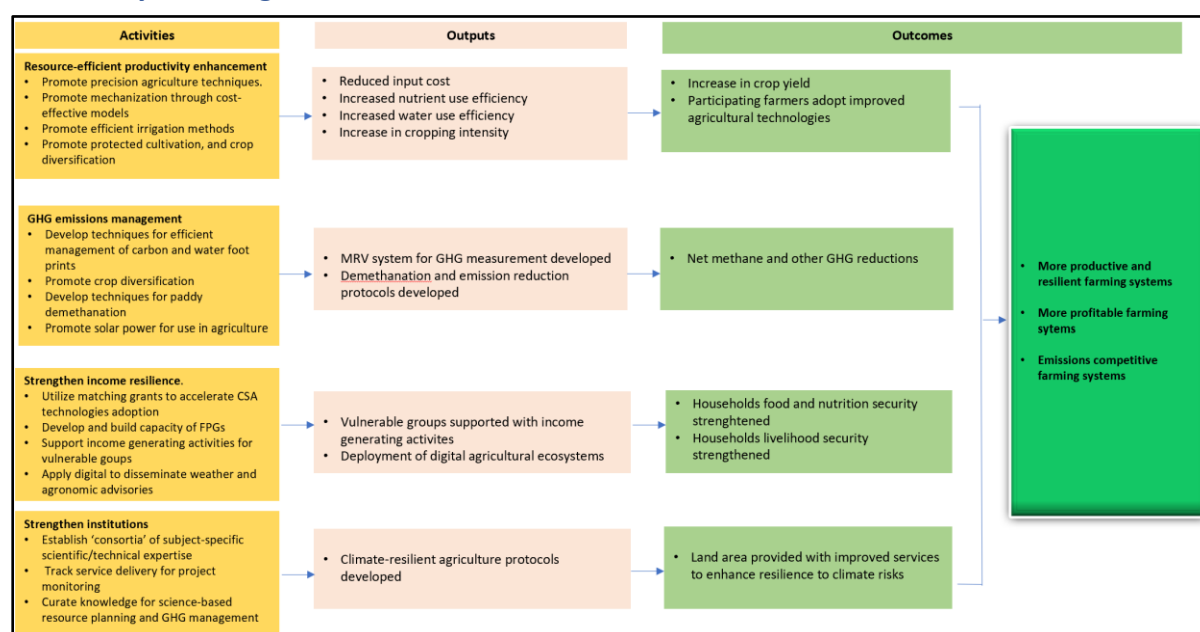


Figure 3: Result Chain of NDKSP-II

The key assumptions that underpin the project's results chain include: (i) GoM's commitment to implement the project; (ii) active leadership of the concerned government department(s) to foster an enabling environment for project implementation including the onboarding of required technical human resources, research institutes, and technical agencies; (iii) smallholders' interest in participation and that they see profit in adoption of productivity-enhancing and climate-smart practices; and (iv) enabling ecosystem for the development and implementation of data-driven digital platforms and beneficiaries interest in adoption of digital services.

1.9 Project Components

The project's components are strategically interconnected, focusing on reducing input costs and increasing crop productivity to drive profitability. By promoting efficient use of natural and external resources and adopting precision farming for emission reduction, the project aims to establish a sustainable production system that strengthens income resilience. Recognizing the critical need for stable farm earnings, the project will support interventions to enhance commodity value chains and empower farmer groups through capacity building and agribusiness expansion. Implementation will be backed by a consortium of technical experts while reinforcing the institutional framework established under PoCRA.

Component A: Resource Efficient Productivity Enhancement

Component B: Promoting Precision Agriculture for GHG Emissions Management

Component C: Improving and Building Income Resilience

Component D: Project Management and Strengthening Institutions

The description of each component is elaborated below:

1.10 Component A: Resource Efficient Productivity Enhancement

The objective of this component is to strengthen and promote a resource-efficient productivity system that is resilient to climatic variability and change. This includes interventions on both arable and non-arable land (catchment treatment, small work, all input pertaining to production). Simply increasing inputs to enhance productivity is counterproductive, as it does not address the underlying input inefficiencies exacerbated by climatic challenges; hence, this approach focuses on building resource efficiency by improving soil health, fertilizer-use efficiency (FUE), and water-use efficiency (WUE) through a series of low-carbon, climate-resilient activities tailored for each cluster of project villages through a data-driven system. (Detailed narrative on different data-driven systems through digital agriculture is outlined in component-D). By efficiently utilizing both natural and artificial/external resources, the component will establish⁹ and sustain productivity at optimal levels. This will serve as the foundation for the implementation of project activities and promote a systems approach to productivity enhancement.

The organic carbon stock in soil is critical for both maintaining ecosystem health and sustaining agricultural productivity. It is estimated that in India majority of the soils (> 50 percent) are deficient in soil organic stock¹⁰ suggesting that proper management and practices can enhance organic carbon levels to support better productivity. Soil carbon stock is critical for maintaining soil temperature¹¹ that is critical for conserving water and essential soil nutrients, ensuring their timely availability through a soil structure. The project will assess the physical characteristics of soil to identify conditions that support a productive agricultural system. This will include soil profiling in selected areas to highlight specific deficiencies and nutrient imbalance in the soil of individual land parcels. Additionally, the project will collaborate with the ICAR-National Bureau of Soil Survey and Land Use Planning (NBSS&LUP), to provide targeted advisories to project clusters, aiming to improve soil health and resource efficiency by promoting site specific nutrient management practices.

Sub-Component A1: Promoting Production Systems Efficiency - The sub-component aims to restore and enhance soil composition to optimize agriculture productivity. The support under this project will include satellite-based advisories, creation of demonstration sites for soil profiling, and appropriate fertilizer/nutrient procurement. Enhanced soil-efficiency will be delivered as a function of activities that (i) augment soil productivity and land use efficiency by investing in land preparation and compartment bunding/ land levelling to improve the physical aspects of soil-system efficiency; (ii) build organic matter through the promotion of agronomic practices that buffer temperature fluctuations to enhance biological (microbial)

⁹ 77 percent of Maharashtra's cropped area is highly vulnerable, ICAR.

¹⁰ ICAR-National Bureau of Soil Survey and Land Use Planning, 2023.

¹¹ Frey S, Lee J, Melillo JM, Six J (2013) The temperature response of soil microbial efficiency and its feedback to climate. *Nature Clim Change* 3:395–398

efficiency of the system; and (iii) restore the ecological efficiency of the landscape through efficient nutrient management that reduces emissions, while also optimizing agricultural production. This sub-component will focus on developing a resource-efficient and climate-resilient production system with respect to soil.

The project will focus on expanding the adoption of climate-resilient agricultural practices to optimize productivity by improving soil health through conservation agriculture practices and resource-use efficiency. A key component of this initiative is switching cultivation towards optimal input usage patterns. The project will implement a series of measures such as soil health assessments, analysis of soil nutrient imbalance, on-farm production of biofertilizers, and the production of organic inputs such as vermicompost, *Beej Amrut*, NADEP compost¹², and others. Additionally, the promotion of conservation agriculture practices, like no-tillage and reduced tillage, will be emphasized to enhance soil fertility and ecosystem health. Critical focus will also be placed on improving Fertilizer Use Efficiency (FUE), ensuring that inputs are utilized more effectively for sustainable crop production. To further enhance soil efficiency, significant investments will focus on expanding controlled irrigation, protected cultivation, and crop diversification, alongside revitalizing abandoned agricultural fallows through restricted irrigation. These combined efforts will help farmers transit to more efficient, resilient, low carbon and cost-effective agricultural practices. Under this sub-component, Village Soil Fertility Index (VSFI) will be developed to assess and monitor soil fertility across project villages. This index will provide a scientific foundation for soil health improvement strategies and allow for the measurement of the impact of interventions over time.

Improving soil productivity and climate resilience requires addressing soil moisture stress – a common issue in project areas due to gaps in monsoon rainfall, which are further exacerbated during drought years. To mitigate this, soil moisture operational products will be used to optimize irrigation system design and inform watershed interventions. Active microwave remote sensing satellite platforms, including Sentinel-1, Risat-1, and the upcoming NISAR¹³, will produce one-hectare scale soil moisture products with high temporal resolution. Additionally, a network of ground-based soil moisture sensors, co-located at Bowen Ratio Energy Balance (BREB) tower sites (erected in selected clusters) will provide calibrated remotely sensed data.¹⁴ A data-driven digital agriculture approach will be adopted by the project to monitor changes in soil properties in real time, offering valuable insights to guide the development of soil management plans, enhancing fertilization recommendations, and optimizing soil amendments and irrigation practices to improve overall productivity.

¹² NADEP composting is a natural process by which biomass wastes, soil wastes and animal wastes are biologically degraded and decomposed into an organic - compost

¹³ The NASA-ISRO Synthetic Aperture Radar (NISAR)

¹⁴ The Bowen Ratio helps track whether energy is used to heat the air or evaporate water, making it crucial for weather, farming, and climate science. In agriculture and water management, it helps farmers estimate water loss from plants, aiding irrigation decisions. For drought monitoring, a high Bowen Ratio (greater or equal to 1) indicates reduced evaporation, signaling dry conditions or potential drought.

Sub-Component A2: Enhancing Water-Use Efficiency - This sub-component will promote the adoption of climate- resilient practices that maintains soil temperature, minimizes soil-water evaporation and enhances water-use efficiency at the farm level. Recent studies predict that while a 1°C rise in ambient temperature will increase irrigation water demand by at least 10 percent, there is an approximately 2 percent change in various parameters present in the soil with every degree change in soil temperature. Achieving water-use efficiency and improving water productivity are critical for conserving this vital natural resource. The project seeks to reduce water dependency by aligning cropping patterns with local water availability and encouraging low-water-consuming crops. This sub-component will invest in enhancing water-use efficiency on the field to improve water security by (i) catchment area and drainage line treatment, (ii) increase groundwater recharge by recharging Open Dug Wells and Bore-wells, (iii) creating water storage structures, and (iv) enhancing efficient input-use through micro-irrigation systems.

Efficient water management is fundamental to boosting agricultural productivity, enhancing sustainability, and building resilience against climate change. Applying water at the right stage of crop growth and in the correct quantity plays a key role in improving production and productivity. This approach increases water use efficiency, thereby reducing the water footprint. The project will promote water-saving agronomic practices, including crop residue management on the farm to reduce evaporation and enhance soil moisture, mulching, and Broad Bed Furrow (BBF) systems to improve water infiltration, prevent waterlogging, and enhance drainage in rainfed areas. Intercropping will also be promoted to optimize water use by growing multiple crops with varying water requirements on the same land. Additionally, the project will deploy the Crop-Water Budget tool, developed in the first phase of PoCRA, to calculate village water balances and assess supply-side conditions. This will be used to develop strategies for demand-side management for soil and water conservation. Crop-water budgeting tools will be integrated with efficient irrigation methods (e.g., drip irrigation with farm ponds) and irrigation scheduling practices. The project will focus on participatory soil and water conservation interventions, primarily in area treatment and drainage line treatment. Area treatment interventions will include continuous contour trenches (CCT), deep CCT, compartment and graded bunding, and repair of old paddy bunds etc. Drainage line treatment will focus on loose boulder structures and water harvesting structures like farm ponds and open dug wells etc. Furthermore, the project will collaborate with the Groundwater Surveys and Development Agency (GSDA) to prepare groundwater recharge plans, complete with cost estimates, and provide technical support, including protocol development for supervising groundwater recharge work and training field staff.

The project will deploy a data-driven consortium approach to achieve the objectives of the sub-component. Soil moisture sensors will be deployed in selected clusters to support irrigation scheduling and water conservation. These integrated practices will not only reduce evaporation and improve transpiration but also contribute to building both below-ground and

above-ground carbon stocks, thus enhancing carbon sequestration and promoting sustainable production systems. Daily and weekly water estimates will be generated using thermal remote sensing products from open-source global satellite platforms (including the upcoming Indo-French TRISHNA¹⁵). Additionally, a network of BREB systems, strategically installed at selected pilot sites, will facilitate the calibration of energy balance components from existing and future thermal satellite platforms. This will help derive spatial-temporal evapotranspiration data for project villages at the desired level of granularity. Efficient groundwater use in the project villages is essential for sustaining crop supplies during periods of low rainfall that result in soil moisture stress during Kharif, as well as for supporting extended Kharif and Rabi crops. Using custom-developed water budget tool, demand-supply gaps will be assessed to target and design soil and water conservation interventions.

Sub-component A3: Technology Transformation for Climate Resilience - The project will promote technologies that integrate innovative practices, tools, and strategies aimed at mitigating climate risks, enhancing productivity, reducing GHG emissions, and improving resource efficiency, thereby fostering resilience. By leveraging scientific advancements in areas such as precision farming, resource-efficient production systems, and climate-resilient innovations, suitable package of practices and Information and Communication Technology (ICT) based products will be developed and transferred to farmers to enhance resilience against climate variability. Investment support under this sub-component will include extensive creation of IT based sub-platforms for decision support systems and precision application of technologies.

This subcomponent will leverage technology to promote systemic change for climate resilience in agriculture by (i) demonstrating and promoting on-farm technologies and climate-resilient practices for farmers to adopt, (ii) using ICT platforms to provide matching grants as incentives for the adoption of demonstrated technologies, and (iii) utilizing technology to collect and analyse dynamic data to provide advisories and digital support for timely management of climate variability. Through training, demonstrations, Farmer Field Schools (FFS), and Digital Field Schools (DFS)¹⁶, the project aims to increase awareness and adoption of these technologies among farmers. Key practices will include appropriate farm mechanization, crop-specific irrigation scheduling and agronomic techniques designed to minimize evaporation, serving as model approaches for carbon sequestration and sustainable agriculture, Integrated Crop Management (ICM)—including field preparation, climate-resilient seed selection, seed treatment, Integrated Pest Management (IPM), Integrated Nutrient Management (INM), mechanization, protective irrigation, and harvest management,

¹⁵ Thermal Infra-Red Imaging Satellite for High-resolution Natural Resource Assessment (TRISHNA) mission, a collaborative endeavor between ISRO and CNES.

¹⁶ The Digital Farmers School expands agricultural education by offering online learning platforms for farmers, overcoming the limitations of traditional Farmer Field Schools (FFS) that can only accommodate a limited number of participants and may not be accessible to female farmers due to time and distance constraints. This approach allows more farmers to attend remotely, enabling direct interaction with experts and fostering knowledge exchange and community engagement.

among others. Additionally, the project will also pilot the use of farm tools and equipment that are ergonomically tailored for women to improve their productivity. Under this sub-component, the conservation and multiplication of climate-resilient seeds will be promoted by providing monetary incentives for production of foundation and certified seeds and by offering a buy-back program. Demand-based matching grant provided through DBT for the farmers to adopt on-farm technologies will be a major investment under this sub-component.

1.11 Component B: Promoting Precision Agriculture for GHG Emissions Management

Promoting precision agriculture will focus on reducing greenhouse gas emissions, sequestering carbon, cutting methane emissions, and maintaining economically viable crop yields. Considering the varying carbon and water footprints of different crops and crop rotations, the project will introduce crop-specific moisture conservation techniques, develop agroforestry modules, and implement new methods to enhance soil organic carbon.

Sub-component B1: Balancing Methane and Paddy Production - Through this sub-component, the project will deploy precision agriculture techniques for promoting low-methane paddy¹⁷, as substantial methane is generated when organic material in flooded rice fields decompose. With the adoption of greater water-use efficiency (WUE) and fertilizer-use efficiency (FUE) practices, the project will also identify and monitor those practices which also help reduce GHG emissions. Suitable pilots will be conducted to ascertain the combination of practices that will generate most competitive methane emission. Investment under this sub-component will include extensive mobilization support, equipment for measurement, reporting and verification (MRV) for CH₄, N₂O emission and Bowen towers and sensors will be installed for evaluation of moisture and evapotranspiration.

Given the diverse agro-climatic zones in the state, the project will expand and deepen technology interventions to sustain crop productivity while keeping emissions competitive, including precision agriculture techniques for promoting low-methane paddy such as Direct Seeded Rice (DSR), System of Rice Intensification (SRI), Alternate Wet and Dry (AWD), water-straw-fertilizer co-management, and Zero Tillage for promoting low-methane paddy. Activities will involve pilot studies and demonstrations of various technologies for methane reduction from paddy, combining innovations in digital agriculture, crop and microbial genetics, and nutrient use efficiency to reduce emissions and enhance soil carbon sequestration. The project will collaborate with ICAR-Central Rice Research Institute (CRRI), Cuttack, a leading research institute, to assess methane emissions from rice fields in the project area, share protocols for low-methane rice production technologies, and build the capacity of extension agents and farmers in adopting low methane rice production. The project will provide targeted extension services to equip farmers with the knowledge and

¹⁷ For example, DSR-Direct Seeding Rice

skills needed to transition from traditional flood irrigation and excessive fertilizer use to more sustainable practices. Additionally, it will promote strong collective action among farmers, ensuring coordinated efforts in water management and sustainable rice production. By fostering collaboration, the project will help farmers adopt best practices that enhance productivity and reduce emissions while preserving natural resources. Additionally, participatory research will explore alternative cereal crops for farmers willing to replace second-season paddy, contributing to methane emission reduction.

Building on the success of PoCRA in reducing agricultural emissions, the project will develop a digital measurement, reporting and verification (MRV) system to measure, report, and verify paddy GHG emissions and carbon sequestration in selected villages while supporting carbon trading pilots that will facilitate smallholders' access to voluntary carbon markets.¹⁸ A monitoring network combining remote sensing and in-situ measurements will be established. The project will leverage digital MRV solutions, integrating innovative remote sensing technology and big data analytics—such as machine learning and artificial intelligence—to assess methane emissions at scale and analyse key factors influencing rice GHG emissions, including production areas, crop rotation, residue management, and water practices. In-situ monitoring will capture direct emissions data alongside variables like climate, soil types, fertilizer use, and agronomic practices. The MRV system will generate insights on best practices for monitoring rice climate mitigation efforts and evaluating their impact on GHG emissions, with a focus on methane reduction and smallholders' access to voluntary carbon markets.

Sub-component B2: Reduced Energy Consumption in Agriculture: To reduce emissions while building climate resilience, the project will support the transition from conventional to non-conventional energy sources for powering agricultural pumps in the State. The sub-component will make investments in promoting (i) use of solar technologies in water pumping, farm operations, and food storage, and (ii) generation and use of bioenergy from agriculture biomass. This will not only ensure proper management of agricultural waste but also generate electricity.

This sub-component focuses on generating alternative energy while promoting reduced energy consumption through precision farming methods. The project will provide support for portable solar-powered irrigation solutions for farmers, aligning with government schemes in the project area. Additionally, pilot interventions will be created to provide insights for expanding the initiative across the state. On the other hand, to promote bioenergy, the project will support biomass briquettes/pallets and biochar manufacturing, as well as

¹⁸ The current methods for reporting GHG emissions and validating or verifying emission reductions are often costly, error-prone, and time-consuming, relying heavily on manual processes and in-person surveys. However, digital MRV technologies offer a more cost-effective solution by streamlining data collection, processing, and quality control. In this project, technologies such as smart sensors, satellites, drones, cloud computing, artificial intelligence, the Internet of Things, and blockchain encryption will be leveraged to enhance the efficiency and accuracy of monitoring, reporting, and verification processes.

biomass-based cogeneration projects. These initiatives aim to improve soil fertility, carbon sequestration through biochar, reduce stubble/trash burning by utilizing surplus agricultural residue while providing an additional source of income to farmers or Farmer Producer Organizations (FPOs) through the sale and/or processing of surplus agriculture residue.

To efficiently manage energy consumption and reduce emissions, the project will establish strategic partnerships with leading research institutes to design crop-specific packages of practices with the goal to effectively manage moisture and temperature variability, reduce carbon and water footprints, and sustain optimal crop productivity. Additionally, the project will develop web and mobile app-based tools to monitor water-use efficiency thereby optimizing energy consumption. The selection and deployment of renewable technologies and their locations will be determined jointly by beneficiaries and consortia partners. However, the project will conduct feasibility assessments to ensure the suitability and effectiveness of the selected technologies.

Sub-component B3: Promoting Crop Diversification to Enhance Carbon Sequestration - There are three key cropping systems followed in Maharashtra; (a) pulses, (b) high value commodities with regional and global demand i.e. cotton, soyabean and sugarcane, (c) high value horticulture including grapes for wine industries. The state agriculture is well diversified, but high climatic variability makes it vulnerable. This sub-component will focus on climate safeguard measures including identifying and supporting fallow lands and areas currently under traditional no-till zones for crop diversification initiatives. In addition, farmers will be encouraged to adopt millets and tasar silk - crops well-suited to specific regions of the State. To promote this transition, the project will invest in pilot programs aimed at demonstrating the benefits of millet cultivation, planting material (carbon sink), catchment treatment, and robust satellite-based monitoring systems. It is estimated that 10-20 percent of the target area will be successfully transitioned to diversified cropping systems under this initiative. Beneficiaries selected for crop diversification and pilot programs will be incentivized, provided that any significant losses are appropriately compensated. The crop cycles will be established in advance, ensuring that the project facilitates full recovery of the harvest.

Carbon sequestration through plantation activities is a crucial strategy for mitigating climate change by capturing and storing atmospheric carbon dioxide (CO₂). This project will enhance carbon sequestration by promoting a variety of plantation practices, including agroforestry, bamboo cultivation, horticulture, along with protected cultivation. Lessons from POCRA indicate that these practices not only capture carbon but also contribute to biodiversity, improve soil health, and sustain local livelihoods. In the drought-prone regions of Marathwada and Vidarbha, agroforestry will be prioritized to build resilience in these areas. The project will also focus on promoting bamboo plantations, particularly on farmers' fields, community land, and arable wastelands. Bamboo will be planted along river banks, farm bunds, and other similar areas. In addition, the project will provide training on bamboo

cultivation and processing to help farmers maximize its potential. Similarly, horticulture will be promoted through subsidies and training, encouraging farmers to grow selected fruits. The project will target vulnerable groups, such as marginal and small-scale farmers, women, tribal farmers, and farmers with physical disabilities, to engage in the cultivation of horticultural crops, agroforestry, and bamboo. Cultivating along contour lines on sloped land will help reduce soil erosion and water runoff, while also increasing soil organic carbon through the decay of crop residue. This practice will lower CO₂ emissions and open up new areas for irrigation, further enhancing the sustainability and productivity of the land.

1.12 Component C: Improving and Building Income Resilience

This component aims to enhance the income resilience of farmers by promoting agri-business models and supporting related income-generating activities, helping to stabilize incomes in the face of climate-related shocks such as droughts, floods, or extreme weather events. The component will design activities that support farmers, especially female farmers and farmers' groups, in adopting climate-resilient practices through a participatory approach in selected locations, encouraging farmers' interest groups to identify, access, and implement suitable practices in alignment with ongoing government initiatives.

To facilitate this, the project will utilize matching grants, based on government modalities, to incentivize eligible farmers to accelerate the adoption of demonstrated technologies and agronomic practices. Substantial investments will be made to develop Farmer Interest Groups (FIGs), formulate packages of practices, and facilitate matching grants for the adoption of piloted interventions.

The strong presence of Farmer Producer Organizations (FPOs) in Maharashtra, including exclusively women-led and women-owned groups offers the project an opportunity to shift farming practices towards promoting optimal input usage patterns using a participatory approach. The project will also equip farmer members of FPO as service providers of knowledge and resources needed to transition from intensive input usage to low-risk and optimal input use, thereby sustaining productivity and returns per unit of land while simultaneously reducing emissions. During this phase, the project intends to pilot few innovations as per the local resources, for example, pisciculture through available farm ponds.

Sub-Component C1: Strengthening Agribusiness Value Chains - This sub-component aims to facilitate the efficient operation of agricultural markets—including input, output, and financial markets—while improving farmers' access to these markets. To this end, the project will support and strengthen a variety of market services, including ones offered via digital platforms. Targeted services, identified through a demand-driven approach, include those which lend support to value chain transactions, promote food safety and traceability, and

strengthen the regulation of agribusiness using innovative practices and tools. The goal is to improve competitiveness of the smallholder by enhancing access to demand driven value chains by leveraging a participatory approach. Investment under this sub-component will be extended at two levels – at the farm gate level, creating small storage assets, and agrologists support.

Sub-Component C2: Agribusiness Growth Centers - Going beyond the farmgate, the project will also concurrently drive interventions in the post-harvest space. This may entail agribusiness interventions such as supporting farmer groups with on-site processing machines and processing centers or creating institutional support by networking with the Marketing Board to drive convergence for post-harvest facilities. Agribusiness growth centers may also be established to address input supply and post-harvest management issues by ensuring timely availability of seeds and other inputs, aggregating produce, perform processing activities, and develop market linkages. The project will finance (i) infrastructure, equipment, advisory, and operating costs up to a predefined limit; (ii) mobilization of farmers or farmer collectives and capacity building; (iii) support for professional management of the Agribusiness growth centers; (iv) value chain and market assessment studies; and (v) capacity building of project staff on preparing bankable business plans for micro-enterprises.

Acknowledging the transformative impact of Artificial Intelligence (AI) on agribusinesses in the near future, PoCRA-II will prioritize capacity building for farmer groups to prepare them for the AI-driven new normal. The project will equip small farmers with the necessary skills and support to access and utilize a wide range of AI-powered services, including commodity-specific input advisories, embedded finance solutions, weather and agronomic insights, smart storage, food traceability, market linkages, and logistics optimization. These efforts aim to foster smart, resilient, sustainable, and profitable agribusinesses at the grassroots level. The project will comply with all World Bank guidelines governing AI.

Sub-Component C3: Promotion of Agri-entrepreneurship - Apart from the focus on groups, the project will also support interventions to foster development of individual agri-entrepreneurship by supporting allied activities such as small ruminants, fisheries, sericulture etc. which provide regular cashflow to the farmers and importantly, reduce their dependence on a single source of income. Support to various activities and interventions under this sub-component will necessarily build on the state-of-art digital agriculture platform already established under the aegis of Phase 1 of the project.

1.13 Component D: Project Management and Strengthening Institutions

PoCRA's robust institutional framework for implementation will be broadened and deepened to achieve a higher level of resilience under PoCRA-II. Institutional arrangements involving government agencies, private sector including AgTech startups and community institutions will contribute requisite level of administrative, technical and advisory support to the project. The Department of Agriculture, together with its machinery at district, block and village levels (including Gram Krishi Vikas Samitis) will drive the implementation at the grassroots. Backed by a Monitoring and Evaluation (M&E) Agency, technical expertise in emissions management, crop-suitability, hydrology, soil profiling, market dynamics, trading, agribusiness will be drawn from the 'consortia' partners to handhold effective management through the course of the project. A detailed institutional framework has been developed in Annex -2. In addition, this component will implement the digital agriculture and ICT domain related activities required for the project.

Sub-Component D1: Consortia of Scientific and Technical Expertise - In addition to PoCRA's initiative in engaging researchers/scientists from leading institutes to support project implementation with desired science/technology, PoCRA-II will take a further step in creating a 'consortia' of subject-specific scientific/technical expertise with clearly defined inputs to assist implementation on: (a) improving soil use efficiency related to soil quality, soil nutrient availability and nutrient uptake potential as effective reference points for improvement of crop productivity in varying soil types; (b) determining WUE and Nutrient Use Efficiency in agricultural production systems as the boundary conditions of Soil Use Efficiency; (c) choice of efficient irrigation methods and irrigation scheduling related to improved productivity with reduced integrated GHG emissions, and (d) develop resilience protocols towards adaptive arrangement of emerging environmental impacts to support adoption of improved production systems. All support under this sub-component will be provided through TA with different consortia partners.

The consortia approach to addressing cross-cutting issues will generate significant data and methodologies on resource-efficient production protocols. It will enhance the capacity of the State Agriculture Department to implement precision agriculture for specific crops. An Innovation Lab focused on efficiency issues related to soil, water, fertilizers, and nutrients will strengthen the department's capacity in climate-resilient agriculture. The data generated by the consortium of partners, through the use of cutting-edge technology and scientific research, will directly inform the Village Adaptation Plans (VAPs) developed by communities in a participatory and well-informed process, prioritizing and customizing practices. The ongoing learning from these partnerships will be reflected at the grassroots level through dynamic VAPs, ensuring real-time, need-based adoption of resilient practices.

Sub-Component D2: Digital Services for Climate Resilient Agriculture - The project aims to enhance resilience, productivity, and profitability for smallholder farmers during PoCRA-II implementation. The project will leverage existing datasets and innovative applications, ensuring the continuance of current platforms while introducing new ones to integrate precision agriculture, emission management, and real-time advisory services. These interventions will provide actionable insights, enabling farmers and government agencies to make informed decisions and mitigate climate impacts. The real-time data driven digital advisory services will also benefit female farmers with decision-making at the farm level. Key digital interventions proposed include standardizing and enhancing the applications built during PoCRA, and addition of new applications such as Interactive Digital Soil Mapping Platform, Precision Agriculture Tools, Climate-resilient Advisory Services, Emission Management Systems, Integrated Farm Management Systems, and Market Linkage Platforms.

Aligned with the "digital by default" principle, PoCRA-II will integrate next-generation digital services to address the dynamic needs of climate-resilient agriculture. Building on the success of Phase I, which established a comprehensive digital ecosystem and applications, Phase II will embed emerging technologies such as Generative AI, blockchain, and machine learning across its interventions. These technologies will drive evidence-based decision-making, real-time monitoring, and proactive service delivery. The Agriculture Data Exchange setup under the project will serve as the central repository for agriculture and allied departments of the state, harmonizing diverse datasets to enable predictive analytics, optimize resource use, and provide precision farming insights. These systems will deliver actionable, hyper-localized advisories to farmers, ranging from irrigation schedules to pest and disease management, ensuring timely and personalized support. By leveraging the India Enterprise Architecture (IndEA) principles, the digital platforms developed will ensure scalability, efficiency, and inclusivity. To further amplify these capabilities, the project will integrate Internet of Things (IoT) sensors and satellite imagery for continuous monitoring of crop health and soil conditions. AI-driven predictive analytics will forecast crop yields and market trends, empowering farmers with actionable intelligence for optimized decision-making. Additionally, blockchain based traceability platform will be developed to ensure transparent supply chains, enhancing market access and building consumer trust in the quality and origin of agricultural products.

The objective of this sub-component is to integrate AI into all resilient agriculture implementations, starting from pre-production to post-harvest management. This will be based on strategic partnerships with national and global experts who have domain experience across both AI and CRA. The AI model will be trained using large-scale data and evidence to develop the required algorithms to support: (i) AI-Driven Decision Support Systems: Utilizing AI and predictive analytics to generate actionable insights for farmers, such as irrigation schedules, pest and disease management, and crop yield forecasts. By leveraging real-time

data from IoT sensors, satellite imagery, and historical datasets, the system will provide personalized advisories, enabling evidence-based decision-making and climate adaptation. (ii) Agriculture Data Universe and Predictive Analytics: Centralizing diverse datasets through the Agriculture Data Universe to facilitate predictive modelling and analytics. This integrated data platform will support precision farming, optimize resource usage, and enable scenario analysis for climate risk management. (iii) Generative AI for Farmer Engagement: Implementing generative AI tools to enhance farmer engagement through conversational AI and customized educational content. This includes AI-powered chatbots with regional language support, delivering timely and relevant information on farming techniques, weather forecasts, and market trends. (iv) Resilient Agriculture with IoT and Remote Sensing: Integrating IoT sensors and remote sensing technologies for continuous monitoring of soil health, crop growth, and environmental conditions. This data-driven approach will enable real-time interventions, optimize input usage, and improve productivity.

Sub-Component D3: Project Monitoring, Evaluation, and Learning - This component will further cover the overall project Monitoring and Impact Evaluation (M&IE) and reporting activities, including the project impact evaluation. The results-based M&E system aims at measuring progress towards the PDO to allow for timely course corrections and evidence-based decision-making.

Sub-Component D4: Capacity Enhancement

The Capacity building initiatives based on Capacity Enhancement Need Assessment (CENA), recommendations are described under this sub-component. Also, training patterners and training areas are mentioned under this component.

Sub-Component D5: Project Management

This sub-component contains human resource management for project implementation at various stages. Finally, under this component, the PMU will implement all activities related to communication, public awareness and outreach, including setting up and maintaining a comprehensive project website that also accommodates an open space for lodging stakeholders' complaints as part of the project's Grievance Redress Mechanism (GRM). All project incremental operating costs are covered under this component.

A component-wise indicative and illustrative list of activities are listed in table 1.

Table 1: An indicative component wise list of activities

Component	Sub-Component	Activities	Sub-Activities
A. Resource Production System Efficiency			
	A1. Promoting Production Systems Efficiency		
		A1.1 Enhancing Soil/ Land Efficiency	
			Preparation of Digital soil maps and for advisories to the farmers
		A.1.2. Soil health assessment	
			Preparation of village fertility index
		A.1.3. On - farm production of biofertilizers	
			Establishment of on-farm production of biofertilizer / biopesticide/Bio-agents
		A.1.4 Production of organic inputs	
			Preparation of Vermicompost/ NADEP compost/Bijamrut/Jeevamrut/Dashparni extract /Bio dynamic compost
		A.1.5 Conservation agriculture	
			Promoting No tillage/ reduced tillage
		A.1.6 Enhancing FUE	
			Assessment of Fertilizer Use Efficiency (FUE) for optimal input advisories
	A2. Enhancing Water Use Efficiency		
		A2.1 Catchment Treatment	
		Area treatment	
			Continuous Contour trenches
			Deep Continuous Contour trenches
			Compartment Bunding
			Graded Bunding
			Terracing
			Paddy Bunding
			Repair of Old Paddy Bunds
		Drainage line treatment	
			Loose boulder structures
			Gabion structures
			Earthen Nala Bunds
			Cement Nala Bunds
		Water harvesting structures	
			Individual Farm Pond
			Group Farm Pond
			Open Dug Well
			New Bodi Structure
		Recharging of Groundwater	

Component	Sub-Component	Activities	Sub-Activities
			Recharge of Open dug well
			Recharge shaft
			Recharge trench with recharge shaft
			Rejuvenation of Water harvesting structures
			Cement Nala Bund (CNB) Rejuvenation
			<i>Bodi</i> structure
			Rejuvenation of water bodies
		A2.2 Reducing water-footprint through water use efficiency	
			Technological Advancements for WUE (Assessment through Bowen Tower)
			Micro-Irrigation
			Drip
			Sprinkler
			Water pumps (SC / ST)
			Water pipes (SC / ST)
	A3. Technology Transformation for climate resilience		
		A.3.1 Climate Resilient Technologies for Field crops	
			FFS for field crops
		A.3.2 Climate Resilient Technologies for Fruit crops	
			FFS for fruit crops
		A.3.3 Conservation and multiplication of climate resilient seeds	
			Seed Production of climate resilient varieties
			Promotion of indigenous/local seed varieties
			Incentive for village seed production programme
			Support to Taluka seed farm
B. Promoting Precision Agriculture for GHG Emissions Management			
	B1	Balancing Methane and Paddy Production:	
			Promotion of improved rice cultivation methods (SRI/AWD/DSR/SRT) (individual)
	B2	Reduced Energy Consumption in Agriculture	
		B2.1	Promotion of alteranate energy in agriculture
	B3	Promoting Crop Diversification to Enhance Carbon Sequestration	
		B3.1	Agroforestry
		B3.2	Bamboo plantation
			Peripheral and Boundary Plantation
		B3.3	Horticulture Plantation
		B3.4	Protected cultivation
			Shed net house (GI/MS pipes)
			Polyhouse (open vent)
			Planting material in Polyhouse/ shed net house-Vegetable/ flowers

Component	Sub-Component	Activities	Sub-Activities
			Planting material in Polyhouse-flowers
		B.3.5 Enhancing millet production	
C. Improving and Building and Income Resilience			
	C1 Strengthening Agribusiness Value Chains		
		C1.1 FPOs as a vehicle of transformation	
			Facilitation of FPCs for establishment of agri-business
			Capacity building of CMRC, CLF, and SHGs
			Facilitation of PESA Gram Sabhas/ Gram Panchayats
			Establishment of value addition units
			Promotion of Farm Machinery Banks (Women Self Help Group)
			Allied activities for FPOs
	C2 Agribusiness Growth Centers		
		C.2.1 Creation of storage infrastructure	
		C2.2 Agro Input and post-harvest Service Centers	
		C2.3 Value Chain and Market assessment studies	
		C2.4 Capacity Building for agribusiness planning and executions	
	C3 Promotion of agri-entrepreneurship (Individual)		
			Goat Rearing (Landless Families)
			Backyard Poultry (Landless families)
			Inland Fisheries
			Sericulture
			Apiculture
D. Project Management and Strengthening Institutions			
	D1 Consortia of Scientific and Technical Expertise		
		D1.1 Consortia Approach	
			MoUs with National and International Institutions
			Analytical studies pertaining to climate resilience
			Internships
			Agri-tech
	D2 Digital Services for Climate Resilient Agriculture		
			Single Window interface for Farmer App
			Management and Integration System (MIS-resources)
			AI driven advisory (Use drone, IoT and Chatbot)
			Cloud services and software
			MIS Hardware (Mobile/ Computer/ apps/ Tab/Laptops)
			AMCs/ CMCs
	D3 Project Monitoring, Evaluation and Learning		
			D.3 Monitoring, Evaluation, and Learning
	D4 Capacity Enhancement		
		D.4.1 Capacity Enhancement Need Assessment (CENA)	
		D4.2 Training	

Component	Sub-Component	Activities	Sub-Activities
			Project officials - International
			Project officials-Within country
			Project Field Staff
			Cluster Assistant (Induction training)
			Other stakeholders
			GKVS,Krushitai & Volunteers- Within District
			Farmers - within district
		D.4.3 Workshops/Seminar/Conference	
			District level (officials, FPOs, Sarpanch etc)
			State level (officials, Scientists, Sarpanch, FPOs etc.)
		D.4.4 Exposure visits	
			Farmers- Within District
			Farmers- Within state
			Farmers- Inter state
	D5 Human Resource Management		

The list of activities is indicative and may include the climate resilient intervention recommended by the Project Steering Committee, provided the implementation mechanism, cost-norms, and pattern of assistance are aligned to the guiding principles of the project.

1.14 Project's Core Guiding Principles

The project planning relies heavily on the microplanning exercise, which will not only identify the suitable resources, but also will align with the farming system requirements, farmer requirements, as well as the village / cluster level requirements.

The key guiding principles under the project are:

- A. Assessment of vulnerabilities of the mini-watersheds in given project districts to prioritize the selection of villages for project interventions
- B. Participatory micro-planning approach to evolve the Village Adaptation Plan aligned to project objectives
- C. Demand-driven approach for selection and implementation of climate resilient interventions
- D. Planning for natural resource management based on water balance assessment of the villages aligned to the hydrological boundaries.
- E. Interventions implemented on saturation mode, by not keeping any maximum limit of financial assistance for individual farmer
- F. End to end Direct Benefit Transfer (DBT)-based intervention financing
- G. Digital by default- Digital architecture as an integral part of project life cycle, from planning to impact assessment
- H. Land-acquisition for any of the project intervention is not supported

1.15 Geographical Coverage: guiding principle / basis of selection of Villages

The Government of Maharashtra resolved to implement the second phase of the project in 21 districts of Maharashtra vide Government Resolution dated: June 30, 2023. The project districts include 11 districts in Vidarbha, 8 from Marathwada and 2 districts of North Maharashtra

Vidarbha region: Akola, Amravati, Buldhana, Washim, Yavatmal, Nagpur, Bhandara, Gondia, Chandrapur, Gadchiroli, and Wardha

Marathwada region: Chatrapati Sambhaji Nagar, Beed, Jalna, Latur, Dharashiv, Nanded, Parbhani, and Hingoli

North Maharashtra region: Nashik and Jalgaon

The most vulnerable village clusters in each block from the selected districts are chosen for implementation of the project interventions. The village selection process and the cluster details are described later in the report.

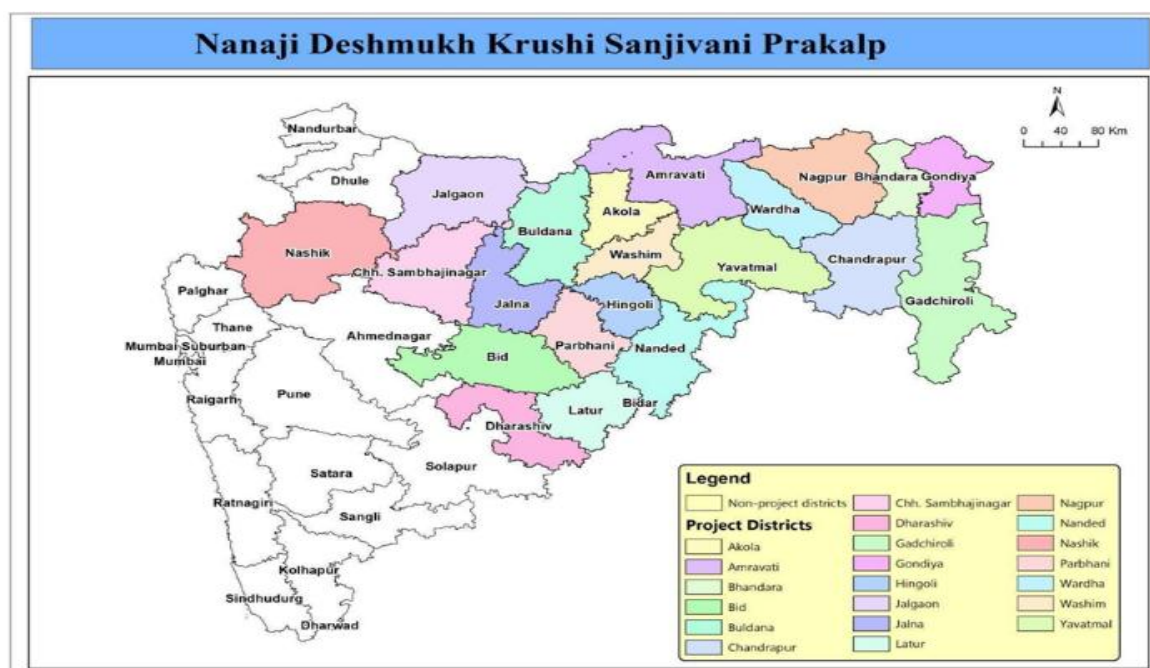


Figure 4: The NDKSP Phase II project districts in Maharashtra

Guiding principle for village selection

The project has followed a scientific assessment-based approach to arrive on selected project area, to ensure the core pillars of the NDKSP-II align with the climatically vulnerable areas in the State, as well as the proposed interventions fetch maximum dividends to the farmers, communities and villages.

The project will follow a mini-watershed, cluster approach for project execution and implementation of proposed interventions. The finalised parameters for geographical area selection have been derived from the indicators mentioned in the ICAR-CRIDA report namely, Risk and Vulnerability Assessment of Indian Agriculture to Climate Change, 2019 with relevant modifications. The Committee indicated that the parameters approved may be applied in the 21 districts to identify approximately 6800 villages. It was also decided that villages included in the first phase of the project shall not be part of the second phase. As per guiding principles of Village Selection Committee, the primary objective of vulnerability assessment is to compute a vulnerability index for clusters and select the most vulnerable.

The key pillars for selection of project areas are based on (i) Vulnerability parameters and assessment for identification of most vulnerable areas, and (ii) Clustering opportunities for collective agriculture practices to achieve project outcomes.

Village Selection Committee

A Village Selection Committee, chaired by the Additional Chief Secretary (Agriculture), was formed to select villages for the project's second phase. The committee established scientific criteria for village selection and decided on a mini-watershed cluster approach for execution. Using modified parameters from the ICAR-CRIDA report, "Risk and Vulnerability Assessment of Indian Agriculture to Climate Change, 2019," they aim to identify 7201 villages across 21 districts. Villages from the first project phase will be excluded. The main goal is to compute a vulnerability index for clusters and select the most vulnerable.

Village Selection Criteria

Villages in the project districts are grouped into clusters based on mini watersheds identified using GIS. This ensures effective treatment from ridge to valley. Clusters are prioritized by the 'Overall Sensitivity Index', as determined by the Village Selection Committee. The detailed selection criteria and other parameters considered are included in annexure 3.

The list of clusters and the villages contained therein was approved by the GoM vide Government Resolution dated 14th October 2024. District-wise list of villages for project interventions [\(link\)](#)

District	No. of Taluka	Total No. of villages	No. of villages in Ph-II
Akola	7	986	149
Amravati	13	1997	454
Chh. Sambhaji Nagar	9	1356	296
Bhandara	7	864	291
Beed	11	1368	401
Buldhana	13	1444	310
Chandrapur	15	1792	561
Gadchiroli	12	1675	532
Gondia	8	942	293
Hingoli	5	711	148
Jalgaon	15	1513	319
Jalna	8	967	177
Latur	10	948	216
Nagpur	13	1859	563
Nanded	16	1603	375
Nashik	15	1922	674
Dharashiv	8	733	138
Parbhani	9	843	173
Wardha	8	1376	383
Washim	6	789	189
Yavatmal	16	2137	559
Total	224	27825	7201

Table 2: Number of project villages in the project districts

1.16 Project Beneficiaries:

Based on the scope of the project and technical requirements of the project activities, the project will have a number of beneficiaries from different categories, i.e., farmers, landless families, farmers groups, Self Help Groups, Farmer Producer Companies, etc., The inclusion of the beneficiaries for the purpose of project assistance will be based on the existing guidelines of the Department of Agriculture and Allied Departments with the customised project approach approved by the State Government. The project will support those individual beneficiary farmers who are holding less than 5 hectares of land (except in seed production intervention where there is no limit on land holding). Besides, the project will also directly benefit public sector institutions and agencies involved in agricultural research, extension, and technical advisory services. Also, the State Agriculture Universities (SAU),

various line departments, and other implementing units at district and sub district level will also benefit through a strategic partnership with the project for mainstreaming climate resilience in agriculture.

The following guiding principles will be followed for inclusion of farmer beneficiaries and for supporting the climate resilient interventions.

- The project will ensure prioritization of the socially deprived classes and women farmers such as ST/SC farmers, women farmers, physically challenged persons, marginal farmers, small farmers
- Prioritization of beneficiaries and interventions (in line with the framed activities) will be built into the IT framework of the project to avoid discretion at the decision-making units at village level.

Guiding Principles for Planning, Activity Prioritization, Sequencing and Beneficiaries Targeting

The following table indicates the overarching principles for project planning, activity prioritization, beneficiary targeting, and sequencing

Table 3: Key guiding principles and process

Guiding Principles	Process
Microplanning as an integral part of Village Adaptation Plan (VAP) for each project village	The village-level project planning will be executed with the active engagement of both the community and institutional stakeholders. The participatory approach will facilitate the mapping of resources, identification of issues, and constraints, while also potential activities for individual benefits, Natural Resource Management (NRM) interventions, Farmer Producer Companies (FPC), and Self-Help Group (SHG) support. Care would be taken to involve all the sections of the stakeholders as well as the vulnerable sections of the society. The village adaptation plans will be formally approved through a resolution by the Gram Sabha. These plans will encompass individual benefit activity plans, NRM work plans, FPO and SHG strategies, environmental management plans, and initiatives for the adoption of climate-resilient technologies. Throughout the planning phase, the project will provide comprehensive support to the community, including capacity building, technical expertise, and IT solutions, to ensure effective implementation and long-term sustainability.

Guiding Principles	Process
Activities determination	The activities to be incorporated in the micro plan will: - be aligned to the PDO - follow participatory planning process principles - adhere to the watershed treatment principles - indicate benefit sharing mechanism - follow inclusive criteria
Individual beneficiary targeting	The farmers in the selected villages and holding land upto 5 ha, as well as the landless families will be targeted for need based climate resilient interventions. The Gram Krushi Vikas Samitis (GKVS) will be empowered to approve the applications of the farmers as per project guidelines The following categories will be given priority for project activities targeted to benefit individual farmers: • ST/SC farmers • Women farmers • Disabled farmers • Marginal and small land holders • Other farmers
FPCs/SHGs	All existing FIGs and SHGs in the project villages , and the existing FPCs in the project districts (though outside of project clusters) would be encouraged to participate in the project to take forward the project objectives of enhancing farmers’ profitability through a collective approach and value addition
Pattern of assistance	Project assistance will be provided for comprehensive watershed treatment, technology adoption for climate resilient agriculture production systems and strengthening of agri-business value chain and agri. entrepreneurships. The interventions meant to natural resource management activities would be fully funded by the project while those benefiting individual farmers or FPOs would be given assistance in the form of back ended subsidy through Direct Benefit Transfer mode. The pattern of assistance for interventions will be largely aligned to various schemes of the central and state government.
Sequencing	Ridge to valley approach would be followed while treating the watersheds. The project activities would be sequenced in a manner so as to fully harness the potential of infrastructure already created as well as keeping in mind the capacities of the target groups and available opportunities.

1.17 Project Approach and Strategy:

The proposed project approach is conceptualized around building resilience in the farming systems vulnerable to the climate variabilities. Salient features of the project are mentioned below.

- The project focuses on resource conservation through natural resource management systems and development of decision support systems for the community by using advanced technologies.
- The project creates an enabling system for the farmers to adopt the climate resilient technologies for production of the commodities suited to the agro-climatic zones with the localized approach considering the spread of the mini-watersheds. The recommended package of practices for improving resilience at the farm level will be further boosted by the real-time farm level advisories which will be generated in consortium mode, considering the local area weather forecast, cropping pattern, crop condition, water balance, and soil health prevailing at the level of the revenue circles, and or villages.
- The resilience package in the project would include diversified cropping systems, agroforestry, horticulture, afforestation, small livestock systems, sustainable agri and allied enterprises.
- Resilience will be enhanced by promotion of micro-irrigation, precision farming, protected cultivation, use of non-conventional energy, and optimum farm mechanization through Farm Machinery Banks which can run on custom hiring basis.
- Improved productivity levels will be supported by efficient post-harvest infrastructure managed by Self Help Groups, and the Farmer Producer Companies, to increase the share of farmers in the final price of agriculture commodities.
- The project will make conscious efforts to bring in the advanced knowledge through enhanced consortia approach involving the research and academic organisations. The project will also have non-financial MOUs with civil society organizations which have common goals of climate resilient agriculture and people's participation.
- The project is aligned with the Government norms for agriculture and allied sector activities.

1.18 Financing:

The Government of Maharashtra, in its resolution dated 30th June 2023, has mentioned the cost of the project to be around INR 6000 Cr. This is based on the Project Preliminary Report approved by the Department of Economic Affairs, GOI, where the project cost in USD is mentioned as 700 million, of which World Bank share will be USD 490 million (which is tentative subject to finalization at the time of financing). The period of the project will be six years from the date of loan approval by the World Bank.

1.19 Expected outcomes:

The expected outcomes will be measured by monitoring the progress of the Key Performance Indicators mentioned below.

The PDO will be measured using the following indicators:

- KPI#1 - Beneficiaries with enhanced resilience to climate risks – CRI (Number) By Gender
- KPI#2 - Enhanced Farm Income: Increase in Farm Income at Household level (by Gender) (%)
- KPI#3 - Increase in water use efficiency (percentage)
- KPI#4 - People using digitally enabled services (CRI)

1.20 Participatory Microplanning

The term micro-planning implies a multi-level and decentralised planning approach to the overall development. In order to carry the benefits of development to the poor, to ensure the continuity of balanced growth and to provide social justice, a micro-level approach to planning is proposed in the project. Also, micro-planning helps in fixing priorities for different regions depending upon their specific needs.

Building on the lessons learned from the NDKSP Phase I, the micro-planning exercise in Phase II will serve as a key entry point during project implementation, facilitating community mobilisation and awareness in the village. The MLP exercise will enable stakeholders to recognize the risks and vulnerabilities posed by climate change, allowing for the development of tailored adaptation strategies at the village level. The Village Adaptation Plans (VAP) will be developed through a participatory process and these plans will be crafted with locally agreed-upon inclusive criteria, fostering greater community participation in building resilience to climate challenges. This approach will also strengthen long-term ownership of the project.

Objectives of MLP :

The objectives of the MLP are as follows:

1. To gather scientific data on the effects of climate change on agriculture at the village level and develop management plans for crop planning and natural resources based on shifting climate patterns and water availability.
2. To compile technological solutions from agricultural universities and research institutes aimed at addressing the impacts of climate change on farming.
3. To analyse the value chain of farm products and recommend measures to strengthen it.
4. To prepare a Village Adaptation Plan (VAP) that incorporates the proposed initiatives and actions.
5. To raise awareness among farmers and other stakeholders about the project's implementation.

Guiding principle for the sub-component:

1. Microplanning exercise would lead to preparation of VAP
2. VAP, through participatory approach, will identify key set of indicative interventions across the production system. Some of the parameters amongst others are listed below:
 - Land holding, Land use Resource Mapping:
 - Seeds – varieties used, source of purchase, seed treatment, etc.
 - Soil and Water Resources – Map of the water conservation structures in the village; description of the watershed and interactions between the sub-zones, flow of rivers; status of surface and groundwater structures; upstream and drainage line treatment; etc.
 - Irrigation – type and coverage, status
 - Soil health – soil testing status and drawbacks if any; usage of fertilizers, pesticides, biofertilizers, biocontrol agents, weedicides, any other protective methods, agroforestry as method of pest management; status of soil health.
 - Cropping pattern – details of farming practices undertaken in the village
 - Livestock & Fodder
 - Farm equipment / mechanization – coverage of mechanization, availability of equipment bank if any

Awareness Building and Mobilization

The MLP exercise will serve as a key entry point for building project awareness and mobilising the community. In addition to the MLP, field officials will carry out an intensive outreach and consultation in each village focused on rapport building, awareness generation, and engaging the farming community. This social mobilisation will enhance community participation in local decision-making, improve access to various project benefits, social services, and productive resources, and increase the efficiency of utilising locally available resources. It will also create opportunities for asset building at the village level and improve accessibility for beneficiaries. The key points for awareness building & mobilisation efforts will be as below-

- Decentralised Approach & Community Participation
- Partnership Building and Networking
- Media Engagement and Special Events for Public Awareness
- Advocacy for Resource Mobilization and Policy Reform

Mobilisation Strategy and Tool:

A seminar will be organised at the district level by the PMU, while the SDAO will host a seminar at the taluka level to orient participants about the mobilisation and awareness campaign. The district seminar will engage stakeholders, including the TAO, while the sub-divisional seminar will involve agricultural assistants, cluster assistants, village volunteers and Krishi Tai. The village functionaries, in coordination with the agriculture assistant, will conduct field visits to implement theme-based mobilisation and awareness activities. The broad themes of the seminars will focus on:

- Efficient use of Water & Water Conservation - Micro irrigation, Water budgeting
- Carbon Sequestration & Soil Health Management and reduction in GHG emission plantation, soil health, IPM & INM, Non-conventional energy sources
- Disruptive Agriculture Technologies- Zero tillage, BBF
- Agribusiness Management - Post-harvest management

The mobilization tools will include-

- Prabhat pheri, Mashal pheri, display boards, meetings with gram panchayats, FGDs, and separate meetings with youth, SHGs, farmers groups, etc.
- Use of Audio-Visual Aids to disseminate the project's information, role play/ small play at village level for active participation from all communities etc.
- Use of various communication modes for announcements, small videos on YouTube, use of twitter, Facebook, Instagram, WhatsApp group etc.

Micro planning process Flow

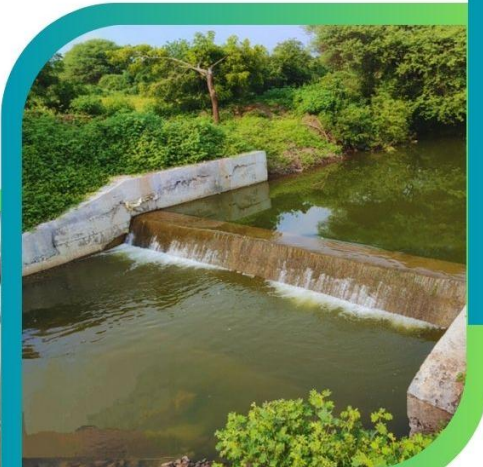
The details of the micro-planning process are provided in Annexure 5.

Table 4: Microplanning Calander - Daywise Activities

Day	Morning session	Afternoon Session	Evening session
Day 0	--	Meeting of Volunteers and the Gram Krishi Vikas Samiti	<i>Mashal pheri</i> and Village meeting
Day 1	Students' Rally	Focused Group Discussion on Key Agricultural Topics and Climate Adaptation	Focused Group Discussion on Climate-Resilient Technologies
Day 2	- Voluntary Community Service (<i>Shramdan</i>) -School activity- Essay/Speech - Transverse walk (1)- marking of NRM works on maps, collection of soil sample and well observation	Water budget exercise FGD about crop planning	Focused Group Discussion on the Value Chain in Agriculture and Allied Enterprises
Day 3	-Transverse walk (2)- marking of NRM works on maps, collection of soil sample and well observation -Resource Mapping	Preparation of draft of Village Adaptation Plan	Women assembly (<i>Mahila sabha</i>)
Day 4	-Value Chain – Livestock -Discussion and finalization of Village Adaptation Plan draft	Village assembly meeting (<i>Gram sabha</i>) and approval to Village Adaptation Plan	

Section 2

Component A - Resource Efficient Productivity Enhancement



Section 2:

Detailed Project Components and Implementation Modalities

The project components are systematically connected to ensure efficiency and resilience in farming. The project aims to reduce input costs and increase crop productivity, leading to higher profits for farmers. By promoting efficient use of both natural and external resources and reducing emissions through precision farming techniques, a sustainable production system is established. This sustainable system, in turn, supports resilient farm income, addressing a key concern. To achieve this, the project will support efforts to strengthen specific commodity value chains and work with farmer groups to build their capacity and explore agribusiness opportunities. Technical expertise will be leveraged through a consortium of partners to implement all components, while also expanding and strengthening the institutional framework created under PoCRA.

Component-wise implementation modalities:

Overall Outline for each component to be done in following way:

1. Brief description of the component / subcomponent
2. Guiding principles
3. Key objectives / outcomes
4. Detailed description of activities and process to be followed (if any)
5. Tabular representation

2.1 Component A: Resource Efficient Productivity Enhancement

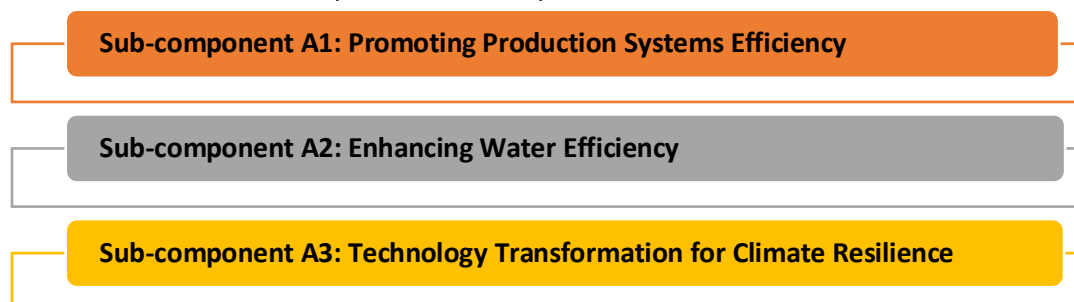
The objective of this component is to strengthen and promote a resource-efficient productivity system that is resilient to climatic variability and change. This approach focuses on improving resource efficiency rather than just increasing inputs, which can be ineffective given climatic challenges. By enhancing soil health, fertilizer-use efficiency (FUE), and water-use efficiency (WUE), it builds a low-carbon, climate-resilient production system tailored to each village cluster using data-driven digital agriculture (detailed in Component-D). By optimizing both natural and external resources, the system reduces crop vulnerability and maintains sustainable productivity, forming the foundation for all project activities.

Guiding principles for Component A will include:

- a) Promoting soil health to boost productivity and enhance climate resilience through multipronged farming techniques and technologies (A1)
- b) Promoting production systems efficiency through enhancing fertilizer use efficiency and integrating practices for improved soil health. (A1)

- c) Increased adoption of practices for both measurable results on nutrient use and water use efficiency (A1, A2)
- d) Scaling-up and adoption of technologies and practices for enhanced climate resilience across the project area. (A3)
- e) Enhancing Carbon sequestration in soil through multiple innovations and practices, targeted advisories and extension services. (A3)

There are three sub-components of component A which are as follows.



2.2 Sub-component A1: Promoting Production Systems Efficiency

This sub-component focuses on restoring and improving soil health to boost agricultural productivity. It will achieve this by: (i) enhancing soil health; (ii) increasing organic matter with agronomic practices that stabilize temperature and support soil microbes; and (iii) promoting efficient nutrient management to reduce emissions while maintaining high yields. The goal is to develop a resource-efficient, climate-resilient farming system focused on soil health.

The project will focus on expanding the adoption of climate-resilient agricultural practices to optimize productivity by improving soil health through conservation agriculture practices and resource-use efficiency. A key focus of this initiative is optimizing input use for better cultivation practices. The project will implement soil health assessments, address nutrient imbalances, and promote on-farm production of biofertilizers and organic inputs like vermicompost, *Bijamrut*, and NADEP compost. Conservation agriculture practices, such as no-till and reduced tillage, will be encouraged to improve soil fertility and ecosystem health. Efforts will also enhance Fertilizer Use Efficiency (FUE) to ensure more effective and sustainable input use.

To further boost soil health, investments will support controlled irrigation (sub-component A2), protected cultivation, and crop diversification, along with restoring abandoned farmland through restricted irrigation. These measures will help farmers transition to resilient, low-carbon, and cost-effective agricultural practices. Additionally, a Village Soil Fertility Index (VSFI) will be developed to monitor soil health across project villages, providing a scientific basis for improving fertility and tracking the impact of interventions over time.

Improving soil productivity and climate resilience requires addressing soil moisture stress – a common issue in project areas due to gaps in monsoon rainfall, which are further exacerbated during drought years. To address this, soil moisture data will be used to improve irrigation planning and watershed management. Satellite platforms like Sentinel-1, Risat-1, and the upcoming NISAR will provide high-resolution, one-hectare scale soil moisture maps. Additionally, ground-based sensors placed at Bowen Ratio Energy Balance (BREB) tower sites (in selected sites) will calibrate satellite data for greater accuracy.

A data-driven digital agriculture approach will enable real-time monitoring of soil conditions, helping refine soil management plans, enhance fertilization strategies, and optimize irrigation and soil amendments. These insights will ultimately improve productivity and resource efficiency.

To promote Production systems efficiency, this component will follow a structured approach through following set of interventions:

Table 5: Activities under component A1

Sr.No.	Activities	Who will do
1	Developing Digital Soil Maps for advisories	NBSS
2	Soil Health assessment	NBSS
3	Preparation of Village Soil Fertility Index	Project
4	Soil health-based advisories	NBSS & Project
5	Establishment of on-farm production units for biofertilizers / NADEP/ others	Project
6	Conservation agriculture	Project
7	Enhancing Fertilizer Use Efficiency (FUE)	IISS, Bhopal

These interventions would be implemented to achieve following outputs / outcome throughout the project duration:

- 1. Reducing Chemical Fertilizer Use:** The project aims to minimize chemical fertilizer dependence by implementing Soil Health Card recommendations for site-specific nutrient management and conducting awareness campaigns on environmental and economic impacts.
- 2. Promoting Alternative Nutrient Sources:** Emphasis will be on bio-fertilizers, organic manures, compost, and nutrient-rich crop residues to enhance soil fertility, reduce chemical inputs, and recycle organic waste for long-term sustainability. **Integrated Nutrient Management (INM):** A balanced approach combining organic and inorganic fertilizers, guided by regular soil testing, will optimize nutrient application while reducing environmental risks.
- 3. Precision Fertilizer Application:** To minimize losses, the project will promote precision techniques such as slow-release fertilizers. Demonstrations of fertigation with drip irrigation and foliar application during crop growth will enhance efficiency.

Detail description of each activity to be followed are elaborated below:

Enhancing Soil Health

Soil is the foundation of the agriculture production system. However, it is under tremendous abiotic stress with additional burden to feed the growing population. The rate of degradation of soil has reached an alarming situation, and conserving the soil with restoration of soil health has become the major challenge. Considering the potential consequences of poor soil health, the project aims at: (i) enhancing soil productive potential by improving physical parameters of soil-system efficiency; (ii) building organic matter to promote agronomic practices that buffer temperature fluctuations and enhance biological efficiency of a productive system, and (iii) restoring ecological efficiency of the landscape by effective nutrient management that realizes 'optimum agricultural production.' As a preliminary step, the project has taken a conscious decision to assess both (i) land resource utility and (ii) soil health prevailing in selected villages.

The organic carbon stock in soil is another critical element for both maintaining ecosystem health and sustaining agricultural productivity. In India, over 50% of soils lack adequate organic carbon, affecting productivity. Maintaining soil carbon is key to regulating temperature, conserving water, and ensuring nutrient availability. This project will analyse soil characteristics, conduct soil profiling to identify deficiencies, and address nutrient imbalances at the farm level. In collaboration with ICAR-NBSS&LUP, it will provide targeted advisories to improve soil health and promote site-specific nutrient management for better resource efficiency.

Preparation of Digital Soil maps for Advisories

With the support of ICAR-NBSS & LUP, the project aims to create high-resolution village-level soil maps that provide detailed insights into soil characteristics. These maps will serve as a foundation for issuing soil health-based advisories tailored to farmers' needs. These Digital Soil maps will provide insights into the soil variations in each of the project districts/geographies with adequate representation of the possible soil variations in each district. LRI can be applied in daily information on the magnitude of available soil moisture and has the potential to assist the farmers in many crop management decisions apart from facilitating the preparedness of the planners and administrators to mitigate the adverse effects of weather variations.

Key Features of Digital Soil Maps:

1. Comprehensive Soil Database:

- Includes soil depth, texture, pH, EC, salinity, organic matter, cation exchange capacity (CEC), available water capacity, and bulk density.
- Helps in determining root depth, water drainage, nutrient availability, and overall soil fertility.

2. Use of Remote Sensing & GIS Technology:

- NBSS&LUP will analyse soil characteristics at a granular level using advanced remote sensing and GIS-based soil mapping techniques.

3. Village-Wise Soil Health Advisories:

- Based on soil characteristics, farmers will receive advisories on:
 - Soil fertility management
 - Crop suitability based on soil depth and water-holding capacity
- These advisories will be delivered through a farmer-centric app and integrated into extension activities.

4. Application in Extension & Training Programs:

- Soil health maps will be used in Farmer Field Schools (FFS) and training programs.
- Extension workers will use these maps to educate farmers on best soil management practices.

By leveraging NBSS & LUP expertise and integrating digital tools, the project ensures efficient soil resource utilization, benefiting farmers and enhancing climate resilience in agriculture. The LRI database will help in preparing optimum land use plans for adaptation to climate change. A few important products of LRI are listed below.

1. Soil Map of each village
2. Interpretation of Climate data
3. LRI data interpretation
4. Available Water Capacity (AWC) map of each village
5. Soil Fertility map of each village
6. Crop suitability map of each village

Agricultural land use planning for the entire project villages will be done through a stage-wise protocol depicted in the figure 15 below:

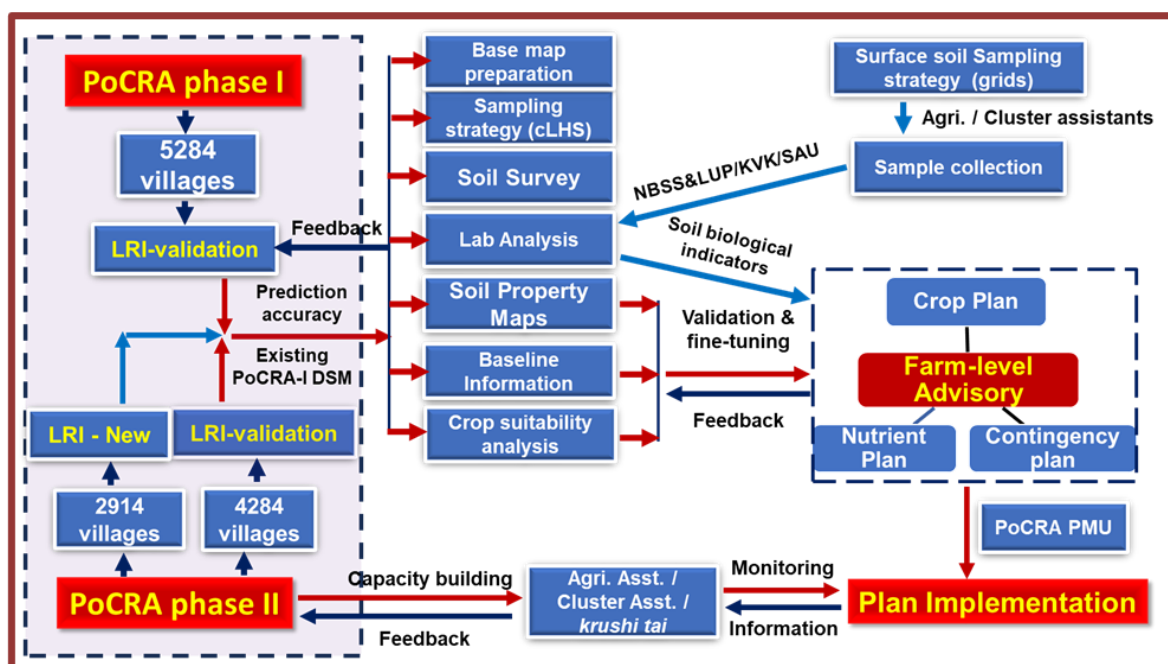


Figure 5: Schema of activities to be taken up in collaboration with NBSS & LUP during NDKSP Phase II

Soil health assessment

Soil health is the ability of the soil to sustain the productivity, diversity, and environmental services of terrestrial ecosystems. It can be characterised through the measurement of biological, chemical, and physical soil properties. Enriching soil enhances life from the microscopic scale to the global scale. Soil health management can increase microbial diversity, improve farmer livelihood, increase agricultural productivity, and build climate resilience.

To assess soil health effectively, the project will focus on key indicators such as soil organic carbon, soil biological activity, soil productivity and soil physical properties that reflect the overall condition of the soil and its ability to support sustainable agricultural practices. These indicators will help measure changes in soil quality and provide insights for improving soil management practices. The project will assess soil health based on the following parameters:

1. Soil Organic Carbon (SOC)
2. Soil Biological Activity
3. Soil Productivity
4. Soil Physical Properties

Soil Health Assessment Approach:

The project will implement regular soil health assessments using a combination of field observations, laboratory testing of soil, and monitoring tools. These assessments will be conducted at various stages of the farming season to track changes in soil health indicators before and after the implementation of soil health improvement practices, such as regenerative agriculture, BBF, conservation agriculture, biofertilizer and biopesticide usage.

The data collected will be used to generate actionable insights, provide feedback to farmers, and refine soil management strategies for long-term sustainability.

Additional to the assessment executed by project (and IISS), capacity-building workshops and Farmer Field Schools (FFS) will include training on soil health assessment techniques, empowering farmers to participate in and understand soil health improvement efforts.

At the baseline, the project will conduct analysis of soil samples collected during the micro-planning process. The analysis results combined with the results of the soil samples tested earlier, and will be used to determine Village Soil Fertility Index.

Preparation of Village Soil Fertility Index

The Village Soil Fertility Index (VSFI) will be developed to assess and monitor soil fertility across project villages. This index will provide a scientific basis for soil health improvement strategies and measure the impact of interventions over time.

Key Steps in VSFI Preparation:

1. Soil Sampling:
 - Soil samples will be collected from different landforms within villages to ensure comprehensive coverage.
2. Testing at State Govt. Labs
3. Data Integration & Index Development
4. Application in Advisory & Planning

This initiative will ensure sustainable soil management, improve land productivity, and support climate-resilient farming practices.

Soil health-based advisories

Project will disseminate the soil advisories based on soil health assessment of the villages. The soil characteristics-based maps will be used by the extension functionaries to educate the farmers during FFS and other training programmes. The maps will provide deep insights to the farmers about crop suitability, soil depth, available water capacity, Soil Organic Carbon,

Establishment of on-farm production units

The project will support on-farm biofertilizer production units for individual farmers, SHGs, and FPCs to promote use of biofertilizers for sustainable agriculture. NIPHM will provide training to master trainers, FPC members, and SHGs, ensuring skill development in biofertilizer production. This initiative will enhance soil health, reduce dependency on chemical inputs, and improve farm productivity through localized, cost-effective biofertilizer solutions. The project will establish such units with scale appropriate investment at individual level in project villages, and at SHG level in project clusters. Also, the FPCs will be promoted to take up this activity as an enterprise at large scale and will be facilitated to follow the regulatory ecosystem for biofertilizers and biopesticides.

Production of organic inputs

Aim of promotion of production of organic inputs is to make farming viable and aspirational by increasing net incomes of farmers on account of cost reduction, reduced risks, similar yields, incomes from intercropping and to cut down production costs by encouraging farmers to prepare essential biological inputs using on-farm, natural and home-grown resources. The Government of India has launched the National Mission on Natural Farming (NMNF) as a separate and independent scheme to promote chemical-free and climate-smart agriculture in 2023-24. NMNF will cover a 7.5 lakh hectare area by developing 15,000 clusters. Aligning to the GOI policy, and to support the farmers in the project villages, the project will promote production of organic inputs such as vermicompost, NADEP compost, Bijamrut, and Biodynamic compost etc.

Conservation agriculture

Conservation Agriculture (CA) is a farming system that can prevent losses of arable land while regenerating degraded lands. It promotes maintenance of permanent soil cover, minimum soil disturbance, and diversification of plant species. It enhances biodiversity and natural biological processes above and below the ground surface, which contribute to increased water and nutrient use efficiency and to improved and sustained crop production. CA principles are universally applicable to all agricultural landscapes and land uses with locally adapted practices. Soil interventions such as mechanical soil disturbance are reduced to an absolute minimum or avoided, and external inputs such as agrochemicals and plant nutrients of mineral or organic origin are applied optimally and in ways and quantities that do not interfere with, or disrupt, the biological processes.

Somasundaram et. al (2024) have reviewed the long-term experiments carried out by various institutes in the world, and concluded that “the long-term NT experiments play a significant role in improving soil health, SOC sequestration, and in-depth understanding of greenhouse gas (GHG) emission, climate change mitigation and optimizing resource use efficiency, to cater for the needs of the present- and future-generations.” Most of the research work about CA in India has been conducted by ICAR-Indian Institute of Soil Science (ICAR-IISS), Bhopal.

The project will tie up with ICAR-IISS, and the SAUs to develop a comprehensive approach towards conservation agriculture focusing on the cropping systems prevailing in the project districts.

Principles of Conservation Agriculture

Conservation Agriculture is based on three main principles adapted to reflect local conditions and needs:

1. **Minimum mechanical soil disturbance:** Minimum soil disturbance involves practices such as low disturbance, zero tillage, and dibbling of the seeds. Zero tillage and reduced tillage are key conservation agriculture practices aimed at minimizing soil disturbance to enhance soil health and sustainability. Zero tillage completely eliminates ploughing, while reduced tillage involves minimal soil manipulation to retain soil structure and moisture. These practices help prevent soil erosion, improve water retention, enhance organic matter buildup, and reduce fuel and labor costs. Periodic tillage that exceeds these disturbance limits should be avoided.
2. **Permanent soil organic cover:** The crop residues retained after the harvest of the crops act as organic mulch. The root biomass of crops as well as of weeds create an organic network beneath the soil surface. Tropical agriculture having heavy dependency on rains and having very poor irrigation facilities, is more vulnerable to soil loss. The organic biomass retained beneath the soil protects soil from such loss, besides creating an ideal ecosystem for microflora.
3. **Species diversification:** Crop rotation involving combination of cereals, pulses, oilseed, fibre crops (cotton), and vegetables crops grown in suitable seasons provides diversification in the farming system.

Benefits of Conservation Agriculture

1. Economic benefits that improve production efficiency. There are three major economic benefits that can result from CA adoption:
 - Time saving and thus reduction in labour requirement.
 - Reduction of costs, e.g. fuel, machinery operating costs and maintenance, as well as a reduced labour cost.
 - Higher efficiency in the sense of more output for a lower input.
2. Agronomic benefits that improve soil productivity, by adopting conservation agriculture.:
 - Soil Organic matter increases.
 - In-situ water conservation.
 - Improvement of soil structure, and thus improving rhizosphere

3. Environmental benefits that protect the soil and make agriculture more sustainable:
 - Reduction in soil erosion, and thus silting of nala, rivers and water reservoirs will be reduced.
 - Improvement of water quality.
 - Improvement of air quality.
 - Biodiversity increases.
 - Carbon sequestration.

The project is taking the principles of CA to the farmers through Farmer Field School (FFS) approach, as well as through Farmer-to-Farmer Extension (F2FE) approach, exposure visits, training, demonstrations, and financial support to encourage adoption among farmers.

Enhancing Fertilizer Use Efficiency (FUE)

Fertilizer Use Efficiency (FUE) refers to the proportion of nutrients applied through fertilizers that are effectively absorbed and utilized by crops. It measures the efficiency of nutrient uptake in relation to how much fertilizer is applied. A high FUE means that more of the nutrients from fertilizers are being used by crops for growth and yield, with less being lost to the environment through runoff, leaching, or volatilization. Achieving high FUE is critical not only for increasing productivity but also for reducing environmental impacts such as water pollution and greenhouse gas emissions.

The NDKSP Phase II aims to address these challenges through integrated approaches that promote the efficient use of fertilizers. Fertilizer use efficiency encompasses various strategies aimed at optimizing the amount of fertilizer applied to crops to meet their nutritional needs while minimizing losses to the environment.

Factors Influencing Fertilizer Use Efficiency

Fertilizer Use Efficiency (FUE) can be influenced by several factors that impact the availability, uptake, and utilization of nutrients by plants.

1. Losses Due to Leaching: Soluble nutrients like nitrogen in the form of nitrates can move below the root zone with water percolation, reducing nutrient availability for plants.
2. Losses in Gaseous Forms: Nutrients such as nitrogen can be lost as gases (e.g., ammonia volatilization or nitrous oxide emissions) under certain conditions, such as high soil pH, wet conditions, or improper fertilizer application methods.
3. Chemical Reactions During Mixing: Chemical interactions between fertilizers or with soil components can lead to nutrient immobilization, precipitation, or reduced availability before they reach the plant roots.

4. **Physical and Chemical Properties of the Soil:** Soil texture, structure, pH, organic matter content, and cation exchange capacity significantly affect nutrient retention, mobility, and availability for plant uptake.
5. **Weather and Irrigation Practices:** Excessive rainfall or irrigation can exacerbate leaching and runoff, while insufficient water limits nutrient solubility and uptake.
6. **Crop Type and Root Architecture:** Different crops have varying abilities to absorb nutrients, depending on their root systems and nutrient demands.

Importance of Soil Health in Fertilizer Efficiency

Soil health is crucial for optimizing fertilizer efficiency. Healthy soils have better nutrient retention capacity and enhanced microbial activity, which helps improve the uptake of nutrients by crops. The NDKSP Phase II supports various soil health initiatives, including soil testing and organic farming practices, to enhance soil fertility and reduce the need for excessive chemical fertilizers.

- Soil Testing and Nutrient Management
- Organic Farming and Integrated Nutrient Management (INM)

Efficient Fertilizer Application Methods

Proper fertilizer application methods are essential to ensure that fertilizers are effectively utilized by plants, reducing waste and environmental pollution. The NDKSP Phase II encourages farmers to adopt efficient fertilizer application techniques such as band placement, fertigation, and controlled-release fertilizers.

Band Placement: Fertilizers applied in narrow bands near the root zone are more effectively absorbed by crops. This method minimizes losses to the environment and reduces the amount of fertilizer required.

Fertigation: The application of fertilizers through irrigation systems (such as drip irrigation) ensures that nutrients are delivered directly to the root zone, improving nutrient uptake and reducing leaching.

Controlled-Release Fertilizers (CRFs): These fertilizers release nutrients gradually, matching the crop's nutrient uptake needs, and reducing the risk of nutrient losses due to leaching or volatilization. A study by FAO (2019) indicates that fertigation and controlled-release fertilizers can reduce nutrient losses by up to 30%, ensuring more efficient use of fertilizers.

Integrated Water and Fertilizer Management

The efficient use of water and fertilizers go hand in hand. Fertilizer application requires an appropriate amount of water to ensure that nutrients are absorbed by plants and not lost due to runoff or evaporation. The NDKSP Phase II promotes integrated water and fertilizer

management practices, ensuring that fertilizers are applied under the right moisture conditions.

- **Integrated Fertilizer and Water Management (IFWM):** This approach emphasizes synchronized management of both water and fertilizers. It ensures that water is available in sufficient quantities to dissolve fertilizers and facilitate their absorption by plants. The use of fertigation systems in Maharashtra's cotton-growing regions has led to significant improvements in both water and fertilizer use efficiency, resulting in higher yields and reduced input costs.
- **Precision agriculture** leverages technology to apply fertilizers more accurately, reducing waste and enhancing efficiency. The NDKSP Phase II incorporates precision farming tools and technologies to improve fertilizer use efficiency, monitor nutrient levels, and optimize fertilizer application.
- **Remote Sensing Technology:** Use of satellite imagery and drones to assess crop health and detect areas with nutrient deficiencies.
- **Soil Sensors:** Use of real-time soil nutrient sensors to measure nutrient availability in the soil.
- **Crop Nutrient Models:** These models predict the nutrient requirements of crops based on climatic conditions, soil types, and crop growth stages, helping farmers make data-driven decisions about fertilizer application.

Assessment of Fertilizer Use Efficiency (FUE) (Report) for optimal input advisories

The project will assess the FUE at the baseline, midline, and at the endline of the project by using the methodology suggested by the ICAR-IISS, Bhopal.

Partial factor productivity as the ratio of yield to the total amount of fertilizer applied

Agronomic nitrogen (or nutrient) use efficiency as the ratio of yield under fertilized plot minus yield under control plot, to amount of applied fertilizer

Nitrogen recovery, as the ratio of nitrogen uptake under fertilized plot minus uptake under control plot, to amount of applied fertilizer.

The nutrient use efficiency for nitrogen, phosphorus and sulphur will be calculated by using nutrient use efficiency measure viz. Agronomic Efficiency

Agronomic efficiency (AE) - It is calculated by considering the yield of fertilized plot, yield in control plot and quantity of fertilizer nutrient applied by using the following formula (Baligar et al., 2001).

$$AE (Kg Kg^{-1}) = \frac{\text{Yield of fertilized plot (kg ha}^{-1}) - \text{Yield of control plot (kg ha}^{-1})}{\text{Quantity of nutrients applied (kg ha}^{-1})}$$

Mobilisation and Training

The FFS will play a key role in promoting FUE by conducting hands-on training and peer learning. Through practical field demonstrations, farmers will observe the benefits of

balanced fertilizer applications and alternative nutrient sources, fostering wider adoption of sustainable practices.

The project's strategy to increase Fertilizer Use Efficiency (FUE) focuses on promoting sustainable and science-based nutrient management practices. Farmers will be educated on the benefits of biofertilizers and efficient fertilizer use through targeted awareness campaigns, Farmer Field Schools (FFS), and exposure visits.

The project will introduce a user-friendly digital application to calculate precise fertilizer doses based on soil health data and soil testing reports, empowering farmers to make informed decisions. Capacity-building programs will equip farmers with knowledge on the synergistic use of chemical and organic fertilizers, irrigation management, and proper application methods. Support for farmer groups and women's self-help groups (SHGs) will enable the production and sale of biofertilizers as an enterprise, promoting local availability and reducing costs.

Table 6: Activities, Beneficiary types and Output Indicators of A1.1

Sub-component	Activity	Beneficiary type (Individual/Farmer Group/ Community)	Output Indicators
A1.1: Enhancing Soil/ Land Efficiency	Preparation of Digital soil maps and for advisories to the farmers	All farmers, groups and Community	Number of digital maps. Integration of data into Advisory module.
A1.2: Soil health assessment	Preparation of village fertility index	All farmers of village/ Community	Number of villages with VFI charts
A1.3: On - farm production of biofertilizers	Establishment of on farm production of units	Individual farmer, SHG, FPC	Number of units installed in project villages
A1.4: Production of organic inputs	Preparation of Vermicompost/ NADEP compost/Bijamrut/Jeevamrut/Dashparni Ark/Bio dynamic compost - Ten drum theory	Individual farmer, SHGs	Number of units installed in project villages
A1.5: Conservation agriculture	No tillage/ reduced tillage	Farmers	No of farmers / adopted Area
A1.6 Enhancing fertilizer use efficiency (FUE)	Assessment of fertilizer use efficiency (report) for optimal input advisories	Farmers, community	Report of analysis. Data integration into advisory module

2.3. Sub-component A2: Enhancing Water-Use Efficiency

This sub-component aims to enhance water-use efficiency and build resilience to climate change by promoting climate-resilient agricultural practices, improving irrigation techniques, and enhancing soil moisture through various agronomic practices. Key interventions include implementing catchment area treatment, increasing groundwater recharge through recharging open dug wells, creating water storage structures, and enhancing efficient input-use via micro-irrigation systems. The project will also promote water-saving agronomic practices such as crop residue management to reduce evaporation, mulching, and the use of Broad Bed Furrow (BBF) systems to improve water infiltration, prevent waterlogging, and enhance drainage in rainfed areas. Intercropping will be encouraged to optimize water use by growing multiple crops with varying water requirements on the same land. Additionally, the project will deploy the Crop-Water Budget tool to calculate village water balances and assess supply-side conditions, integrating these tools with efficient irrigation methods and scheduling practices. Participatory soil and water conservation interventions will focus on area treatment, including continuous contour trenches (CCT), deep CCT, compartment and graded bunding, and repair of old paddy bunds etc. as well as drainage line treatment like Cement Nala Bandh and water harvesting structures like farm ponds and open dug wells etc. Collaboration with the Groundwater Surveys and Development Agency (GSDA) will support sustainable groundwater management and recharge efforts through groundwater recharge plans and technical support. The project will use a data-driven consortium approach, deploying soil moisture sensors to assist with irrigation scheduling and water conservation. Daily and weekly water estimates will be generated using thermal remote sensing products from global satellite platforms, including the upcoming Indo-French TRISHNA.

A network of BREB systems will be installed at selected pilot sites to calibrate energy balance components from thermal satellite platforms, providing detailed spatial-temporal evapotranspiration data for project villages. Efficient groundwater use is essential for sustaining crop supplies during periods of low rainfall and supporting extended Kharif and Rabi crops, with custom-developed groundwater recharge tools assessing demand-supply gaps for targeted intervention design. Ultimately, this sub-component aims to improve water security, boost agricultural productivity, and enhance sustainability through the adoption of innovative and integrated water management practices.

To promote Water use efficiency, this component will follow a structured approach through multiple set of interventions, which is largely driven by microplanning exercise.

Promoting Efficient Water Use: Water Security at the Farm

This sub-component focuses on climate-smart practices to regulate soil temperature, reduce evaporation, and improve water-use efficiency at the farm level. Studies show that a 1°C rise in temperature increases irrigation demand by 10% and alters soil properties by 2%. To

conserve water, the project will promote water-efficient cropping patterns and encourage low-water-consuming crops.

Efficient water use at the right crop stage is crucial for boosting productivity and reducing the water footprint. This project will promote water-saving practices like crop residue management, mulching, and Broad Bed Furrow (BBF) systems to retain moisture, prevent waterlogging, and improve drainage in rainfed areas. Intercropping will be encouraged to maximize water use efficiency.

The Crop-Water Budget tool from Projects first phase will help to assess village water balances and develop demand-side water management strategies. This will be integrated with efficient irrigation methods like drip irrigation with farm ponds and scheduled irrigation practices.

Water Budget of the village:

The water budget computation of the village provides real time information about the amount of water received through rainfall and its distribution based on the soil characteristics, population, livestock and crops as well as the balance at the end of monsoon. Also, the computation reveals total runoff generated, runoff impounded in the existing structures, and balance runoff.

To enhance the water security at farm level through watershed approach, water budgeting will play a significant role in scientific planning of various soil and water conservation measures that need to be adopted.

The project, in its first phase, has developed an advanced water budgeting tool in partnership with IIT-Bombay, serving as a critical foundation for micro-planning initiatives. This tool uses a mass balance approach, considering various water sources and sinks that impact soil water storage. The water budgeting process focuses on three core components: inflow, outflow, and changes in soil moisture storage, which together determine the amount of water entering and leaving a specific area over time.

Uses of the Water Budget Tool in Project Phase II:

1. To assess the village's supply-side conditions, based on soil type and available water and existing cropping pattern.
2. Tool will be used to calculate the village water balance, which gives insight about rainfall, total runoff generated, total impounded runoff and available runoff for the impounding in the village area.
3. Analysis of historical data on rainfall, drought conditions, and related factors for the village.
4. This tool will help extension workers and village communities to plan kharif and rabi crops.
5. Facilitates the micro-planning process for planning soil and water conservation measures tailored to the specific water balance needs of the village."

6. Develop demand-side management strategies focusing on water use efficiency and productivity.
7. Educate the community on hydrogeological and climatological factors, soil and water conservation, and water use efficiency.
8. Utilize the real-time Water Budget dashboard to help extension staff and community to assess and plan for crop water deficits, dry spells, and protective irrigation needs.

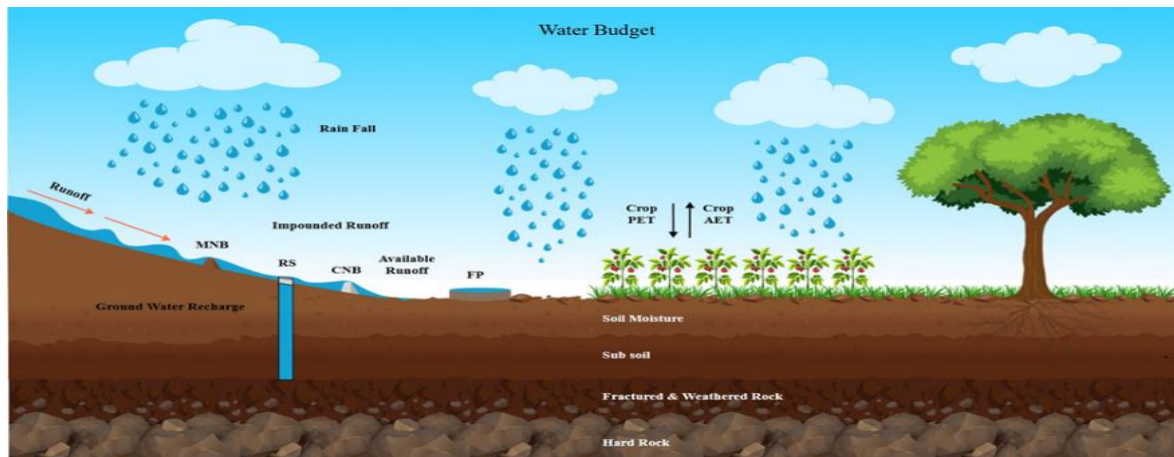


Figure 7: Water Balance Concept

Watershed approach

Watershed management is essential for maintaining watershed functions that support plant, animal, and human communities. Ensuring water security for agriculture, especially with variable precipitation, has become a key policy focus.

A watershed is the area defined by topography where all water drains to a common outlet, acting as the dividing line between drainage regions. It is both a hydrological unit for analysing the water cycle and a socio-political-ecological entity vital for food, social, and economic security, providing life support services to rural communities.

In watershed management, the “Ridge to Valley” (*Matha te Paytha, in Marathi*) approach is adopted. This approach ensures various treatment works in the upper reaches of the catchment area which reduce the rate of erosion, control the velocity of runoff, recharge the groundwater in the lower reaches and help in raising the ground water table.

The different objectives of watershed management programs are:

1. To control runoff, degradation and thereby conserve soil and water.
2. To manage and utilise the runoff water for useful purposes.
3. To protect, conserve and improve the land of watershed for more efficient and sustained production.
4. To protect and enhance the water resource originating in the watershed.
5. To check soil erosion and to reduce the effect of sediment yield in the watershed.
6. To enhance the groundwater recharge.

Situation analysis

Maharashtra's rainfall patterns are highly variable, leading to moisture stress during critical crop growth stages and increasing agricultural risks. With limited irrigation potential, soil and water conservation are crucial, especially as 96.4% of the land is degraded to varying degrees, and soil erosion rates are among the highest in India. Poor watershed management accelerates soil degradation, and excessive dependence on groundwater further depletes resources. Prioritizing watershed development is essential to sustain agriculture, improve rural livelihoods, and mitigate land productivity decline (ICAR-Indian Institute of Soil and Water Conservation, Dehradun, 2023).

According to the National Compilation on Dynamic Ground Water Resources of India, 2023 by the Central Ground Water Board, groundwater extraction levels have been assessed. Across the Maharashtra, groundwater was evaluated in **353 talukas**, out of which **224 talukas** fall within the project area. Among these **9 talukas** are classified as **Over-exploited**, **2 talukas** are **Critical**, **39 talukas** are **Semi-critical**, **174 talukas** are considered **Safe**, **one taluka** is categorized as **Saline**.

This classification highlights the varying groundwater conditions across the project area. The watershed development programme is the only option to bring stability in the production of dry land. Considering the importance of watershed management the Government of Maharashtra has implemented various schemes of watershed management viz.

- Integrated Watershed Management Programme (IWMP).
- Rashtriya Krishi Vikas Yojana (RKVY).
- Mahatma Phule Water and Land Conservation Campaign.
- Construction of Cement Nala Bandh in chain mode (check dam)
- Adarsh Village Scheme.
- Mega Watershed Development Programme under NABARD's Rural Infrastructure Development Fund (RIDF).
- Pradhan Mantri Krishi Sinchai Yojana (PMKSY) 2.0,
- Jalyukt Shivar Abhiyaan 1.0 & 2.0

Guided by the Microplanning exercise and preparation of VAPs, the project will invest in several interventions which will lead to:

- i) Catchment Area and Drainage Line Treatment
- ii) Creation of Water Storage Structures

The details of soil and water conservation interventions are provided in the annexure 4.

The project will also collaborate with the Groundwater Surveys and Development Agency (GSDA) to develop groundwater recharge plans, provide cost estimates, and offer technical support, including training field staff for effective implementation. There Recharge shaft and recharge trench works will help to increase ground water table.

Guiding Principles for NRM activities:

- The NRM activities to be undertaken in the project will be derived from the Water budget exercise and will be executed for catchment areas relevant to the project requirement including crop planning, suitability and soil characteristics, among others.
- Project will use existing technical specifications and cost norms of the Department of Agriculture for implementation of NRM works.
- The project will encourage Gram Krushi Vikas Samiti (GKVS) to take the lead in both initiating and executing NRM Works.
- Training will be provided to project officials and community representatives to build capacity among them for efficient and quality of works and execution.
- Project will ensure people's participation and information dissemination at every stage- need assessment, MLP, Village Plan, technical stages, and execution- so that ownership of the village over NRM assets is well and truly established.
- The project will use an integrated and seamless digital platform for Natural Resource Management (NRM) works which ensures accuracy, transparency and real time monitoring,

Mechanism for implementation of NRM Works

- Village Adaptation plan- Comprehensive village Adaptation plan will be prepared according to needs and demand assessment arising during the process of micro planning.
- Water budgeting – During the process of micro planning water budgeting will be computed and as per the water budget NRM activities will be proposed.
- Village Adaptation Plan Approval: - Prepared village plans will be technically vetted by the Agriculture Department and administrative sanction will be given by a District Committee headed by the District Collector.
- Estimate Preparation: - Field staff of Agriculture Department/ empanelled technical expert will prepare the work estimate.
- Technical Sanction: - Officials of the Agriculture Department will verify, estimate and issue technical sanction for the work.
- E-Tendering and Work order: - GKVS/GKVS supported by Department of Agriculture will do e-tendering process at Maha tender portal. After the e-tendering process work order will be issued.
- Work Execution: - After getting a work order, work execution will be done by a contractor under the guidance of the Agriculture Department. Before start of work and after completion of work geo tagged photos will be captured which will ensure transparency in execution.
- Checking and recommendation: - Department personnel will take all measurements and record detailed measurement books (MB).

- Payment Approval & Subsidy release: - After verification from relevant authorities' payment will be done to the concerned vendor
- All above mentioned steps will be implemented digitally through online portal and mobile app.
- Relevant authorities from department will carry out necessary checks at various levels which ensures transparency and accountability in works.

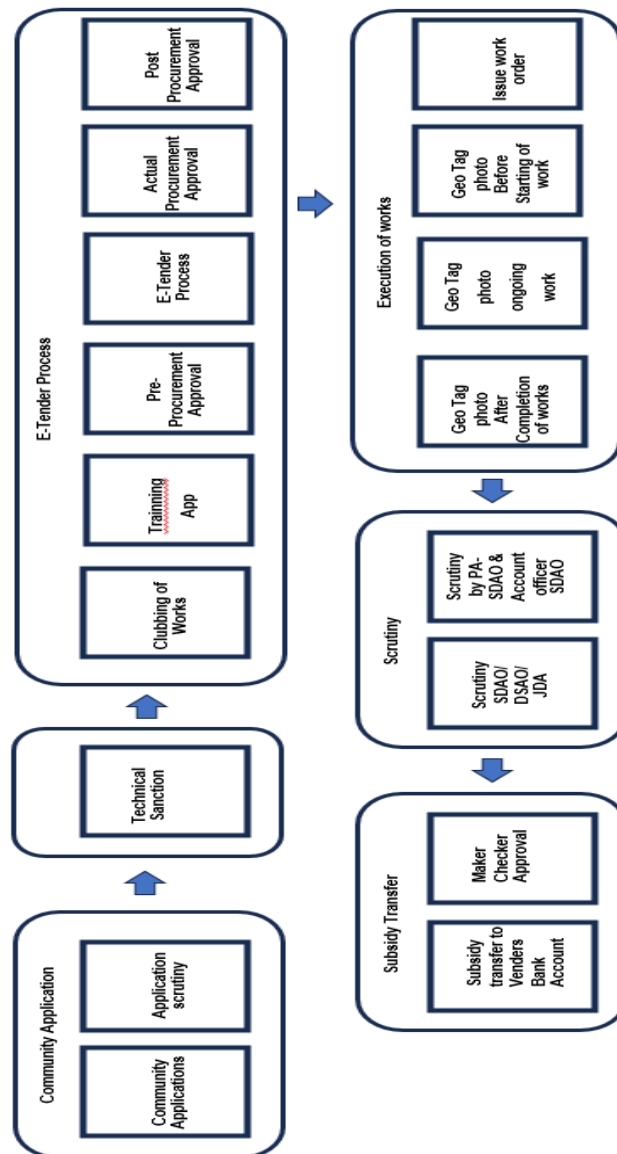


Figure 8: Flow chart for NRM Works

Table 7: Activities to achieve project objectives

Sub-component	Activity	Beneficiary type (Individual /Farmer Group/ Community)	Output
A.2.2 Promoting Efficient Water Use Water Security at the Farm	CCT/Deep CCT	Individual/ Community	Area promoted with in-situ soil conservation measures.
	Compartment Bunding	Individual	
	Graded bunding	Individual	
	Paddy Bunding/ Terracing (Majagi)	Individual	Area promoted with in-situ soil conservation measures.
	Repairs of old paddy bunds	Individual	No. of structures Constructed
	Loose boulder structures	Individual/ Community	No. of structures Constructed
	Gabion structure	Community	Number of structures constructed, and water storage capacity created
	Earthen Nala Bunds	Community	Number of structures constructed, and water storage capacity created
	Cement Nala Bunds	Community	Number of structures constructed, and Water storage capacity created
	Farm Pond	Individual/Community	Water storage capacity created
	Bodi Structure	Individual	
	Open Dug well	Individual	
	Recharge of open dug well	Individual	No of structures constructed
	Recharge Shaft / Recharge Trench with Recharge Shaft.	Community	No. of structures constructed
	Cement Nala Bund (CNB) Rejuvenation	Community	No. of structures rejuvenated
	Rejuvenation of Water bodies	Community	

Reducing water-footprint through water use efficiency

Water is an indispensable resource for agriculture, especially in regions grappling with water scarcity and climatic variability. Efficient water management is fundamental to boosting agricultural productivity, enhancing sustainability, and building resilience against climate change. The strategic focus on improving water use efficiency (WUE) within the NDKSP Phase II addresses critical concerns for agricultural productivity, farmer livelihoods, and the sustainable management of water resources.

The water footprint in agriculture is the total volume of freshwater used to produce agricultural products. This includes water used for irrigation, livestock, and food processing. To reduce the water footprint in agriculture for irrigation purposes, application of water in a precise way is necessary.

Water Use Efficiency Assessment

Water is one of the most limiting natural resources in farming systems, and its requirements vary significantly depending on plant species. Water use efficiency (WUE) in cropping systems is defined as the total biomass or yield produced per unit of water used by plants and the soil surface. It can also be expressed as the ratio of photosynthesis to transpiration. Crop growth and development are influenced by factors such as rainfall, evaporation, transpiration, and drainage systems. Consequently, improving WUE in both irrigated and rainfed agricultural systems is essential to enhance crop quality and yield. The researchers have highlighted that mulching techniques significantly improve both yield and WUE by reducing evapotranspiration. For instance, plastic mulches have been shown to enhance WUE by 20–60% by reducing evaporation rates. This approach also improves soil water retention and infiltration while creating favourable conditions for root proliferation and seed germination (Osuji, 1990; Liu et al., 2010).

Water-use efficiency can be based either on evapotranspiration ('ET efficiency') or on crop transpiration ('T efficiency'). The difference is important, because suppression of soil-water evaporation and prevention of weed transpiration can improve ET efficiency. However, it need not improve the efficiency, which is a measure of crop performance. These two water-use efficiencies may also be based on either the total dry-matter production or the marketable yield, and the basis should always be stated.

In dry regions where the proportion of water returned to the atmosphere by evaporation may be 90% or more, it is necessary to increase water-use efficiency. That is, we want to produce more plant material with smaller amounts of water. However, increasing water-use efficiency may be the best way to provide enough water for crop production in areas where evapotranspiration exceeds rainfall.

How to assess Water Use Efficiency:

To evaluate water, use efficiency (WUE), several approaches are commonly employed. These include calculating the ratio of crop yield (biomass) to the total water utilized (comprising irrigation and rainfall), utilizing weighing lysimeters for precise water measurement, monitoring soil moisture levels, and applying remote sensing techniques to estimate vegetation indices. Additionally, WUE assessment involves analysing crop coefficients based on plant phenology, while accounting for factors such as irrigation scheduling, climatic conditions, and soil characteristics.

Key Aspects of Water Use Efficiency (WUE) Assessment:

Direct Measurement Methods:

- **Field Measurements:** Determining WUE by harvesting crop biomass and dividing it by the total water used (irrigation + rainfall) during the growing season.
- **Weighing Lysimeters:** Accurately measuring water use by tracking weight changes in a soil column under controlled conditions.
- **Soil Moisture Monitoring:** Utilizing probes to measure soil moisture content at various depths throughout the crop growth cycle.

Indirect Methods:

- **Crop Coefficients (Kc):** Estimating water requirements based on established Kc values specific to the crop type and growth stage.
- **Remote Sensing:** Employing satellite imagery to derive vegetation indices, such as NDVI, which are correlated to plant biomass and water use.
- **Evapotranspiration (ET):** Quantifying water loss through evaporation and transpiration using approaches like the Penman-Monteith equation.

In Phase I of the project, localized Crop coefficient (Kc) values have been developed for major crops. These values are integrated into the Water Budget tool to determine crop water requirements. This helps assess crop water stress during dry spells and minimizes water losses under normal conditions.

Bowen Towers as a tool to assess AET

The Bowen Ratio Energy Balance (BREB) tower, or Bowen tower, is a scientific instrument used to measure the energy fluxes at the Earth's surface. These fluxes include latent heat flux (energy used for evapotranspiration) and sensible heat flux (energy that heats the air). The tool is particularly important for understanding water and energy exchanges in ecosystems, such as agricultural fields.

The BREB method assumes that the vertical transport of energy in the atmosphere (sensible and latent heat) is proportional to the measured gradients in temperature and vapor pressure.

Applications

- The Bowen tower helps estimate evapotranspiration, which is critical for understanding how much water crops lose through evaporation and transpiration.
- Bowen towers measure energy fluxes, which are crucial for understanding the microclimate around agricultural fields which is useful for planning crops and farming practices as per local climatic conditions.
- Helps in identification of water needs of different crops at various growth stages to plan irrigation strategies.
- Combined with other sensors, Bowen towers can track soil moisture depletion and alert farmers to potential drought stress in crops.

Network of Bowen Ratio Energy Balance Tower sites for PoCRA 2

The Bowen Ratio Energy Balance (BREB) method is one of the widely used method for the estimation of evapotranspiration (ET) from ground observations. The BREB tower is suitable for vegetated surfaces, especially croplands. This method requires the measurement of meteorological variables such as air temperature, relative humidity, wind speed and direction at two (or) three different heights above the vegetation canopy to estimate Bowen ratio (defined as the ratio of sensible to latent heat fluxes). Combining Bowen ratio with surface radiative fluxes (from the four-component net radiometer) and the soil heat flux (from the soil heat flux plates buried underground below the tower), we will be able to estimate ET. Typically, the measurements will be carried out at intervals of 1 minute to 5 minutes and will be averaged at half-hourly intervals. The BREB towers are easy to install and maintain and are relatively less expensive than eddy covariance observations. ET can be reliably estimated from BREB tower at daily and other longer time scales (e.g. weekly, fortnightly etc.) offering critical insights about the ET process, which are relevant for micro-watershed water budget, crop water budgeting, and water & crop stress Decision Support System tools proposed in PoCRA 2. The ground observed ET fluxes from BREB towers are suitable for validating ET modelled from satellite observations with pixel area ranging from 0.25 sq. km to 1 sq. km. Further, the data from a network of BREB towers in various clusters in different districts will help in the estimated ET from globally available satellite remote data sets for the entire project areas of PoCRA 2.

Distribution of network of BREB tower sites for ground ET observations

The state of Maharashtra has divided into eight meteorological regions (based on rainfall and temperature Figure 1). However, it may be seen from Figure 1 that these regions are somewhat mixed and not contiguous, and moreover the dominant crops will form additional zones in these regions. Hence, it is desired to have representative BREB tower sites in the PoCRA 2 clusters that will cover the meteorological and dominant crop regions of the project. These sites can be located in croplands that are managed by the state agricultural universities and KVKs, this will ensure safety of the instrument. These sites can be ideally co-located with soil moisture network observation sites proposed in PoCRA 2. Further, agrometeorologists in

the state agricultural universities can manage these sites, ensuring recording of continuous data at each of the BREB sites, which will be utilised and processed for generating ET through remote sensed platform for all the PoCRA 2 clusters at near real time.

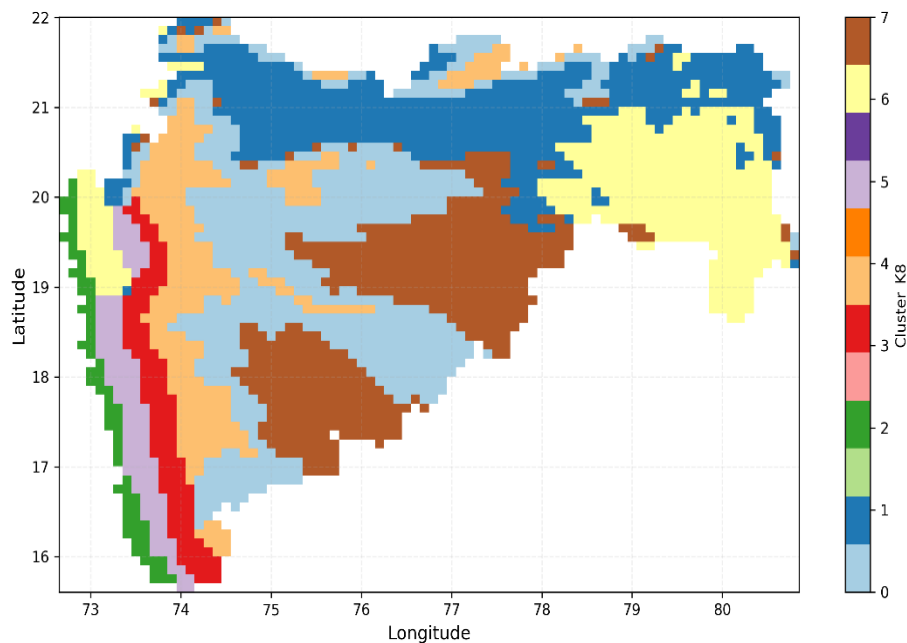


Figure 9: Meteorological clusters in Maharashtra (Prepared by Prof. Eswar, IIT Bombay)

Reference note of steps to be followed for BREB Tower Network

1. Identification of sites and number of Bowen Towers to be installed.
2. The selected sites will be in the premises of State KVKs/ SAUs.
3. Technical specifications of the sensors of the Bowen Towers may be taken from renowned institutions such as Indian Institute of Science, Bengaluru and the like. in
4. Teams from the SAUs will be given the responsibility to gather, maintain and manage the data.
5. Once the project is completed these SAUs will continue to maintain and manage Bowen Towers.
6. The data from Bowen Towers will be analyzed and the data from the Towers will be calibrated and validated with remotely sensed data (Ex.ModIS, Ecstress and Trishna Satellites)
7. Temporal ET data at the cadastral/ farm scale will be archived for further use with DSS tools (e.g., crop water budgeting) at the innovation lab of PMU.
8. The remote sensed ET at the farm scale level will be stored at the PMU innovation lab for further use in agro-hydrology advisory.

Reference note of steps to be followed for Soil Moisture Sensor Network

1. Some of the soil moisture station sites may be co-located at Bowen tower sites. Additional sites will be selected based on soil-crop combinations with the help of NBSS LUP Or similar public institutions.
2. The soil moisture stations will have multi depth sensors upto 1 meter.
3. NBSS LUP Nagpur Or similar public institutions will be given the responsibility to gather, maintain and manage the data for the soil moisture stations.
4. The data from the soil moisture stations will be analysed by NBSS and LUP Nagpur Team Or similar public institutions. NBSS and LUP Team or Or similar public institutions will use the data from the soil moisture stations to calibrate and validate remotely sensed data (Ex. Sentinel-1, RiSat-1, and NISAR). NBSS and LUP will be responsible to generate daily/weekly soil moisture products for all the PoCRA-II project area. NBSS and LUP Or similar public institutions will also be responsible to train Partner Universities and PMU team.
5. The temporal soil moisture data at cadastral/ farm scale will be archived for further use with DSS tools (ex. Irrigation advisory, Crop Water Budgeting) at the innovation lab at the PMU.
6. The remote sensed soil moisture at the farm scale level stored at PMU innovation lab will be used for agro-hydrology advisory.
7. Specific manual surveys for soil moistures will be conducted periodically at some benchmark sites by NBSS and SAU teams Or similar public institutions to evaluate the project interventions.

The project will deploy a data-driven consortium approach to achieve the objectives of the sub-component. Soil moisture sensors will be deployed to support irrigation scheduling and water conservation. These integrated practices will minimize evaporation, improve transpiration, and boost carbon sequestration, supporting sustainable farming. Water estimates will be generated daily and weekly using thermal remote sensing, including data from the upcoming Indo-French TRISHNA satellite. A network of BREB systems at pilot sites will help refine evapotranspiration data for precise water management in project villages.

Water Use Efficiency Enhancement Interventions

Application of water at the proper stage of crop and in proper quantity also plays an important role in enhancing production and productivity. This ensures an increase in water use efficiency and thereby reducing the water footprint. Water use efficiency can be achieved through efficient use of water with the help of following techniques.

Water-Saving Agronomic Practices

- **Crop Residue Management:** Retaining crop residues on the field to reduce evaporation and enhance soil moisture.

- **Mulching:** Covering the soil with organic or synthetic materials to retain moisture and reduce evaporation.
- **Broad Bed Furrow (BBF):** Improves water infiltration, prevents waterlogging, and enhances drainage in rainfed areas.
- **Inter cropping:** Optimizes water use by growing multiple crops with varying water requirements on the same land
- **Drought-Resistant Crops:** Growing crops like millets, sorghum, or pulses that require less water.
- **Crop Rotation:** Alternating crops with high and low water requirements to prevent over-extraction of water.
- **Capacity Building and Farmer Engagement:** The project emphasizes building farmers' capacity through training programs on efficient water use, climate-smart agriculture, and collective water governance strategies. This participatory model strengthens community ownership and accountability in managing water resources.

The project will promote all these agronomic practices through FFS, advisories and various communication tools.

Water-Saving Technologies

Micro-irrigation is an advanced agricultural technique that delivers water directly to the root zones of plants through a network of pipes, tubes, and emitters. Unlike traditional flood or surface irrigation, it applies water in small, precise amounts, ensuring efficient water use and minimizing wastage

- **Drip Irrigation:** Water is delivered drop by drop directly to the plant roots through emitters.
- **Sprinkler Irrigation:** Water is sprayed over the crop area, simulating natural rainfall.

Guiding principle for Micro irrigation

This intervention will be Demand driven, and benefits will be given to all eligible beneficiaries. Micro-irrigation systems with requisite BIS standards would be eligible for project support. Assistance for installation of drip and sprinkler irrigation system shall be limited to 5 ha. per beneficiary

For subsidy calculation cost norms will be adopted which are given as standard unit costs for different components under different spacing in the scheme guidelines issued by GOI.

The vendors shall train the eligible farmers on the proper use and maintenance of the system.

Water Pump and Pipes:

Efficient use of water in agriculture is critical for maximizing crop yield while conserving water resources. Water pumps and pipes play a crucial role in delivering water efficiently, reducing waste, and optimizing irrigation.

Many projects' areas inhabited by SC and ST landholders face unique agricultural challenges, particularly in accessing and managing water for irrigation. The process of pumping water

from natural water sources and transporting it to the fields is often hindered by limited infrastructure and resources, affecting agricultural productivity.

To address these challenges, the Project will provide dedicated support to farmers from SC and ST communities only to enhance water accessibility and management. This assistance will specifically include provisions for water pumps and pipes, ensuring a more efficient and reliable irrigation system. By facilitating better water availability, the initiative aims to improve agricultural productivity and support the livelihoods of these farmers.

Approach:

Water pumps and pipes with requisite BIS standards would be eligible for project support. For subsidy calculation cost norms of State Govt./ Central Govt. guidelines will be adopted. Beneficiaries from SC and ST communities only will be eligible for support.

Table 8: Activities, Beneficiary types and Output indicators of Subcomponent A2

Sub-comp onent	Activity/sub-activities	Beneficiary type	Output Indicators
A2.1 Micro-planning process at Village Level			
	Preparation of Village Adaptation Plan (VAP)	Individual/Farmer Group/Community	Number of VAPs prepared
	Krishi Tai	Individual	Number of Krishi Tai onboarded
A2.2 Catchment Treatment			
	Area treatment		
	Continuous Contour trenches Model 5-8 (0.30/0.45 meters)	Individual/Community	Total area treated (hectares)
	Deep Continuous Contour trenches	Individual/Community	Total area treated (hectares)
	Compartment Bunding	Individual/Community	Area covered under compartment bunding (hectares)
	Graded Bunding	Individual/Community	Area covered under graded bunding (hectares)
	Terracing (Majagi)	Individual/Community	Area covered under terracing (hectares)
	Paddy Bunding	Individual/Community	Area covered under paddy bunding (hectares)
	Repair of Old Paddy Bunds	Individual	Area of paddy bunds repaired (hectares)

Sub-comp onent	Activity/sub-activities	Beneficiary type	Output Indicators
	Drainage line treatment	Community	Number of structures constructed, and Water storage capacity created
	Loose boulder structures	Community	Number of loose boulder structures constructed
	Gabion structures	Community	Number of structures constructed, and Water storage capacity created
	Earthen Nala Bunds	Community	Number of structures constructed, and Water storage capacity created
	Cement Nala Bunds	Community	Number of structures constructed, and Water storage capacity created
	Water harvesting structures	Individual/Farm er Group	Total number of water harvesting structures created
	Individual Farm Pond	Individual	Storage Capacity Created (TCM)
	Group Farm Pond	Farmer Group	Storage Capacity Created (TCM)
	Open Dug Well	Individual	Number of open dug wells constructed
	New Bodi Structure	Individual	Storage Capacity Created (TCM)
	Recharging of Groundwater	Individual/Com munity	Number of groundwater recharge structures constructed
	Recharge of Open dug well	Individual	Number of structures constructed
	Recharge shaft	Community	Number of recharge shafts constructed
	Recharge trench with recharge shaft	Community	Number of recharge trenches with shafts constructed
	Rejuvenation of Water harvesting structures	Individual/Com munity	Number of rejuvenated structures
	Cement Nala Bund (CNB) Rejuvenation	Individual/Com munity	Number of CNBs rejuvenated
	<i>Bodi</i> structure	Individual	Number of <i>bodi</i> structures rejuvenated
	Rejuvenation of water bodies	Community	Number of water bodies rejuvenated
A2.3 Reducing water-footprint through water use efficiency			

Sub-comp onent	Activity/sub-activities	Beneficiary type	Output Indicators
	Technological Advancements for WUE (Assessment through Bowen Tower)	Community	Number of Bowen Towers Installed
	Micro-Irrigation	Individual	Area covered under Micro-Irrigation
	Drip	Individual	Area covered under Drip irrigation
	Sprinkler	Individual	Area covered under sprinkler irrigation
	Water pumps	Individual	Number of beneficiaries supported
	Water pipes	Individual	Number of beneficiaries supported

2.4 Sub-component A3: Technology Transformation for Climate Resilience:

The project will promote innovative technologies and climate-smart practices to boost productivity, cut GHG emissions, and improve resource efficiency. By leveraging precision farming and ICT-based solutions, farmers will gain access to tailored strategies that enhance resilience against climate challenges.

This subcomponent will drive climate resilience in agriculture by promoting on-farm technologies, ICT-based incentives, and data-driven advisories. Through training, Farmer Field Schools (FFS), and Digital Field Schools (DFS), farmers will adopt climate-smart practices like mechanization, efficient irrigation, and Integrated Crop Management (ICM) including field preparation, climate-resilient seed selection, seed treatment, Integrated Pest Management (IPM), Integrated Nutrient Management (INM), mechanization, protective irrigation, and harvest management, among others. Key initiatives include seed conservation, a buy-back program for climate-resilient seeds, and demand-based matching grants via DBT. By integrating precision farming and resource-efficient systems, the project empowers farmers to sustain productivity despite climate challenges, ensuring long-term agricultural sustainability.

Activities under this sub-component

Climate Resilient Technologies dissemination and adoption

1. Climate-Resilient Technologies for field and fruit crops
 - FFS for field crops
 - FFS for fruit crops
 - CRT through FPOs
 - Digital Field School (DFS)

2. On-farm production of Bio-pesticides and Bio-agents
 - On-farm production of Bio-pesticides
 - On-Farm Production of Predators and Parasites
3. Seed Production of Climate Resilient Varieties

Farmer Field School dissemination of Climate Resilient Technologies:

The Farmers Field School (FFS) is an extension method in agriculture, developed in the late 1980s by the Food and Agriculture Organization (FAO). The project aims to implement FFS at the village level, connecting progressive farmers with others, thereby promoting climate-resilient technologies through peer learning and knowledge exchange. This approach marks a departure from traditional programs, where generalised advice was provided by external experts, emphasising instead localised and farmer-led solutions.

The objectives of the Farmer Field School (FFS) focus on introducing and promoting good agricultural practices for selected crops, enhancing farmers' knowledge in key areas such as Integrated Pest Management (IPM), Integrated Nutrient Management (INM), sustainable cultivation methods, and preparedness for weather contingencies. FFS aims to empower farmers to make informed, independent decisions about their farming practices, helping them reduce cultivation costs, restore soil fertility, boost productivity, and cope with the emerging weather contingencies. Through these objectives, the FFS seeks to create a resilient and sustainable farming ecosystem that benefits both the farmers and the environment.

Key guiding principles of Farmer Field School:

- Field-based program lasting for the entire cropping season.
- Experiential, participatory, and farmer-centred learning approach.
- Each session includes three core activities: agro-ecosystem analysis, special topic discussion, and group dynamics activity.
- Comparative studies conducted between IPM and non-IPM treated plots, with additional studies focused on local field problems.
- Use of non-formal education (NFE) methods, such as sharing, case studies, role play, problem-solving exercises, panel discussions, group dynamics, brainstorming, and simulation games.
- Involves around 30 farmers in hands-on activities aimed at improving decision-making capacity.
- Designed to address specific crop issues, ensuring FFS is well-planned for a stronger impact.

Approach for implementation of FFS activity

1. The primary objective of the Farmers Field School (FFS) is to enhance crop productivity by promoting the transfer of climate-resilient agricultural technologies and good agricultural practices at the farm level. FFS plays a pivotal role in increasing crop yields,

improving produce quality, reducing costs of pesticides and fertilisers, lowering overall production costs, and ultimately increasing farmers' income.

2. By promoting the adoption of climate-smart practices via on-farm demonstrations using the FFS approach, the project focuses on the following:
 - Introducing drought- and salinity-tolerant crop varieties like Cotton, Chick Pea, Pigeon Pea, Soybean, Rabi Sorghum, and high-value vegetables.
 - Supporting crop diversification to better cope with climate variability.
 - Promoting Integrated Pest Management (IPM), Integrated Nutrient Management (INM), and related sustainable practices aimed at enhancing farm productivity.
 - Encouraging appropriate farm mechanization.
3. Farm School Planning and Execution: Cluster Assistants (Agriculture Graduates deployed by the project) with the help of Agriculture Assistants, and *Krushis Tais* will execute this activity. Potential host and guest farmers for organizing FFS in villages will be identified. The plans for FFS in each season will be discussed and approved by the GKVS, and accordingly the schedule will be prepared by the SDAOs.
4. The integration of the Farmer Field School (FFS) with the NDKSP's digital advisory system offers multiple benefits to farmers. Through the Farmer App, they receive real-time, weather-based crop and disease advisories, field observation data, and expert insights, which are critical for timely decision-making. Additionally, soil health card information allows for precise, crop-specific fertiliser recommendations, ensuring balanced nutrient management.

Implementation and Monitoring of FFS

1. The GKVS, in consultation with farmer interest groups, will prepare a list of progressive and achiever farmers who grow major crops and are willing to share the climate-resilient technologies they have implemented. Farmers representing diverse sections of the community will be selected and enrolled in the FFS with the assistance of Cluster Assistants.
2. Each village will have at least one FFS, focusing on either major crops or relevant climate-resilient technologies. The selected farms will remain active in the FFS for three consecutive years to ensure consistent learning and technology adoption.
3. Each FFS plot or host farmer plot will serve as a focal point for climate-resilient agriculture. These farms will be recognized as "Climate Resilient Farms," and a visible board will be placed on-site to identify them as key learning hubs.
4. A minimum of 50% of host farmers selected for the FFS will be women, with both host and guest farmers being women in women FFS. Women host farmers will be selected from actively working women within farming families. Women-led FFS will focus on technologies such as the preparation of Neem Seed Kernel Extract, On-farm production of biofertilizers, production of organic inputs, conservation of local resilient seed, drudgery reduction approaches in farming, post-harvest handling of produce etc.

5. The Cluster Assistant will facilitate the FFS and ensure that all information, including activities, results, and observations, is uploaded in the dedicated mobile application for FFS.
6. Each FFS will receive continuous technical support from KVKs, with scientists and experts from SAUs, NRCs, and other relevant institutions conducting at least six field visits during the crop season. This ensures comprehensive guidance on implementing climate-resilient practices.
7. Documentation of results, including yield and crop harvesting, will be conducted on farmers' fields in the presence of project staff and experts from KVKs and SAUs. Additionally, a Field Day will be organised during the harvest to showcase the benefits of climate-resilient technologies to other farmers in the community.
8. The entire FFS process will be closely monitored by Technology Coordinators, ensuring adherence to project goals and successful adoption of climate-resilient practices.

FFS for fruit crops

Farmer Field Schools (FFS) for fruit crops will be conducted in project areas where major fruit crops such as mango, pomegranate, citrus, banana, guava, custard apple, and sapota are grown. These FFS sessions will focus on building farmers' capacity through hands-on learning and participatory approaches to adopt climate-resilient practices and improve productivity. Below are the climate resilient technologies that the project will promote among horticultural farmers:

The project will promote these climate-resilient horticultural practices through an extensive outreach program that includes Farmer Field Schools (FFS), on-farm demonstrations, workshops, and exposure visits. By engaging farmers directly through practical demonstrations and peer learning, the project aims to ensure widespread adoption of these techniques. Custom hiring centres (CHCs) will provide access to necessary tools and equipment, enabling farmers to implement these practices efficiently. The integration of these practices into horticultural systems will not only improve crop productivity but also enhance the resilience of farmers to the uncertainties posed by climate change, ensuring sustainable horticulture for the future.

Climate Resilient Technologies: Technical Features

A. Field crops

Climate variability poses significant challenges to crop productivity, necessitating the adoption of climate-resilient technologies to ensure sustainable agriculture. These technologies focus on enhancing crop tolerance to drought, heat stress, and erratic rainfall, improving soil health, and optimizing resource use efficiency. The project will promote proven climate-resilient practices such as drought-resistant seed varieties, precision water management, soil conservation techniques, integrated nutrient management, and pest control strategies for field crops. Details of climate resilient technologies is given in annexure 6.

Table 9: Features and Benefits of Climate-Resilient Technologies

Agronomic Practice	Main resilience feature and contribution to the climate agenda	Main agronomic benefits expected	Potential impact under the NDKSP
Broad Bed and Furrow (BBF) Cultivation	In situ water conservation practice that strengthens the farming system's capacity to cope with dry spells during rainy season	Encourages moisture storage in the soil profile. Disposes safely of surplus surface run-off without causing erosion. Provides a better drain and more easily cultivated soil in the bed. Possibility to reuse runoff water stored in water tank	Increased crop productivity Enhanced soil health Improved water security Coping with weather contingencies
Raised bed & Dibbling technology	Adaptation to adverse climate change like dry and wet spells Enhances absorptive capacity of soil Reduced soil erosion and sustainable soil health	Higher crop productivity Reduced production costs	Increased crop productivity Enhanced soil health Improved water security Coping with weather contingencies
Minimum or zero tillage	Enhances carbon sequestration and help reduce GHG emission Enhances soil organic carbon thereby promoting soil biodiversity Reduced soil erosion and sustainable soil health	Less soil compaction Less moisture evaporation Higher crop productivity Reduced production costs Save time for sowing by 10-15 days compared to conventional tillage.	Increased crop productivity Enhanced soil health Improved water security Reduced production cost Coping with weather contingencies
Improved seed varieties	Adaptation to adverse climate conditions through use of climate-smart seed varieties (short maturity duration, stress-tolerant)	Reduces risk of major crop losses due to adverse weather. Helps preserve crop mix even with late onset of monsoon.	Increased crop productivity Improved water use efficiency

Agronomic Practice	Main resilience feature and contribution to the climate agenda	Main agronomic benefits expected	Potential impact under the NDKSP
Intercropping	Climate-risk management strategy to help farmer cope with adverse climate events that may have a different impact on 2 crops grown simultaneously on the same plot	Reduces likelihood of total crop failure due to adverse weather.	Reduces likelihood of total crop failure due to adverse weather. Coping with weather contingencies
Integrated Pest Management	Enhanced absorptive capacity of the farming system to manage pest infestations induced by high levels of humidity (pests) or prolonged droughts.	Reduces infestations. Reduces crop losses. Reduces use of chemical pesticides. Improves crop quality.	Increased crop productivity and reduce cost
Integrated Nutrient Management	Enhanced absorptive capacity of the farming system to manage soil nutrient deficiency induced by imbalanced fertilizer use and soil degradation.	Reduces nutrient deficiencies. Mitigates soil salinity. Reduces crop losses. Improves crop quality. Enhances crop health. Higher yields. Resistance to biotic & abiotic stresses. Helps to fetch better market prices.	Increased crop productivity Enhanced soil health
Integrated Weed Management	Enhances crop resilience by reducing weed competition for nutrients, water, and sunlight, particularly under erratic rainfall and rising temperatures.	Reduces weed pressure. Minimizes yield losses. Lowers dependency on chemical herbicides. Improves soil health and moisture retention.	Increased crop productivity, Reduced input costs, Improved farm sustainability.
Incorporation of Crop diversification	Enhances ecosystem stability by reducing climate vulnerability, improving soil fertility, and breaking pest and disease cycles.	Improves yield stability. Enhances soil fertility. Reduces pest and disease outbreaks. Promotes efficient resource use.	Higher farm resilience, Diversified farmer income, Long-term sustainability. Coping with weather contingencies

Agronomic Practice	Main resilience feature and contribution to the climate agenda	Main agronomic benefits expected	Potential impact under the NDKSP
Retention of weed biomass in the soil	Promoting carbon sequestration and enhancing soil organic matter	improves soil fertility and structure. Increased Soil Moisture Retention prevent soil erosion Suppression of Future Weed Growth	Enhances soil health and carbon sequestration Improves crop yields Environmental sustainability
Reducing methane emission in paddy cultivation	Reduce methane emission Reduce water footprint	Improves soil fertility Reduces soil erosion Increases productivity Reduces cost of production	Enhances soil health and carbon sequestration Reduction in water footprint Improves crop yields Environmental sustainability through reduction in methane emission

B. Fruit Crops

Horticulture, a critical sector in many agricultural systems, faces numerous challenges due to climate variability. Shifts in temperature, erratic rainfall patterns, droughts, and other climatic changes can severely impact fruit and vegetable production. To mitigate these impacts, the project emphasises the promotion of climate-resilient horticulture practices aimed at enhancing productivity, conserving natural resources, and ensuring long-term sustainability. These practices are designed to optimise water and nutrient use, improve plant health, and increase resilience to environmental stresses.

Climate-resilient technologies for fruit crops focus on enhancing productivity, conserving resources, and mitigating climate risks such as erratic rainfall, temperature fluctuations, and prolonged droughts. These technologies aim to improve the sustainability and profitability of fruit cultivation while ensuring environmental conservation.

Table 10: Horticulture Practices

Horticulture Practices	Main resilience feature and contribution to the climate agenda	Main benefits expected	Potential impact under the NDKSP
Plantation on Ridges	Improves drainage in high rainfall areas, prevents waterlogging, and enhances root aeration during dry periods.	Reduces root diseases, improves plant growth, and enhances nutrient uptake.	Increased plant survival rate, improved productivity, better soil health.
Mulching	Conserves soil moisture, reduces soil erosion, regulates soil temperature, and suppresses weeds.	Enhances water-use efficiency, prevents soil degradation, and improves crop yield.	Reduced water stress, lower irrigation costs, improved soil organic matter.
Micro irrigation and fertigation	Optimizes water and nutrient use, reduces evaporation losses, and ensures efficient input application.	Enhances water-use efficiency, minimizes nutrient leaching, and improves plant health.	Increased crop productivity, reduced input costs, better soil fertility management.
Soil moisture management in rhizosphere	Help maintain adequate soil moisture and nutrients within root zone of newly planted saplings	Improves plant resilience against extreme weather, and maintains ecosystem balance.	Higher crop sustainability, reduced climate risk of high temperature, enhanced farmer incomes.
Canopy Management (Training, Pruning, Staking)	Enhances light penetration, air circulation, and disease prevention.	Increases fruit quality, prevents pest infestations, and improves overall plant health.	Higher yield per plant, better-quality produce, reduced disease incidence.
Integrated Pest and Nutrient Management (IPNM)	IPNM boosts soil health, cuts chemical use, and supports biodiversity, aiding climate resilience.	IPNM enhances crop yield, soil health, boosts crop productivity, enhances nutrient use efficiency, reduces input costs, and minimizes environmental impact.	IPNM promotes sustainable agriculture, enhances food security, improves crop resilience, conserves natural resources, and supports climate adaptation efforts.
Bagging of Fruits	Protects fruits from pests, diseases,	Improves fruit quality, reduces pesticide use,	Increased market value,

Horticulture Practices	Main resilience feature and contribution to the climate agenda	Main benefits expected	Potential impact under the NDKSP
	sunburn, and excessive moisture loss.	and prevents physical damage.	reduced post-harvest losses, improved food safety.
Good Harvesting Practices	Reduces post-harvest losses by adopting scientific harvesting and handling methods.	Maintains fruit freshness, minimizes mechanical damage, and extends shelf life.	Better market prices, improved farmer income, reduced food wastage.

Capturing FFS data through Mobile App

The activities of each FFS are captured by the facilitator through an FFS mobile application, which acts as a vehicle for demonstration and adoption of climate resilient agriculture technologies in the village. The features of FFS App include

- Geo-fencing of FFS & Control plots
- Registration of Host and Guest farmers
- Scheduling of FFS classes
- Curriculum of each session to be implemented during life cycle of the crops
- Functionality to capture all activities of FFS viz. Photos of attendance, technologies demonstrated, observations of crops and pests, diseases, weeds etc.
- FFS class conclusion with action plan based on observations and advisories for next week/ fortnight
- Functionality for Coordinators and SDAOs to review the quality of FFS conducted
- Functionality for facilitators to capture adoption of the technologies on other farmers' fields
- Crop yields and cost of cultivation data can be captured
- Creates repository of crop related data analysis to be used for generation of advisories through AI/ML
- FFS App will have integration with other APIs and existing apps to ensure that the curriculum of FFS can respond flexibly to weather changes, pest outbreaks, and field conditions
- The App will be supplemented by easily retrievable SOPs and reading material which can be accessed by the field staff as well as the host and guest farmers. This will help knowledge transmission on a variety of best practices in standardised fashion
- The FFS also serves as an important data repository for use by various organisations for research purposes

Digital Farmers School (DFS)

The concept of the Digital Farmers School expands the reach and accessibility of agricultural education by providing online learning platforms for farmers. Unlike traditional Farmer Field Schools (FFS), which can only accommodate a limited number of participants, Digital Farmers Schools allow many farmers to attend sessions remotely, overcoming physical space limitations. This innovative approach facilitates direct interaction between farmers and experts or scientists, enhancing knowledge exchange and fostering community engagement. Additionally, sessions are recorded, enabling farmers to revisit the content at their convenience, which is particularly beneficial for reinforcing learning and accommodating diverse schedules. The project proposes to conduct DFS for major crops and technologies in the project area targeting both the farmers, SHG members, project functionaries, and GKVS members.

Involvement of FPOs in CRT dissemination

The farmer-members of these FPOs need agro-met advisories and up-to-date information about the climate resilient technologies. The project will involve the FPO directors in the extension process through the project functionary. Training and exposure visit activities will be implemented for these target groups, who will disseminate the information to the rest of the officials. The commodity groups in the FPOs will be involved in the Physical and or Digital FFS.

On-farm production of Bio-pesticides and Bio-agents

On-farm production of Bio-pesticides

The excessive use of chemical pesticides has led to ecological imbalances, pest resistance, environmental pollution, and food safety concerns. To address these challenges, promoting bio-pesticides as a sustainable alternative is crucial. Bio-pesticides, derived from natural materials such as bacteria, fungi, and plant-based extracts, offer effective pest and disease management while preserving soil health, biodiversity, and environmental sustainability.

Globally, up to 42% of potential crop yields are lost due to plant pathogens. While chemical pesticides are widely used for pest and disease control, their indiscriminate application results in resistance, resurgence of pests, and environmental contamination. In contrast, bio-pesticides provide an eco-friendly solution, ensuring sustainable crop protection and food safety. They are an essential part of Integrated Pest & Disease Management (IPDM), complementing other sustainable pest & disease control practices.

Common microbial biopesticides include:

- *Bacillus subtilis*, *Bacillus pumilus* – Controls bacterial and fungal diseases.
- *Trichoderma harzianum*, *Trichoderma virens* – Antagonistic fungi effective against soil-borne pathogens.
- *Pseudomonas fluorescens*, *Pantoea agglomerans* – Enhances plant disease resistance.

- *Streptomyces lydicus*, *Coniothyrium minitans* – Controls fungal pathogens in crops.
- *Bacteriophages* – Used against bacterial pathogens like *Xanthomonas* spp. and *Pseudomonas syringae*.

To promote on-farm production of bio-pesticides, the project will support:

- Farmers, SHGs will receive training on the production, application, and benefits of bio-pesticides.
- Practical demonstrations will be conducted in Farmer Field Schools (FFS) to showcase the effectiveness of bio-pesticides.
- Farmers adopting on-farm bio-pesticide production will receive financial support through Direct Benefit Transfer (DBT).
- Support will be provided for setting up biopesticide production units at the farm level for individual farmers, SHGs.
- Collaboration with institutions like NIPHM for technical guidance and access to research-backed bio-pesticide formulations.

On-Farm Production of Predators and Parasites

The indiscriminate use of synthetic chemicals has led to pest resistance, resurgence, and environmental pollution. Promoting on-farm production of natural enemies such as predators and parasites can help restore ecological balance while reducing dependency on harmful pesticides.

Predators and parasites are two distinct biological agents that play important roles in ecosystems and can have significant impacts on the populations of other organisms. Predators help control prey populations, preventing them from overpopulating and depleting resources. This, in turn, can have cascading effects on the entire ecosystem, affecting the abundance and behaviour of various species. Parasites are organisms that live in or on another organism (the host) and derive their nutrients and shelter from the host. Unlike predators, parasites do not typically kill their host immediately. Parasites can have a variety of effects on host populations. They can reduce the fitness of the host (weakening but not necessarily killing them), influence host behaviour, and sometimes even regulate host populations indirectly. Example - *Bracon* spp., *Trichogramma* spp., bee fly, *Apis* spp., salvinia weevil etc.

To promote on-farm production of beneficial predators and parasitosis, the project will implement the following interventions:

- Training for farmers, SHGs on mass rearing and release techniques of predators and parasitosis with focus on identification, breeding, and utilization of bio-control agents.
- On-farm demonstrations through Farmer Field Schools (FFS) to showcase biological control effectiveness.

- Exposure visits to successful biocontrol farms and agriculture research centers.
- Direct Benefit Transfer (DBT) support for individual farmers, SHGs to establish on-farm bio-control units.
- Technical guidance from National Institute of Plant Health Management (NIPHM) and linkages with government agencies and biological control labs for access to cultures of natural enemies.

By promoting on-farm bio-pesticide and bio agents' production, the project aims to reduce input costs, enhance farm productivity, minimize chemical dependency, and promote climate-resilient and eco-friendly farming.

Conservation and multiplication of climate resilient seeds

The conservation and multiplication of climate-resilient seeds play a crucial role in ensuring agricultural sustainability and food security in the face of climate change. These seeds, which include traditional, drought-tolerant, flood-resistant, and disease-resistant varieties, help farmers adapt to changing climatic conditions while maintaining stable crop yields.

Objective:

1. Enhance climate-resilient agriculture by increasing the availability of resilient seeds for cereals, pulses, and oilseeds.
2. Increase smallholder farmer participation in the seed production of resilient varieties.
3. Empower farmers and FPCs to create economic stability.
4. Encourage the conservation and multiplication of indigenous seed varieties.
5. Preserve agricultural biodiversity.
6. Strengthen food security.

To accomplish these objectives, the project will implement the following sub-activities across the project area.

1. Seed production
2. Project support for conservation of local seeds
3. Incentive for village seed production programme
4. Support to Taluka seed farm
5. Seed Production of climate resilient varieties

The project is actively promoting the adoption of climate-resilient crop varieties in the project area. These varieties, identified in Section 3.4.3.1 are specifically selected to withstand climate-related challenges such as erratic rainfall, rising temperatures, and increased pest pressure. To ensure their widespread availability, the project supports farmers in the production of quality seeds at the local level.

By facilitating the production of foundation and certified seeds, this initiative enhances the resilience of farming systems while strengthening agricultural sustainability and food security.

A robust seed production system for climate-resilient varieties plays a crucial role in stabilizing food production amidst changing agro-ecological conditions. This approach not only empowers farmers but also ensures long-term adaptation to climate change by promoting locally available, high-quality seeds.

Situation Analysis:

Nucleus and breeder seeds are produced by ICAR, SAUs, and other research institutes, forming the foundation for certified seed production. The project collaborates with FPOs/FPCs in seed production, with Mahabeej providing technical expertise and buyback support, ensuring availability and distribution of resilient varieties.

The classification of seeds ensures quality and genetic integrity in agriculture. The main seed classes are:



Figure 10:Classes of seeds

Mahabeej has been a key player in seed production for over 40 years, producing, processing and marketing seeds for over 50 crops and 225 varieties. It collaborates with 50,000 seed growers across 2.5 lakh acres during kharif and rabi seasons, ensuring quality through effective monitoring, distribution, and grower incentives.

In Maharashtra, the Government of India and the Government of Maharashtra are promoting Farmer Producer Companies (FPCs). FPCs are actively involved in producing certified seeds of climate-resilient varieties, supported by programs like NFSM and NMOOP. Over 60 FPCs produce foundation and certified seeds under the guidance of the Seed Certification Agency and Mahabeej, contributing to seed availability and adoption of new varieties.

Seed Authentication, Traceability and Holistic Inventory (SATHI) is a Government of India seed portal that automates the entire seed supply chain, covering production, certification, licensing, inventory, sales, and traceability. All seed producer organizations must use it for registration and processes. The portal streamlines seed management and ensure transparency. The portal is useful in

1. Enhancing seed quality and purity through digital management of the seed life cycle.
2. Ensuring accountability with seed traceability and advanced modules.

3. Improving efficiency by integrating technology into seed certification and reducing errors.
4. Streamlining processes with faster registration, approvals, and reporting.
5. Offering an eco-friendly approach to seed operations.

The project aims to strengthen the supply chain for climate-resilient seed varieties, including drought-resistant, short-duration, and salinity-tolerant crops. By utilizing MAHABEEJ and FPC networks, it facilitates seed production, processing, storage, and certification.

Guiding principle for project support on Seed Production of Climate-Resilient Varieties

Steps followed for implementation and providing assistance to seed growers

1. Identification of the climate resilient varieties of crops dominant in project area
2. Identification of seed production agencies
3. Registration of farmers as seed growers
4. Provision of breeder and foundation seed to the registered growers
5. Inspection by Seed certification agency
6. Verification by project officials
7. Processing of seed-by-seed production agencies
8. Disbursement of financial assistance

The details of the procedure followed is narrated below.

1. **Identification of the climate resilient varieties of crops dominant in project area:** - The project identifies climate-resilient crop varieties through approval by the Technical Advisory Committee, chaired by the Principal Secretary (Agriculture) and including SAU Vice Chancellors and breeders. Varieties developed by SAUs, ICAR, and other institutes, recommended for regions like Marathwada, Vidarbha, and North Maharashtra, are promoted for seed production by growers. Recommended varieties from SAUs, ICRISAT, CICR, Mahabeej, and other institutes are listed in Table.
2. **Identification of seed certification and seed production agencies:** - Seed production is regulated by the Seed Act and Maharashtra Seed Certification Agency (SCA), Akola. The project has identified Mahabeej as the lead seed production agency since Kharif-2018, promoting farmers in project villages. Additionally, licensed Farmer Producer Companies (FPCs) are encouraged to organize seed growers for foundation and certified seed production.
3. **Registration of farmers as seed growers:** - Under the Seed Act, 1968, and Indian Seed Certification (1988 & 2013), the Maharashtra State SCA announces the seed production registration process each season. Government bodies, cooperatives, FPCs, farmer groups, or individuals can implement the program. Farmers, regardless of landholding size, can register through seed production agencies and must pay registration and inspection fees to the certification authority.

4. **Provision of breeder and foundation seed to the registered growers:** - The MSSCL and FPCs provide Breeder and/ or Foundation seed to the registered seed growers as per the requirement of the approved programme. Project facilitates both the MSSCL and FPCs as well as the seed growers to avail the breeder and foundation seed before the season.
5. **Inspection and hand holding by Seed certification agency and project official:** - Processing and purchase of seed-by-seed production agencies: -The seed production agencies (MSSCL and FPCs) process the seed to ensure the quality as per the standards mentioned in the Seed Act. Seeds conforming to the requisite standards are purchased by the agencies.

Promotion of indigenous/local seed varieties

Indigenous seeds, native to a region and adapted to local climates, soils, and pests, are crucial for biodiversity, sustainability, and resilience to climate change. These seeds hold economic, ecological, and cultural value and contribute to sustainable agriculture and food security. Conserving them preserves traditional knowledge and enhances adaptability to environmental changes.

The Protection of Plant Varieties and Farmers' Rights Act, 2001 is an act to provide for the establishment of an effective system for protection of plant varieties, the rights of farmers and plant breeders and to encourage the development of new varieties of plants. It recognizes farmers' contributions to conserving plant genetic resources, promoting agricultural growth, and stimulating investment in research and development, ensuring access to quality seeds for farmers.

The above Act provides for the conservation of indigenous varieties. In line with this, the conservation of indigenous varieties is being implemented through the project.

The salient features of chapter VI titled 'Farmers Right' under the said Act are as follows:

- A farmer who has bred or developed a new variety shall be entitled for registration and other protection in like manner as a breeder of a variety under this Act.
- The farmers' variety shall be entitled for registration if the application contains declarations as specified in clause (h) of sub-section (1) of section 18 in this Act.
- A farmer who is engaged in the conservation of genetic resources of land races and wild relatives of economic plants and their improvement through selection and preservation shall be entitled in the prescribed manner for recognition.
- A farmer shall be deemed to be entitled to save, use, sow, resow, exchange, share or sell his farm produce including seed of a variety protected under this Act in the same manner as he was entitled before the coming into force of this Act.

Box: Rahibai Soma Popere, born in 1964, is a renowned Indian farmer and seed conservationist, widely known as the "Seed Mother" of Maharashtra. She has dedicated her life to preserving indigenous seeds and promoting sustainable agriculture in the Akole block of Ahmednagar district. Through her efforts, she has conserved hundreds of native seed varieties, including paddy, millets, pulses, oilseeds, and vegetables, enhancing biodiversity and food security.

Popere has established in-situ germplasm conservation centers and nurseries, sharing 43 landraces of 17 crops with over 210 farmers in seven villages. She also promotes organic farming, perennial kitchen gardens, and the use of wild food resources, emphasizing their role in seed sovereignty and nutritional security. Recognized as a woman leader in agriculture, her work inspires farming communities to preserve agro-biodiversity and adopt sustainable practices.

Project support for conservation of local seeds:

- Encourage farmers to conserve local and native seeds through awareness and support initiatives.
- Facilitate the distribution of conserved seeds to other farmers for their use.
- Provide guidance and training to farmers on increasing the production of local and native seeds and organise study tours, exposure visits.
- Raise awareness about the advantages and quality of existing indigenous seeds.
- Conduct field visits to demonstration plots showcasing conserved seeds to share best practices.
- Assist farmers with the registration process for conserved local and native seeds.
- Ensure that District Agriculture Officers accept applications from farmers for registration.
- Forward the applications to the Director (Research) of the relevant Agricultural University for further processing and collaborates with organizations dedicated to seed conservation.
- Incentivise the registered farmers with recognition and monetary assistance.

The conservation and multiplication of indigenous or local seed varieties is a vital initiative under the NDKSP project. The project focuses on conserving and multiplying indigenous seed varieties to preserve biodiversity and enhance food security.

Project will promote individual farmers, SHGs and FPCs to create seed bank in project villages with an objective to be prepared for any weather contingency, especially in the need of resowing due to late onset of monsoon or during early dryspell.

Incentive for village seed production programme

To encourage the maximum participation of villages in the rural livelihood program, it is essential to provide incentives to participating villages. The villages eligible to produce a substantial quantity of high-quality seeds by implementing the project supported seed production program will be granted special incentives. The incentivization model will be based on Government of India guidelines in coordination with MSSC.

The incentive system for such villages producing higher quantities of CRV seed will be implemented after the willingness of the villages through GKVS resolution. Such villages will be treated as seed hubs, and the registered seed growers' groups in such villages will be supported with seed supply chain infrastructure.

Support to Taluka seed farm

The Department of Agriculture has established the Taluka Seed Farms (TSF) at Taluka places to undertake the seed production programmes in coordination with the MSSC. The Department of Agriculture implements seed production programs by following the national standards and protocols, which are established by the Seed Certification Agency. These protocols are applicable for production of foundation and certified seed of those varieties developed by the public sector research institutes. It is necessary to support the production programs at these TSFs to make them more efficient by preparing foundation seeds. For this purpose, encouragement will be provided to the production programs at the TSFs through the project. There are 122 TSFs in the project area with 2270 ha of cultivable area for seed production. The Project will support TSFs by bearing the cost of foundation seed and certification charges. Also, the technological support with the SOPs of climate resilient technologies will be rendered. Moreover, the climate resilient technologies will be demonstrated on the farms of TSFs in the project area.

Table 11: Activities, Beneficiary types and Output Indicators for A3

Sub-component	Activity	Beneficiary type (Individual/Farmer Group/ Community)	Output Indicators
Climate Resilient Technologies for Field crops	FFS for field crops	Individual Farmer	No of FFS No of Farmers benefited
Climate Resilient Technologies for Fruit crops	FFS for fruit crops	Individual Farmer	No of FFS No of Farmers benefited
On-farm production of Bio-	On-farm production of Bio-pesticides	Individual farmer FPC SHG	Number of units installed in project villages

Sub-component	Activity	Beneficiary type (Individual/Farmer Group/ Community)	Output Indicators
pesticides and Bio-agents	On-Farm Production of Individual farmer	Individual farmer FPC SHG	Number of units installed in project villages
Conservation and multiplication of climate resilient seeds	Seed Production of climate resilient varieties	Individual Farmer	No of Farmers benefited Area covered
	Promotion of indigenous/local seed varieties	Individual Farmer	No of Farmers benefited
	Incentive for village seed production programme	Community	No of villages
	Support to Taluka seed farm	Individual	No of TSF

Section 3

Component B - Promoting Precisions Agriculture for GHG Emissions Management



Section 3

Component B: Promoting Precision Agriculture for GHG Emissions Management

3.1 Component overview:

Promoting precision agriculture will focus on reducing greenhouse gas emissions, sequestering carbon, cutting methane emissions, and maintaining economically viable crop yields. Considering the varying carbon and water footprints of different crops and crop rotations, the project will introduce crop-specific moisture conservation techniques, develop agroforestry modules, and implement new methods to enhance soil organic carbon.

GHG Accounting:

The quantification of GHG emissions is an important step in managing and ultimately reducing emissions, as it provides an understanding of the project's GHG mitigation potential. Further, the Paris Agreement also mandates reporting of assumptions and methodological approaches including those for estimating and accounting for anthropogenic greenhouse gas emissions to achieve the goals of Article 2.

It is proposed to incorporate agriculture, water and nutrient management, agroforestry, etc., practices to build resilience in agriculture while reducing the GHG emissions and enhancing carbon stocks.

The World Bank has adopted the Ex-Ante Carbon-balance Tool (EX-ACT), developed by FAO in 2010, to estimate the impact of agricultural investment lending on GHG emissions and carbon sequestration in the project area. EX-ACT is a land-based appraisal system that allows the assessment of a project's net carbon-balance, defined as the net balance of CO₂ equivalent GHG that are emitted or sequestered because of project implementation compared to a no project or without project scenario.

Objectives

- To assess the spatial and temporal variations in soil physical, chemical and microbial properties in relation to GHG emission from project area
- To quantify the effect of Climate-smart agricultural practices like Direct Seeded Rice, Alternate Wetting and Drying, Precision nitrogen management on GHG emissions (methane and nitrous oxide) from important growth stages of rice at certain intervals in NDKSP sites of Maharashtra state
- Training and capacity development of different stakeholders on GHG monitoring, reporting and verification and Climate-smart agricultural practices like DSR, AWD and Precision N management in rice production.

Methodology for GHG Accounting:

Data Collection: GHG samples will be collected weekly from selected plots using the Closed Chamber Technique and analysed via Gas Chromatography for methane and nitrous oxide emissions. Soil samples will also be collected to assess properties like organic carbon, nutrient levels, and microbial activity.

Data Analysis and Scaling: The FAO model, The EX-Ante Carbon-balance Tool (EX- ACT) renamed the Environmental externalities Accounting Tool will be applied to upscale GHG emissions data across NDKSP sites for spatial and temporal trends. This tool provides a consistent framework for estimating and tracking the impact of agricultural interventions on greenhouse gas (GHG) emissions. It uniquely covers the entire agricultural sector, including Agriculture, Forestry, and Other Land Use (AFOLU). EX-ACT's identifies climate mitigation outcomes of agricultural interventions at all stages (ex-ante, during, ex post) and levels (local, regional, national). The tool relies heavily on the Intergovernmental Panel on Climate Change (IPCC) methodology for GHG emissions inventories. It will serve as a vital resource for promoting sustainable agricultural practices and achieving emissions reduction goals in the NDKSP project sites.

The project will partner with a technical institution to implement this activity.

Sub-component B1: Balancing Methane and Paddy Production - Through this sub-component, the project will deploy precision agriculture techniques for promoting low-methane paddy¹⁹, as substantial methane is generated when organic material in flooded rice fields decompose. With adoption of greater water-use efficiency (WUE) and fertilizer-use efficiency (FUE) practices, the project will also identify and monitor those practices which also help reduce GHG emissions. Suitable pilots will be conducted to ascertain the combination of practices that will generate most competitive methane emission. Investment under this sub-component will include extensive mobilization support, equipment for measurement, reporting and verification (MRV) of CH₄, N₂O and installation of Bowen towers and installation of sensors in KVKs and Agriculture Universities for moisture measurement and evapotranspiration.

Given the diverse agro-climatic zones in the state, the project will expand and deepen technology interventions to sustain crop productivity while keeping emissions competitive, including precision agriculture techniques for promoting low-methane paddy such as Direct Seeded Rice (DSR), System of Rice Intensification (SRI), Alternate Wet and Dry (AWD), water-straw-fertilizer co-management, and Zero Tillage for promoting low-methane paddy. Activities will involve pilot studies and demonstrations of various technologies for methane reduction from paddy, combining innovations in digital agriculture, crop and microbial

¹⁹ For example, DSR-Direct Seeding Rice

genetics, and nutrient use efficiency to reduce emissions and enhance soil carbon sequestration. The project will collaborate with ICAR-Central Rice Research Institute (CRRI), Cuttack, a leading research institute, to assess methane emissions from rice fields in the project area, share protocols for low-methane rice production technologies, and build the capacity of extension agents and farmers in adopting low methane rice production. The project will provide targeted extension services to equip farmers with the knowledge and skills needed to transition from traditional flood irrigation and excessive fertilizer use to more sustainable practices. Additionally, it will promote strong collective action among farmers, ensuring coordinated efforts in water management and sustainable rice production. By fostering collaboration, the project will help farmers adopt best practices that enhance productivity and reduce emissions while preserving natural resources. Additionally, participatory research will explore alternative cereal crops for farmers willing to replace second-season paddy, contributing to methane emission reduction.

Building on the success of PoCRA in reducing agricultural emissions, the project will develop a digital measurement, reporting and verification (MRV) system to measure, report, and verify rice GHG emissions and carbon sequestration in selected villages while supporting carbon trading pilots that will facilitate smallholders' access to voluntary carbon markets.²⁰ A monitoring network combining remote sensing and in-situ measurements will be established. The project will leverage digital MRV solutions, integrating innovative remote sensing technology and big data analytics—such as machine learning and artificial intelligence—to assess methane emissions at scale and analyse key factors influencing rice GHG emissions, including production areas, crop rotation, residue management, and water practices. In-situ monitoring will capture direct emissions data alongside variables like climate, soil types, fertilizer use, and agronomic practices. The MRV system will generate insights on best practices for monitoring rice climate mitigation efforts and evaluating their impact on GHG emissions, with a focus on methane reduction and smallholders' access to voluntary carbon markets.

Sub-component B2: Reduced Energy Consumption in Agriculture: To reduce emissions while building climate resilience, the project will support the transition from conventional to non-conventional energy sources for powering agricultural pumps in the State. The sub-component will make investments in promoting (i) use of solar technologies in water pumping, farm operations, and food storage, and (ii) generation and use of bioenergy from agriculture biomass. This will not only ensure proper management of agricultural waste but also generate electricity. Under this sub-component a sizeable amount of money will be spent on alternative energy – particularly solar panels and subsequent gravity-based drainage systems.

²⁰ The current methods for reporting GHG emissions and validating or verifying emission reductions are often costly, error-prone, and time-consuming, relying heavily on manual processes and in-person surveys. However, digital MRV technologies offer a more cost-effective solution by streamlining data collection, processing, and quality control. In this project, technologies such as smart sensors, satellites, drones, cloud computing, artificial intelligence, the Internet of Things, and blockchain encryption will be leveraged to enhance the efficiency and accuracy of monitoring, reporting, and verification processes.

This sub-component focuses on generating alternative energy while promoting reduced energy consumption through precision farming methods. On the other hand, to promote bioenergy, the project will support biomass briquettes/pallets and biochar manufacturing. These initiatives aim to improve soil fertility, carbon sequestration through biochar, reduce stubble/trash burning by utilizing surplus agricultural residue while providing an additional source of income to farmers or Farmer Producer Organizations (FPOs) through the sale and/or processing of surplus agriculture residue.

To efficiently manage energy consumption and reduce emissions, the project will establish strategic partnerships with leading research institutes to design crop-specific packages of practices with the goal to effectively manage moisture and temperature variability, reduce carbon and water footprints, and sustain optimal crop productivity. Additionally, the project will develop web and mobile app-based tools to monitor water-use efficiency thereby optimizing energy consumption. The selection and deployment of renewable technologies and their locations will be determined jointly by beneficiaries and consortia partners. However, the project will conduct feasibility assessments to ensure the suitability and effectiveness of the selected technologies.

Sub-component B3: Promoting Crop Diversification to Enhance Carbon Sequestration: There are three key cropping systems followed in Maharashtra; (a) pulses, (b) high value commodities with regional and global demand i.e. cotton, soyabean and sugarcane, (c) high value horticulture including grapes for wine industries. The state agriculture is well diversified, but high climatic variability makes it vulnerable. This sub-component will focus on climate safeguard measures including identifying and supporting fallow lands and areas currently under traditional no-till zones for crop diversification initiatives. In addition, farmers will be encouraged to adopt millets and tasar silk - crops well-suited to specific regions of the State. To promote this transition, the project will invest in pilot programs aimed at demonstrating the benefits of millet cultivation, planting material (carbon sink), catchment treatment, and robust satellite-based monitoring systems. It is estimated that 10-20 percent of the target area will be successfully transitioned to diversified cropping systems under this initiative.

Carbon sequestration through plantation activities is a crucial strategy for mitigating climate change by capturing and storing atmospheric carbon dioxide (CO₂). This project will enhance carbon sequestration by promoting a variety of plantation practices, including agroforestry, bamboo cultivation, horticulture, along with protected cultivation. Lessons from POCRA indicate that these practices not only capture carbon but also contribute to biodiversity, improve soil health, and sustain local livelihoods. In the drought-prone regions of Marathwada and Vidarbha, agroforestry will be prioritized to build resilience in these areas. The project will also focus on promoting bamboo plantations, particularly on farmers' fields,

community land, and arable wastelands. Bamboo will be planted along riverbanks, farm bunds, and other similar areas. In addition, the project will provide training on bamboo cultivation and processing to help farmers maximize its potential. Similarly, horticulture will be promoted through subsidies and training, encouraging farmers to grow selected fruits. The project will target vulnerable groups, such as marginal and small-scale farmers, women, tribal farmers, and farmers with physical disabilities, to engage in the cultivation of horticultural crops, agroforestry, and bamboo. Cultivating along contour lines on sloped land will help reduce soil erosion and water runoff, while also increasing soil organic carbon through the decay of crop residue. This practice will lower CO₂ emissions and open new areas for irrigation, further enhancing the sustainability and productivity of the land.

Detailed description of Subcomponents

3.2 Sub-component B1: Balancing Methane and Paddy Production:

Key activities:

This initiative will promote low-methane paddy cultivation by using precision agriculture to minimize methane emissions from flooded fields. By improving water and fertilizer-use efficiency, the project will identify and monitor practices that also cut greenhouse gas emissions. Considering the state's diverse agro-climatic zones, the project will scale up technology interventions to sustain crop productivity while reducing emissions. Precision agriculture techniques like Direct Seeded Rice (DSR), System of Rice Intensification (SRI), Alternate Wet and Dry (AWD), zero tillage, and water-straw-fertilizer co-management will be promoted for low-methane paddy cultivation. Pilot studies and demonstrations will integrate digital agriculture, crop genetics, and nutrient efficiency to cut emissions and enhance soil carbon.

In collaboration with ICAR-CRRI, the project will assess methane emissions, share protocols, and train extension agents and farmers in sustainable rice production. Targeted extension services will support farmers in shifting from flood irrigation and excessive fertilizers to efficient practices. Collective action will be encouraged for better water management, while participatory research will explore alternative cereal crops to reduce second-season paddy methane emissions.

Building on NDKSP's success in reducing agricultural emissions, the project will develop a digital measurement, reporting and verification MRV system to track rice GHG emissions and carbon sequestration in select villages while supporting smallholders' access to voluntary carbon markets. A monitoring network will combine remote sensing and in-situ data to assess methane emissions, crop practices, and environmental factors. Using AI, machine learning, and big data analytics, the system will generate insights to refine climate-smart rice farming and enhance carbon trading opportunities for farmers.

Balancing Methane in Paddy cultivation refers to the process of reducing methane (CH₄) emissions produced in flooded rice fields. Methane is a potent greenhouse gas (GHG) that forms in anaerobic (oxygen-lacking) conditions typical of conventional flooded paddy fields. De-methanation focuses on altering farming practices to reduce or prevent these anaerobic conditions, thereby minimising methane emissions during rice cultivation.

In Maharashtra, paddy area occupies between 9 and 10% of total crop area. The data from 2020-21, 2021-22, 2022-23 and 2023-24 (tentative) is presented below.

Table 12: Paddy area in Maharashtra

Crop	Area ('000ha)			
Maharashtra	2020-21	2021-22	2022-23	2023-24 (tentative)
Rice	1473	1473	1522.93	1529.02
Total crop area	15948	16379	15859.6	15564.1
%	9.2	9	9.6	9.8

Rice cultivation is being explored as a critical area to improve technical efficiency and reduce agricultural emissions intensity.

Promotion of improved rice cultivation methods

Methanation occurs in flooded rice fields when organic material in the soil decomposes in the absence of oxygen. This leads to the production of methane, which escapes into the atmosphere and contributes to global warming. Rice farming accounts for a significant portion of global methane emissions from agriculture, making it crucial to mitigate this effect. The high methane output not only impacts the environment but also indicates inefficiencies in water and nutrient management in rice fields.

The project will expand and deepen technology interventions to sustain productivity of Paddy while keeping its GHG emissions competitive, including precision agriculture techniques such as Direct Seeded Rice, Systemic Intensification of Rice, Alternate wet and dry method (AWD) and Zero Tillage for promoting low-methane paddy. ICAR-Central Rice Research Institute (CRRI), Cuttack is a leading institute in research. Project will have formal tie-ups with CRRI in the areas of assessment of Methane emission from rice fields in project area, sharing the protocol for rice production with less methane emission, and building capacities of the extension functionaries in project area.

Direct Seeded Rice (DSR): Direct Seeded Rice is an establishment method where rice seeds are sown directly into the main field rather than transplanting seedlings from a nursery. This system promotes more efficient use of water and nitrogen, reduces greenhouse gas emissions, and lowers labour demand, thereby avoiding the drawbacks associated with the

transplanted system. The DSR method can be implemented either by sowing pre-germinated seeds into puddled soil (wet DSR) or into a well prepared non-puddled seedbed (dry DSR).

1. **Dry DSR:** In dry direct-seeded rice (D-DSR), dry seeds are sown directly into the main field without prior germination. This method is commonly used in rainfed uplands, medium lowlands, lowlands, and deepwater ecologies during the wet season. This method offers several benefits, including up to 30% water savings compared to conventional transplanted rice and a reduction in methane (CH₄) emissions by 18-20%. Additionally, D-DSR reduces labour requirements, enhances seedling emergence, and decreases the risk of lodging.
2. **Wet DSR:** This practice is recommended when a late onset of monsoon rains delays timely sowing. By applying irrigation water and using sprouted seeds, farmers can sow the crop on time through the direct seeding method, which uses less water compared to traditional transplanting. When proper management practices are followed, the yield of Wet-DSR is comparable to that of transplanted rice. Additionally, Wet-DSR increases water productivity by 0.3 to 0.4 kg rice m⁻³ water.

Advantages of direct seeding

No significant reduction of yield under optimal conditions

- Savings on irrigation water by 12-35% under efficient water management practices
- Reduces labour and drudgery by eliminating seedling uprooting and transplanting
- Reduces cultivation time, energy, and cost
- No plant stress from transplanting
- Faster maturation of crops
- Lower GHG emissions
- Mechanised DSR provides employment opportunities for youth through service provision business model
- Increases total income by reducing cost of cultivation

3. **DSR on permanent beds with no tillage:** This is a conservation agriculture technique adopted by innovative farmers in the state. The protocol for no-till rice includes dibbling of rice seeds on the raised beds followed by weed management by using herbicides and retaining the roots of weeds in the soil. The harvesting of rice is done by cutting the crop near to the ground, retaining the roots in the soil. The same beds are used for the next crop without ploughing the land.

Box: Innovative DSR practice- SRT

Innovative farmer Shri Chandrashekhar Adavale from Raigad district is practising DSR on permanent beds on his 20-ha farm since 2011. The technique is popular by the name 'Saguna Rice Technique- SRT' and more recently as 'Saguna Regenerative Technique- SRT'. SRT is being adopted slowly by some small and marginal farmers cultivating rice on medium to light soil in Konkan and Vidarbha region. There are many advantages of SRT including resource conservation, reducing water footprint, reducing women drudgery, reducing cost of cultivation, reducing gap between two crops, enhancing soil fertility, and increasing crop yield.

Alternate wetting and drying (AWD): This is a water-saving technology that lowland (paddy) rice farmers can apply to reduce their water use in irrigated fields. In AWD, irrigation water is applied to flood the field a certain number of days after the disappearance of ponded water. (AWD) is a simple and inexpensive way of reducing water consumption in rice production by 30%, thus enabling farmers to cut down on production costs without yield penalty. AWD entails periodic draining of the field to a certain threshold, usually 15 cm below the soil surface, and re-flooding. A perforated tube placed in the soil enables the farmer to monitor the water level below the soil surface to determine when to irrigate. AWD technology has also been proven to effectively mitigate greenhouse gas (GHG) emissions, specifically methane (CH₄), from rice production by 30-70%, without causing a yield reduction. During the dry phases, the methane-producing bacteria are inhibited, thus, setting a condition to reduce GHG emission. It reduces water use by up to 30% and reduces GHG emissions by up to 50% while maintaining yields.

Alternate Wetting and Drying (AWD) represents a critical advancement in water and methane management for rice cultivation, offering a pathway to more sustainable agriculture. By reducing water use and methane emissions, AWD not only helps conserve resources but also contributes to global efforts in climate change mitigation. With proper implementation, AWD can provide a "triple win" by enhancing water efficiency, maintaining yields, and lowering the environmental footprint of rice production.

Systemic Intensification of Rice (SRI): The System of Rice Intensification (SRI) is an innovative agricultural methodology designed to increase the productivity of rice while minimising inputs and enhancing sustainability. The traditional practice of continuous flooding of the rice field produces a significant amount of greenhouse gas (GHG), specifically methane (CH₄). With SRI, soil moisture is kept at an optimum level, thus, reducing methane emission from rice production. Opting for organic fertilisers over the synthetic nitrogen fertilisers, which emit the GHG nitrous oxide (N₂O), also reduces the GHG contribution from rice production. This system is focusing on efficient water management, improved soil fertility, and optimal planting techniques.

The System of Rice Intensification (SRI) utilises several innovative practices to enhance rice production. It begins with transplanting young seedlings (10-15 days old) at wider spacings (around 25 cm x 25 cm), allowing greater access to sunlight, nutrients, and water. SRI emphasises intermittent irrigation instead of continuous flooding, promoting aerobic soil conditions and reducing water use by 30-50%. Enhanced soil aeration is achieved through less flooding and soil loosening practices, which improve root development.

Benefits of SRI

1. By promoting aerobic soil conditions through reduced flooding, SRI significantly cuts methane emissions by up to 50% or more since methane producing anaerobic conditions are minimised.
2. SRI can increase rice yields by 20-100% compared to traditional practices, depending on local conditions and the farmer's experience with the system.
3. SRI practices enhance soil fertility and structure, leading to increased organic matter and microbial activity, which are vital for sustainable farming.
4. The adoption of SRI techniques leads to substantial water savings, making it particularly valuable in regions facing water scarcity.
5. Farmers practising SRI often experience lower costs of production due to reduced water and fertiliser use, alongside higher yields, which can lead to increased income.
6. SRI systems can reduce methane emissions by 35-48% compared to conventional cultivation systems.

Table 13: Beneficiaries and output indicators of activities B1.1 and B1.2

Activity/Sub-activities	Beneficiary (Individual/Farmer Group/Community)	Type	Output Indicators
B.1.1 Promotion of improved rice cultivation methods (SRI/AWD/DSR/SRT) (individual)	Individual		Area covered under improved rice cultivation methods (hectares)

3.3 Sub-component B2: Reduced Energy Consumption in Agriculture:

To cut emissions and boost climate resilience, the project will support shifting from conventional to renewable energy for agricultural activities, wherever appropriate and relevant. It will explore opportunities to promote solar-powered water pumps, farm operations, and food storage, along with bioenergy generation from agricultural waste, ensuring efficient waste management while producing electricity.

This sub-component promotes alternative energy while reducing energy use through precision farming. To boost bioenergy, it will support biomass briquettes, biochar production, among other options. These efforts will enhance soil fertility, sequester carbon, reduce crop residue burning, and create income opportunities for farmers and FPOs.

To optimize energy use and cut emissions, the project will collaborate with leading research institutes to develop crop-specific practices for managing moisture, temperature, and resource efficiency. Web and mobile tools will track water-use efficiency to reduce energy consumption. Renewable technology selection will be guided by beneficiaries and partners, with feasibility assessments ensuring effectiveness.

To enhance climate resilience in agriculture, integrating alternative energy sources into farming practices which reduce dependency on non-renewable resources, optimize water management, and improve livelihoods.

NDKSP-II support:

1. Solar-powered PH Infrastructure
2. Promotion of bioenergy from agriculture biomass

1. Solar-powered PH Infrastructure

a. Solar-Powered Processing Units

Solar-powered processing units hold immense potential to revolutionize agriculture in Maharashtra. By reducing post-harvest losses, enhancing value addition, and ensuring energy sustainability, these units can significantly improve farmers' incomes and contribute to the state's agricultural growth. With supportive policies and widespread awareness, Maharashtra can become a leader in integrating solar technology into agriculture.

b. Solar Refrigeration Systems:

Solar refrigeration system is a cooling system powered by solar energy, designed to reduce reliance on conventional electricity or fossil fuels. This eco-friendly technology is particularly beneficial in areas with limited access to reliable power and is gaining traction in applications such as agriculture, healthcare, and domestic use.

2. Promotion of bio-energy from agriculture biomass:

India has embarked upon an ambitious energy transition journey with a target of fifty percent cumulative electric power installed electricity capacity from non-fossil fuel-based energy resources by 2030 and achieving net zero by 2070. To attain the ambitious renewable energy targets and to achieve self-reliance in the energy sector it is imperative that domestically available renewable energy alternatives are optimally utilised. One such alternative is modern bioenergy. With a large surplus of biomass and other waste available in the country, energy recovery from these resources is a viable solution. Modern bioenergy is unique as it provides

several social and environmental benefits apart from providing clean fuels. For example, bioenergy applications can help to mitigate air, water, and land pollution. (*Ministry of New and Renewable Energy, Dec 23, 2024*).

Biochar production

Biochar is a carbon-rich material produced by the thermal decomposition of organic biomass, such as field crop residues, under limited oxygen conditions (a process called pyrolysis). Biochar stores carbon in the soil for hundreds to thousands of years and thus, the level of greenhouse gases like CO₂ and CH₄ can be reduced significantly from the atmosphere. To make biochar from biomass material continuous biochar production machines or modified portable metallic kilns. It is used primarily as a soil amendment to improve soil fertility, enhance water retention, and sequester carbon, making it a valuable tool in sustainable agriculture.

Sources of field crop residues for biochar production

Farmers can use a variety of crop residues to produce biochar, including Rice husk and straw, Wheat straw, Maize stalks and cobs, Sugarcane trash, Cotton stalks, Groundnut shells, Soybean and Lentil residues.

Benefits of Biochar in agriculture production

Soil Improvement: Biochar enhances soil fertility and structure, promoting better root growth and nutrient retention. It increases water-holding capacity and reduces soil erosion, improving overall soil health.

Carbon Sequestration: Biochar is a stable form of carbon that can remain in the soil for hundreds or even thousands of years, helping to mitigate climate change by storing carbon dioxide.

Waste Management: Biochar can be produced from various organic materials, such as agricultural residues and forestry by products, providing a sustainable solution for waste management.

Demethanation: Biochar application helps in reduction of the process of methanogenesis due to its slow release of nutrients action and which is possible for slow decomposition, so it stops methanation.

Project support: Project will support farmers and FPOs for small to medium units of biochar production like 'biochar kiln' for pyrolysis of organic waste or crop residue on the farm itself which will be useful, for application in all types of crops, as fertilizer.

3.4 Sub-component: B.3 Promoting Crop Diversification to enhance Carbon Sequestration

Guiding principles:

- Both individual and communities shall be supported.
- Project will facilitate technical support to design and financial support shall be provided through DBT mechanism, wherever appropriate and relevant.
- Training and mobilisation activities shall be supported.
- Technical partners will be onboarded to develop SOPs, MRVs and related guidelines.

Activities to be implemented under these subcomponents are as follows:

- Agroforestry
- Bamboo plantation
- Horticulture crops/ plantation
- Protected cultivation
- Enhancing Millet Production

Maharashtra in 2023 held 8% of India's overall forest and tree cover, or 65,383 sq. km of 8,27,357 sq. km. The tree green cover area under agroforestry has changed from 1,06,303 km² (ISFR 2013) to 1,27,590 km² (ISFR 2023) at the national level, i.e., there has been a net increase of 21,287.05 km² of tree green cover area in agroforestry. In percentage terms, it reveals that the tree green cover in agroforestry has increased by 20.02% during 2013-2023. Maharashtra (14,524.88 km²) has the highest increase of tree cover followed by Rajasthan and Uttar Pradesh. In general, twenty-one States and UTs have shown an increasing trend in tree cover, which indicates that agroforestry is being promoted in these States.

Tree plantation programme is mainly implemented in the State through the Forest Department, FDCM and the Directorate of Social Forestry.

Table 14: Area under tree plantation programme

Year	Area under tree plantation programme		(Area in ha)
	Forest Dept.	FDCM	Social Forestry
2019-20	1,02,987	8,987	21,032
2020-21	7,234	2,361	3,302
2021-22	10,698	11,835	7,413
2022-23	18,164	2,523	5,557

Activities Planned by project under the Agroforestry Plantation

Under Agroforestry Plantation, the project will support activities aimed at encouraging farmers to plant a diverse range of locally suited tree species. These species are selected for

their economic viability, ecological benefits, and cultural significance. The planned activities include various types of plantations, carefully designed to optimize land utilization, enhance productivity, and promote the sustainable preservation of natural resources. Under this scheme farmers are expected to plant locally suited tree species, such as Saag, Ain, Shisam, Moh, Jambhul, Khair, Shivan, Hirda, Beheda, Amla, Dhavada, Kadu Nimb, Bell, Sonchafa, Gulmohar, Turmeric, Jatropha, Tamarind, Karanj, Jarul, Anjan, Arjun, Babul, Cirrus, Nilgiri, and Shevaga etc.

Type of plantations in Agroforestry

Following are the various types of plantations under Agroforestry activity, which the project will suggest and support.

Table 15: Various types of plantations under Agroforestry activity

Sr. No.	Type of plantations	Magnitude (no of plants/ha)
I	Peripheral and Boundary Plantation	-
II	Low Density Plantation	Less than 100
		>100 up to 500
III	High Density Block Plantation	500 to 1000 (3.5mX3.5m)
		>1000 to 1200 (3mX3m)
		>1200 to 1500 (2.5mX2.5m)
		>1500 (less than 2.5mX2.5m))

Peripheral and Boundary Plantation: To make potential use of the area occupied by these bunds around the periphery of the farmers' fields, tree species can be grown as peripheral/boundary plantations to add more income to the farmers' basket.

Low Density Plantation on Farmlands: Low Density Block Plantation without sacrificing the yield of the existing crops/cropping systems, shall be incentivized at the proportionate rates as applicable to per plant expenditure.

High Density Block Plantation: High-Density Block Plantations (HDBP) will be supported as a complementary income source for farmers. Farmers can establish block plantations on waste and degraded lands unsuitable for crops, turning them into productive assets. These plantations can serve as intermediate blocks, strip plantations, or windbreaks. The initiative aims to create livelihood and income opportunities for farmers.

Detailed description of activities under B.3:

Bamboo plantation

Across the world there is a search for eco-friendly materials, which can be harvested and used sustainably. The climate change threat has only intensified the search for alternatives. Bamboo is the "The Green Gold" of the 21st century. It has multiple uses and is a gift of mother nature.

Uses of Bamboo:

Bamboo is an eco-friendly resource that helps sequester carbon, improve soil quality, and prevent erosion, making it valuable for stabilizing hilly areas. Its versatility extends to construction, where it is used for building houses, bridges, and furniture. Bamboo poles serve as trellises and fences in gardens, while its stems are processed into paper, biofuel, and various household items. Additionally, bamboo is used in clothing, accessories, and musical instruments. Young shoots are consumed as vegetables, and it also holds significance in traditional medicine and craftsmanship for making bows, arrows, and swords.

Situation analysis:

At the national level the National Bamboo Mission is implemented for promoting holistic growth of the bamboo sector. This is done by adopting area-based, regionally differentiated strategies to increase the area under bamboo cultivation and its marketing. In Maharashtra, the forest department has set up a centre of excellence for bamboo at Chichpalli, Chandrapur and in a short span of time two hundred artisans have received skill enhancement training on bamboo. Realising its importance and its future value, the Government has set up a Maharashtra Bamboo Development Board in August 2016. Outside the forests, bamboo is also grown widely in farm bunds, and it is always seen in bunches and clumps in rural Maharashtra. The total bamboo bearing area of the country (ISFR 2023) has been estimated to be 1,54,670 km². There is an increase of 5,227 km² in the bamboo bearing area of the country as compared to the previous assessment reported in ISFR 2021. Madhya Pradesh has a maximum bamboo bearing area of 20,421 km², followed by Arunachal Pradesh (18,424 km²), Maharashtra (13,572 km²).

Activities Planned by project under the Bamboo Plantation

The bamboo plantations will be promoted predominantly in farmers' fields, community and arable wastelands. Project proposes to support plantation of Bamboo in the project area especially on the banks of rivers/ river lets, wastelands, farm bunds and similar areas. Also, the focus will be on the area aligning to the forest belt and having a good river network. Project will support training related to cultivation and processing of bamboo.

Following are the various types of plantations under Bamboo plantation activity, which the project will suggest and support.

Table 16: Bamboo plantation activity

Sr. No.	Type of plantations	Magnitude (no of plants/ha)
I	Peripheral and Boundary Plantation	5 M distance per plant
II	High Density Planting	400 (5 x 5 M)
		280 (6 x 6M)

Peripheral and Boundary Plantation: In order to make maximum use of the peripheral area and bunds of farmers' fields, bamboo can be planted on the peripheral area and bunds. This initiative will help stabilise bunds and reduce soil erosion, while also creating additional income opportunities for farmers.

High Density Planting: Central block planting/strip planting/ high density planting of windbreak plants can be done on farmers' fields. In order to make maximum use of the available land, unfit and barren land can be utilized for crop production.

Horticulture

Horticultural practices, which include the cultivation of fruits, vegetables, flowers, and trees, enhance carbon storage in plant biomass and soils. Perennial plants and trees significantly contribute to soil organic matter, increasing the soil's capacity to sequester carbon over the long term. Moreover, horticulture supports sustainable land management practices, such as agroforestry and organic farming, which further enhance carbon storage. By integrating horticulture into climate action strategies, we can achieve multiple benefits, including improved biodiversity, enhanced ecosystem services, and increased agricultural productivity, making it a crucial component in the fight against climate change

Benefits of horticulture

Horticulture plays a vital role in modern agricultural systems, offering a wide range of benefits that span environmental, economic, and social spheres. As global populations rise and natural resources face increasing pressure, horticulture provides a sustainable solution to meet the growing demand for food, livelihoods, and ecological stability.

Horticulture supports food security, enhances urban and rural landscapes, fosters sustainable farming innovations, and aids in combating climate change. With diverse applications like cultivating fruits, vegetables, ornamental plants, and medicinal herbs, it has become central to traditional and modern agriculture.

Activities planned by project under the horticulture plantation

The project presents a significant opportunity to enhance carbon sequestration through strategic fruit crop plantation activities. By promoting the planting of fruit orchards, the project aims to stabilize carbon emissions and improve farmer incomes, thereby contributing to both environmental sustainability and economic resilience.

Eligible Fruit Crops

The following fruit crops will be eligible for plantation under the scheme:

Table 17: Fruit crops will be eligible for plantation under the scheme

Sr.no.	Fruit	Planting Distance (m.)	No. of trees	Sr. no.	Fruit	Planting Distance (m.)	No. of trees
1.1	Mango	10 X 10	100	10	Tamarind	10 X 10	100
1.2	Mango (intensive cultivation)	5 X 5	400	11	Jamun	10 X 10	100
2	Cashew	7 X 7	200	12	Kokum	7 X 7	200
3.1	Guava	6 X 6	277	13	Jackfruit	10 X 10	100
3.2	Guava (intensive cultivation)	3 X 2	1666	14	Fig	4.5 X 3	740
4	Pomegranate	4.5 X 3	740	15	Sapota	10 X 10	100
5	Kagzi Lime	6 X 6	277	16	Coconut		
6	Orange/ Mosambi	6 X 6	277		Coconut Seedlings Banavli (with bag)	8 X 8	150
7	Orange (Indo-Israel technology)	6 X 3	555		Coconut Seedlings – Banavli (without bag)	8 X 8	150
8	Custard Apple	5 X 5	400		Coconut Seedlings - T X D (With Bag)	8 X 8	150
9	Aonla	7 X 7	200		Coconut Seedlings - T X D (With Bag)	8 X 8	150

The list of the plant species given here is subject to change with the updating in the guidelines of the DoA under horticulture development schemes.

Activities and guiding principle under the subcomponent: The subcomponent will promote horticulture plantation in each of the project cluster with focus on the following key points:

- **Financial Support:** The project offers 100% financial assistance for planting high-value fruit crops, including digging pits, selection of varieties suited to local climates, crop protection, gap filling and chemical fertilizers
- **Training and Resources:** Farmers receive training on best practices for horticultural production, including pest management and irrigation techniques, CRTs interventions.
- **Sustainability Focus:** Emphasis is placed on sustainable practices such as organic inputs and water management to enhance productivity while addressing climate challenges.
- **Targeted Beneficiaries:** The project primarily supports small-holder farmers, particularly in drought-affected areas like Marathwada and Vidarbha.
- **Potential clusters** for Mango, Citrus (Orange, Sweet Lime & Kagzi Lime), Guava, Custard, Aonla and Pomegranate, etc. will be identified by the department preferably in fallow land, with patches identified during micro planning and having reasonable water sources for perennial crops as well as market potential.
- Beneficiaries with assured irrigation facility shall be selected for horticulture production.
- Project will ensure adoption of the package of practices developed by the centres of excellence (Citrus and Mango) and SAU through continuous technical support and advisory dissemination

Protected Cultivation:

Protected cultivation offers crops a controlled environment and safeguards them from adverse climatic conditions through innovative structures such as polyhouses, shade net houses, and crop covers including anti-hailnet. This approach ensures more efficient use of inputs like fertilizers, pesticides, and water compared to open-field farming. It also enhances productivity and quality, resulting in higher returns for the produce. Additionally, protected cultivation allows farmers to grow crops off-season, enabling them to secure better market prices. It also supports seed production of vegetable crops and the cultivation of flowers.

Objectives

1. Protection of plants from abiotic stress (physical or by non-living organisms) such as temperature, excess/deficit water, hot and cold waves, and biotic factors such as pest and disease incidences, etc.
2. Efficient water uses with minimum weed infestation.
3. Enhancing productivity per unit area.
4. Minimising the use of pesticides in crop production
5. Promotion of high value, quality horticultural produce.
6. Propagation of planting material to improve germination percentage, healthy, uniform, disease free planting material and better hardening.
7. Year-round and off-season production of flower, vegetable crops.
8. Production of disease-free and genetically better transplants.

Situation analysis

In India, small and medium farmers have increasingly adopted flower and vegetable cultivation using various modular protected structures, tailored to their investment capacity and local market availability. Among the different methods of protected cultivation, greenhouse farming offers the highest benefits.

The Government of India (GOI) is implementing the Mission for Integrated Development of Horticulture (MIDH), previously known as the National Horticulture Mission (NHM). This scheme aims to increase the area and productivity of horticulture crops while promoting post-harvest management practices.

Guiding principles

1. Project will support erection of a polyhouse, shade net house, vegetable planting material in shade net house and polyhouse for crops such as Capsicum, Tomato, Broccoli, Cucumber, Cherry, Red cabbage, Lettuce, Chilli and Muskmelon etc. and flower cultivation in polyhouse only such as Carnation, Gerbera, Anthurium & Orchid crops.
2. Existing structures where support is not provided for planting material by any Government scheme will also be eligible for planting material subsidy.
3. The crop cover and anti-hailnet measures will be demonstration a pilot basis, adopting light-house approach.
4. Protected cultivation structures with requisite BIS standards as per state government guidelines would be eligible for project support.
5. Farmers will be trained at notified training centres like Horticulture Training Centre (HTC), Talegaon, KVK Baramati, etc.
6. The technical guidelines and financial assistance are to be calculated as per the prevalent unit cost norms given under the scheme guidelines and policies issued by Govt. of Maharashtra (GoM) & Govt. of India (GoI).

Approach

The project will implement this component on a limited and demonstrative scale rather than adopting a demand-driven approach. The maximum eligible area for project support in constructing structures such as shade net house and polyhouse will be restricted to 0.50 hectares per village.

Enhancing Millet Production

Project will support the following set of interventions:

- Promotion of Millet Cultivation
- Strengthening seed supply chain of millets
- Strengthening commodity value chains with a special emphasis on millets

Promotion of Millet Cultivation

The project area has significant potential for enhancing millet cultivation, with major millets like sorghum, pearl millet, and finger millet already grown extensively. Traditionally, small millets are cultivated in the western parts of Nashik and the northern areas of Vidarbha, offering a strong base for expansion. Favourable agro-climatic conditions and growing awareness of millets' nutritional and climate-resilient benefits present opportunities to increase their area and productivity. The project will focus on clusters with a history of millet cultivation, combining local knowledge with improved seeds, agronomic techniques, and market linkages to boost profitability and sustainability.

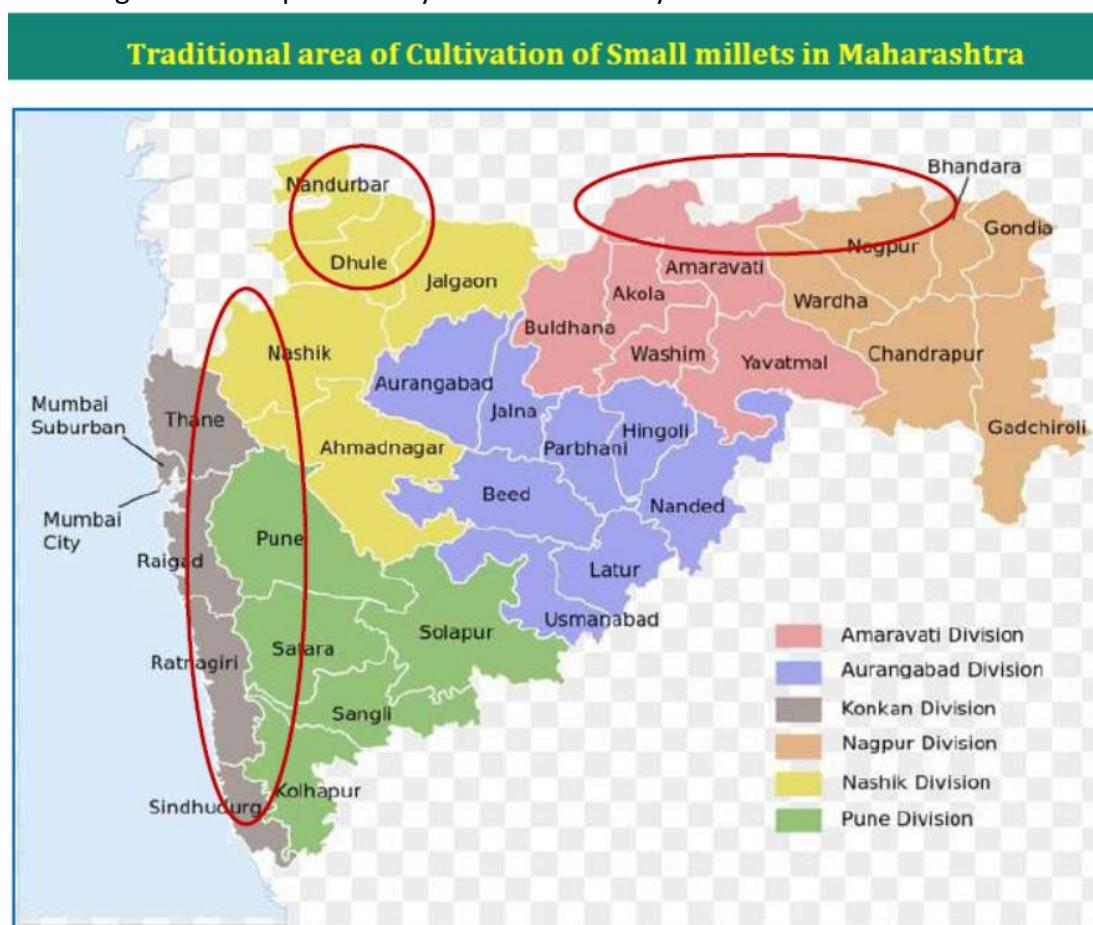


Figure 11:: Traditional are of cultivation of small millets in Maharashtra

The project will integrate indigenous practices with advanced methods to promote millet cultivation, particularly in rainfed and semi-arid areas. Key interventions include Farmers Field Schools (FFS) to demonstrate best practices, seed production programs, and agribusiness support for processing, storage, and marketing. Additionally, awareness campaigns will highlight the nutritional and environmental benefits of millets. By aligning with government initiatives like PMFBY and PMKSY, the project aims to address climate change, malnutrition, and food security challenges while ensuring sustainable livelihoods for millet farmers.

Strengthening seed supply chain of millets

To successfully promote the cultivation and consumption of nutritious, drought-resistant, and climate-resilient millets, it is essential to build a robust seed supply chain. Below are key strategies to strengthen this chain:

1. Promote the use of high-quality, disease-free seeds with climate-resilient traits to ensure that farmers adopt millet varieties capable of withstanding environmental stresses like droughts and floods.
2. Establish and promote community-based seed banks to ensure easy access to quality millet seeds. Encourage farmers to practise seed saving and sharing, especially for traditional and local millet varieties, to preserve biodiversity and reduce dependence on external sources.
3. Offer incentives to seed producers to scale up the production of millet seeds. This could include financial support, subsidies, or technical assistance to seed-producing organisations and individuals. Such incentives will help boost the availability of quality millet seeds in the market.

The project will prioritise the promotion of millets, recognizing their nutritional value, climate resilience and economic potential for farmers in rainfed areas. Efforts will focus on improving seed production, introducing appropriate machinery, and strengthening market linkages to ensure fair pricing. By leveraging traditional knowledge and advanced agricultural practices, the NDKSP aims to increase millet cultivation and consumption, addressing challenges related to climate change, food security, and sustainable agriculture. Collaboration with government schemes and agribusiness initiatives will further support millet farmers, ensuring a sustainable and resilient future for these vital crops.

Strengthening commodity value chains with a special emphasis on millets

The project will be implementing the following strategies:

1. The project will focus on strengthening the millet value chain with an emphasis on increasing productivity and production through seed production and FFS. The technology transfer to farmers about the harvesting, storage and marketing through FFS.
2. **Promotion of Scale-Appropriate Machinery:** Introducing cost-effective machinery for primary processing of millets such as cleaning, grading, and dehulling to improve processing efficiency.
3. **Support for Infrastructure Development:** Facilitating the establishment of small-scale processing units in project areas to ensure decentralized processing and reduce post-harvest losses.
4. **Training and Capacity Building:** Organizing workshops and training for farmers and FPOs on advanced processing techniques and value addition.
5. **Value Chain Development:** Encouraging agribusiness models for producing millet-based processed products, ensuring better market reach and profitability.

6. **Support to FPOs:** The project will support FPOs in primary and secondary processing to enhance consumption and marketability of millet products.
7. **Market Linkages:** Strengthening connections with retailers and consumers by promoting processed millet products through branding and certification, ensuring quality standards.
8. **Shelf-Life Extension Solutions:** Promoting research and adoption of technologies for improving millet storage, such as vacuum packing or chemical-free preservation methods.

The NDKSP aims to create a robust processing ecosystem, enhancing millet value chains and ensuring higher economic returns for farmers. Further more details about processing or agribusiness strategy are provided in the agribusiness section.

Table 18: Activities, Beneficiary types and Output indicators for sub component B2

Sub-component	Activity/ activities	sub- Beneficiary type (Individual/Farmer Group/Community)	Output Indicators
Enhancing carbon sequestration	Agroforestry	Individual/ Community	Area covered under Agroforestry, Bamboo and Horticulture plantation
	Bamboo plantation	Individual/ community	
	Horticulture	Individual	
	Protected Cultivation	Individual	Area covered
	Enhancing millet production	Individual/ Groups	No of farmers

Carbon-credit facilitation through project activities:

India, the world's third-largest greenhouse gas (GHG) emitter, recorded 420.96 MMtCO₂e emissions from its agriculture sector in 2019, accounting for 13.44% of the country's total emissions, a 3.22% increase from 2016 (India's National Communication, NC3). Under a business-as-usual (BAU) scenario, these emissions are projected to rise to 515 MMtCO₂eq by 2030. However, adopting mitigation practices would reduce emissions by 85.5 MMtCO₂eq, with about 80% of this potential achievable through cost-saving measures. Three key strategies—efficient fertiliser use, zero-tillage, and rice-water management—alone could contribute significantly to mitigation efforts. To enable these reductions, significant awareness-building among farmers and local governing bodies, as well as large-scale investments, are crucial.

Maharashtra GHG emission profile 2021-22 says that the agriculture, forestry and other land use (AFOLU) sector contributes to 8.2% of the total GHG emissions in the state. Within the agricultural sector, 21% of emissions are attributed to agricultural land. Globally, around 8% of agricultural greenhouse gas emissions are produced by growing rice. Methane is 28 times more potent as GHG than carbon dioxide.

India ratified the Paris Agreement in 2016, committing to reducing the emissions intensity of GDP by 45% by 2030. The Indian Carbon Market (ICM) was established to support decarbonization by pricing emissions reductions through carbon credit trading.

The Emerging Carbon Market: Opportunities, Scope, and Framework

As the world transitions toward a low-carbon economy, the carbon market presents several opportunities for different stakeholders

Carbon Credits in Indian Agriculture: Mechanisms and Pricing Framework

Carbon credits are generated when farmers adopt practices that reduce or capture carbon emissions. These credits can be sold in carbon markets to industries looking to offset their emissions. In India, agriculture-based carbon credits can come from:

1. **Regenerative and Sustainable Farming** – Practices like no-till farming, crop rotation, organic composting, and cover cropping improve soil health while storing carbon in the soil.
2. **Agroforestry and Afforestation** – Planting trees alongside crops helps absorb carbon dioxide (CO₂) and provides additional ecological benefits.
3. **Methane Reduction from Livestock** – Improved feeding practices and biogas digesters can significantly cut methane emissions from cattle.
4. **Efficient Water and Fertilizer Use** – Precision irrigation (like drip irrigation) and reduced fertilizer use lower nitrous oxide emissions.

Assessing scope of Carbon Credit Generation

The first step is to assess which farming practices under the NDKSP project can contribute to carbon credit generation. For that, farmers will be encouraged to take a comprehensive approach to carbon farming that improves both climate resilience and carbon sequestration. Some of the agricultural practices with potential to earn carbon credits through agroforestry, organic farming/promotion of bio fertilizers units, precision farming, regenerative and sustainable farming, water management practices, diversified climate-resilient crops, watershed management, water-efficient irrigation, management of crop residue, Demethanation of paddy, methane reduction from livestock, biochar application/reduction in fertilizer application, promotion of bio and solar energy, reduced fertilizer application, crop residue management and mulching, improved water management/irrigation, drip irrigation, improvement crop planting and harvesting.

These sustainable practices followed by FPOs to generate carbon credits will be verified, with the credit facilitation support provided by the NDKSP, which will be valuable in both

compliance and voluntary markets, allowing farmers to receive financial benefits for their environmental efforts.

Challenges that may arise in implementing carbon credits in agriculture

1. Lack of Awareness and Capacity Building of the farmers and FPOs
2. Measurement, reporting, and verification (MRV) challenges about carbon credit generation as soil testing remains a major challenge.
3. The process of registering projects, verifying carbon savings, and trading credits is expensive, often making it unviable for small-scale farmers.
4. Carbon credit programs often require multiple years of implementation and verification before payments are received.
5. The Indian carbon market is still evolving, with unclear policies and inconsistent pricing mechanisms for agricultural carbon credits.
6. Data Accuracy and Standardization
7. Market Access and Price Volatility

India's Carbon Credit Mechanisms for Agriculture

The Indian government is working towards a structured carbon market that includes agriculture, integrating both compliance and voluntary carbon markets wherein the government is promoting carbon farming projects under the National Mission for Sustainable Agriculture (NMSA)

Scope of Carbon Credits in Agriculture under NDKSP Phase II

NDKSP Phase II enables carbon credit generation by promoting regenerative agriculture and climate-resilient practices. Key activities like soil health improvement, efficient water use, agroforestry, and emissions reduction align with global carbon sequestration standards. Practices such as zero tillage, cover cropping, and organic enrichment enhance soil carbon storage, making farmers eligible for carbon credits. Precision irrigation and renewable energy solutions further contribute to emissions reduction.

All kind of project interventions include RA, CA, are likely to be undertaken under the PoCRA-II. In addition, soil conservation and long-rotation horticulture practices are also likely to be included in the project. This will be done in assistance with technical partners and with the project's MRV model.

While the roadmap is not very clear as of now, the Carbon Credit would be explored as one of the income generating activities for the vast project areas. In view of all these, the project in due course of time, is likely to have a technical partnership with a specialized agency to establish MRV, and to help the project.

Section 4

Component C - Improving and Building Income Resilience



Section 4

Component C: Improving and Building Income Resilience

4.1 Component Overview:

This component strengthens farmers' income resilience by promoting agri-business models and climate-smart practices to mitigate climate shocks like droughts and floods. Using a participatory approach, it will help farmer groups identify, access, and implement sustainable practices aligned with government initiatives.

The project will use matching grants, following government guidelines, to support farmers in adopting proven technologies and agronomic practices. Investments will focus on developing Farmer Interest Groups (FIGs), creating practice packages, and funding piloted interventions to boost income generation.

Maharashtra's strong network of Farmer Producer Organizations (FPOs), including women-led groups, provides an opportunity to promote optimal input use through a participatory approach. The project will empower FPO members as knowledge and resource providers, helping farmers transition from intensive to efficient input use while maintaining productivity and reducing emissions.

Situation Analysis

The Department of Agriculture has been bringing the farmers together in the form of Farmer's Interest Group (FIG), Commodity Interest Group (CIG) or farmer's Self-Help Group (SHG). These groups are federated either into a cooperative or a Farmers Producer Company to give desired scale either for input buying or output marketing. There have also been attempts to have tie ups with processing industries to whom raw material can be catered. Chapter XI A of the Companies Act 2002 enables registration of producer's companies. There are several agencies promoting FPOs in the state. FPOs are increasingly being strengthened and nurtured under various projects in the state such as, World Bank assisted MACP project, small farmer agri-business consortium (SFAC), NABARD, NDKSP (Phase 1), SMART project, PMFME and MAGNET.

Guiding principle for each subcomponent

- Any farmer producer organisation (FPO) which includes Farmer producer company, Women SHGs, ATMA registered FIGs and PESA Gram Sabha will be eligible under project.
- Farmer Producer Company which are registered and are operating in project districts will be eligible while women SHGs, ATMA FIGs and PESA Gramsabhas of project villages will be eligible.

- Project will tie up with the MSRLM and MAVIM institutions to promote agribusiness activities among the women farmers, which will help empower rural women and will also ameliorate the gender gap in sustainable development and climate change mitigation efforts.
- The PESA gram Sabha will be supported for agriculture related infrastructure as per Government cost norms.
- FPOs will be assisted to prepare comprehensive business plan and tying up of various FIs for credit facilitation.
- Project will develop digital and web-based IEC material, documentaries for various services such as digital advisories, market intelligence, capacity building of FPOs and DBT.
- The project will support promotion of Climate Smart Technologies that will help in embedding the ethos of climate resilience within community action.
- The project will guide the FPOs in adoption of climate resilient technologies through FFS, establishment of on-farm bio production units, millet processing and seed production units. The necessary training and technical guidance will be provided through the various national level institutes.

Project Approach and supported activities -

Project Approach -

The project will support the farmers' collectives or Farmer producer organisation (FPOs) to improve accessibility of various agribusiness activities. These farmers collectives or FPOs may include farmer producer companies, Women self-help group, ATMA registered farmers interest groups (FIGs) and Gram Sabha in PESA Areas

1. **Farmer Producer Company** - In Maharashtra, as of October 2023, 10,517 Farmer Producer Companies (FPCs) were registered, with 7,683 situated within the project districts. The Eastern Vidarbha region accounts for 545 FPCs, while the remaining project areas host 6,840. While most FPCs currently focus on adding value to crops like rice, cotton, and pulses, this project will specifically prioritize the Eastern Vidarbha region for enhancing the value of millets and forest produce through processing and marketing initiatives.

2. **Women Self Help Group**: This project will collaborate with the Maharashtra State Rural Livelihoods Mission (MSRLM) and the Mahila Arthik Vikas Mahamandal (MAVIM) to empower rural women and reduce gender disparities in sustainable development and climate change mitigation.

MSRLM aims to improve rural women's livelihoods through self-managed community organizations, with over 650,000 SHGs registered in project districts. (source: MSRLM) MAVIM, a state corporation, has mobilized over 1.3 million women into SHGs, including 50,276 in rural areas as of December 2022. This partnership will leverage the existing networks of these state programs to support women's involvement in project activities.

3. ATMA registered Farmers Interest Group (FIG): Farmers groups are registered under the Agriculture Technology Management Agency (ATMA) are more than 81000 in project districts. The component on value chains will largely focus on the FPCs, SHGs and formally registered Farmer's groups for strengthening climate resilient value chains in Maharashtra.

4. Gram Panchayats/Gram Sabha in PESA Area: There are 7 districts, and 892 revenue villages are covered under PESA from the project. The project wishes to support the PESA GPs and Gram Sabhas with funds to create infrastructure such as warehouses and other storage spaces. Gram Sabhas and GPs may also be supported to kickstart ancillary activities wherever found feasible.

The component "Improving and building income resilience of the project" is centred on strategies to improve the economic resilience of farmers, safeguarding livelihoods against climate-induced uncertainties. The district wise details of FPOs are given in the annexure 9.

This component is aimed at improving the income resilience by achieving following goals:

1. Generating Livelihood Opportunities
2. Enhancing Food Security
3. Value Addition
4. Unlocking Market Opportunities
5. Enhancing Income Security Job Creation / Employment Generation
6. Technology-Driven Adaptation
7. Reducing GHG and Carbon Footprints
8. Renewable Energy in the Sector
9. Forward Linkages
10. Focusing on Millets
11. Losses Reduction
12. Climate Smart Resilient Technology
13. Capacity building of community Organisations

Supported Activities (indicative List)-

The component of "**Building and improving income**" is divided into two subcomponents-

Sub-component C1: Strengthening Agribusiness Value Chains

Sub-component C2: Agribusiness Growth Centers

Sub-component C3: Promotion of Agri-entrepreneurship (Individual)

The proposed strategies of the project to build and improve income resilience of the farmers through implementation of various activities under each of the subcomponents are described as follows –

Table 19: Activities under component C

C. Improving and Building and Income Resilience			
	C1 Strengthening Agribusiness Value Chains		
		C1.1 FPOs as a vehicle of transformation	
			Facilitation of FPCs for establishment of agri-business
			Capacity building of CMRC, CLF, and SHGs
			Facilitation of PESA Gram Sabhas/ Gram Panchayats
			Establishment of value addition units
			Promotion of Farm Machinery Banks
			Allied activities for FPOs
	C2 Agribusiness Growth Centers		
		C.2.1 Creation of storage infrastructure	
		C2.2 Agro Input and post-harvest Service Centers	
		C2.3 Value Chain and Market assessment studies	
		C2.4 Capacity Building for agribusiness planning and executions	
	C3 Promotion of agri-entrepreneurship (Individual)		
		C 3.1 Goat rearing (Land less families)	
		C 3.2 Backyard Poultry (Land less families)	
		C 3.3 Inland Fisheries	
		C 3.4 Sericulture	
		C 3.5 Apiculture	

4.2 Sub-Component C1: Strengthening Agribusiness Value Chains:

This sub-component focuses on improving agricultural markets—covering inputs, outputs, and finance—while enhancing farmers' market access. The project will strengthen market services, including digital platforms, to support value chain transactions, food safety, and agribusiness regulation through innovative tools. A demand-driven, participatory approach will be used to boost smallholder competitiveness and connect them to value chains more effectively.

This subcomponent will build on existing Farmer Producer Organisations (FPOs) as a major driver of change in the selected value chains. The activities under this subcomponent will

contribute to strengthening of the FPOs. The FPOs will be enabled to identify the business opportunities, to prepare the business proposals, and to explore the marketing opportunities for their products or services. The status of the FPOs and nature of their structures differ from organization to organization. In order to mainstream the federation of farmers for agri-business transformation under this project, conscious efforts will be taken to support the FPCs, CMRCs, CLFs, SHGs as well as PESA Gram Sabhas.

Table 20:Activities for sub component C1

Sub-component	Activity	Beneficiary type (Individual/ Farmer Group/ Community)	Output Indicators
FPOs as a vehicle of transformation	Facilitation of FPCs for establishment of agri-business	Farmer Group	Training plan of FPCs
	Capacity building of CMRC, CLF, and SHGs (FPOs)	Farmer Group	Training of CMRC, CLF
	Facilitation of PESA Gram Sabhas	Farmer Group	Training plan for PESA Gram Sabha
Strengthening Value Chain	Establishment of production, post-harvest, value addition units	Farmer Group	Number of units
	Promotion of Farm Machinery Banks	Farmer Group	Number of units
	Allied activities for FPOs	Farmer Group	Number of beneficiaries

C1.1: FPOs as a vehicle of transformation

Guiding Principles

Assessment of Farmer Producer Companies (CENA 2.0)

FPO will be assessed since previous business activity and will be strengthened on the basis of gaps found in assessment. Project will assess the FPOs based on a few assessment parameters and training needs assessment will be done with the help of a project assessment tool developed by the project during Phase I.

Based on the observations as per the score obtained in assessment of FPOs in phase II, the issues of good governance and capacity building efforts for financial and business management as well as generating better awareness about climate resilience technologies

will be a part of the strategy for phase II interventions. Linkages with the FIs will be facilitated through MOUs and other modes.

C1.1.1: Facilitation of FPCs for establishment of agri-business

Project will develop a mechanism to facilitate the FPCs for business development through a consortia approach involving the institutes of excellence in the sector of agri-business development such as NABARD-BIRD, IIMs, VAMNICOM, MANAGE, PDKV Akola, CFTRI Mysore, CIPHET, Ludhiana ANAMATI, RAMETI, SAUs, KVKs, etc. and the like. It involves the knowledge of Business Selection, Detailed Project Report preparation and handholding support to business /activity supported under the project.

C1.1.2: Capacity building of CMRC, CLF, and SHGs – (FPOs)

Project will support women led FPOs i.e. MAVIM and MSRLM promoted SHGs, VO, CMRCs and CLFs to increase the women participation in the project for promotion of farm based agri enterprises through mobilization and capacity building of women farmers.

C1.1.3 Facilitation of PESA Gram Sabhas-

Project villages in Scheduled Areas will need a separate structure of support in line with the PESA Act and due to less potential for individual entrepreneurship. Accordingly, where private entrepreneurship is not willing to come forward, PESA gram sabha and Gram Panchayats will be supported for various activities such as infrastructure support, agro-based enterprises, NTFP based value chains, inland-fishery etc.. The capacity building and handholding support will be provided in tribal areas as per need.

Capacity Building and Skill Enhancement of FPOs

All FPOs need handholding support to build their skills and capacities making them capable of efficiently managing their business ventures. This entails building the technical, managerial, financial, skills and capacities of the community organisations. For achieving this, the project has envisaged a multipronged strategy that has a long term of building the institutional capacities beyond the project period.

Orientation and Capacity Building of the FPOs:

The selected members of FPOs shall be trained, their skills can be enhanced through various capacity building and incubation efforts. This will also help in demonstrating and showcasing innovative technological solutions offered by both state-led R&D institutions and privately owned start-ups and entrepreneurs.

The representative members of the FPOs will get access to continuous handholding support by way of participation in periodic workshops arranged at the district/sub-district levels. Specialised residential trainings will be also organised for the FPO representatives at premier

training institutes such as VAMNICOM, MANAGE, PDKV Akola, CFTRI Mysore, CIPHET, Ludhiana VANAMATI, RAMETI, SAUs, KVKs, etc.

Special batches for Women SHGs and FPOs will ensure better interaction, team building and problem solving amongst the women groups.

Specialised On-site Skills Workshop: The District level functionaries will arrange specialised on-site skills development workshops for FPO representatives for skilling the members in the required areas such as warehouse management, equipments maintenance and repairing, inventory management, allied business skills such as sericulture, goat rearing etc, soft skills and IT skills. These On-site Skills Workshops will be organised at local level by KVKs, FPOs workshop Demonstration plots of universities, departments or research stations etc.

Handholding support The FPOs need continuous handholding support at all the stages that include- project conceptualisation and business planning, project execution and finally the post-project planning for sustainable operationalisation. The expenditure for the capacity building under agribusiness will be borne by the project.

C1.2 Strengthening Value Chain

As part of the value chain development the interventions under this subcomponent will extend beyond a production cluster. Therefore, creating value chains around crops and agro based commodities in the project area is an important element of this project. The climate change impact and resilience are also intertwined elements in this value chain. A value chain promotion intervention can range from improving business operations at production, processing, storage level, the access to knowledge, information and innovation and the linkages to financial and technological services. Successful value chain interventions that achieve poverty reduction goals can be beneficial to climate change adaptation, as they build farmers' assets and institutional linkages. Gender and social dimensions of value chain strengthening will be focussed upon, thus improving the visibility and income opportunities of women and marginalised sections

The sustainability of the food value chain revolves around three dimensions.

- **Economic dimension**, which focuses on activities that each actor or service provider provides that is commercially feasible (profitable) or services that are financially viable.
- **Social dimension**, which refers to social and cultural aspects regarding societal acceptance of the distribution of benefits and costs associated with the increased value that has been created.
- **Environmental dimension**, which refers to the sustainable use of inputs and resources throughout the chain as well as any impact on biodiversity, the amounts of greenhouse gas (GHG) emissions released, and the carbon sequestration and reduction potential of GHGs in

the process of value creation. This could be by the way of reduction in post-harvest loss and efficient systems of handling the produce.

C1.2.1 Establishment of Agro based Value addition units

FPOs shall be encouraged to actively participate in value chain development of selected crops like Cotton, Paddy, Chickpea, Pigeon Pea, Soya bean, Rabi Sorghum, Capsicum, Mango, Guava, Custard apple and Citrus.

For climate smart post-harvest management, emphasis shall be given for establishment of primary processing and aggregation centres/ common facility centers with cleaning, sorting, grading, weighing, packaging facility. Primary processing units like mini dal mill/ oil expellers/ vegetable and fruit based primary processing units/ Cotton bale making units shall be established in the project area for value addition of the farm produce which shall help farmers for fetching better price, long shelf life and better marketability of their farm output.

The project will also support secondary processing units such as spices units, food processing units, animal and poultry feed units, milk processing units, ripening chambers, small cold storages, cotton ginning and processing units, silage making units etc.

For secondary processing unit support, the project will prioritise FPOs that have well-developed backward linkages, a robust raw-material supply network of shareholder/member farmers for the commodity that is to be processed. Identified forward linkages shall help in marketing of the processed products.

Guiding Principles

1. Existing FPOs shall be strengthened and encouraged to actively participate in value chain development.
2. Project will promote establishment of primary processing, aggregation /common facility centres with cleaning, sorting, grading, weighing, and packaging facilities for millets and other commodities grown in the project districts.
3. Efforts will focus on strengthening the production/commodity clusters and the FPOs that are formed around these production clusters. The production clusters that were earlier supported in the phase I of the project or through any other efforts of the state shall be prioritised for further value chain linkages through FPOs.
4. The project will extend support to allied agri-business activities of FPOs which are based on- agro-forestry (bamboo processing, lac processing, Bhilawa, Chironji, Moh, Jamun, Amla processing etc), processing and value addition of medicinal and aromatic plants, sericulture units including cocoon processing, reeling etc. & inland fisheries and other allied businesses that aid income generation and are identified to have a promising potential in the project area.

C1.2.2 Promotion of Farm Machinery Banks

Project districts have significantly low farm power availability. To increase mechanization in regions such as Eastern Vidarbha and Marathwada, Farm Machinery Banks (FMBs) will be established to provide affordable access to machinery for small and marginal farmers. A climate resilient FMB model will be promoted to include the equipments required for CRT such as BBF Technology, zero tillage, mulching, contour cropping etc. The custom hiring services will be promoted to offset the adverse economies of scale arising due to small landholding and high cost of individual ownership.

Guiding Principle

1. The cost of FMB will be in accordance with the cost norms and subsidy structure outlined in the existing Government of Maharashtra (GoM) guidelines for mechanisation.
2. Support will be provided on a "one FMB per village" basis. The eligibility criteria for FPOs will be based on a first-come, first-served basis.

C1.2.3 Allied activities for FPOs

Animal husbandry, dairy, fishery, sericulture, medicinal & aromatic sectors, and agro-forestry are allied activities to agriculture. These activities supplement family nutrition, income and employment opportunities on an individual level in the rural areas, particularly among landless labourers, small & marginal farmers and women. The allied agri-businesses, especially agro-forestry, apiculture and sericulture, production of bio pesticides, biofertilizers, organic manures, vermicompost and other bio-agents is a markedly useful CSA technology which has proven results for reduction in cost of cultivation and augurs' environmental health.

Guiding Principle

1. Goatery and poultry activities will be supported according to the AHD guidelines for MAVIM and MSRLM only women Self-Help Groups (SHGs).
2. The FPO business plans will be holistic, end-to-end and technically vetted and supported by technical specialised partners such as KVIC, SAU, Directorate of Sericulture, Bamboo Mission, KVKs, Animal Husbandry etc.
3. The project will extend support to the admissible components of this business plan, however the other important aspects such as market linkages, capacity building and technical hand holding will have to be assured by the partners.
4. The project will support promotion of Solar Powered Cold Storage, renewable energy technology-based infrastructure like Solar Units, rain water harvesting structures, biogas and biofuel generation units, recycling of agricultural waste units (biomass), Compost Units & Vermicompost & Bio Agent/Bio Fertilizer Production Units.

4.3 Sub-Component C2: Agribusiness Growth Centers:

Agribusiness growth centers will serve as a hub that provides support to access all kind of services to all stakeholders involved in it. It involves capacity building support, resources and infrastructure support to enhance the productivity, innovations and market access. Beyond the farmgate, the project will strengthen post-harvest interventions by supporting farmer groups with on-site processing units and linking them with the Marketing Board for better post-harvest facilities. Agribusiness growth centers will be established to ensure timely input supply, aggregate produce, facilitate processing, and create market linkages. The project will fund infrastructure, equipment, advisory services, and operational costs within set limits, along with farmer mobilization, capacity building, and professional management support. Additionally, it will conduct value chain studies and train project staff to develop bankable business plans for micro-enterprises.

In addition to supporting farmer groups, the project will promote individual agri-entrepreneurship by encouraging allied activities such as small ruminant farming, fisheries, and sericulture. These ventures provide farmers with a steady income while reducing reliance on a single source of earnings. All initiatives under this sub-component will be integrated with the advanced digital agriculture platform established in Phase 1 of the project, ensuring seamless support and implementation.

Table 21: Activities for sub component C2

Sub-component	Activity	Beneficiary type(Individual/ Farmer Group/ Community)	Output Indicators
Agribusiness Growth Centers	Creation of storage infrastructure	Farmer Group/ Community)	Number of units
	Agro Input and post-harvest Service Centers	Farmer Group/ Community)	Number of units
	Value Chain and Market assessment studies		Number of Case studies
	Capacity Building for agribusiness planning and execution		Training plan for officials.
	Promotion of agri-entrepreneurship (Individual)	Individual	Number of units

C.2.1 Creation of storage infrastructure (small godowns, warehouse infrastructure, seed storages and agri malls)

Small godowns, warehouses and seed storages

Warehouses play a crucial role in the Agricultural supply chain. Agri Warehouses provide storage facilities for agricultural inputs and agricultural products including grains, seeds. Adequate storage facilities address post-harvest challenges such as post-harvest losses, market linkage and fair price realisation. The project will support the construction and strengthening of the godowns and small warehouses and other storage structures such as silos and onion storage units. The project expects the following benefits to be derived by using these warehouses.

- Efficient storage system and better pricing–
- Optimal use of storage capacity
- Protection against spoilage

The Warehouse will be constructed as per WDRA (Warehousing Development Regulatory Authority) norms which will provide a scientific storage facility at farm gate level. This will reduce post-harvest losses to a 1 percent or less minimum percentage. SOPs will be developed for the construction and scientific management of godowns and warehouses.

Guiding Principles

1. An assessment of the project area through secondary data sources, to identify surplus production, existing storage facilities and additional requirements will be undertaken.
2. Technical Authorities such as WDRA, Maharashtra state warehousing corporations (MSWC), etc. shall be on board to ensure the designing and construction of technically sound storage structures and their further scientific management.
3. Handholding support to FPOs during construction and further management of the godowns for effective and efficient use of warehouse infrastructure will be done.
4. Strengthening warehouse receipt finance - Warehouse Receipt Finance is a financing model that enables farmers to access credit by using their stored agricultural products as collateral. The constructed warehouse under the project will be encouraged to be registered under WDRA to enable them to issue warehouse receipts.
5. This sub-component shall also support other suitable storage facilities such as silos, onion storage structures, provided the designs of the structures have been vetted by competent authority.
6. The project will support creation of seed hubs linking with FPOs, SAUs and MAHABEEJ and other public sector players.

Infrastructure support to existing FPOs for promotion of climate smart Technology.

Under this sub-component FPOs will be promoted and supported to create infrastructure that is based on climate smart technologies. The project will provide support to the existing FPOs to establish special projects listed below. This is an indicative list.

- Use of Solar power for Agri-business unit, Warehouse or Farm Machinery Bank

- Biogas and biofuel generation unit
- Recycling of agricultural waste units like Biochar

Guiding principles

This component will be guided by a technical advisory authority that will assess the techno-economic feasibility of the proposed climate smart technology-based infrastructure creation. Examples of such TAs are MEDA, IITs, forest department, MNRE etc.

The emphasis will be on low to medium scale technological solutions that will help the FPOs to become resource efficient and climate resilient.

C2.2 Agro Input and Post-harvest Centers

The project will support the agriculture input supply and post-harvest services through FPOs for the ease of farmers for timely cultivation and harvesting operations. It includes maintenance of Farm equipment, sale of input and other services in agriculture and allied areas including post-harvest management and market linkages for income generation and entrepreneurship development.

Guiding principle

1. The project will guide about the various services such as facilitation of input supply to the input supply centres established.
2. The technical guidance about the repair & maintenance of all types of farm machinery will be provided through KVKs, SAUs etc.
3. FPOs will get supported to establish the post harvest service centres for various services required from harvesting to marketing.

C2.3 Value Chain and Market assessment studies-

The Project will focus on analysis of all the steps involved in producing and delivering a product or service, from raw materials to the final customer, with the goal of identifying areas for improvement, potential bottlenecks, and opportunities to maximize value at each stage of the process by examining the interactions between different actors within the chain. It will help to develop business strategies, supply chain optimization, product development and market analysis.

Guiding Principle

The commodity wise value chain assessment through training and capacity building.

The project will prioritise FPOs that have well-developed backward linkages, a robust raw-material supply network of shareholder/member farmers for the commodity that is to be processed.

C2.4 Capacity Building for agribusiness planning and execution-

Selection of Master Trainers - The master trainers will be selected from the field level officials for the capacity building and handholding of the FPOs as per the training need assessment.

For each district, 5-6 Master Trainers will be selected among Dy PD ATMA, SDAOs, DDA – DSAO Level, Nodal Officer – DSAO Level, Technical Officer – DSAO/SDAO Level, TAOs, PS – Agribusiness and PS – Procurement. These master trainers will train a minimum of 2-4 members from the Board of Directors of each entity.

Orientation and Training of Master Trainers: The Master Trainers will be trained in various leading organisations such as BIRD, Mangalore, BIRD, Lucknow, VAMNICOM, MANAGE, PDKV Akola, CFTRI Mysore, CIPHET, Ludhiana, VANAMATI, RAMETI, and concerned institutions in the Animal Husbandry Department, etc. so as to make them aware about the various facets of business planning and execution. The various topics covered in these trainings would include DPR preparation, project evaluation, agriculture value chain concept, quality production, finance management, supply chain management and post-harvest management, world bank procurement processes, technical training (such as warehouse management, processing and value addition of agri-commodities, marketing and branding of agri-commodities etc). This trained human resource will act as facilitators for supporting the FPOs in the project area.

Market Advisory

Project will take up the activity that will support agricultural marketing such as E-commerce, online reports from agriculture markets and provide information to farmers and FPOs on daily basis. It will help the farmers to take the market related decisions based on market price of commodity. It will help the farmers to cope with both market and climate vagaries.

Guiding Principle

- 1 Providing IT based services to farmers e.g. crop advisories, DBT portal (Market Advisory).
- 2 Guiding about the online web applications supporting marketing and market intelligence.
- 3 Registration on digital marketing platforms such as NCDEX or social stock exchanges.
- 4 Registration for trademark, patent and brand registration.
- 5 Support for accreditation, certification of processes, testing and systems. Example, ISO/GAP/Organic /FSSAI/HACP certification.
- 6 WDRA registration, GI tagging etc.
- 7 Support for digital and web-based IEC activities.
- 8 Any other intervention deemed suitable

Financial Support under Project

- 1.The project proposals of a single FPO amounting up to Rs. 1.00 Cr. will be supported under the project with back ended subsidy 60% (maximum up to 60 lacs) in project area except PESA. The support to FPO will be available during project period only.
- 2.The FPOs will be supported in the Scheduled PESA areas (governed by the PESA Act) with higher back-ended subsidy for all the interventions under Project for the proposals up to Rs.1.00 Crore.
- 3.The Gram panchayat/ Gramsabha in PESA villages will be supported for the common infrastructure such as small godowns and storage structures up to 250 metric tons with 100 % project support.

4.4 Sub-Component C 3: Promotion of Agri-entrepreneurship among the individual farmers and landless families

The promotion of agri-entrepreneurship in the project area focuses on empowering farmers, rural youth, and aspiring entrepreneurs to establish diverse agri-based businesses such as crop production, livestock farming, agro-processing, and value-added products. This initiative aims to enhance economic stability, foster self-reliance, and reduce dependence on traditional farming practices within rural communities.

Agri-entrepreneurship offers a pathway to diversify income, create sustainable businesses, and contribute to local economic growth and environment friendly agriculture. With adequate training, resources and support, individuals can transform agricultural expertise into profitable ventures that are resilient to climate change.

Agri-allied enterprises like goat rearing, backyard poultry, sericulture, apiculture, and inland fisheries integrate seamlessly with primary agriculture, enhancing climate resilience. These enterprises help mitigate risks, boost incomes, support rural livelihoods, and promote sustainable resource use, ensuring financial and ecological sustainability

C3.1 Goat Rearing

Goat rearing is a key strategy for climate-resilient agriculture in Maharashtra, particularly in drought-prone regions. Goats' adaptability to arid conditions, minimal resource requirements and ability to produce milk, meat, and manure make them an accessible and sustainable livelihood option especially for landless beneficiaries. This practice diversifies income sources, enhances soil fertility through organic manure and provides economic stability in the face of

climate challenges. By integrating goat farming into traditional agriculture, rural communities can achieve greater resilience, food security and ecological sustainability.

Key Objective

1. To support landless families for generating additional source of income.
2. To utilize goat manure as a natural fertilizer, enhancing soil health.
3. To enhance the availability of meat, milk, and other goat-derived products, improving food security in rural areas.
4. To generate self-employment, especially for rural women and youth.

Implementation Approach

1. People belonging to landless families and widow, separated and divorced women farmers in the project area are eligible for support.
2. Selection of beneficiaries will be done by GKVS based on relevant guidelines.
3. The financial assistance is to be calculated as per the prevalent unit cost norms given under the scheme guidelines and policies issued by Govt. of Maharashtra (GoM) & Govt. of India (GoI).

C3.2 Backyard Poultry

Backyard poultry farming has become an integral part of climate-resilient agricultural practices in the project area of Maharashtra state. These areas are characterized by high vulnerability to climate change with erratic rainfall, water scarcity, and rising temperatures significantly affecting traditional farming practices. In this context, backyard poultry offers a sustainable, low-cost solution that supports both agricultural diversification and food security.

Key Objective

1. To provide landless families with a reliable and additional source of income.
2. To integrate backyard poultry with other farming activities, utilizing farm waste for poultry feed and manure for crops.
3. To improve access to protein-rich food for rural communities.
4. To generate self-employment, especially for women and youth.

Implementation Approach

1. People belonging to landless families and widow, separated and divorced women in the project area are eligible for support.
2. Selection of beneficiaries will be done by GKVS based on relevant guidelines.
3. The financial assistance is to be calculated as per the prevalent unit cost norms given under the scheme guidelines and policies issued by Govt. of Maharashtra (GoM) & Govt. of India (GoI).

C3.3 Inland Fishery

Inland fisheries can be effectively integrated into Integrated Farming Systems (IFS) to build climate resilience, diversify incomes and ensure sustainable livelihoods for farmers. The project area contains numerous water bodies that serve as a vital livelihood source for villagers through fishing activities.

Project areas particularly in Eastern Vidarbha have a large number of water bodies such as Malgujari Talav. These water bodies serve as a livelihood source through fishing activity. The project will support fisheries in water bodies owned by the Gram Panchayat in the PESA area.

Key Objective

1. To provide small scale and marginal farmers with an additional and stable income source.
2. To enhance farm resilience to climate change by diversifying production systems and reducing dependence on a single crop.
3. To use fish pond waste as natural fertilizer for crops, promoting nutrient cycling and enhancing soil fertility.
4. To create new livelihood opportunities and improve food security in rural communities.

Implementation Approach

1. Small scale and marginal farmers having farm ponds in the project area are eligible for support.
2. Selection of beneficiaries will be done by GKVS based on relevant guidelines.
3. Water bodies in the PESA area which are owned by the Gram Panchayat are also eligible for support.
4. The financial assistance is to be calculated as per the prevalent unit cost norms given under the scheme guidelines and policies issued by Govt. of Maharashtra (GoM) & Govt. of India (GoI).

C3.4 Sericulture

Sericulture plays a crucial role in climate-resilient agriculture, especially in Maharashtra, where it can be integrated into farming systems for added economic stability and sustainability. Mulberry plantations, essential for silkworm feeding, are cultivated ensuring optimized use of land and water resources. The establishment of rearing houses and provision of quality rearing material supports healthy silkworm growth. Sericulture enhances farm income through diversification, promotes sustainable agricultural practices and strengthens resilience to climate variability by providing an additional revenue stream and reducing dependency on a single crop.

Under this component the project will support mulberry plantation, construction of silkworm rearing house and rearing material for cocoon production.

Key Objective

1. To provide small scale and marginal farmers with an additional, stable income source through cocoon production.
2. To enhance farm resilience to climate variability by promoting biodiversity and reducing dependency on a single crop.
3. To create additional employment opportunities, especially for women and youth.

Implementation Approach

1. Small scale and marginal farmers in the project area are eligible for support.

2. Selection of beneficiaries will be done by GKVS based on relevant guidelines.
3. The financial assistance is to be calculated as per the prevalent unit cost norms given under the scheme guidelines issued by Govt. of Maharashtra (GoM) & Govt. of India (GoI).

C3.5 Apiculture

Apiculture plays a vital role in climate-resilient agriculture by supporting biodiversity and enhancing crop productivity through pollination. In Maharashtra, integrating beekeeping with an integrated farming system (IFS) optimizes the use of resources like land and water, while providing farmers with additional income from honey, wax and pollination services. The practice enhances food security, reduces dependency on single crops and promotes sustainable farming by encouraging organic and eco-friendly methods. With proper training, apiculture can become a valuable component of climate-resilient farming, benefiting both the environment and rural livelihoods by creating new employment opportunities, particularly for women and youth.

Key Objective

1. To provide small and marginal farmers with an additional source of income through honey, wax and pollination services.
2. To improve crop yields by using bees for pollination, benefiting both agricultural productivity and biodiversity.
3. To reduce dependence on single crops by diversifying farm activities and creating a buffer against climate-induced losses.
4. To promote organic farming by using bees as natural pollinators, enhancing crop growth without chemical inputs.
5. To create employment opportunities, particularly for women and youth.

Implementation Approach

1. Small and marginal farmers in the project area are eligible for support.
2. Selection of beneficiaries will be done by GKVS based on relevant guidelines.
3. The project will focus exclusively on areas with year-round favourable conditions for beekeeping for the implementation of apiculture.
4. The financial assistance is to be calculated as per the prevalent unit cost norms given under the scheme guidelines and policies issued by Govt. of Maharashtra (GoM) & Govt. of India (GoI).

Table 22: Activities, Beneficiary types and Output indicators for C3

Sub-component	Activity/sub-activities	Beneficiary type (Individual/ Farmer Group/ Community)	Output Indicators
Promotion of Agri-entrepreneurship	Goat Rearing	Individual	Number of units supported
	Backyard Poultry	Individual	Number of units supported
	Inland Fishery	Individual and Farmer Group	Number of units supported
	Sericulture	Individual	Number of units supported
	Apiculture	Individual	Number of units supported

IT Initiatives for Agri-Business Value Chain

To have a seamless process in place, PMU of NDKSP had developed a registration, application, and approval mechanism for the FPC/SHG/Farmers group. The project will use the direct benefit transfer mechanism to provide hassle free & transparent service to beneficiaries.

Process Flow for Implementation of Agribusiness Proposals -

The project will implement the DBT model for all agribusiness activities, providing an online and seamless registration and application system for both beneficiaries and project staff. The process flow is described as under.

Registration and primary application: FPOs will register on DBT portal of project with all required documents. After registration FPOs will apply for activity with primary information. **Registration and primary application scrutiny:** a concern officer will scrutinize the registration details and will check the eligibility of FPOs as per upload documents. **Detailed project report (DPR) submission:** DPR will be uploaded by FPOs with all necessary document as per project guidelines.

Registration & Application Scrutiny - The registration and application done by concerned FPO will be scrutinized by the district level project official.

Pre-Work Site Inspection -

- After scrutiny, application will be accepted and then, field officials will visit the proposed site for the requested component.

- The officials will confirm the site /location and other eligibility criteria as per the guidelines to implement the component and will recommend for conditional pre-sanction approval.

Conditional Pre-Sanction Letter:

- The relevant officer will verify the application documents and issue a conditional pre-sanction letter for the approved component.

Procurement Process Approval –

FPO must complete the entire procurement process as per the world banks procurement guidelines (for work or goods)

The district level officer will scrutinize all the documents and give approval to the entire procurement process of FPOs.

Activity Implementation/ Demand for subsidy:

- After procurement process approval, the FPO will give work order to the vendor selected for the supply of goods or construction work.
- After getting work order, concern vendor will complete the task and FPO will claim the subsidy with all related documents.

Site Inspection (Post work Implementation):

- After submission of the subsidy claim in the DBT system, the concerned officer will perform a post-implementation site inspection.
- The officer will geo-locate the site, verify necessary documents, and upload all relevant records to the system.
- Based on the subsidy calculation, the officer will recommend the eligible subsidy amount.

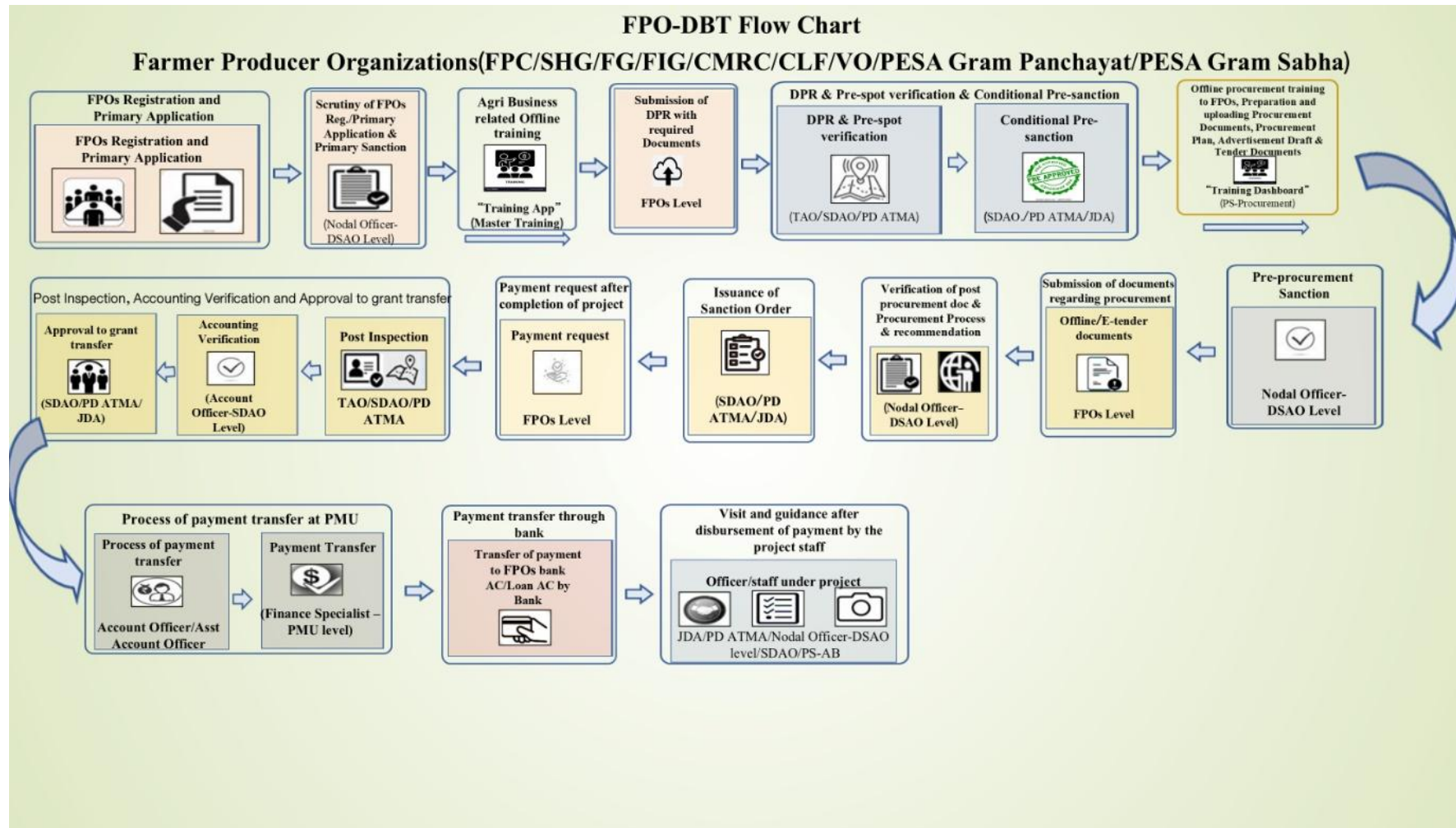
Financial Verification:

- The accounts officer at the sub-division level will verify the FPO's application with the uploaded documents and he will recommend the subsidy amount according to the financial guidelines for approval.

Final Approval and Disbursement:

- After completing the financial verification, the concerned officer will grant final approval for the subsidy.
- The subsidy will be processed online and will directly transfer to the FPOs self-bank account or loan account of FPOs.
- Necessary offline trainings will be organized by the project as per need

Figure 12: FPO -DBT flow chart



Section 5

Component D -Project Management and Strengthening Institutions



Section 5

Component D: Project Management and Strengthening Institutions

5.1 Component overview

NDKSP-II will build on its strong institutional framework to enhance resilience by expanding its scope and depth. Implementation will involve government agencies, private sector players, including AgTech startups, and community institutions, ensuring comprehensive administrative, technical, and advisory support. The Department of Agriculture, along with its district, block, and village-level networks (including Gram Krishi Vikas Samitis), will lead on-the-ground execution. A dedicated Monitoring and Evaluation (M&E) agency, supported by expert consortia partners, will provide guidance on emissions management, crop suitability, hydrology, soil profiling, market trends, and agribusiness. Additionally, this component will oversee the integration of digital agriculture and ICT-based solutions essential for the project's success.

Sub-component D1: Consortia of Scientific and Technical Expertise

Sub-component D2: Digital Services for Climate Resilient Agriculture

Sub-component D3: Project Monitoring, Evaluation and Learning

Sub-component D4: Capacity Enhancement

Sub-component D5: Human Resource Management

5.2 Sub-Component D1: Consortia of Scientific and Technical Expertise :

Building on NDKSP's efforts to engage researchers and scientists from leading institutes, NDKSP-II will establish a specialized consortium of experts to provide targeted scientific and technical support for project implementation. This consortium will focus on: (a) enhancing soil use efficiency by assessing soil quality, nutrient availability, and uptake potential to optimize crop productivity across different soil types; (b) evaluating Water Use Efficiency (WUE) and Nutrient Use Efficiency as key factors in sustainable soil management; (c) promoting efficient irrigation methods and scheduling to boost productivity while minimizing greenhouse gas emissions; and (d) developing resilience protocols to help farmers adapt to emerging environmental challenges and adopt improved production systems.

The consortium-driven approach will generate valuable data and methodologies for resource-efficient farming, strengthening the State Agriculture Department's ability to implement precision agriculture for key crops. An Innovation Lab will focus on optimizing soil, water, fertilizer, and nutrient efficiency, enhancing the department's capacity in climate-smart agriculture. Insights from cutting-edge technology and scientific research will directly shape Village Adaptation Plans (VAPs), developed through a participatory process to tailor resilience strategies to local needs. This continuous learning will ensure that VAPs remain dynamic, enabling real-time adoption of sustainable practices at the grassroots level.

To ensure sustainability in building climate resilience through transformative management of agriculture, soil, and water resources, the project is fostering collaboration and convergence by creating social infrastructure in the form of GKVS and FPOs, supported by the Department of Agriculture. By bridging knowledge gaps and enhancing stakeholder capacities through strategic partnerships and targeted training, the project empowers farmers, women, and various groups to improve their efficiency in resilience building, thereby ensuring the sustainability of investments. Institutional support for research, training, and workshops is provided in collaboration with organisations such as Indian Council of Agriculture Research (ICAR) institutes viz Central Research Institute for Dryland Agriculture (CRIDA) Hyderabad and National Bureau of Soil Survey and Land Use Planning (NBSS&LUP), and other Government and semi-government institutions such as IIT Mumbai, State Agriculture Universities, Groundwater Survey and Development Authority (GSDA) National Institute of Plant Health Management (NIPHM), Tata Institute of Social Science, Krishi Vidyan Kendra, NIPHT, Vanamati, ICAR institutes, State Agriculture Universities, and various research foundations, enhancing the collective efforts toward climate resilience.

Inter departmental Approach:

The project approach will be departmental convergence top-down and bottom-up. The high-level project steering committee establishes a strong convergence mandate, positioning this project as a catalytic investment within the broader drought mitigation initiative of the Government of Maharashtra. The state has developed comprehensive plans, including irrigation, agri-business, watershed development, crop, and contingency plans for all districts. Additionally, specific crop and varietal advisories are provided through university extension services. The implementation of the Agri stack project will also create three core registries: a Farmer Registry, a Digital Crop Survey, and a Geo-Referenced Land Parcel, all under the Central Government's Digital Public Infrastructure for Agriculture initiative.

By emphasising collaboration and convergence, the project will serve as an ideal platform for relevant departments including agriculture, water conservation, rural development etc. it aims to implement interventions in various sub components and activities like in the post-harvest sector, supporting farmer groups with on-site processing machines and forming

partnerships with the Marketing Board linked with Cooperation and Marketing Department to enhance facilities. The Women and Child Development (MAVIM, MSRLM), Rural Development, Revenue and Forest Department, GSDA, (Water Supply and Sanitation) will be approached to foster the project's objectives to improve resilience and profitability of small farmers by promoting resource-use efficiency and emission-competitive production systems with data-driven decision tools.

Strategic Partnerships

The project will make conscious efforts to engage the researchers/scientists from leading institutes to support project implementation with desired science/technology. The project will create a 'consortia' of subject-specific scientific/technical expertise with clearly defined inputs to assist implementation on a) improving soil efficiency related to soil quality, soil nutrient availability and nutrient uptake potential; b) determining water-use efficiency in agricultural production systems; c) choice of efficient irrigation methods to reduced integrated GHG emissions, and d) develop resilience protocols towards adaptive arrangement of emerging environmental impacts. The project will have strategic partnerships around the thematic areas with organisations of national and international repute.

An indicative list has been given in the figure below:

Table 23: List of the knowledge partners

Sr. No.	Name of Organization	Areas of Collaboration
1	National Bureau of Soil Survey and Land Use Planning, (NBSS&LUP), Nagpur	Digital soil mapping (DSM) and farm level advisories for climate resilient agriculture
2	Central Rice Research Institute (CRRI), Cuttack	To estimate methane emissions from rice fields and overall GHG emission, prepare a Package of Practices (PoP) to enhance rice productivity while reducing methane emissions, and build the capacities of Project and DoA functionaries for better extension of resilient practices.
3	Indian Institute of Soil Science, Bhopal	To develop a strategy to enhance Nutrient Use Efficiency (NUE) based on soil analysis, create a protocol for Conservation Agriculture, and build the capacities of Project and DoA functionaries.
4	Indian Council of Agricultural Research - Indian Agricultural Statistics Research	Transferring the information available in the existing systems of ICAR-IASRI to Project on Climate Resilient Agriculture, Govt. of Maharashtra for Development of AI-ML based Digital Advisory System for Farmers.

Sr. No.	Name of Organization	Areas of Collaboration
	Institute (ICAR-IASRI), New Delhi	
5	State Agricultural University (SAU)	To generate crop advisories, conduct participatory research for Climate Resilient Agriculture, provide breeder/foundation seed of CRV to FPCs for multiplication, and build the capacities of Project and DoA functionaries.
6	Groundwater Surveys and Development Agency, (GSDA), Pune	Preparation of Village wise Groundwater recharge plan, technical handholding and impact assessment for project village.
7	Bharat Rural Livelihoods Foundation, New Delhi	To achieve the shared goal of improving the lives and livelihoods of tribals and farmer communities in Maharashtra.
8	Watershed Organisation Trust, Pune	Mutual sharing of information for the development and upgradation of digital agricultural advisory services.
9	Paani Foundation, Mumbai	Paani Foundation will conduct 'Digital Paani Shalas' or 'Digital Schools on Watershed and Water Management' for villages associated with NDKSP while NDKSP will mobilize villages to participate in the Digital Paani Shalas.
10	Maharashtra State Rural Livelihoods Mission (MSRLM) , Mumbai	To support the Women Self Help Groups (SHGs)/Village Organisations (V.O.) / Women Farmer Producer Company (FPC) of MSRLM to enhance participation of women in the project.
11	Mahila Arthik Vikas Mahamandal (MAVIM), Mumbai	To support the Women Self Help Groups (SHGs)/Village Organizations (V.O.) Community Managed Resource Centers (CMRCs)/Women Farmer Producer Company (FPC) of MAVIM to enhance participation of women in the project.
12	Gokhale Institute of Politics and Economics (GIPE), Pune	Economic and Financial Analysis of the project

Sr. No.	Name of Organization	Areas of Collaboration
13	Maharashtra State Seed Corporation (MAHABEEJ)	Climate Resilience Seed Production and bi-fertilizer production and distribution

Table 24: Consortia of Scientific and Technological expertise

Subcomponents	Activity/Sub-activities	Beneficiary Type (Individual/Farmer Group/Community)	Output Indicators
Consortia of Scientific and Technological expertise	MoUs with National and International Institutions	Community	Number of MoUs signed with institutions
	Analytical studies pertaining to climate resilience	Community	Number of analytical studies conducted
	Internships	Individual	Number of interns placed
	Agri-tech	Individual/Farmer Group/Community	Number of Agri-tech solutions developed/adopted

5.3 Sub-Component D2: Digital Services for Climate Resilient Agriculture :

NDKSP-II aims to boost resilience, productivity, and profitability for smallholder farmers by leveraging existing data and innovative technology. While maintaining current platforms, the project will introduce advanced tools to integrate precision agriculture, emission management, and real-time advisory services. These solutions will provide actionable insights, helping farmers and government agencies make informed decisions to mitigate climate risks. Key digital initiatives include upgrading existing applications and introducing new ones such as an Interactive Digital Soil Mapping Platform, Precision Agriculture Tools, Climate-Smart Advisory Services, Emission Management Systems, Integrated Farm Management Systems, and Market Linkage Platforms.

NDKSP-II will adopt a "digital by default" approach, integrating advanced technologies to support climate-smart agriculture. Building on Phase I's digital ecosystem, Phase II will incorporate Generative AI, blockchain, and machine learning to enhance decision-making, real-time monitoring, and service delivery. The project's Agriculture Data Exchange will serve

as a centralized hub, streamlining datasets across state departments to enable predictive analytics, optimize resources, and provide precision farming insights.

Farmers will receive hyper-localized advisories on irrigation, pest control, and disease management, ensuring timely, personalized support. Aligned with India Enterprise Architecture (IndEA) principles, these digital platforms will be scalable, efficient, and inclusive. The project will also deploy IoT sensors and satellite imagery for continuous crop and soil monitoring, while AI-driven analytics will forecast yields and market trends, helping farmers make informed decisions. Additionally, a blockchain-based traceability system will enhance transparency in supply chains, improving market access and consumer trust in product quality and origin.

This sub-component aims to integrate AI into every stage of climate-resilient agriculture, from pre-production to post-harvest management. The initiative will leverage strategic partnerships with national and global experts in AI and climate-resilient agriculture (CRA) to develop data-driven solutions.

Key AI-driven interventions include:

1. **AI-Powered Decision Support:** Using predictive analytics, AI will provide real-time insights on irrigation, pest control, and yield forecasts. By analyzing data from IoT sensors, satellite imagery, and historical trends, farmers will receive personalized, evidence-based recommendations for climate adaptation.
2. **Agriculture Data Universe & Predictive Analytics:** A centralized data platform will aggregate diverse datasets, enabling precision farming, resource optimization, and climate risk scenario analysis.
3. **Generative AI for Farmer Engagement:** AI-powered chatbots with multilingual support will offer farmers real-time advice on farming practices, weather updates, and market trends, enhancing accessibility and learning.
4. **Smart Agriculture with IoT & Remote Sensing:** IoT sensors and remote sensing will continuously monitor soil health, crop growth, and environmental conditions, enabling timely interventions, efficient resource use, and improved productivity.
By integrating AI into agricultural practices, the project aims to enhance resilience, sustainability, and profitability for smallholder farmers.

This section provides a bird's eye view of the purpose of the NDKSP IT Ecosystem, conceptual design of the IT Ecosystem, existing applications and new digital interventions for Phase II, integration with external applications and expansion of the Innovation lab.

Purpose of the NDKSP IT Ecosystem

The NDKSP IT Ecosystem is envisaged to cater to the following two broad objectives:

- Support implementation, monitoring and evaluation of the project to enable efficient service delivery
- Provide advisory support to the beneficiaries of the project to enhance climate-resilience and profitability of smallholder farming systems

Besides, the NDKSP IT Ecosystem is required to be designed such that it is innovative, integrated, user friendly and robust digital and new media platform comprising applications for multi-channel service delivery and data analysis. The NDKSP IT Ecosystem will cater to the following broad themes.

- Implementation of Innovative Technologies
- Development of Fool-proof IT Ecosystem
- Usage of Enhanced Data Analytics for making Informed decisions
- Dissemination of Digital Advisory Services
- Integration with External Stakeholders
- Solutions for holistic development of Agriculture Value Chain

Conceptual Design of the NDKSP IT Ecosystem for Phase II

The figure below depicts a conceptual design of the NDKSP IT Ecosystem architecture for Phase II of the project.

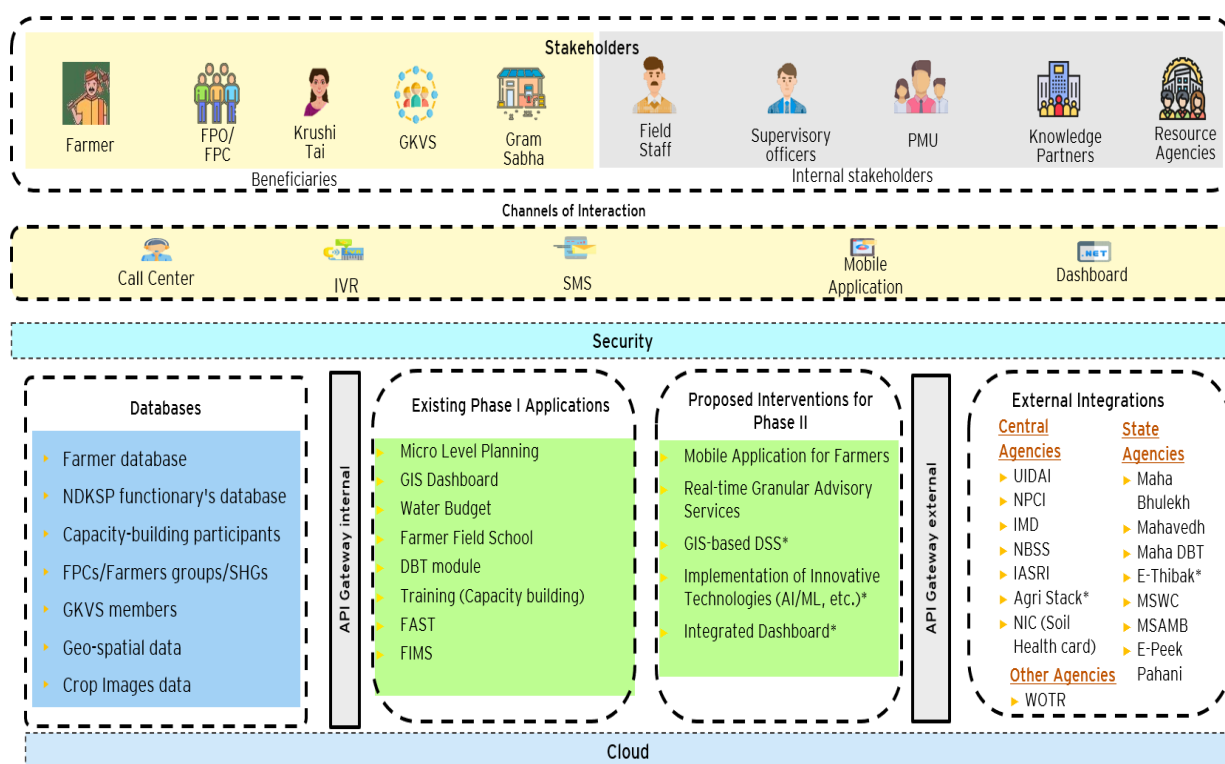


Figure 13: Conceptual design of the NDKSP IT Ecosystem architecture for Phase II

The following description briefly highlights the functionality of each layer that is an integral part of the conceptual design.

The key stakeholders of this IT Ecosystem include beneficiaries (Farmers and FPO/FPC/SHG), GKVS, Gram Sabha and Krushi Tai, PMU officials, Supervisory officers, Field staff, Knowledge partners and resource agencies.

The seamless exchange of information between stakeholders is through multiple service delivery channels like mobile application, SMS, dashboard, call centre & IVR.

Databases and External Integrations: The following are the various databases that will be integral to the project in Phase II.

- Farmer database – captures list of farmers and their registration details acquired through DBT
- The NDKSP functionary's database – captures list of PMU and field staff and their details that includes designations and responsibilities
- Capacity-building participants – captures details of farmers who were trained through the NDKSP
- FPCs/Farmers groups/SHGs – captures list of FPOs/FPCs and their registration details acquired through DBT
- GKVS members – captures list of GKVS members in the NDKSP villages and their contact details
- Geo-spatial database – captures geo-spatial data during inspections and mapping them with respective activities for multiple purposes
- Crop images database – captures images during inspections and crop health monitoring

All applications will be hosted on cloud with API based integration with multiple databases and service delivery channels. Cloud infrastructure would include data centres (DC) and data recovery (DR) mechanisms. Also, integration will be achieved with multiple central agencies, state agencies and third-party agencies primarily for user authentication, payment disbursement, land verification and advisory generation.

Enhancement and Standardization of Existing Applications

Following are the existing applications which are developed in Phase I and will function as key digital enablers for implementation of PDOs during Phase II of the project.

Table 25: Enhancement and Standardization of Existing Applications

Sr. No	Application	Description
1	Direct Benefit Transfer (DBT)	It includes three modules: a Farmer Module for processing applications by farmers, an NRM Module for processing village communities' applications for natural resource management works, and an FPO Module for processing applications from FPOs/FPCs/SHGs.
2	GIS Dashboard	It acts as a single-window platform for visualising and monitoring project activities, helping in decision-making and dissemination of information. It offers key features like location-based weather updates, crop monitoring, soil health data, and drill-down functionality from region to village level, using GIS and remote sensing technology to track crop cycles, rainfall, crop health, and vegetation cover.
3	Farmer Field School (FFS)	The app enables real-time documentation of field observations, geo-referencing of training plots, and managing and monitoring of climate-resilient farm field school sessions.
4	Field Activities Supervision and Tracking System (FAST)	This GIS-based app manages attendance, leave, and salary calculations for the NDKSP field officials, requiring a selfie for attendance marking and serving as a centralized monitoring platform for PMU officials.
5	Training Management	This application facilitates the planning, conducting, and monitoring of all capacity building initiatives, serving as a centralised platform to track activities and verify farmer training completion status required for fund disbursement.
6	Micro Level Planning (MLP)	This application aids communities in a 3-day planning exercise to develop a village plan by capturing and analysing socio-economic status, natural resource utilisation, cropping patterns, and value chain integration, while an automated engine calculates water balance for the creation of Village Development Plan (VDP).
7	Water Balance	The application is used to compute water balance at village and watershed level. The tool takes into consideration the input data such as rainfall, soil, land use, cropping pattern, existing water bodies and other infrastructure. The Water Balance output is used to plan the NRM interventions and Crop advisories.
8	Finance Information Management System (FIMS)	This application enables budget allocation from PMU to GKVS -level offices and facilitates real-time tracking and monitoring of allocated funds across categories, generating seamless reports.

Although these applications are well developed, they need to be enhanced to make them more robust, and to be standardised to make them more compatible when integrated with

external platforms. Following are the key activities to be performed for enhancement and standardisation of existing applications.

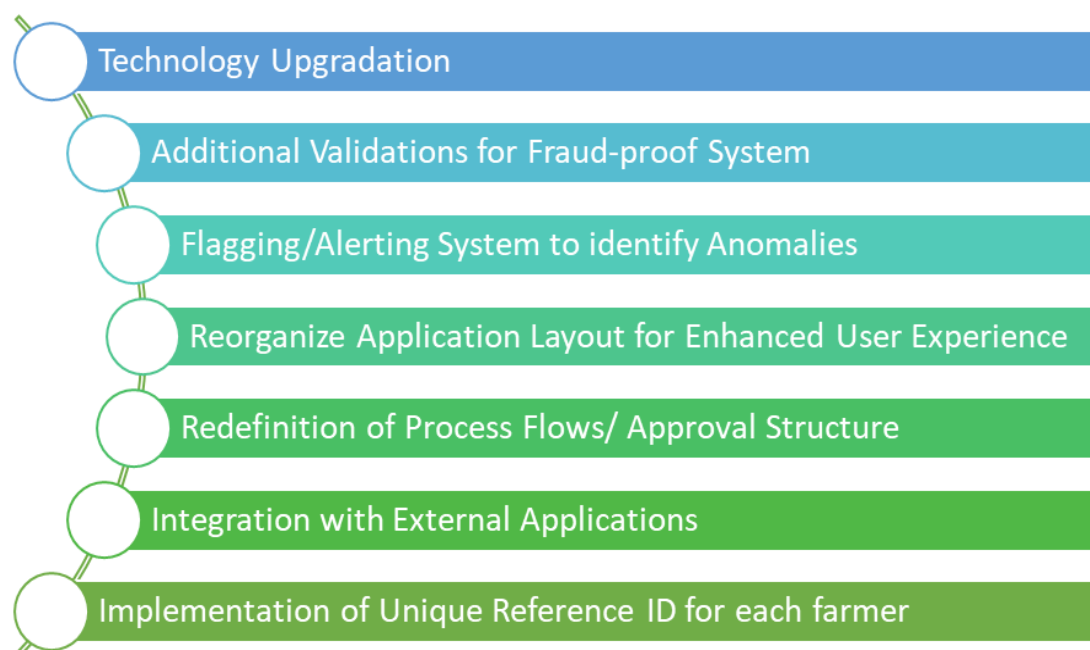


Figure 14: Key activities in existing applications

Single Window Interface for farmers

A Single Window Interface (SWI) for Farmers is an open, interoperable and federated digital platform that is designed to become a one-stop destination for small and marginal farmers, by providing digital access to essential services like crop advisories, weather forecasts, fertilizer calculations, crop management, pest and disease information, real-time market prices, crop budgeting, community forum and government schemes and policies. This platform would bridge the gap between farmers and policymakers, promote transparency, and facilitate efficient decision-making. For the convenience of farmers and other stakeholders, this interface will be available in the form of a Web Portal and a Mobile App, with same features to be available on both the products.

To leverage the strength of the large databases available across multiple agencies in the Agriculture Dept., which are often credible and ground-truthed, the single window interface would be linked to the VISTAAR Network.

A new pan-India DPI-effort, called VISTAAR (“Virtually Integrated System to Access Agricultural Resources”), is an open, interoperable and federated digital network that is designed to increase the incomes of small-scale farmers, by digitizing the social networks that farmers and those that work with them use to share knowledge with each other.

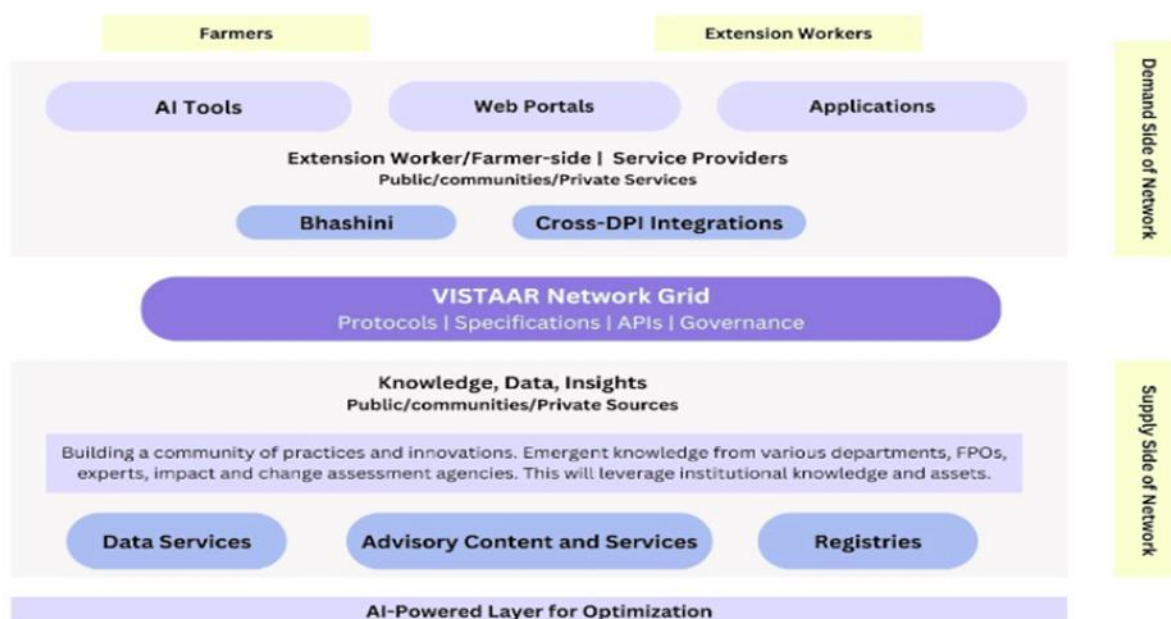


Figure 15: VISTAAR Architecture

The network is supported by AI/ML Layer which acts as a digital bridge between information providers and information seekers (ex: farmers) through implementation of standardized specifications (such as Beckn Protocol) for seamless communication.

VISTAAR Network based AI-Powered chatbot would be the key feature of the Single Window Interface. This chatbot which is integral to both web portal and mobile app, will fetch data from diverse information providers and provide real-time services related to weather forecast, crop advisory, best agricultural practices, market prices, scheme information, etc., in multiple languages with the support from Bhashini or other language models.

GIS-based Decision Support System:

The NDKSP Phase II would elevate GIS Dashboard to a Decision Support System (DSS) which combines geographic information with sophisticated analytical tools to support decision-making processes. GIS Dashboard for Phase II comprises all existing features of the NDKSP Phase I Dashboard along with enhanced features, Improved performance, UI enhancements and Designed for Scalability. Below are key new features of the GIS Dashboard for the NDKSP Phase II.

- Along with existing DBT Farmer Module services which display application details in geographical format, services for NRM and FPO/FPC Modules also should be developed.
- Real-time weather information along with weather advisory, crop advisory and soil-based advisory
- Other features include enhancement of water budget tool and monitoring status of FFS and MLP activities

Along with above features, the new GIS-based DSS comprises integration of multiple layers of spatial data, analytical tools for spatial analysis, visualisation tools, and tools for evaluation of different scenarios.

Along with above features, GIS-DSS comprises various Analytical and Visualisation Dashboards which are Use Cases of Other NDKSP applications. These use case related dashboards are explained in further sections.

Integrated Dashboard as Decision Support System (DSS):

The integrated dashboard as a Decision Support System (DSS) for the NDKSP consolidates and visualises data from various applications, aiding in decision-making by presenting real-time insights through charts, graphs, and maps. It facilitates scenario analysis, allowing officials to simulate different outcomes based on varying inputs, which enhances risk management and strategic planning. Additionally, the dashboard helps track responsibilities and pending actions across the NDKSP applications, streamlining workflows and ensuring accountability. This functionality allows officials from district to village level to monitor progress efficiently, ensuring timely decision-making and alignment with project timelines. Ultimately, the dashboard improves operational efficiency and promotes sustainable practices while enhancing individual and collective accountability within the organisation.

Implementation of Innovative Technologies

AI-Driven Advisory Systems:

- a) ***Pest/Disease Image Identification and Advisory:*** An AI-based pest and disease identification system will identify pest/disease in the uploaded image and generate necessary advisory to mitigate the risk of pest/disease attack.
- b) ***Enhancing Resource Allocation and Preventing Redundancy:*** AI for Location-Based Benefit Application streamlines resource allocation and prevents redundancy by identifying existing activities in proposed project locations. It optimises the application process, ensuring benefits are directed where needed, enhancing the fairness and effectiveness of programs for the benefit of farmers and the government.
- c) ***Image Recognition through AI/ML:***
 - i) **Enhancing Vendor Oversight with Image Recognition and GSTIN Extraction:** Streamlining vendor analysis with AI-driven GSTIN extraction promotes transparency and efficient procurement in the NDKSP.
 - ii) **Ensuring Accountability and Verification in Agricultural Infrastructure Claims:** Visual documentation and AI-driven serial number extraction boost accountability, curbing fraudulent claims and ensuring transparency.
 - iii) **Strengthening Post-Disbursement Oversight:** Ensuring accuracy in post-disbursement inspections through image recognition and geotagging, enhancing transparency and project management.

- d) **The NDKSP's Innovative AI Chatbot for Agriculture:** The NDKSP can collaborate with the Generative AI model providers to develop a specialised chatbot for agriculture. It leverages cognitive search capabilities to enhance contextual accuracy and employs AI models and Bhashini for Marathi conversational interactions. With a training model on curated agricultural data, this AI chatbot delivers real-time, precise information on farming techniques, crops, schemes, pests, and diseases, revolutionising farmers' access to vital insights and resources.

IoT Integration:

Smart Agriculture

- a) **IoT based farming:** Integrating IoT into the NDKSP offers real-time data collection and analysis for climate-resilient agriculture, focusing on soil conditions, water management, weather forecasting, and supply chain tracking. IoT-based farming implementation can be initiated on a Proof-of-Concept (PoC) basis in select areas to empower farmers with informed decisions, resource efficiency, and climate-resilient practices.
- b) **Drone-based observations:** Drones equipped with sensors are vital for the NDKSP, offering crop monitoring, precision agriculture, pest control, mapping, and disaster management. A PoC for drone-based applications, collaborating with drone tech start-ups, can guide project expansion for more climate-resilient agriculture.

GIS-DSS and Remote Sensing Integration

GIS-based DSS:

The GIS-based DSS comprises integration of multiple layers of spatial data, analytical tools for spatial analysis, visualisation tools, and tools for evaluation of different scenarios. Also, GIS-DSS comprises various Analytical and Visualisation Dashboards which are Use Cases of Other NDKSP applications.

Farmer Field School (FFS) Use Cases:

- Sowing Date Dashboard - Visualise the sowing patterns for various crops within a specified date range
- Pest & Disease Dashboard - Analyse the prevalence of pests and diseases affecting crops
- Technology Demonstration & Adoption Dashboard - Visualise the adoption of agricultural technologies in FFS and control plots
- Soil Testing Dashboard - Track changes in soil nutrients over time
- Yield & Cost of Cultivation Dashboard - Compare yield and cost of cultivation between FFS and control plots

NRM Analytical Dashboard for Site Suitability:

Use of satellite imagery of 10m resolution in the planning process to identify existing structures before implementation and new structures post-implementation.

Micro Level Planning (MLP) Use Case - Runoff Water Impoundment and Management Dashboard:

The dashboard will incorporate the following key metrics that are crucial for understanding the dynamics of runoff water management:

- **Runoff Available:** The total amount of runoff water available in a given area.
- **Currently Impounded:** The volume or percentage of runoff water that is currently being stored or utilised.
- **Runoff Available for Impounding:** The remaining runoff water that can still be impounded or utilised.
- **Runoff Impounded by Proposed Structures:** The expected volume or percentage of runoff water that will be impounded by future or planned structures.
- **Runoff Available after Proposed Structures:** The remaining runoff water available after the proposed structures have been implemented.

Training Application Use Case:

Monitoring and Tracking of the ground staff on daily basis for their status of work completion.

Integration of Remote Sensing:

Synergistic use of technologies such as Remote Sensing could lead to a robust system for building agricultural intelligence in order to address various issues linked to the performance of crops at village, taluka, and district level. The main objective of use of remote sensing in agriculture is for real time monitoring of crops and interpreting / inferring the diagnostics to mitigate the stress due to adverse conditions or calamities. Also, quantitative indicators pertaining to weather, soil and crops supported by intensive field information/observation collected through digital tools on a real-time basis are used in the analysis to generate strong agricultural intelligence information for specific crops throughout the season.

Use of GIS and Remote Sensing in Village Planning:

- **Village Micro Planning** - For better village level planning it is necessary for the NDKSP to do natural resources mapping. Using geospatial technology, the NDKSP can prepare village-wise natural resources maps.
- **Village Base map** – With help of GIS, a base map of the village can be prepared which gives the basic information about the village, e.g. village drainage pattern, farm details with survey number, reference layer such as road network, and village micro hydrological units.
- **Land Use Land Cover** - This map will give the information about village land use. It helps us to understand the cropping season like kharif and rabi, and the indicator for

ground water availability for agriculture. Also, this map will help the NDKSP in understanding the cropping pattern.

- **Watershed Potential Treatment (WPT) Map** - This map is very useful for watershed management and soil and water conservation planning. Using multiple GIS and RS databases e.g. Topography, soil, water resources structures, LULC, Soil Land capability, Drainages, Groundwater prospect etc. A WPT map will be prepared. It will also help us in understanding the existing water resources structures, area of the village which should be given priority for soil and water conservation works.

Use of GIS and Remote Sensing in Agricultural Activities:

Satellite data can also be used to monitor various agricultural activities implemented in the project region using the geo-coordinates and random sampling and help improve the monitoring of these activities during and after the project implementation period. Some of the activities where satellite data can be used are as below:

- Crop Health Monitoring
- Crop Mapping and Classification
- Yield Estimation and Forecasting
- Pest and Disease Management
- Soil Analysis and Fertility Mapping
- Land-Use Planning and Monitoring
- Natural Resource Management

Use of GIS and Remote Sensing in Carbon Sequestration:

Carbon sequestration helps mitigate the effects of climate change by reducing the concentration of greenhouse gases in the atmosphere. GIS plays a crucial role in carbon sequestration efforts by helping in site selection and assessment, carbon stage estimation, devising emission reduction strategies, spatial data visualisation, monitoring land use changes, and managing activities related to capturing and storing carbon dioxide (CO₂) from the atmosphere.

Integration with External Applications

While one of the main objectives of the NDKSP Phase II revolves around the data, it is obvious to coordinate with external agencies to integrate our system with their applications for enhancing the NDKSP dataset knowledge and decision-making capabilities.

Also, integration of external applications with the NDKSP IT Ecosystem offers a multitude of benefits such as enhancing the efficiency and productivity of our system, and availability of real-time data for better decision making.

The following figure depicts the key applications/activities that are to be integrated with the NDKSP Digital Ecosystem in the NDKSP Phase II.

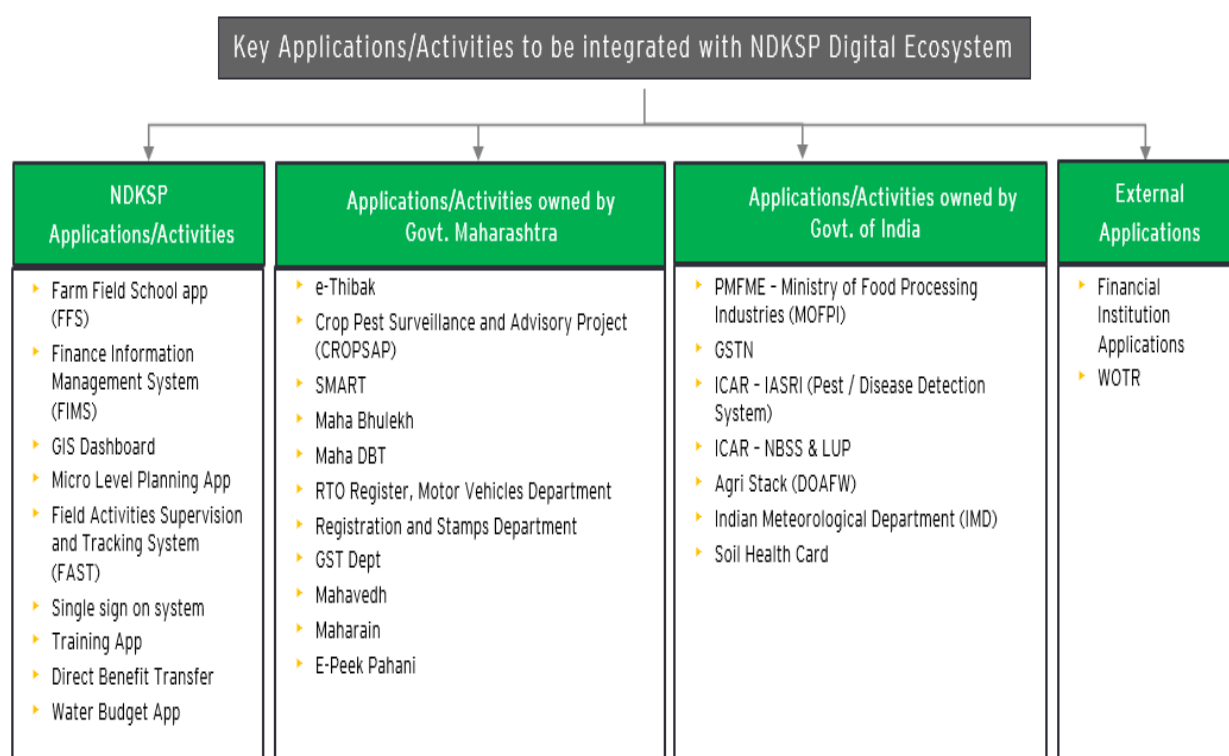


Figure 16: Key application / activities to be integrated with NDKSP digital ecosystem

Integration with Agri-Stack of Ministry of Agriculture, Govt. of India

Integrating the NDKSP with the Agri-Stack of the Ministry of Agriculture, Govt. of India offers several advantages, including improved data access for climate-resilient practices, enhanced traceability, better coordination among stakeholders, and fostering innovation and market access for climate-resilient agriculture products. This integration can significantly benefit the agriculture sector. Following are the key areas where Integration with Agri Stack benefits the NDKSP.

- **Data Exchange**
- **Enhanced Analytics**
- **Targeted Interventions**
- **DBT Implementation**
- **Market Linkages**
- **Elimination of Duplicate Subsidy Requests**
- **Data Management and Integration System**

Dataset Expansion for the NDKSP Phase II

The process of capturing and integrating data from a multitude of sources is a pivotal component of the NDKSP Phase II. This process facilitates the harmonisation of one-time, periodic, and continuous data feeds from central government agencies, various departments of the Government of Maharashtra, external entities, and the NDKSP's internal datasets.

Data Utilisation and Output

The harmonised data undergoes transformation and analysis to provide actionable insights for various stakeholders within the NDKSP. The resulting data serves as the backbone for Interactive Dashboards, Analytical Reports, Custom Reports, and GIS Maps.

Data Transformation through Agriculture Data Universe

The NDKSP is at the forefront of this agricultural revolution, placing farmers at the epicentre of data-driven decision-making. This section outlines the concept of the "Agriculture Data Universe," which represents a comprehensive strategy to harness data from diverse sources and empower farmers across Maharashtra.

Key Components

1. **Data Integration:** The heart of the Agriculture Data Universe is the seamless integration of data from multiple sources. APIs are employed to aggregate data on weather conditions, land records, soil health, and other critical factors, creating a comprehensive and dynamic repository.
2. **Farmer-Centric Design:** The system places farmers at the centre, ensuring that the information and support they receive are tailored to their specific needs. This includes personalised advisory services, crop management insights, and real-time alerts based on data analysis.
3. **Climate Resilience:** The data-driven approach aids in building climate-resilient agriculture by providing farmers with tools to adapt to changing weather patterns, make informed planting decisions, and optimise resource utilisation.
4. **Collaborative Ecosystem:** The Agriculture Data Universe fosters collaboration with various stakeholders, including government bodies, start-ups, and agricultural experts. This collective effort aims to drive innovation and enhance the agricultural landscape of Maharashtra.

Impact

The Agriculture Data Universe has the potential to revolutionise agriculture in Maharashtra:

- **Enhanced Productivity:** Farmers gain access to real-time information on weather, soil conditions, and best practices, leading to increased crop yields and improved income.
- **Resource Optimization:** Efficient use of resources such as water and fertilisers are promoted, contributing to sustainable and eco-friendly agriculture.
- **Risk Mitigation:** Data-driven insights enable farmers to proactively address climate-related risks and crop diseases, reducing losses and improving livelihoods.
- **Economic Growth:** The implementation of the Agriculture Data Universe will drive the economic growth in the agriculture sector and improve the overall well-being of farmers in Maharashtra.
- **Cost Reduction:** Farmers can get access to a wide range of information including advisories at free of cost which would reduce their overall farming costs.

Agriculture Data universe to Decision Support System – Transforming Data into Actionable Insights

This roadmap from the Agriculture Data Universe to a Decision Support System (DSS) outlines the key stages and milestones for this transformation, where data becomes an asset in driving informed decision-making for the benefit of farmers across Maharashtra.

Stage 1: Data Aggregation and Integration

This stage involves the collection of data from diverse sources, including but not limited to Agri stack, Land Records, IMD, Soil Health Card, and other relevant agencies. The data is aggregated through APIs, ensuring real-time updates and a comprehensive dataset.

Stage 2: Data Processing and Analytics

Once the data is aggregated, the next stage focuses on processing and analytics. Data processing techniques are applied to clean, transform, and prepare the data for analysis.

Stage 3: Farmer-Centric Design

The success of the roadmap relies on a farmer-centric approach. In this stage, the focus shifts to designing the Decision Support System with farmers in mind.

Stage 4: Real-Time Decision Support

The core objective of this roadmap is to provide real-time decision support to farmers. In this stage, the Decision Support System is put to use.

Stage 5: Ongoing Improvement and Expansion

The roadmap is a dynamic process that never truly ends. It continuously seeks to improve and expand the Decision Support System.

Expansion of Digital Innovation Lab

The Digital Innovation Lab at the NDKSP is one of the key drivers in successful implementation of the project. To take advantage of applications developed under the lab and utilise them for a greater cause, it is necessary for the Innovation Lab to expand its services and leverage the technology to develop and deliver innovative solutions to resolve wider-range of problems. To make Innovation Lab more robust and effective in the NDKSP Phase II, it is much needed for the inclusion of external partners as stakeholders of the lab for improving its capabilities.

Objectives:

Implementation of the Hub and Spoke model at the NDKSP with Digital Innovation Lab becoming the central hub and external partners such as Agri-tech Start-ups and Agricultural Universities as spokes for the creation of innovative solutions would be the key focal point for its expansion.

Following are the key objectives of Digital Innovation Lab as Central Innovation Hub during the NDKSP Phase II.

- ***Enhance Research and Development Capabilities:*** Collaboration with external partners to enhance its lab's infrastructure and resources to support advanced research and development projects which would in turn increase the ability to develop and test new technologies and solutions.

- **Drive Digital Transformation:** Lead the development of initiatives that accelerate digital transformation across the Agriculture Industry by developing and implementing advanced digital technologies which would improve operational efficiencies for the overall agriculture industry.
- **Support Startups and Entrepreneurs:** Digital Innovation Lab will act as Incubator/accelerator to support Agri-tech start-ups and innovators to scale their operations and bring innovative products to market.
- **Foster Collaboration and Partnerships:** Creation of opportunities for collaboration with industry experts, academic institutions, and Agri-tech startups which would strengthen partnerships and drive innovation through collective expertise and shared resources.
- **Support Emerging Technologies:** Leverage the in-house resources and external partners expertise to drive research and development of emerging technologies such as AI, blockchain, IoT and remote sensing. This would place the lab at the forefront of technological advancements and create pioneering solutions.

Implementation Model:

The NDKSP provides valuable datasets that can drive innovation and solutions tailored to the specific needs of Maharashtra's diverse agricultural landscape. This implementation model outlines a strategic approach for leveraging these datasets to foster effective partnerships and achieve impactful outcomes. Following are the key components that drive the implementation of this model.

- **Data Access Management:** The NDKSP should create a comprehensive dataset catalogue with a secure access portal for startups, ensure data quality, and provide regular updates for reliable analysis
- **Partnership Framework:** The NDKSP should develop an MoU for collaboration, form joint working groups with experts, and hold regular meetings to review progress and address challenges.
- **Innovation and Development:** Collaborate to identify key agricultural challenges related to climate resilience, develop solutions using the NDKSP datasets, and launch pilot projects to test and validate these solutions in local farming communities.
- **Capacity Building and Training:** Conduct workshops and mentorship sessions between PMU officials, Agri experts, and startups to bridge knowledge gaps and drive innovation, while also offering ongoing technical support to optimise data-driven agricultural solutions.
- **Monitoring and Evaluation:** Define performance metrics to evaluate collaboration effectiveness, implement a feedback mechanism for stakeholder input, and regularly assess the impact of solutions on agricultural resilience, productivity, and sustainability.

- **Dissemination and Scaling:** Publish case studies and reports to share insights with the agricultural community and explore opportunities to scale successful solutions to other regions and sectors based on lessons from pilot projects.

Expansion of lab services to state and national levels:

Further to the development of innovative and scalable solutions, the NDKSP can promote these solutions to the other departments of Government of Maharashtra for resolving wider issues, or take it to other state governments in the country or central government through customization of solutions.

5.4 Sub-Component D3: Monitoring & Evaluation and Learning:

The project aims to mainstream climate resilience in the farming community through a bottom-up approach. The project will develop a result based robust monitoring and evaluation system (M&E) to track and measure the progress towards achieving the project development objective (PDO). A robust monitoring & evaluation framework and project management information system (MIS) would capture the activity wise all the important outputs and outcomes. The project will develop both internal and external monitoring mechanisms to consistently assess the progress of the project, which will also cover ESMF activities under the project.

Additionally, it will also provide PMU with real time information of physical progress (outputs) and financial milestone vis-à-vis outcome indicators. The M&E system intends to provide real time data and insights allowing identification of challenges or any deviation from planned outcomes. The M&E system will ensure that timely adjustments and course corrections can be made as needed by monitoring progress of project components. The learning from M&E of indicators would be built into project implementation processes for improving project performance.

The key building blocks of the M&E system will comprise of:

- 1) Inception Report
- 2) Baseline survey for sampled clusters
- 3) Concurrent Process & Progress Monitoring
- 4) Mid-term evaluation
- 5) End-term Impact Evaluation
- 6) Assessment of Environmental & Social Management Framework
- 7) Result Framework Indicators (RFID) for the project



Figure 17: The key building blocks of the M&E system

The PMU will have the responsibility for planning and coordinating M&E activities for the entire project. The M&E process in the project will be managed in the following manner:

- A baseline study of relevant indicators shall be conducted by the PMU through an M & E service provider who has adequate experience in planning, monitoring and impact assessment.
- The team of technical specialists, finance and procurement specialists in PMU with M&E coordinator will be responsible for the quarterly consolidation and reporting.
- The M&E specialist from the PMU will concurrently monitor and evaluate the identified indicators with help of M&E agency.

Internal Monitoring Mechanism

Monitoring and Evaluation Dashboard

The project will design and create a dashboard to track the physical and financial progress of the activities implemented under the project and report. The dashboard will be designed and developed with the help of MIS and GIS systems. All the applications developed under the project, such as the DBT portal, FPO portal, NRM portal, FFS app, MLP app, etc., will be integrated with the M&E dashboard to capture the progress of the activities. Also, the M&E various reports formats and Result Framework Indicators (RFID) will be included in the M&E dashboard

Performance Monitoring System

The PMU will use the performance monitoring index (PMI) to systematically track and assess the activity-wise progress toward achieving PDO. This system allows PMU to monitor performance of field functionaries and make informed decisions based on real-time data to achieve project objectives. The PMU will define and identify the key performance indicators (KPIs) that align with PDO to monitor the performance of Districts, Subdivisions, Talukas, and gram panchayats against key project activities. The purpose of using PMI is to create an environment of competition amongst districts, sub-divisions, talukas, and Gram panchayats to improve project implementation performance. This system will include graphical representations of results, breakdown into individual component-wise reports, monitoring, and review of progress of project activities. The PMU will design a dynamic and flexible PMI system, which includes new parameters periodically, such as every month/quarter. The ranking of districts, subdivisions, talukas, and Gram panchayat based on their performance index in project activities will help their objective assessment. This system will help identify areas needing attention, enabling necessary corrective actions.

Result Framework and Indicators

Considering the scope of the project which includes a large number of interventions, it is envisaged that there will be large-scale generation of data on a recurring basis during the entire length of the project period. This data is expected to provide support to analysis, research, and mid-course correction of the project and most specifically, concurrent monitoring and evaluation during the project period. Learnings generated from the project would need to be imbibed and changes will be made in operational procedures, as and when required. There would also be a need of spreading good practices beyond the boundaries of the targeted intervention area to ensure cross-learning.

A robust monitoring and evaluation system aids in assessing the performance of the project against predetermined project indicators. The system, in addition, will help PMU and policy makers in assessing the trends in important parameters arising out of the project. Imaginative use of M&E framework can help in understanding some of the core issues in a public sector development program such as the functioning of new community level organisation such as Farmer Interest Groups (FIGs) and their associations in the form of Farmer Producer Organizations (FPOs), governance issues, empowerment of the people, inclusiveness, democratic principles, principles of equity, etc.

Thus, there is an imperative need for a systematic assessment and understanding of the key project indicators at project conceptualization / starting phase, implementation phase, project completion, and post project completion phase. The M & E system would help in improving the accountability and transparency, meeting timeframes and project milestones, maintaining quality of goods/works/services, and meeting the targets that contribute to Project Development Objective (PDO) level indicators.

Objectives:

Keeping the above in view, objectives of the Monitoring, Learning & Evaluation Component include the following.

- To develop an integrated system to track the progress of activities, outputs outcomes and impacts of the project
- To understand the effects of development interventions and the progress in comparison to the baseline situation
- To establish a comprehensive system for baseline data collection, analysis, evaluation, and report generation, which will enable continuous monitoring of measurable indicators throughout the project lifecycle. This system will also include before and after project evaluations, along with the generation of Concurrent Monitoring Reports to ensure timely insights and adjustments
- To set up a standardised learning and evaluation process for stakeholders and dissemination of learning from the development process for use by the stakeholders as well as the wider community
- To conduct implementation audit, monitoring, tracking, impact and outcome analysis and building the capacity of implementation partners

Approach:

To achieve the above-mentioned objectives the key approach is to have an internal and external Monitoring and Evaluation system, a participatory monitoring, learning and evaluation process, an MIS and GIS product designed to store, track, analyse and present data and information.

The main focus of the project M&E system will be on

- Results-based management (timely monitoring, analysis and feedback on project activities)
- Evaluation (of project outputs, outcomes and impacts, using appropriate baseline and controls)
- Self/participatory monitoring and learning on a concurrent basis
- Periodic Benefits tracking on sample basis to understand status of various activities and project processes
- The system will also be used for
- Planning and Defining Course of Action
- Learning for all Stakeholders
- Empowerment and Capacity Building of the community-based institutions

The introduction of the external M&E agency will help the Project Management Unit (PMU) to gain an external perspective of the work being carried out under the project. The MIS and GIS will provide a quick and easy understanding of data and information not just at the project management level but also at other levels involving direct and indirect stakeholders as it would be web enabled. Additionally, the IT-MIS system will promote transparency and accountability in procurement and the financial system. It

is planned to design a multi-channel IT system, which would also include a mobile application to collect and aggregate information at various levels.

Component of M&E System

(a) The Management Information System (MIS) will serve as a crucial tool for effective project management, focusing primarily on monitoring both inputs and outputs. Standardised reports on inputs will be generated to help project teams evaluate progress. Additionally, the MIS will be integrated into a comprehensive IT system with interactive dashboards that provide real-time insights. The system will also include various subsystems, outreach documentation, and special orders or guidelines issued periodically. These dashboards will enhance decision-making by offering a visual representation of key project metrics and performance indicators.

(b) **Third Party or External Monitoring:** External monitoring will be carried out by a third-party agency at three key stages of the project: a baseline study at the start, a midline study at the mid-term, and an endline study once implementation is complete and the consolidation phase begins. Additionally, six-monthly concurrent monitoring rounds will be conducted to assess both progress and processes. The project will also ensure random asset verification by the external agency to maintain oversight.

The project will also have internal and external audits carried out by third party agencies and social audit at village level by the community. Drawing lessons from social, internal, and external audits can significantly enhance project monitoring. Together, these audits provide a comprehensive monitoring framework that strengthens governance, identifies risks, and drives corrective actions in a timely manner.

Baseline Survey: A baseline survey will be undertaken by the project to understand the pre- project situation on key parameters covering socio-economic dimensions as well as the environmental aspects. The baseline survey will not only cover the project areas but also relevant Control sites which will be used to assess the (incremental) impact of project interventions vis à-vis generic growth influences over time.

Concurrent Monitoring: Field level implementation would be independently monitored by identified external M&E agencies. This will be done by selecting representative clusters within each major area for intensive periodical visits. These reviews will focus on understanding the critical processes as well as the progress of performance indicators. Field visits will also include random verification of the information of progress reports as well as understanding the sequencing of events. The progress reports data prepared by the M & E agency with a half yearly cycle (including sample and control areas) will be used to understand overall progress.

- Six-monthly progress reports would be prepared covering the following
- up-to-date physical and financial expenditure data compared to annual and end project targets
- updated Key Performance Indicators (KPI) compared to annual and end-project targets
- successes and problems encountered during the reporting period with suggested remedial actions
- Socio-economic and environmental impacts of the project.
- Joint reviews of the project will be undertaken on a six-monthly basis wherein a number of stakeholders will participate to discuss progress and achievement of results compared to the plan.
- Impact Assessment: Two full-scale impact evaluation studies will be undertaken at midpoint (Mid-Term Assessment Report) and at completion (Final Assessment Report) of project implementation.

Mid Term Assessment: The study would include an impact assessment of the project to date and also focus on implementation processes and recommend adjustments in the project design and/or implementation arrangements to overcome identified bottlenecks. The Assessment Report would be a comprehensive overall impact assessment including quantitative and qualitative assessment of progress against project development objectives. The assessment will include socio-economic and environmental impacts of the project.

End term Assessment: The Final Assessment will be taken up towards the end of the project. It will focus on understanding the outcomes of project interventions and effect of the same on the target population and compare these with the baseline situation to assess the effectiveness of the project in terms of physical infrastructure development, socio economic changes, environmental impacts as well as institutional strengthening (of FPOs/FPCs, CHCs, etc.)

Both the impact assessments will update the financial and economic analysis of project returns undertaken at the start of the project. These assessments will also undertake analysis of issues relating to sustainability of project outcomes and impacts.

Social and Environmental Management Audits: At the project's conclusion, the Monitoring & Evaluation (M&E) agency will conduct an audit to assess the implementation of the Environment and Social Management Framework (ESMF). These audits will evaluate the effectiveness of the social and environmental management measures applied during the project and the outcomes achieved.

Result Framework Indicators (RFID) –

The proposed project results framework indicators are presented in annexure 7.

Feedback and learning mechanism

In addition to the responsibilities assigned to the M&E agency, the PMU, along with field functionaries from the Agriculture Department at various levels, will conduct regular field visits. These visits will complement the work of the M&E agency, enabling a more integrated and thorough review of project activities. The 360-degree feedback gathered through this process will provide valuable insights and create opportunities for continuous learning and improvement.

The project will build on the strong and robust IT infrastructure established during Phase 1, enhancing its capacity to support efficient and transparent operations. The IT system includes multiple internal checks and validation mechanisms designed to ensure transparency, accuracy, and accountability. These checks help prevent duplication of data and activities, reducing the potential for errors or manipulation. This system will use geotagged images captured before and after project interventions, ensuring a clear and verifiable record of on-ground work. This process will act as a safeguard against fraud, enabling real-time monitoring and validation of project activities. The integration of such technologies will significantly strengthen oversight and improve confidence in project implementation.

We prioritise farmers at the heart of our efforts, adopting a "farmer-first" approach to improve their livelihoods. To support this, we will gather direct feedback from farmers via our Farmer Mobile Application. A dedicated Call Centre will be established within the PMU to address farmer queries promptly and effectively. Additionally, a comprehensive feedback and grievance management system for all stakeholders will be implemented to ensure transparency and continuous improvement in service delivery.

5.5 Sub-Component D4: Capacity Enhancement

The Nanaji Deshmukh Krushi Sanjeevani Prakalp (NDKSP) phase II will be implemented in 21 districts in 7196 villages. To assess the capacity building requirement on various topics for various stakeholders a "Capacity Enhancement Need Assessment" (CENA 2.0) has been conducted by the VANAMATI, Nagpur, an apex training centre under the aegis of the Department of Agriculture, Government of Maharashtra. Apart from the recommended CENA trainings, the need-based training programs will also be imparted by the project on selective thematic areas to the project stakeholders such as Project Management Unit (PMU) level officials and staff, District and Sub divisional Project Officials, Project Specialists, Field Functionaries, Farmers, members of FPO/FPC/SHGs, women entrepreneurs, field level implementing agency Gram Krishi Sanjeevani Samiti's members, individual stakeholders etc. The project will also impart need based training to project agricultural extension functionaries

and other project supportive individuals/ groups as appropriate. The capacity building program will include Training Programs, Workshop, Seminars, Conferences, Dialogues, Webinars, Exposure visits and various online events on key areas and prepare extension functionaries to cope with new challenges and for effective implementation of the project.

This project is complex in nature and requires partnerships with various specialised agencies and networks. Additionally, institutional development and the transfer of technology and processes to community-based organisations will be crucial components. A capacity development program is defined as a series of activities aimed at strengthening the knowledge, skills, abilities, and behaviours of individuals, while also improving institutional structures and processes to achieve project objectives sustainably.

Given the project's focus on enhancing climate resilience, the capacity-building efforts will prioritise raising awareness about climate change issues and strengthening community capabilities to address them.

To achieve this, the project will offer a range of online and in-person training programs in collaboration with national and state-level training institutes such as the Vaikunth Mehta National Institute of Cooperative Management (VAMNICOM) in Pune, the Vasant Rao Naik State Agricultural Extension Management Training Institute (VANAMATI) in Nagpur, the Yashwantrao Chavan Academy of Development Administration (YASHADA) in Pune, Krishi Vigyan Kendras (KVKs), Regional Agriculture Extension Management Training Institutes (RAMETI), the National Institute of Post Harvest Technology (NIPHT) in Pune, the National Institute of Agricultural Extension Management (MANAGE), the Bankers Institute of Rural Development (BIRD-NABARD), the National Institute of Plant Health Management (NIPHM), and other Centers of Excellence.

In addition to the aforementioned training institute and arrangements, the project will collaborate with specialized training institutes and implement capacity-building initiatives as needed, in alignment with the project's implementation strategy and guidelines.

Capacity Building

A key aspect of the NDKSP Phase II is its emphasis on boosting the productive potential of villages and clusters while strengthening their natural resource base and community capabilities. The initiative prioritises empowering local communities to engage in participatory planning, implementation, and monitoring with a sustainability focus. Acknowledging the multidisciplinary nature of the NDKSP Phase II, the project recognizes that capacity building should be a continuous and needs-based endeavour at all levels. In addition to training farmers, the project aims to enhance the capacity of the project management unit (PMU) and organisations involved in supporting farmers' development (FPO/FPC/SHG, CBOs, farmer interest groups etc.) through training and exposure visits. This capacity development for farmers will encompass a wide range of skill areas, including natural resource

management, farm mechanisation, and agricultural system development in the context of climate change and adaptation. To facilitate this, the project will employ various strategies, collaborating with different organisations and institutions for effective capacity building.

Capacity Development Objectives

The primary goal of capacity building in this project is to enhance skills for implementing climate-smart agriculture, performing essential functions, and promoting adaptation and resilience through various IT-enabled methodologies and technologies. This initiative aims to create a solid foundation for achieving project objectives by increasing community participation, encouraging positive engagement in managing common property resources, integrating Environmental and Social Management Framework (ESMF) principles into all project activities, and improving the socio-economic conditions of marginalised groups, particularly women.

The objective of this component is to promote the adoption of strategies for building climate resilience through long-term adaptive management of agriculture, soil, and water resources. It also focuses on skill development for small farmers to enhance resource use efficiency and create emission-competitive production systems, supported by data-driven decision-making tools. The project envisions large-scale capacity development programs for small farmers and other stakeholders in drought- and flood prone regions. Additionally, action research and analytical studies will contribute to the knowledge base, with a strong emphasis on innovation, particularly in resilient technologies within the agricultural sector.

Capacity Development Strategy

The capacity building understanding goes beyond 'training' programs and involves a holistic approach that includes human resource development, organisation development, system / institutional development and cooperation, and network development. All these processes are seen as a continuous process enabling stakeholders, functionaries, implementers, and policymakers to enhance their knowledge and skills and to develop the required orientation and perspectives thereby becoming more effective in performing their roles and responsibilities.

During the project implementation phase, a variety of activities and campaigns will be carried out, including Farmer Field Schools (FFS), the Kharif Hangam (season) Campaign, and Krishi Sanjeevani Saptah (Agriculture week). These on-field initiatives will also enhance the capacity of stakeholders involved.

In the NDKSP-II, four-dimensional strategies have been created to enhance the use of human capital, social capital, and existing institutions in the project area. These strategies focus on:

- Developing human resources and personnel.
- Enhancing the effectiveness and efficiency of organisations.

- Fostering cooperation between organisations and building networks.
- Promoting institutional frameworks for development.

The project acknowledges the need to support and train farmers and village communities responsible for project implementation and execution at the field level so that they can reach their potential capacity. The project envisaged the capacity development program keeping in view the different need-based requirement of various stakeholders at various stages, capacity building will be conducted in three phases viz Orientation Training, Refresher Training cum Workshop and Advance Training either or both physical and online mode.

▪ **System / Institutional Development:** The project will focus on establishment and capacity development of GKVS members which would play a pivotal role as the main community institution in the project planning and implementation. It will also build capacity of the PMU at the state level and units at DPIU and SPIU level for bringing in the change in their approach to address the issue of climate resilience through all interventions

▪ **Cooperation / Network Development:** The project at different phases of its implementation will seek the support from a number of organisations through necessary collaborations. The basic purpose behind network development is for developing knowledge, tools, models, policies that facilitate the workability and scalability of the project. This will be a network of research institutions around specific themes.

▪ **Human Resources Development:** Keeping in view the specific goals of the project, project staff and stakeholders would be groomed regularly through proper training support on all the aspects of the project, including climate resilience, planning process, activity implementation, reporting, procurement, financial, and safeguards requirements.

▪ **Organisation Development at the community level:** The project is aimed at capacitating certain community-based organisations like FPO/FPC, Self Help Groups (SHGs), farmers Interest Groups (FIGs), and agri-business-based cooperatives.

▪ **Farmer Development:** Ultimately the farmers are to be sufficiently trained under the project on various interventions that address climate stress and brings resilience by doing certain things differently.

Most of these activities will be carried out by partners, resource agencies and ATMA. the NDKSP will make a conscious attempt to identify master trainers and extension workers to get oriented with resilience aspects and technical issues/challenges in the project. While some of the regular activities in agricultural extension and research are already part of earlier components with cost inbuilt therein. Specialised training and research initiatives as required for demystifying climate resilience actions (i.e. ex -ante assessment of GHG as well as extent

of carbon sequestration, climate resilient variety/technology adoption, etc.) in the field will be covered in this component.

Capacity Enhancement Needs Assessment (CENA 2.0)

A Capacity Enhancement Needs Assessment (CENA) is currently ongoing exercise by VANAMATI to:

- Develop designs, methodologies, tools, and techniques for conducting CENA exercises for identified project stakeholders, including individual farmers and farmer groups.
- Create capacity development plans based on the findings from the CENA for these stakeholders.

Capacity building initiatives will be implemented according to CENA recommendations. Training programs will be organized, and the types of knowledge partners (resource agencies) involved, along with all necessary training arrangements, will be determined in consultation with state and regional government-supported training agencies, specifically the Vasant Rao Naik State Agricultural Extension Management Training Institute (VANAMATI) and the Regional Agricultural Extension Management Training Institute (RAMETI).

To mainstream the capacity development programs in the project area, ATMA will prepare a training calendar in collaboration with VANAMATI and RAMETI. Training will be delivered in phases, with Krishi Vigyan Kendras (KVKs) serving as the next level of interface, leveraging their technical expertise and understanding of local issues. KVKs will conduct significant extension and training programs through Farmer Field Schools (FFS), with the Project Director of ATMA overseeing extension and capacity development efforts.

Capacity building for master trainers and Sarpanch's for project orientation and the micro-planning exercise will be conducted by YASHADA in coordination with VANAMATI and RAMETI. YASHADA will train master trainers for the Micro-Planning (MLP) initiative, who will then coordinate MLP efforts and support the offices of the Sub-Divisional Agricultural Officers (SDAOs). These master trainers will also serve as resource persons, providing MLP training to Agriculture Assistants and Cluster Assistants, who are essential for facilitating the entire micro-planning process. Additionally, YASHADA will conduct a two-day orientation and MLP training specifically for Sarpanch.

One of the project's key objectives is to develop a series of learning materials for smallholder farmers that communicate strategies for building effective climate-resilient agricultural systems. These materials will raise awareness of climate change and adaptation requirements, laying the groundwork for the capacity development process. Yashwantrao Chavan Academy of Development Administration (YASHADA) will be included as a knowledge partner alongside VANAMATI and RAMETI.

The project will implement a comprehensive communication strategy using (a) mass media, (b) social media, and (c) audio-visual materials highlighting good practices, case studies, and thematic knowledge products.

Stakeholders and topics of the potential trainings:

The tentative capacity building process for the NDKSP Phase II is designed in a series of orientation and refresher training for their better understanding and these trainings, workshops, symposiums, Seminars, Conferences, Dialogues, Webinars, Exposure visits etc. will be imparted in phase manner.

Orientation cum Induction Training

The orientation training will be conducted for GKVS members, including Sarpanch, Dy. Sarpanch, progressive farmers, and members of FPOs, FPCs, and SHGs. This training will cover the project's objectives, components, and implementation strategy. The training will consist of project orientation, focusing on micro-planning exercises to empower Sarpanch to lead and take ownership of the project. Additionally, orientation or induction training will be provided for field officials and functionaries (along with various specialists at DPIU level- procurement specialist, Agribusiness specialist, HRD) PMU members, and other stakeholders involved in the project.

Refresher training will also be imparted to project stakeholders and project implementing officials, staff and specialists at appropriate intervals on updated project implementing strategies, technological interventions, agriculture practices either online or offline as appropriate.

Training cum Workshop

The training cum workshop module will focus on the effective implementation of the Social Audit mechanism, monitoring processes, and conflict resolution strategies etc.

Building Gender into Training

The project has made significant efforts towards Gender Mainstreaming and adopting a gender sensitive approach in the implementation process, including a plan to provide women empowerment training for the agriculture sector.

Training will be imparted at VANAMATI to mid-senior level officers/ consultants as AO, Circle AO, TAO, PS-HRD, PD/DPD ATMA, and other extension workers of the project. The objective and outcome of the 'Women Empowerment for Agriculture Sector' training will be-

- Explaining concepts pertaining to Women Empowerment.
- Spreading awareness of laws protecting women's rights.
- Making women members aware of their roles and responsibilities in development.
- Enhancing participation of women members in planning climate resilient agriculture activities

- Encouraging women's leadership skills.
- Familiarising them with rural/women/child development programmes to reduce poverty.
- Identifying cultural barriers and taking activities to help improve the social-economic condition of women.

5.6 Sub-component D5: Human Resource Management

This component will oversee project monitoring, impact evaluation, and reporting to ensure measurable progress toward the Project Development Objective (PDO). The results-based M&E system will track key indicators, enabling timely adjustments and data-driven decision-making.

Additionally, this component will manage communication, public awareness, and outreach efforts. A dedicated project website will be developed to share updates and provide a platform for stakeholders to voice concerns through the Grievance Redress Mechanism (GRM). All incremental operating costs associated with the project will be covered under this component.

Project Implementation Management

The project implementation framework emphasises collaboration with government departments and maintains a streamlined staffing approach to avoid creating parallel structures. A significant portion of the human resources implementing the project at the PMU, Division, District, Sub-division and cluster levels will be from the agriculture department itself. This will lead to the twin advantages of efficiency and sustainability. The project will benefit from the experience of the long-serving departmental staff and will also help transfer the new lessons of climate resilience to the human resources. To enhance staff efficiency, the project will leverage a robust IT-driven Management Information System (MIS). Additionally, project partners will be integrated into this IT system to support effective monitoring.

The overall project structure is given in the diagram below:

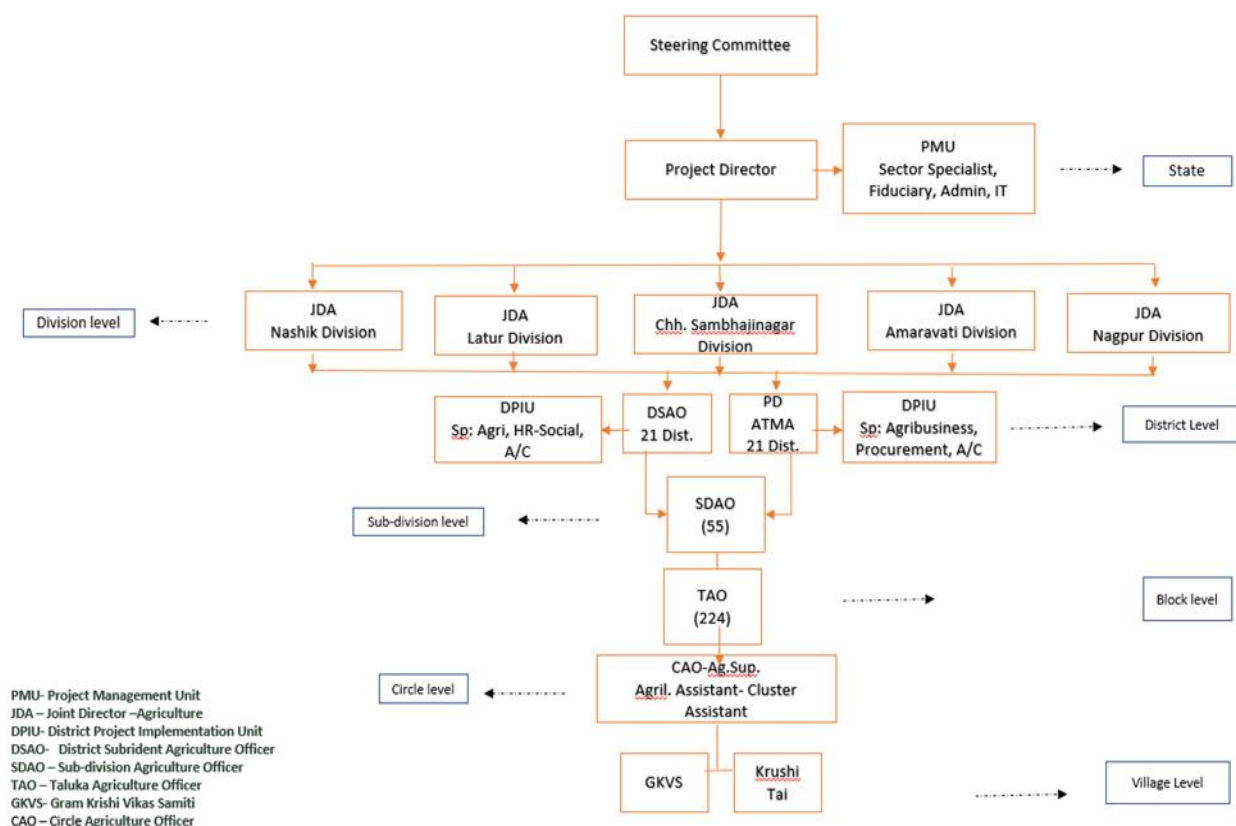


Figure 18: Organogram of the Institutional Structure

Project Governance

Institutional coordination is ensured through the project's governance structure, which addresses the multisectoral nature of climate resilience. The high-level Project Steering Committee and the Project Management Unit (PMU) play crucial roles in securing top-level ownership of the project and facilitating the coordinated involvement of relevant technical departments in its implementation.

Project Steering Committee

A high-level steering committee will function under the Chairmanship of the Chief Secretary. The members of this committee and the responsibilities have been outlined below.

Table 26: Steering committee

Designation	Role
Chief Secretary	Chairperson
Secretary, Agriculture Department	Member
Secretary, Water Conservation Department	Member
Secretary, Animal Husbandry Department	Member
Secretary, Marketing Department	Member
Secretary, Planning Department	Member
Secretary, Finance Department	Member
Vice Chancellors (PDKV, VNMKV and MPKV)	Members
Commissioner, Agriculture	Member
Project Director, NDKSP	Member Convener
Any other person	Invitee

The roles and responsibilities of the Project Steering Committee shall be as follows:

- Provide conceptual, strategic and policy guidance for the design and implementation of the project activities.
- Review Annual Work Plan and Budget, ensuring adequate budget provision for the activities.
- Review the progress of Programme Implementation based on monitoring reports, impact assessment studies and make recommendations.
- Guide and ensure inter-departmental convergence and coordination.
- Review and resolve the issues arising out of field implementation.
- Undertake such tasks that may help the effective implementation of the project

Project Management Unit

The State has set up a Project Management Unit at State level The structure of the PMU has been given below:

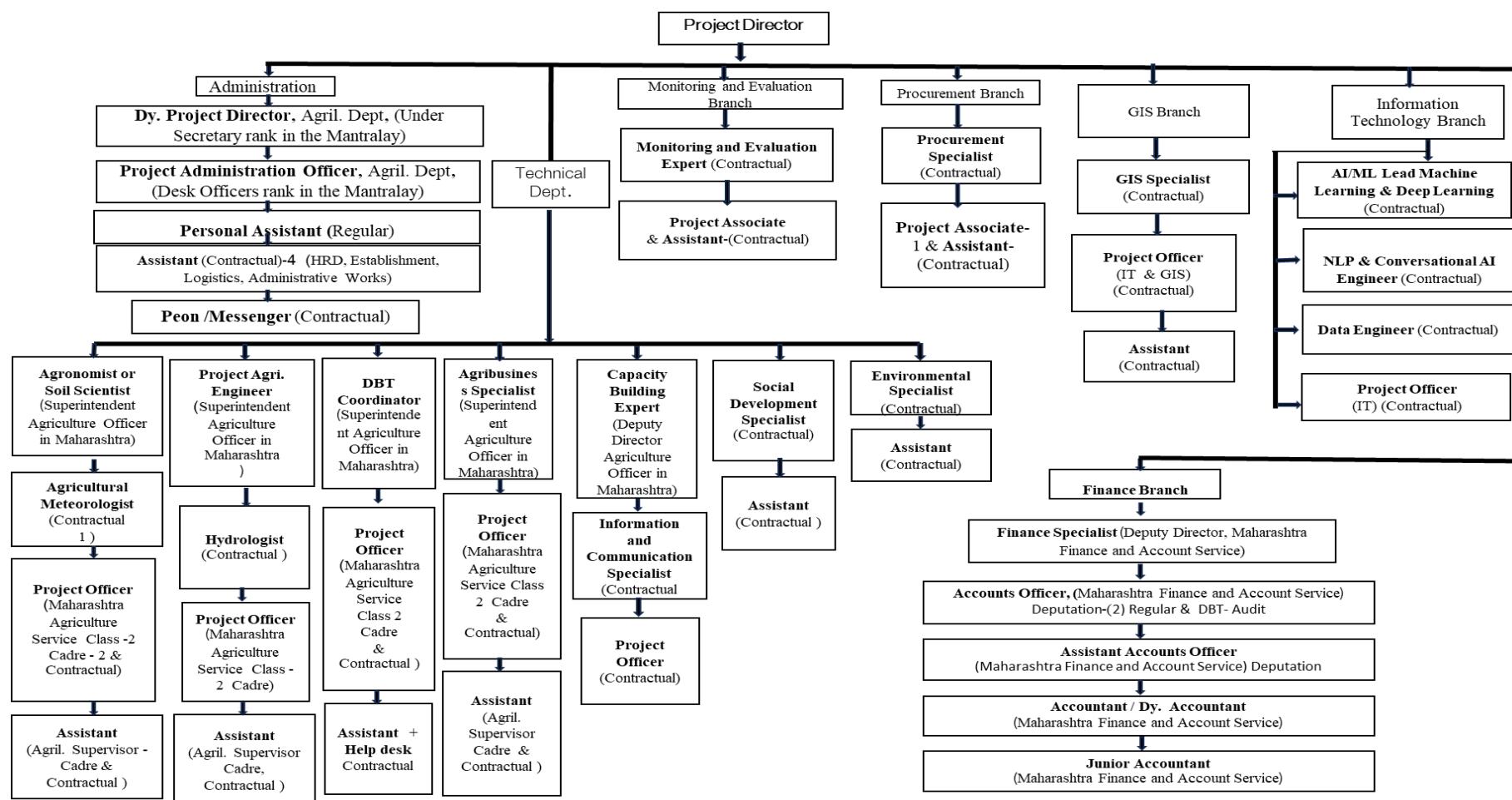


Figure 19: Project Management Unit

The PMU is headed by a Project Director who is responsible for overall management, supervision and implementation of the project activities.

Table 27: Details of the staff about the qualification and their role in the project

Position (to be aligned with pad)	Qualification	Role
Project Director	Senior Officer from Indian Administrative Service in the rank of Secretary to the State Government	1. Overall leadership for the project. 2. All administrative and financial powers shall be vested with the project director for smooth implementation of the project. 3. Important role to interchange the financial limitations assigned to the activity under approved plan. 4. Issue guidelines for effective implementation of the scheme and has authority to make changes in the guidelines wherever required for effective and smooth implementation of the scheme. 5. To coordinate with the World Bank as per the directives of the Steering Committee.
Project Deputy Director	Under Secretary Rank of Mantralaya cadre, on Deputation	Administrative support to the Project Director.
Project Administrative Officer	Desk Officer Rank of Mantralaya cadre, on Deputation	Administrative support to the Project Deputy Director.
Agronomist- cum-Soil Scientist	SAO Rank officer of Agriculture Department, on Deputation	To plan, coordinate with district and field implementation units on crop production and management practices and to plan and coordinate soil conservation measures, soil improvement, soil reclamation.

Position (to be aligned with pad)	Qualification	Role
Project Agriculture Engineer	SAO rank officer of Agriculture Department/ SAU / Specialist with Post graduate degree in Agricultural Engineering	To provide technical support, planning, coordination of watershed development, water use efficiency, farm mechanization, agro-processing activities.
DBT Specialist	SAO Rank officer of Agriculture Department, on Deputation	Control DBT portal and dashboard of Soil and water conservation work and also control the same for farmer / FPO / FPC.
Agri-Business specialist	SAO Rank officer of Agriculture Department, on Deputation	To design agribusiness components, prepare guidelines, promote agribusiness activities, support & monitor preparation of agribusiness plans, promote producer groups / companies.
Finance specialist	Deputy Director Rank officer from Maharashtra Finance and Accounts Service, on Deputation	In charge of financial management, project accounting standards, internal and external audit, fund flow and subsidy disbursement through DBT portal.
Capacity Building Specialist	Deputy Director Rank officer of Agriculture Department, on Deputation	To analyse the need and accordingly plan, coordinate and prepare modules for capacity building required for scheme, beneficiary and implementing agencies.
Hydrologist	Specialization in Hydrology	To conduct hydrological assessment of Clusters.
GIS specialist	Specialist with experience in GIS / Geo-informatics and / or Remote Sensing	Geo-referencing & Fusion of remote sensing Data & Preparation of cluster & village wise maps, analyse remote sensing data, provide IT support to PMU.

Position (to be aligned with pad)	Qualification	Role
AI/ML Lead - Machine Learning & Deep Learning Engineer	Master's/Ph.D. in Computer Science, AI, ML, or related fields.	Architect and deploy scalable AI/ML models. Optimize models for real-time inference. Implement MLOps pipelines for monitoring & retraining. Develop AI- driven automation for data classification. Collaborate with engineers for model deployment.
Social Development Specialist	Specialist with post graduate degree in Social Work preferably in Community / Rural Development and experience in social aspects of agriculture or rural development	To ensure social inclusion, to identify social inequity and genders issues and safeguard issues relating to vulnerable segments in the project. Prepare strategy and follow up for social inclusion.
Monitoring and Evaluation Specialist	A master's degree in economics or agriculture from a recognized university	To prepare a results framework to achieve the final object of the project and considering the objects required for the project, conduct the baseline survey. To plan and prepare the operations to achieve mid-term and final project evaluations.
Procurement specialist	Statutory university's Master's degree in Engineering, Law, Finance, Business Management, Commerce, Agriculture, and A certificate in procurement procedure, issued by any reputed national / international	To manage centralised procurement functions as per the guidelines and manuals, work with district level agencies to assist them in the decentralized procurement functions.

Position (to be aligned with pad)	Qualification	Role
	agency.	
Environment specialist	A Master's degree in environment science/forestry from a recognized university	To identify the perishes of changing the environment and find the solutions to help, guide, train or educate the farmers / FPO / FPC and to study the carbon credit on national or international level and to guide the project on modalities of curb emissions and sequestration.
IEC Content Creator	1.Statutory University's graduate degree in mass media and communication. or journalism or public relation or equivalent to it. 2.Must possess skill of excellent interpersonal dialogue and mastery in languages of Marathi and English	To publish the success stories on social platforms and give publicity the component of the projects. To forward the project success for different authorities or institutes to achieve an award.
NLP & Conversational AI Engineer	B.Tech/M.Tech/ M.S./Ph.D. in Computer Science, NLP, AI, or related fields.	Develop and optimize NLP & Conversational AI models. Implement multi-language AI models for text and voice processing. Fine tune LLMs (GPT, BERT, T5, LLaMA, Whisper AI). Develop real-time chatbot & voice assistant frameworks. Implement sentiment analysis, intent detection, entity recognition.

Position (to be aligned with pad)	Qualification	Role
Data Engineer	B.Tech/M.Tech/ M.S./Ph.D. in CS, Data Engineering, Information Systems.	Build & optimize real-time and batch data pipelines. Ensure scalability and fault tolerance of data infrastructure. Develop data warehouses for AI-driven insights. Implement real-time data streaming solutions. Integrate multi-modal data sources (IoT, satellite, drones).
Agriculture Meteorologist	A master's degree in agriculture meteorology / meteorology /atmospheric science / space science from a recognized university	To participate in the discussions or meetings of agriculture weather advisory and specially work on climate resilience required for the project. To coordinate with the institutes like ICAR, IMD, IITM etc. and prepare strategic research and extension plans for village groups in the project.

Role of PMU

1. Overall leadership, control, monitoring and supervision of the project activities and staff.
2. Ensuring implementation of Government orders and to make funds available to the field units / functionaries as per project requirements.
3. Ensuring timely submission of financial reports to GoM and World Bank.
4. Ensuring deployment of field staff, resource agencies, service providers and other functionaries
5. Coordination with line departments at the state level.
6. Ensuring consolidation of the annual work plan and budget of various units for submission to the state for sanction of the Budget.
7. Ensuring approval of annual work plan and budget by Project Steering Committee and Government.
8. PMU has the financial authority as per finance manual. Project allowance is allowed for the regular staff deployed at PMU and subsequent levels at division, district, sub division.
9. Ensuring timely distribution of the funds to various field units and service providers as per the physical and financial targets and other deliverables.
10. To submit Consolidated Statement of Expenditure (SOE) to the World Bank for reimbursement of claims for preceding financial year and to IUFRs periodically.

11. To monitor the activities and progress of works in the project and send reports at initial level, mid-term and end term and obtain Aide- Memoire from the World Bank.
12. To provide necessary guidance and coordination and ensure proper implementation of the Project.
13. Grievance redressal.

Divisional Implementation Unit

The project districts are divided into 5 Agriculture divisions viz. Chhatrapati Sambhaji Nagar, Latur, Amaravati, Nagpur and Nashik and these divisions are headed by the Joint Director of Agriculture.

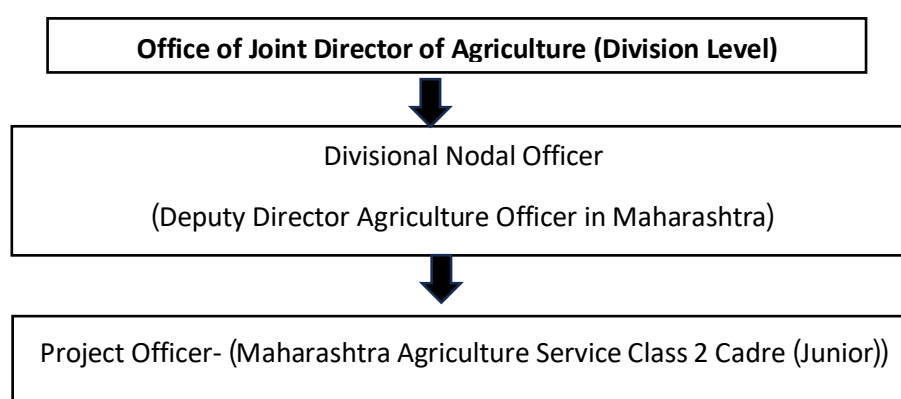


Figure 20: Divisional Implementation Unit

Role of Divisional Implementation Unit

1. JDA will be the administrative head at the division level and responsible for smooth implementation of physical & financial targets of the project in the division.
2. JDA will be responsible for implementation of the annual work plan, regular monitoring of the project activities and to monitor the funds made available to the field functionaries.
3. JDA will give pre-sanction to the FPO proposals of appropriate value as per financial powers delegated to them.
4. The JDA will be responsible for granting technical sanction, approving E-tenders and authorizing payments for NRM work within the pre-defined limits set by the project guidelines and as approved by the state government through Financial Powers.
5. Coordinate with line departments at cluster, block and district level and office of ATMA.
6. To provide technical guidance regarding technical matters.
7. Ensure formulation of village micro-plan, review and appraisal of Village adaptation Plan, preparation of Annual Action Plan.
8. Field appraisal and suggesting suitable amendments to the action plan.
9. Provide guidance for smooth and effective implementation of Annual Action Plan.
10. Timely submission of all reports to Project Director and monitor the progress.

11. Assist offices under them in procurement of goods, works and services, wherever required.

12. Grievance redressal

District Coordination Committee

A district level coordination committee has been proposed under the Chairmanship of the District collector. The proposed members of this committee and the responsibilities have been outlined below.

Table 28: Proposed committee

No.	Officials	Designation in committee
1	Collector	Chairman
2	Chief Executive Officer of Zilla Parishad	Joint Chairman
3	Deputy Commissioner, Animal Husbandry	Member
4	Project Officer, Integrated Tribal Development Project	Member
5	District Social Welfare Officer	Member
6	District Deputy Registrar, Co-operative Societies	Member
7	District Manager, MAIDC	Member
8	District Manager, MAHABEEJ	Member
9	District Marketing Officer, Markfed	Member
10	Nominated Progressive Farmer	Member
11	Representative of Self-Help Group	Member
12	A representative of a non-governmental organisation	Member
13	Senior Scientist & Head, KVK	Member
14	Project Director (ATMA)	Member
15	District Manager, NABARD	Member
16	Manager, District Leading Bank	Member
17	Agriculture Development Officer, Zilla Parishad	Member
18	District Superintendent Agriculture Officer	Member Convener

This Committee will facilitate coordination of project activities amongst agencies represented at district level including DSAO and PD (ATMA). The role of the district level committee to facilitate, collaborate, supervise and bring convergence. The District Collector shall resolve and take decisions regarding any grievances received during the implementation of the project and also redress/ enquire into the complaints or disputes arising even after the completion of the project.

District level Project Implementation Unit (DPIU)

The district level structure for the project has been given below:

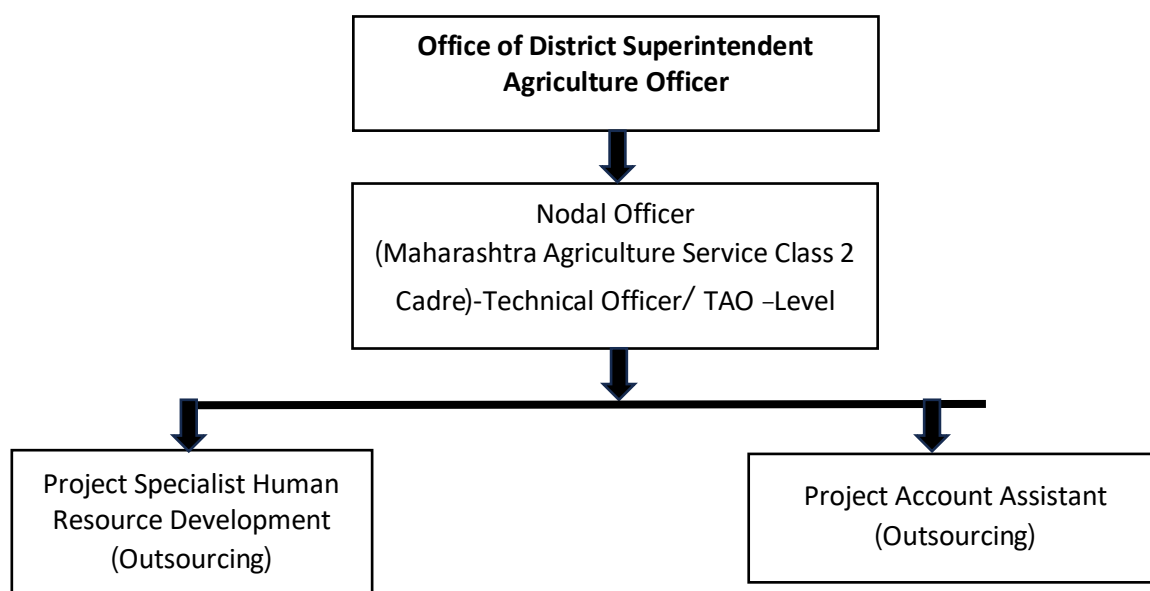
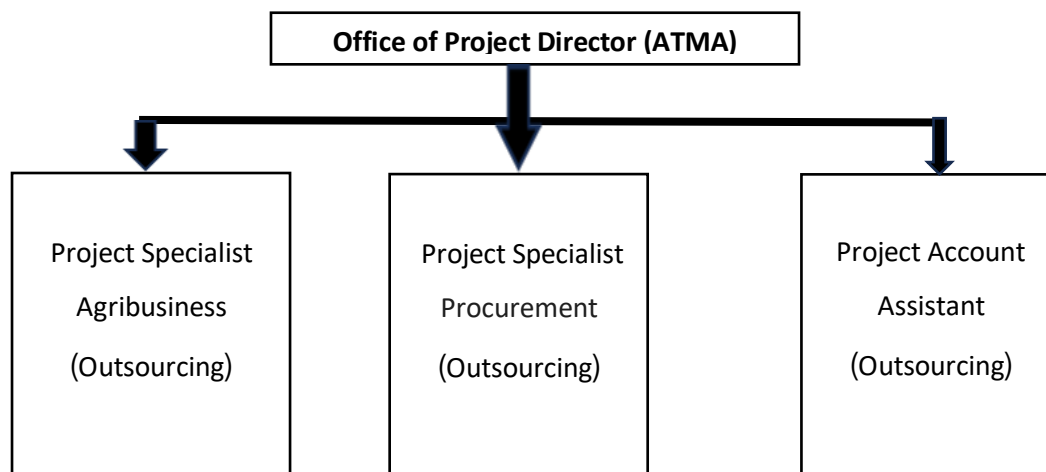


Figure 21: District level structure for the project

Role of DPIU

1. DSAO will be the administrative head at the district level and responsible for smooth implementation of the project in the district.
2. DSAO will be responsible for implementation of the annual work plan, regular monitoring of the project activities and to make funds available to the field functionaries.
3. DSAO will be responsible for sanctioning payments for certain individual beneficiary components, granting technical sanction, approving E-tenders and authorizing payments for NRM work within the predefined limits set by the project guidelines.
4. Coordinate with line departments at cluster, block and district level and office of ATMA.
5. Provide technical guidance regarding technical matters.
6. Ensure formulation of village micro-plan, review and appraisal of Cluster Development Plan.
7. Consolidate expense statement.
8. Timely submission of all reports to higher officers and monitor the progress.
9. Field appraisal and to suggest amendments to the action plan.

10. Provide guidance for effective implementation of Annual Action Plan.
11. Assist SDAO office in procurement of goods, works, and services, wherever required.
12. Grievance redressal



Role of PD, ATMA

1. Liaison with SAUs, VANAMATI, RAMETIs, and line departments for training & technology dissemination activities.
2. PD ATMA will give pre sanction to the FPO proposals of appropriate value and will be responsible for the post work inspection and sanction of subsidy of the FPOs projects as prescribed in project guidelines
3. Effective implementation & coordination of FFS, as well as strengthening of FPOs, FIGs, SHGs etc.
4. Use of audio-visual aids for awareness, publicity & effective program implantation.
5. Review progress of the project.
6. Consolidate expense statement.
7. Timely submission of all reports to higher officers and monitoring the progress
8. Field appraisal and to suggest amendments to the action plan.
9. Provide guidance for effective implementation of Annual Action Plan.
10. Grievance redressal.

Subdivision level structure

The districts are further divided into sub-divisions. One Sub-Divisional Agriculture Officer (SDAO) is normally in-charge of about 3-5 talukas in a district. The project area is spread over 55 subdivisions. The SDAO would also be the drawing and disbursing officer for the sub-division. The project will provide accounts personnel to support proper utilisation of funds at the level of SDAO as well as GKVS.

The subdivision level implementation structure has been given in the diagram below

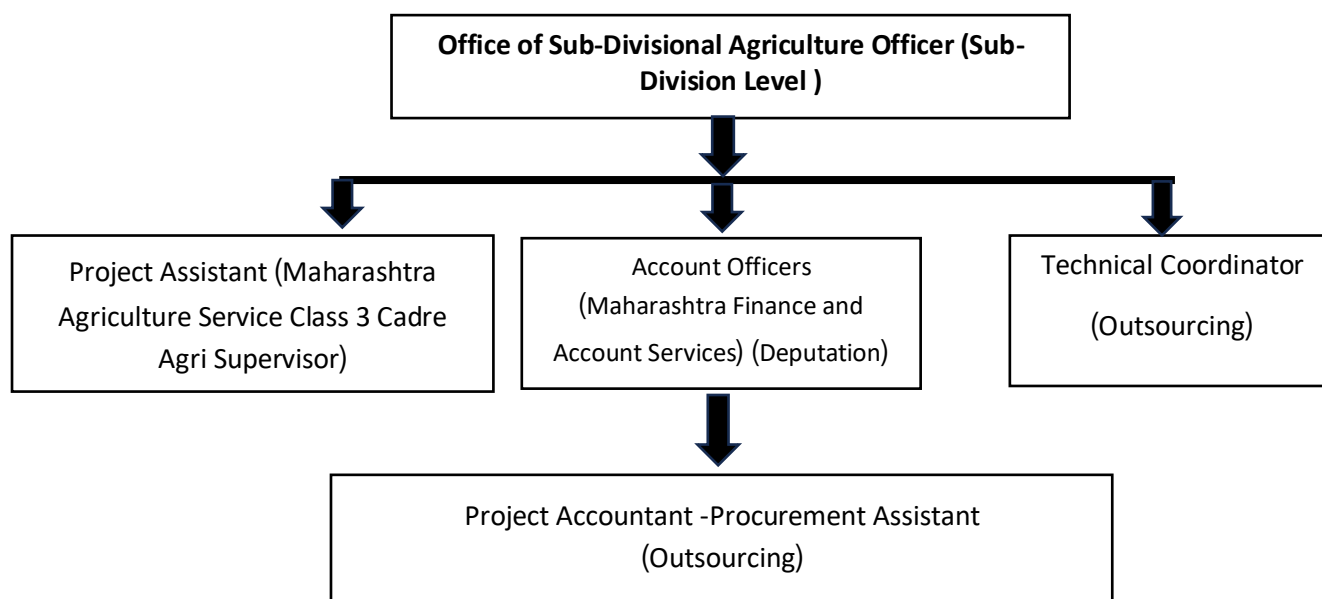


Figure 22: Subdivision level implementation structure

Role of SDAO

1. SDAO will be the administrative head at the subdivision level and responsible for smooth implementation of the project in the sub-division.
2. SDAO will be responsible for regular monitoring of the project activities and to make fund available to the field functionaries.
3. SDAO will plan, execute and monitor the dissemination of CRTs effectively by conducting FFS, exposure visits, training & workshops.
4. SDAO will give pre sanction to the FPO proposals of appropriate value and will be responsible for the post work inspection of the FPOs projects
5. The SDAO will be responsible for granting pre-sanction approval, sanctioning payments for specific individual beneficiary components.
6. The SDAO will also provide technical sanction, approving E-tenders, and authorizing payments for NRM work within the predefined limits outlined in the project guidelines.
7. The SDAO will extend necessary support to GKVS in facilitating the procurement of NRM works, whenever required
8. Coordinate with line departments at cluster & block level and office of ATMA
9. To provide technical guidance regarding technical matters
10. Ensure formulation of village micro-plan,
11. To prepare and submit administrative budget
12. To consolidate expense statement (IUFR)
13. Timely submission of all reports to higher officers and monitor the progress
14. Field appraisal and to suggest alternatives in the action plan.

15. Provide guidance for effective implementation of the project.
16. Grievance redressal

Technology Coordinator at SDAO office (55):

1. Gather recommendations from agricultural universities and the Indian Council of Agricultural Research (ICAR) for key crops in project villages, focusing on climate-resilient varieties and developing updated crop-wise packages of practices.
2. Create an extension action plan to facilitate the widespread adoption of climate-smart technologies among farmers.
3. Enhance participation of farmer producer companies in promoting climate-smart technologies and develop tailored extension plans for these companies.
4. Record audio and video testimonials from farmers successfully adopting climate-smart technologies to create impactful success stories.
5. Present insights on the benefits and challenges faced by farmers in adopting these technologies during monthly district discussion forums.
6. Maintain regular communication with agricultural universities and Krishi Vigyan Kendras (KVKs) to obtain timely crop advisories.
7. Prepare compilation reports on effective technologies implemented within the project.
8. Conduct a technical review of the technology-related components of the village adaptation plan post micro-planning.
9. Organise training sessions addressing environmental issues and ensure compliance with relevant requirements.
10. Plan crop schedules based on water budgeting to optimise resource use.
11. Provide crop-specific advice based on soil properties and coordinate necessary follow-ups.
12. Train Cluster Assistants on crop-wise climate-smart technologies each season.
13. Monitor Farmers Field Schools facilitated by CA.
14. Plan and coordinate training and study tours for farmers, farmer groups, and field staff.
15. Participate in taluka-level Kharif and Rabi season meetings to guide agricultural assistants in developing and reviewing action plans for climate-smart technologies.
16. Provide guidance to field staff on carbon sequestration, emissions, and credits.
17. Oversee initiatives related to the conservation of local seed varieties.
18. Train Krishi Tai on climate-smart technologies for effective implementation.

Taluka level structure

The Taluka Agriculture Officer is in charge of the project at Taluka level. He is the bridge between the Sub-Divisional Agricultural Officer and village level machinery.

Role of the Taluka Agriculture Officer

1. TAO will be the administrative head at the tehsil level and responsible for smooth implementation of the project in the tehsil.
2. TAO will be responsible for regular monitoring of the project activities.
3. TAO will be responsible for granting pre-sanction approval for certain individual beneficiary components as per the project guidelines.
4. TAO will be responsible for the post work inspection of the FPOs projects of appropriate value in the taluka.
5. Coordinate with line departments at cluster & block level
6. To provide technical guidance to the field functionaries and beneficiaries.
7. Ensure formulation of village micro-plan.
8. Ensure maximum adoption of CRT in the project clusters.
9. Timely submission of all reports to higher officers and monitor the progress.
10. Field appraisal and to suggest amendments to the action plan.
11. To resolve conflicts and grievances.

Village level structure

The village level structure is given below:

a. Gram Krishi Vikas Samiti (GKVS):

The Gram Panchayats are constituted and regulated as per Bombay Act no. III of 1959, 'The Maharashtra Village Panchayats Act'. Through the Maharashtra Government order under section 49 (4) Gram Sabha in consultation with the panchayat, from amongst the members of the panchayat, representatives of community-based organisations working in the panchayat area, village level functionaries of panchayat and voters, constitutes a Village Development Committees known as Gram Krishi Vikas Samiti. Out of total 14 members in GKVS, there are 11 members from farming, and agri-allied communities, in which 50% women representation is mandatory.

The project will work in close partnership with GKVS. This is vital for planning, monitoring, and coordinating the implementation of initiatives such as the development and execution of the 'Village Adaptation Plan (VAP).' Additionally, GKVS is instrumental in anchoring climate interventions within the community. The GKVS composition is as given below-

Table 29: GKVS composition

Sr.No.	Post Name	Post
1.	Sarpanch	Ex-officio-President
2.	Deputy Sarpanch	Ex-officio-Member
3.	Gram Panchayat member	Member
4.	Progressive farmer (3 member-one female)	Member

5.	Cooperative Society President (one)	Member
6.	Farmer Producer Company/ Farmer Producer Group representative (one)	Member
7.	SHG member (one)	Member
8.	Agri-Allied Activity farmer (Two)	Member
9.	Talathi	Member
10.	Agriculture Assistant	Joint Secretary
11.	Gram Sewak/ Gram Vikas Adhikari	Member Secretary

{Female social mobiliser 'Krishi Tai' (village level) + cluster assistant (cluster level) + Technology Coordinator (sub divisional level) will support as extension wings}

Roles & Responsibilities of Gram Krishi Vikas Samiti (GKVS)

The GKVS will be responsible and broadly undertake the following works:

1. The Gram Sewak in assistance with Agriculture Assistant will organise GKVS meetings. At least one meeting every month will be organised.
2. Conduct participatory micro-planning process for preparation of Village Adaptation Plan (VAP)
3. Social inclusiveness in benefit distribution and ESMF compliance in various project activities during VAP preparation.
4. Create awareness among the vulnerable farm families about the project interventions suited to their farming systems.
5. Selection of the beneficiaries based on project guidelines.
6. Assisting in the conduct of the Micro- Planning process in the village and developing the Village Adaptation Plan in a participatory manner.
7. Awareness generation and dissemination of climate resilient technologies, IPM & INM, CR seed varieties, bio fertilisers and pesticides, integrated farming, farm mechanisation, protective cultivation etc.
8. Planning of various crops depending upon the local agro-climatic conditions as weather, rainfall, availability of water and soil texture etc., and guide the farmers regarding.
9. Guiding the farming community for agri-allied livelihood activities such as goat rearing, backyard poultry, sericulture, inland fisheries, apiculture etc.
10. Assist the village level institution like SHGs/ Commodity based group/ Farmer Interest Group etc. for entrepreneurship development and guidance about post-harvest management and market related information.
11. Verification of community assets and conduct social audit periodically and execution of grievance redressal mechanism at gram panchayat level.

12. Ensuring beneficiary contribution as well as assistance from commercial banks, wherever necessary
13. Planning and execution of community works as per approved annual action plan
14. Maintenance of assets created through project assistance on community land including books of accounts.

b. Agriculture Assistant (AA)

1. The Agriculture Assistant will update various information in project applications from time to time.
2. The AA will review and verify the documents uploaded by the farmer, conduct a pre-site inspection, assess technical eligibility as per the project guidelines, and submit the details on the DBT Portal.
3. Facilitate Self Help Groups to initiate project activities
4. The AA will be responsible for preparing estimates for NRM works, conducting inspections before and after completion, geo-tagging NRM works, and preparing the Measurement Book (MB) for uploading on the NDKSP-NRM platform.
5. Active participation in microplanning; social audit; grievance redressal
6. Actively promote climate-resilient agricultural technologies to farmers.
7. Organise training programs, workshops, and exposure visits to enhance farmers' knowledge and capacity in sustainable farming practices.
8. Ensure timely communication of government programs, subsidies, and schemes related to sustainable agriculture to the farming community.
9. Provide real-time agro-advisories to the farmers.
10. Help monitor the progress of technology adoption by farmers and provide regular updates to the CA and higher authorities.
11. Collaborate with CAs to ensure the spread of best practices across different villages and clusters.
12. Serve as the key link between farmers and the NDKSP project, ensuring clear communication and the effective execution of project activities.

c. Cluster Assistant (CA)

1. Conduct Farmers Field School (FFS) as per project guidelines
2. Enhance adoption of climate resilient technologies among all sections of farmers in project villages
3. Monitor and measure the adoption of climate-resilient technologies on the farmers in the project villages.
4. Guide farmers on drought-resistant crops and water management to prepare for adverse weather.
5. CA will facilitate farmers for registration and application on DBT-NDKSP
6. CA will work in-synchronisation with GKVS decisions
7. Act as focal interface of project with the farmers, FPOs and the community

8. Supervise and conduct inspections for project activities and maintain records of social audit

d. Female Social Mobiliser 'Krishi Tai'

1. The primary responsibility of female mobiliser would be of mobilisation and facilitation of the farming community, especially women farmers about climate resilient agriculture technologies and package of practices (PoP).
2. She should facilitate SC, ST, women, and PWD farmers and landless people to register on farmer's applications or DBT applications for suitable interventions.
3. She ought to encourage the farming community to implement four to five climate-resilient technologies that are appropriate for their area. She should visit their farm and provide CR technology related information.
4. She should attend the FFS demonstrations of agriculture practices and ensure at least 10 guest female farmers' participation.
5. She should conduct/ participate in farmer interest group/ SHG meetings.
6. She should attend GKVS meetings and present concerns or issues of vulnerable groups to seek assistance from GKVS members.
7. She ought to start coordinating, and creating synergies with different departments and the village's business development groups.

Section 6

Environmental and Social Safeguard



Section 6

Environmental and Social Safeguard

6.1 The Environmental and Social Management Framework (ESMF)

POCRA phase II complies with World Bank's Environmental and Social Framework (ESF), incorporating the Environmental and Social Standards (ESS) along with Environment and Social Policy and Directives. Project has prepared an Environmental and Social Management Framework (ESMF) in order to: (a) efficiently manage environmental and social responsibilities by integrating them into the project design; (b) address environmental and social risks and impacts and (c) handle environmental and social concerns by allocating resources, assigning responsibilities, and implementing E&S procedures. The framework aligns with the relevant statutory provisions of the Governments of India and Maharashtra including environmental and social laws and policies at both national and state levels, covering forests, biodiversity, pollution, waste management, agriculture, stakeholder engagement, gender and social inclusion, tribal development, and grievance redressal.

Environment and Social Risks: The project's climate-resilient and environmentally sustainable interventions are largely benign and aim to improve soil health, resource efficiency, and reduce emissions. There are potential small scale, negative impacts associated with post-harvest processing, use of farm machinery, soil and water conservation works, input supply, minor civil works, waste generation, food safety, emissions, vegetation clearing, and possible use of hazardous chemicals and plastics. Most of these impacts are expected to be site-specific, reversible, and manageable with conventional mitigation measures. Social risks are moderate, as civil works will be small-scale, on government or community land, with no anticipated land acquisition, physical relocation, or economic displacement, and most labour will be locally sourced. Special attention and targeted stakeholder engagement would be needed in Tribal and Left Wing Extremism (LWE) Areas through stakeholder engagement, participatory planning, inclusive beneficiary criteria, and community mobilisation. Risk of exclusion of other disadvantaged and vulnerable socioeconomic groups, especially scheduled castes, particularly vulnerable tribal groups, landless and marginal farmers, tenant farmers, women farmers, farm workers and people with disability will be mitigated through inclusive targeting criteria.

Environment and Social Management Framework. To manage potential environmental and social (E&S) risks associated with POCRA II activities, an Environmental and Social Management Framework (ESMF) has been prepared. The ESMF systematically identifies potential E&S risks and impacts and recommends planning and implementation of risk mitigation measures, and good practices. ESMF include ES plans/frameworks, checklists,

guidelines, and tools such as: a) List of Excluded Activities; b) Legal and Regulatory compliances; c) Environment and Social screening; d) Environment and Social mitigation measures; e) Training and Capacity Building Plan; f) Stakeholder Engagement Plan; g) Labor Management Procedures; h) Integrated Pest and Nutrient Management Plan; i) Indigenous Peoples Planning Framework; j) Grievance Redressal Mechanisms; k) Monitoring, Reporting and Auditing Mechanisms etc.

Exclusion List. The ESMF comprises a list of excluded activities with the potential to induce significant adverse impacts on the surrounding environment and people. The activities excluded by the project are listed below:

- Activities that are non-compliant with country and state laws/regulations;
- Any activity which is not permissible by existing State or Central legislation within National Parks and Sanctuaries, and in critical wildlife habitats declared by due process.
- Digging of deep wells (borewells/tubewells) that are 60 meters or more in-depth in notified areas.
- Construction or repair of check dams or embankments more than 10 meters in height.
- Any activity which is not permissible by existing legislation within a notified Eco Sensitive Zone (ESZ)
- Any activity involving pesticides that are banned by the Government of India.
- Any activity involving pesticides that are in Classes IA, and IB WHO Recommended Classification of Pesticides by Hazard.
- Any activity involving construction within 100 meters of an archaeological site/monument.
- Any activity involving use of Asbestos Containing Materials (e.g., AC pipes for irrigation)
- Activities involving acquisition of private land, restrictions on land use, forced eviction, physical relocation, and economic displacement.
- Activities that do not meet the criteria for Free, Prior, and Informed Consent under ESS7
- Any activity that could involve or lead to use of child labour and/or forced labour;
- Activities with the potential to exacerbate local law and order and security situation.

Environment and Social Screening Checklist. The Village Adaptation Plans prepared by the GKVS will undergo the environment and social screening. The screening is conducted using the standard checklists/guidelines included in the ESMF (which will be translated into local language). The screening will ensure compliance with exclusion list; Identification of necessary permissions and clearances, as applicable; identification of potential environmental and social risks and risk mitigation measures with required budget (if any).

Environment and Social Risk Mitigation Measures. The POCRA project comply with all applicable ESS of the World Bank and Acts, Rules, Regulations, Policies, Guidelines of GOI and GOM that are relevant to project activities. The ESMF identified the potential risks and mitigation measures for all the activities under different components that are presented in the ESMF. The generic/standard checklists and guidelines are also included in the ESMF for common activities. The activity wise risk and mitigation measures are provided in the ESMF in detail.

6.2 Stakeholder Engagement Plan. A Stakeholder Engagement Plan (SEP) has been developed providing guidance on stakeholder engagement throughout the project's preparation and implementation phases. The SEP specifies how and when stakeholders—including marginal and small farmers, landless laborers, women farmers, agricultural workers, gram panchayat members, tribal beneficiaries, relevant government departments, NGOs and partners will be involved in project decision-making and implementation. Special emphasis is placed on ensuring information reaches all beneficiaries, particularly vulnerable and disadvantaged groups such as SC/ST, women farmers, marginal and landless and stakeholders in tribal areas. The SEP includes measures to strengthen the existing Grievance Redress Mechanism (GRM), making it more accessible and responsive through both online and offline channels. The PMU will oversee SEP implementation, including community mobilization, information dissemination, capacity building, and regular reporting as outlined in the Environmental and Social Commitment Plan (ESCP). A comprehensive communication strategy will also be prepared to disseminate information in a format accessible to relevant stakeholders.

6.3 Labor Management Procedures (LMP). POCRA II involves: a) direct workers employed specifically by the implementing agencies for the project; b) contracted workers engaged in consultancy service firms for specific deliverables and technical support (training, monitoring, and evaluation, etc.); c) community workers for small-scale civil works to be undertaken; and d) primary supply workers associated with suppliers of goods and equipment necessary for the execution and monitoring of the project. LMP includes risk mitigation measures related to occupational health and safety, equal and non discriminatory terms and conditions of employment, workers' organization, prohibition of child and forced labor, provision for a worker's grievance mechanism, and sensitive handling of grievances related to sexual exploitation and abuse (SEA), sexual harassment (SH) and gender based violence (GBV). Effective risk mitigation measures for primary supply workers in supply chains include contractual conditions and bidder declarations that prohibit child and forced labor, labor abuse in supply chains and ensure safe, fair working conditions and grievance mechanisms. Labor involve in the supply chain made aware about this Labor Management Procedures.

6.4 Integrated Pest and Nutrient Management (IPNM) Plan. The ESMF includes an Integrated Pest and Nutrient Management Plan (IPNM) practices for the major crops in the project area – Paddy, maize, cotton, soyabean, red gram, sorghum etc. This will be part of the crop productivity enhancement interventions and will be part of all demonstrations the project will be taking up and part of advisory/extension/awareness initiatives. The focus will be on IPM to the extent possible and only the permissible pesticides as per WHO categorization will be allowed. The details are available in the ESMF.

6.5 Indigenous Peoples (Scheduled Tribes) Planning Framework. Scheduled Tribes (STs) will be among key beneficiaries of NDKSP II interventions. The project has prepared an Indigenous People's Planning Framework (IPPF), which outlines the measures to ensure that project minimizes risks and enhances opportunities for ST beneficiaries to participate in, and benefit from the project. Key interventions include: meaningful consultations and participation of tribal communities in village adaptation planning and implementation; mechanisms for consultations, information sharing and capacity building in local tribal language; preparation of indigenous peoples (IPP) or Tribal Development Plan (TDP) that supports additional enhancement measures in tribal areas; including greater convergence with tribal department/agencies and NGOs working in accessible GRM. Free Prior and Informed Consent (FPIC) is not envisaged as NDKSP II will not cause relocation or adverse impacts on customary land and natural resources and/or cultural heritage of tribal communities.

Land acquisition, Land use restrictions and physical and/or economic displacement is not anticipated. Civil works for soil and water conservation and post-harvest activities will be small in scale and conducted mostly on government or community owned/leased lands without any physical encumbrances. The ESMF includes: a) safeguard screening criteria to identify and address any potential adverse impacts on land, land use, assets and livelihoods; b) exclusion list to exclude any activities that could cause adverse impact on land, land use, livelihoods and assets; and c) procedures for community leasing of land from community members.

6.6 Grievance Redressal Mechanism (GRM). The project will establish, publicise, and operate a functional and accessible grievance mechanism, consistent with ESS10. The grievance mechanism will be able to receive, register, and facilitate the resolution of all NDKSP-II relevant grievances in a timebound and transparent manner. The existing GRM will be strengthened for this purpose. For SEA/SH related complaints, GRM will include provision for referral of survivors to relevant gender-based violence service providers, in a safe, confidential, and survivor-centered manner. The SEP includes details of GRM.

The Grievance Redressal Mechanism (GRM) is proposed in the SEP to help address the grievances; if any, in real time so as to avoid discontent and enhance ownership and belongingness of all the stakeholders towards the project.

Tribal communities may have limited access to formal systems due to geographical distance, technological barriers, and varying levels of awareness. Strengthening knowledge about grievance mechanisms and rights can enhance accessibility and engagement. Cultural and language diversity presents opportunities to develop tailored approaches that improve navigation of formal systems for village communities. Streamlining processes and incorporating efficient resolution mechanisms can foster trust and responsiveness. Inclusive frameworks that ensure representation for all groups, including Particularly Vulnerable Tribal Groups (PVTGs), women, and smallholder farmers, can create equitable participation. Establishing clear follow-up systems offers a valuable means to keep stakeholders informed and engaged throughout the grievance resolution process.

In light of the above background, a grievance redressal mechanism is proposed herewith.

Existing Grievance Mechanisms

- **Village-Level Informal Systems:** Many grievances are addressed through consultations with local leaders or elders. Village Panchayats also serve as a platform for raising and addressing community concerns.
- **Government-Run Mechanisms:** These include online portals, like Maharashtra's AAPLE SARKAR, where citizens can submit complaints about public services, as well as block and district administration offices where farmers can file complaints about schemes or services. Moreover, the members of farmer organizations, women SHGs and cooperatives can avail the services in online mode on a platform Centralized Public Grievance Redress and Monitoring System (CPGRAMS). This facility is available to citizens 24x7 to lodge their grievance to the public authorities on any subject related to subject delivery.
- **Farmer Organizations, Women SHGs and Cooperatives:** Farmer producer groups or cooperatives handle their members' grievances related to inputs, marketing, or pricing through collective discussions. Women-led Self-Help Groups (SHGs) also act as informal support structures for their members.

The grievance redressal mechanism ensures transparency and accountability at multiple levels. Project staff grievances will be addressed as per their contracts and government rules. Village-level conflicts will first be resolved by the Gram Krishi Vikas Samiti, with support from the Agriculture Assistant, Cluster Assistant, and Krushi Tai. If unresolved, the matter will be escalated to the Gram Sabha. The further escalation matrix is given in table.

Table 30: Proposed Grievance Mechanism

Step	Description of Process	Mode	Responsibility
GM implementation structure	At the Gram Krushi Vikas Samiti (GKVS) level, all grievance records are maintained at the GKVS offices. Complaints are reviewed and addressed during scheduled GKVS meetings	GKVS meetings, written complaints submitted to the GKVS, or in-person complaints made during scheduled GKVS meetings.	GKVS.
	Taluka Agriculture Officer serves as the Nodal Officer. Taluka Agriculture Officer (TAO) will monitor through his office the resolution of complaints and also monitor the transmission channels post the resolution of complaints.	in person, by post, or through email.	Taluka Agriculture Offices (TAOs) in all project districts
	The Sub-Divisional Agriculture Officer (SDAO) serves as the Nodal Officer and is assisted by the Project Assistant. Sub-Divisional Agriculture Officer (SDAO) will monitor through his office the resolution of complaints and also monitor the transmission channels post the resolution of complaints.	in person, by post, or through email.	Sub-Divisional Agriculture Offices (SPIU) in all project districts
	District Superintendent Agriculture Officer (DSAO) serves as the Nodal Officer, assisted by the Project Specialist (Human Resource Development - HRD). District Superintendent Agriculture Officer (DSAO) will monitor through his office the resolution of complaints and also monitor the transmission channels post the resolution of complaints.	in person, by post, or through email.	District Superintendent Agriculture Offices (DPIUs) in all project districts
	Divisional Joint Director of Agriculture (JDA) serves as the Nodal Officer, assisted by the nodal Officer. Divisional Joint Director of Agriculture (JDA)	in person, by post, or through email.	Divisional Joint Director of Agriculture – Amravati, Nagpur,

Step	Description of Process	Mode	Responsibility
	will monitor through his office the resolution of complaints and also monitor the transmission channels post the resolution of complaints.		Chhatrapati Sambhajinagar, Nashik, and Latur
	Deputy Project Director serves as the Nodal Officer and is assisted by the Specialists (Social and Environmental). Deputy Project Director will monitor through his office the resolution of complaints and also monitor the transmission channels post the resolution of complaints.	Grievances may be submitted in person, by post, or through email.	Project Management Unit (PMU), Mumbai
		By phone (from 9:45 AM to 6:15 PM)	
		Through a WhatsApp message	
		Chief Minister's Helpline and Aaple Sarkar Portal (Government of Maharashtra)	
		The Centralised Public Grievance Redress and Monitoring System (CPGRAMS) is an online platform available to citizens 24/7 for lodging grievances. Any grievance related to the project is referred to the Project Management Unit (PMU) in Mumbai.	
Grievance uptake	Grievances can be submitted through the following channels: • Dedicated phone line • Dedicated WhatsApp number for receiving messages • Email or by post • Letter addressed to Grievance Focal Points at local facilities • Complaint box • Grievance Redressal Portal A complaint form may be submitted via any of the above channels.		

Step	Description of Process	Responsibility
Sorting, processing	Any complaint received is forwarded to GP, Logged in Grievance Register, and categorized according to the complaint types	Local grievance focal points
Acknowledgment and follow-up	Receipt of the grievance is acknowledged to the complainant by GP	Local grievance focal points
Verification, investigation, action	Investigation of the complaint is led by GP. A proposed resolution is formulated by GP and communicated to the complainant by GP	Complaint Committee composed of GP
Monitoring and evaluation	Data on complaints are monitor by the SDAO, DSAO, and the Nodal Officer (JDA) in grievance registers and not reported to the PMU.	Concern Authority
Provision of feedback	Feedback from complainants regarding their satisfaction with complaint resolution is collected by GP	
Training	Training needs for staff/consultants in the PIU, Contractors, and Supervision Consultants are mapped and conducted on topics such as, a) awareness on GRM, b) procedures for GRM and c) Recording and reporting on Grievances	

Institutional Arrangements for ESMF Implementation. The Department of Agriculture will be responsible for overall implementation and reporting of ESMF. The PMU will ensure that ESMF is duly integrated into the project activities and processes. The ESMF implementation responsibilities at various levels of the project structure include.

- a. Environment and Social Specialists: The PMU will include qualified and experienced environment (1) and social specialists (1). He/She will supervise E&S specialists, Project Executives (PE) HR and Agribusiness and provides technical guidance to the E&S Nodal officers. Also facilitates coordination with Agribusiness, CB, M&E, HOD and specialists in PMU.
- b. ES Project Assistant- The Project Executive at PMU will support ESMF (including all instruments) documentation, compilation of reports, action plans, data collection and management etc. and any other responsibility that may emerge or allocate during the project. The Project Executives will report to E&S specialists at PMU.
- c. Project Assistant (HRD) and Project Specialist (Agribusiness) will be responsible for ESMF implementation at District. The PE (HR and AB) will Provide undertake ESMF implementation in the districts, provide training to FPOs, ensure ESMF compliances at FPO level.
- d. Cluster Assistants at the Taluka level, the Cluster Assistants will manage the responsibility of screening of Village Adaptation Plans, integrating the ESF aspects into the plans etc. by guiding the GKVS.

Monitoring and Reporting: The PMU (E&S specialists) will prepare and share half yearly monitoring reports on ESMF Implementation and ESCP compliance. Monthly field visits will be conducted to supervise and support ESMF implementation. In addition, the DPIU staff (PE HR and PE agribusiness) and CAs will cover the ESMF aspects as part of their regular visits and document the progress/status in standard formats. The project will also capture the key compliance requirements as part of MIS. External studies/audits will be undertaken at mid-term and end term or in between based on the need.

2. **Trainings:** The E&S specialists at the PMU will design/develop the training modules and will provide the trainings to the staff responsible for ESMF at DPIUs (PE HR and PE agribusiness) who will further train the Cluster Assistants (CAs). The Cluster Assistants will in turn train the GKVS on the ESMF. A main training will be organized after the inception of the process in year 1 and a refresher training will be organized around the mid-term. In addition, dedicated ESMF trainings will be organized based on need and the relevant modules will be integrated into regular project trainings. Relevant Information Education and Communication Materials (IEC) will support the trainings. The trainings will include exposure visits, demonstrations etc (e.g., IPNM).

Section 7

Financial Management



Section 7

Financial management

7.1 Introduction

A sound Financial Management system is critical for the efficient and effective decision-making for the implementation of the project. This includes proper planning, budgeting, accounting, financial reporting, internal control, auditing and physical performance of the project with the aim of managing the project resources properly for achieving the project objectives. Since the financial transactions of the NDKSP Phase - II will be done at different levels viz. the State, PMU, JDA, DPMU and Sub-division level, it is imperative that the financial management framework captures the details of all transactions at every level and reflects them through proper accounting systems.

The objective of the financial management system of the NDKSP Phase – II is to ensure effective management and utilisation of project resources for achieving the project objectives. The primary objectives of sound financial management are -

1. To ensure smooth flow of funds to the different project implementing levels/agencies, so that there is no delay in implementation of project activities.
2. All financial transactions are as per the rules and procedures and in line with the established norms of the project approved by the competent authority.
3. All such transactions are duly accounted for in the prescribed formats given by the competent authority.
4. All payments due to be made to any service providers are done in an efficient, speedy and transparent manner.

Since the implementation is at different levels, it is important that there is a coherence and standardisation in project budgeting, accounting, financial reporting and audit, etc. aligned with the norms of the World Bank. An important objective of this document and the Financial Management Manual is to guide all project implementing authorities and participants in general and the finance persons in particular, in the financial issues in the project.

The financial management of the NDKSP Phase II aims at producing real time, relevant and reliable financial information that would allow the project executives and implementing agencies to plan and implement the project, monitor compliances with agreed procedures and guide the project progress towards the set objectives.

7.2 Fund Flow

The State Government prepares its own budget, which comprises a Receipts section and an Expenditure section. All incoming receipts of loans and grants through the central government (called ACA - additional central assistance) are included in the receipts budget. Estimated expenditure on all externally aided projects (comprising both the shares of the state government as well as the external agencies) is included in the expenditure budget.

The flow of funds in this project shall be at GOI level, at State level, at PMU level, at District level and at Sub-divisional level. It is mandatory for controlling and implementing agencies to make expenditure on activities related to project management and project components decided by competent authority. Roles and responsibilities of all finance personnel in all levels of this project will be mentioned in detail in the Financial Management Manual of Phase II of this project. Competent authority at all levels of designated account centres will be given definite administrative as well as financial powers after approval from the Finance Department, Government of Maharashtra through Delegation of Powers.

Government of India level

The World Bank shall transfer funds into a Special Account opened for the NDKSP Phase II with the Reserve Bank of India. This account will be operated by the Controller of Aid, Accounts and Audit (CAAA), Department of Economics Affairs (DEA) (MoF), GoI. The Department of Economic Affairs (DEA) under the Ministry of Finance acts as the administrative authority in respect of all fund flows originating under Externally Aided Programmes (EAP). Under the proposed arrangements, there shall be back-to-back transfer of funds from GOI to the State Government on receipt of funds from the World Bank.

State Level

The Government of India will transfer funds to the Loan Account of Government of Maharashtra opened in Reserve Bank of India. The Government of Maharashtra will pass on the funds (World Bank finance - 70% and own contribution - 30%) to project implementing agencies through Annual budgetary grants. The Finance Department shall allocate a budget (with dedicated cash flow) to the Department of Agriculture for implementation of the project. Cash flow will be maintained on BEAMS using the PAO and Treasury system of the state government.

Fund Flow to PMU

The Department of Agriculture will further allocate the budgetary grants through BEAMS (Budget Estimation, Allocation & Monitoring System) to the Controlling Officer i.e. Project Director, PMU, NDKSP Phase II. The Project Director will act as Head of Department and will have administrative as well as financial powers and supervise financial activities to check if there is any deviation of

funds without proper sanctioning authority to other items not initially intended. The budget allocation for Project Director, NDKSP will specify the funds under various major and minor account heads provided for activities to be carried out at PMU Level. The DDO in the office of PMU i.e. Finance Specialist will draw funds through Pay and Account Office (PAO) and incur expenditure for the following purposes:

1. Project Management expenses
2. Project Component expenses

Fund Flow to JDA

The Controlling Officer in PMU will further allocate the budgetary grants through BEAMS to the Joint Director, Agriculture (JDA). The regular DDO in the office of JDA will incur following expenses through District treasuries.

1. Project Management expenses
2. Project Component expenses

Fund Flow at District level

Fund flow to DSAO

The Controlling Officer in PMU will allocate the budgetary grants through BEAMS to the District Superintending Agriculture Officer (DSAO). The DDO in the office of DSAO will incur following expenditure through District treasury for following purposes:

1. Project Management expenses
2. Project Component expenses

Fund Flow to ATMA

The funds sanctioned for the expenditure at the level of ATMA shall be released on BEAMS to DSAO by the Controlling officer in PMU. The DSAO will draw funds from the District Treasury and deposit in a separate account of Project Director, ATMA through core banking. The Project Director, ATMA will make payments from the funds deposited in his account for following expenditure:

1. Project Management expenses
2. Project Component expenses

Fund Flow at Sub-Divisional level

The DSAO will further allocate the budgetary grants through BEAMS to the Sub-Divisional Agriculture Officer (SDAO) after verifying demands from Sub-Divisions under their jurisdiction. The DDO in the office of SDAO will incur the following expenditure through treasury:

1. Project Management expenses
2. Project Component expenses

The actual fund flow for the project & units under the project will be given in detail in the Finance Manual.

Fund Flow at Cluster / Village Level

In Phase II, there is no accounting centre designated at cluster level. Fund flow for cluster/village level expenses will be routed through SDAO. The Funds, if required, and sanctioned to Gram Krushi Vikas Samiti for meeting project related expenses shall be released by the office of SDAO through the Core banking system. The SDAO will draw funds from District Treasury/Sub-Treasuries by presenting Grants-in aid bills and deposit in a bank account of village committees. GKVS will make payments from the funds deposited in the bank account against the project expenses and SDAO will submit Utilisation certificate to PMU regarding those grants after monitoring expenses at village level.

The PMU is also examining feasibility of bringing GKVS into the DBT system of the project, where PMU will directly send funds to them for project related expenses, so that field-level accounting centres will be less burdened and monitoring of funds by PMU and their utilisation will be effective.

The funds for project implementation will be channelled to the project implementation agencies as per the flow as envisaged in Financial Management Manual for this project.

Fund Flow Mechanism

The Annual Work Plan of the project will be approved by the Steering Committee formed by the state government for the project with the Chief Secretary of the state as its Ex-Officio Chairman. All policy decisions regarding financial management will be approved by this committee.

7.3 Audit

There shall be external & internal audit arrangements for the project. Utilisation Certificates will be verified by the audit system to make sure funds released by the state government via PMU to subsequent accounting centres at field levels are utilised for the purpose they are meant for and to check unregulated deviation of funds.

The following types of audits are proposed for the project:

1. External Audit
2. Internal Audit

External Audit

The external audit of the project will be conducted by CAG through the State Principal Accountant General's office, Mumbai for all implementing levels in the project. The statements of expenditure

(SOE) incurred at all levels under the project will be submitted to the State AG latest by June 30th for the preceding financial year. This will provide a three month's period for AG to undertake the audit.

The complete audit of PMU, JDA, DSAO, PD ATMA and SDAO shall be carried out to check effectiveness of project implementation and financial statements shall be audited by the State Accountant General.

Audit by State AG will be conducted in accordance with TOR of Externally Aided Projects (EAP) agreed with CAG for audit of World Bank financed projects. The audit reports will be submitted by the AG team within nine months from the end of each financial year and within three months from the date of receipt of financial statements from project PMU office.

Internal Audit

The overall objective is to provide the project management with independent assurance that:

1. The internal controls established by project management are designed appropriately and
2. The controls are operating effectively and as intended.

Also, the internal audit should play a role in assisting project management

- i. Bringing up a systematic, disciplined approach to evaluate and improve the effectiveness of risk management.
- ii. Identification of control lapses.
- iii. Taking corrective measures in a timely manner in case of any anomaly by implementing machinery.

The internal audit of PMU, JDA, DSAO, SDAO, ATMA (half yearly)) will be conducted by a firm of Chartered Accountants selected by standard procurement methods in line with World bank guidelines and prevalent state government norms.

7.4 Direct Benefit Transfer Mechanism

Objective

Direct benefit transfer (DBT) scheme aims to improve the efficiency and timeliness/prompt delivery of benefits of various schemes and programmes. The Government of India has advised States to adopt this method for transfer of various subsidies and grants under different schemes. Maharashtra is one of the pioneers in adopting DBT under various schemes. In fact, the first phase of this project i.e. the NDKSP Phase- I was instrumental in the development of a streamlined DBT process which involved end-to-end integration starting from the registration of beneficiaries and ending with disbursement of matching grants through the NPCI- enabled Aadhar- enabled

Payment System (AePS). The NDKSP Phase II will adopt the same process for transferring matching grants to beneficiaries adopting this DBT framework. This will be built into the project MIS.

In the NDKSP Phase II, farmers register, apply for listed schemes, and receive back-ended subsidies through a streamlined, desk-wise process that ensures accountability at each step, from registration to fund disbursement.

Partnership with Banks -

The project PMU is working towards tying with Banks/Financial Institutions to make credit facilities easily available to farmers with lower capital in hand. It will help farmers get formal credit lines to carry out project activities in an economic and efficient way. It will also reduce the gap between sanction, execution by recipient and disbursement of subsidy.

VPDA System -

Government of Maharashtra, Finance Department has introduced Virtual Personal Deposit Account (VPDA) to check effective distribution of subsidies to beneficiary accounts without unnecessary holding/ diverting of government money in any other commercial bank account. DDO for this Project, Finance Specialist is declared as VPDA Administrator by the Agriculture Department with the concurrence of Accountant General's Office, Maharashtra State, Mumbai. Pay & Accounts Office has issued authority in the name of Finance Specialist to run activities as VPDA administrator.

Under proposed VPDA system in Phase 2, grants released to projects by the state government under the head Subsidies will be virtually placed in VPDA accounts opened for this purpose after passage of Grants-in-Aid bill by Pay & Accounts Office/ Treasuries till the time of actual transfer of subsidy to beneficiaries. When the subsidy batch is ready to be processed through the DBT portal, Grants virtually placed in the VPDA account will be transferred to the DBT account of the project and within 24 hrs, the amount will be credited to beneficiaries.

Onboarding of Bank for DBT

Separate bank will be selected and onboarded on DBT portal by issuing Expression of Interest (Eoi) along with ToR of tasks required to be performed by this bank from the empanelled banks for doing Government Business as per GR notified by Finance Department after following World Bank procurement guidelines. Current account will be opened in this selected bank and DBT transactions will be carried out using this account as per project guidelines.

Central Bank Digital Currency

Integration of a new payment method of P-CBDC (Programmable Central Bank Digital Currency) to streamline and enhance the subsidy distribution process is under consideration for the phase

II. exploring to offer front-ended benefits to small farmers, while seeking to ensure that the funds are used for the intended purpose.

Project is exploring P-CPDC integration in the DBT portal to offer front-ended benefits to small farmers and to ensure that the funds are used for the intended purpose.

7.5 Workflow of DBT for Individual farmers:

Step 1: Individual Farmer Registration

Farmer Module: Farmers register on the DBT portal using their Farmer Id / Aadhaar number. This registration is mandatory to ensure identification and linking of bank accounts for seamless fund transfers. After registration, the farmer can apply for the desired scheme from the schemes available to him.

Step 2: Application Submission & Verification

Desk 1 (Cluster Assistant - CA): Once the farmer applies for a particular activity, the Cluster Assistant (CA) verifies the farmer's registration details and submits the application for the Gram Krushi Vikas Samiti (GKVS) approval.

Step 3: Site Visit and Inspection (Post-GKVS Approval)

Desk 2 : After the application is approved in the GKVS meeting, pre site visits and inspections are conducted to ensure technical feasibility.

Step 4: Pre-Sanction Issuance

Desk 3 (Pre-Sanction Desk) -After the Desk 2 clearance, the documents and details are scrutinized by the Department Functionaries (Sub divisional Agri Officer /Taluka Agri Officer). Once the verification is complete, a pre-sanction letter is issued to the farmer digitally, allowing them to proceed with the approved activities.

Step 5: Farmer Completes Activities & Submits Bills

Farmer Module: Upon receiving the digital pre-sanction letter, the farmer completes the approved activities on their land. Once the work is done, the farmer submits the bills and invoices for reimbursement through the DBT portal.

Step 6: Post-Site Work Inspection

Desk 4 (SDAO/TAO/Agri Supervisor/CAO): Once the farmer submits the invoices, a post-site work inspection is conducted by SDAO, TAO, Circle Agriculture Officer (CAO), Agriculture Supervisor or Agriculture Assistant as prescribed in guidelines.

This inspection ensures that the activities were completed as per the approved project guidelines for said components before proceeding with the payment. Every activity initiated is geo-tagged for record and verification process, and is collated at project MIS level.

Step 7: Financial Review & Recommendations

Desk 5 (Account Officer - AO): The Account Officer (AO) performs a detailed financial review of the farmer's application and invoices. The Account officer verifies proposal and other financial documents before giving their recommendation for the subsidy payment.

Step 8: Payment Approval

Desk 6 (SDAO/DSAO): Based on the recommendations provided by the Account Officer (AO), the SDAO /DSAO gives final payment approval for the subsidy disbursement to the farmer as per project guidelines.

Step 9: Payment Verification

Desk 7 (Checker - Account Officer, PMU): The Account Officer at the Project Management Unit (PMU) performs verification of financial aspects related to the payment. This check ensures that the payment aligns with the approvals and recommendations received as per project guidelines.

Step 10: Subsidy Disbursement

Desk 8 (Finance Specialist): The Finance Specialist sends a request to the bank for subsidy transfer using the Aadhaar-enabled Payment System (AePS). The subsidy is then transferred directly to the farmer's Aadhaar-linked bank account.

Throughout the entire process, from application submission to fund disbursement, farmers receive digital notifications at each stage, providing real-time updates on their application status, thus eliminating confusion and ensuring transparency.

7.6 Mechanism for post-subsidy disbursement verification –

A comprehensive, multi-layered mechanism will be implemented to ensure the appropriate use of project assets and the integrity of matching grant disbursements. The goal is to ensure that subsidies reach the intended beneficiaries and are used effectively, while also addressing potential issues like corruption or misuse of funds. This will consist of following.

1. Random Visits by Department Officials: Department officials will make unannounced visits to project locations. This allows for a firsthand look at how things are progressing, whether assets are in place and being used as intended, and if there are any immediate red flags. These visits act as a deterrent against misuse and provide real-time feedback to the department.

2. Internal and Procurement Audits: Periodic internal and procurement audits (yearly as well as six-monthly) will be conducted. The internal auditor will verify that sanctioned funds are utilized according to the terms of the sanction order and for their intended purpose. Both the internal and procurement auditors will conduct field visits (using random sampling) to confirm the existence of project assets, and to ensure they are being used for their intended purposes.

3. Social Audit: Social audit acts as a powerful tool to ensure benefits reach the targeted group and are used properly. Social audit shifts the power dynamic by empowering community to actively participate in monitoring and evaluating project benefits. This direct involvement, coupled with transparency and the demand for accountability, makes it a potent tool for ensuring that the intended benefits truly reach the targeted group and are used for their intended purpose, ultimately leading to more effective and equitable development outcome.

A social audit of post-subsidy disbursement will involve community participation in verifying the accuracy and fairness of subsidy payments, ensuring transparency and accountability. This process includes comparing official records with actual ground realities, seeking oral testimonies from beneficiaries, and presenting findings at a public platform.

In essence, this multi-pronged approach combines top-down monitoring by Department officials and auditors with bottom-up accountability through community involvement. This creates a robust system designed to maximize the proper use of project assets and ensure the integrity of the financial assistance provided to the farmers.

Section 8

Procurement Management



Section 8:

Procurement management

8.1 Introduction

The World Bank Procurement Management under the NDKSP (Project on Climate Resilient Agriculture) phase II initiative follows the Bank's Procurement Framework to ensure transparent, fair, and efficient use of resources in climate-resilient agricultural projects. NDKSP phase II, implemented in 21 districts of Maharashtra, focuses on improving farmers' resilience to climate change by funding sustainable water and soil management practices. The procurement process under NDKSP phase II adheres to the principles of value for money, transparency, efficiency, and competition, ensuring that goods, works, and services are procured effectively to benefit Gram Krishi Vikas Samiti (GKVS), FPO/FPC/SHG and their federations.

8.2 Procurement Arrangement and Overview:

The Project shall adhere to World Bank's New Procurement Framework comprising policies and World Bank Procurement Regulations for IPF Borrowers" September 2023 ("Procurement Regulations").

Accordingly, a separate Procurement Manual is prepared for the project describing procurement strategy, procurement approach, methods of procurement and procedures of procurement of Goods, Works, non-consulting services and consulting services.

The procurement Manual will serve as a reference document for undertaking any type of procurement as might be necessitated under the NDKSP phase II. The procurement manual defines the constitution and scope of the Procurement Cell under the aegis of PMU of NDKSP phase II. It also chalks out a procurement plan and risk mitigation plan.

The rights and obligations of the NDKSP phase II, Government of Maharashtra (the Borrower) and the service providers of Goods, Works, Non-consulting Services and Consulting Services for IPF operations are governed by the relevant request for bids/request for proposals document and by the contracts signed by the NDKSP phase II not by procurement regulations or the legal agreement. No party other than the parties to the legal agreement shall derive any rights from, or have any claim to financing proceeds.

8.3 Procurement Implementation Arrangement

The NDKSP phase II is primarily a community demand driven project. The project proposes to develop and implement mini-watershed plans in 7196 villages.

The planning process for the project starts at the village level and mini-watershed based planning in clusters of villages in the project area will play an important role.

Based on the outcome of the participatory micro plan, a detailed Village Adaptation plan (VAP) will be developed. VAP will be the basic unit of planning for this project and it will give an idea of CR technologies to be adopted in Phase II NDKSP villages in a broader sense. GKVS will approve VAP and it will be verified by SDAO. The District Level coordination committee headed by District Collector will give administrative approval to NRM works mentioned in VAP.

Therefore, the procurement requirements are substantial in quantity but are of small value.

8.4 Project Procurement Strategy Development (PPSD)

The PMU has developed the PSD and Procurement Plan for the first 18 months, which will be reviewed and cleared by the Bank. PSD provides rationale of the procurement strategy and optimum procurement approaches to be adopted for carrying project procurement. The PSD defines the procurement methods, procedures and processes applicable for the project including prior review and post review thresholds, risk mitigation measures, etc., which shall be adopted for the project.

Community Driven Development

As per the approved Village Adaptation Plan (VAP) following activities shall be carried out at community level.

- a. NRM works shall be implemented by Gram Krishi Vikas Samiti (GKVS)
- b. Business plan shall be carried out by the Farmer Producer Organisation (FPO)/Farmer Producer Companies (FPC)/Self Help Groups (SHG) and their federations /Farmer Interest Groups (FIG). Also, Farmer Producer Group/Organization engaged in village level agri-business/agri-entrepreneurship are expected to procure necessary Goods and Works for establishing agri-enterprises themselves.
- c. Individual Beneficiary level activities shall be carried out by respective farmers as per the VAP.

Community participation in procurement of works/goods activities from planning to post implementation can help in ownership, smooth operation and maintenance and successful implementation of the project.

The project follows World Bank Regulations (September 2023 version) and uses the World Bank's model procurement documents for these NRM works. For works estimated above INR 3,00,000, The entire procurement process will be carried out on the dedicated E-tender Portal of the Government of Maharashtra by the community with technical support of concerned SDAO this will help in ensuring transparency and adherence to Bank regulation.

Community Work Consideration:

Since NRM works fall under community procurement, for which Gram Krushi Vikas Samiti (GKVS) shall take the lead in both initiating and executing these efforts. GKVS's dedicated Procurement Committee will conduct the entire procurement process independently on the Maha Tender portal, ensuring transparency and accountability. However, in instances where the community may face challenges managing the work autonomously, technical assistance will be provided by Sub-Divisional Agriculture Officers (SDAO) specifically for navigating the Maha Tender platform. Importantly, the evaluation process will remain community-centric, with GKVS members working closely with SDAO officials. This strategy is designed to strengthen community ownership and build local capacities for sustainable resource management. Further details and the Procurement committee structure of GKVS and SDAO will be provided in the Procurement Manual.

FPO/FPC/SHG:

For Phase II, Procurement for Farmer Producer Organisation (FPO)/Farmer Producer Companies (FPC)/Self Help Groups (SHG) and their federations /Farmer Interest Groups (FIG) will be carried out through offline mode up to INR 20 Lakhs. For contracts exceeding INR 20 Lakhs Maha Tender platform shall be used. Since these entities are private bodies they do not have direct access to the Mahatender platform therefore the concerned FPO/FPC/SHG shall approach the officials at the PD ATMA level for providing technical support and floating tenders on Mahatender portal on the behalf of FPO/FPC/SHG.

The evaluation of all procurement processes will be conducted by the respective Procurement Committees within the FPO/FPC/SHG. The role of the PD ATMA level officials is limited to offering technical assistance and support throughout the procurement process.

The procurement process for Farmer Producer Groups/Organizations engaged in agri-business or agri-entrepreneurship at the village level under the project outlines different procurement methods based on the estimated project cost.

Table 31: Different procurement methods

Sr. No.	Activity wise value as per Business Proposal	Advt. Level	Procurement Platform (Offline/Online)	Method of Procurement
1	Up to 20.00 lakh	Local Newspaper	Offline	Request for Quotation (Model RFQ and contract document for small procurement to be usedk) Request for Bid (World Bank SPD)
2	Above 20.00 to 40.00 lakh	District level newspaper	E-Tender	
3	Above 40.00 to 100000 USD	Regional level newspaper		
4	Above 100000 USD	National Level newspaper		

* The above advertisements will be published as per the threshold in the respective newspaper as per the guidelines of DGIPR.

Contract Information Maintenance:

For contracts below USD 100,000 Information will be maintained in the Procurement MIS (Management Information System). For contracts above USD 100,000 Information will be entered into the STEP portal through district level Procurement Specialist (Systematic Tracking of Exchanges in Procurement), which is managed by the World Bank.

Procurement Guidelines:

The procurement Manual and Procurement Approval Desk guidelines in English as well as in local language i.e. Marathi provides simple step by step instructions to the GKVS and FPO/FPC/SHG and their federations on how they will go about procuring the various goods and works required for implementation of the project and includes simple forms and formats to be used during procurement procedures applicable in the project. This is intended to enhance the capacity of the GKVS and FPO/FPC/SHG to undertake procurement to meet project needs.

This provision of the Bank's Procurement Guidelines has been specifically designed to address procurement in Community beneficiary involved projects. It stipulates that procedures, specifications and contract packaging shall be "suitably adapted" to reflect the conditions and capacity of the community, provided they are "efficient and acceptable to the Bank." This provision allows flexibility in defining acceptable procedures at the community level. As a general rule, procurement should be designed to:

- Be simple enough as to be understood and implemented by local staff and the community;
- Be sufficiently transparent to permit real competition among suppliers and to facilitate control in the selection of contractors and use of funds;
- Use simple, standardised documentation; and
- Balance risk versus control/management with efficiency considerations

8.5 Procurement Plan

The PMU with involvement of District PIUs will prepare a procurement plan based on the projected activities in the Project Implementation Plan. The PMU will be the nodal agency with the responsibility of preparing the procurement plan and getting it cleared from the Bank through the online STEP system, the contracts above USD 100,000 at community level will be submitted on STEP portal as Procurement plan and the contracts below USD 100,000 at the community level needs to be maintained in Procurement MIS as a Procurement Plan. The Procurement Plan will include description of goods, works and consultant services to be procured (year-wise over the implementation period) along with their values which are consistent with technically and administratively approved cost estimates and milestones for all procurement activities. List of goods, works and services to be procured under the project year-wise, with estimated cost and method of procurement, Bank's Model procurement documents and review types have been mentioned in the format specified by the Bank.

During preparation of the procurement plans, packaging shall be done appropriately indicating realistic dates. Items of similar nature, which can be supplied by the same set of firms, should be packaged together to achieve economies of scale. Aggregate value of the total package will form the basis for determining the procurement method as well as the review requirement of the World Bank. For procurement of goods, works, equipment etc. suitable packages shall be framed which shall be determined by below noted factors, namely time limits, geographical location (dispersal of works), etc.

Following points should be borne while preparing the Procurement plan

- (a) All procurement shall follow the agreed processes and procedures as specified in Bank Regulations and is elaborated in the procurement manual.
- (b) Bidding documents as agreed with the Bank shall be used.
- (c) Procurement plan shall be based on activities in the agreed cost tables. All Procurements to be carried out on the basis of core principles i.e. Economy, Efficiency, transparency.
- (d) The prepared 18-month plan shall be revised/updated at least once in 12 months and sent to the bank for prior clearance through STEP. After the clearance the Procurement Plan will be published on the project website and Bank website.

Methods of Procurement

Table 32: Method of procurement

Method of Procurements for Goods and Works	Approved Selection Methods for Consultancy Services
Request For Proposal	Quality Cost Based Selection
Request For Bids	Fixed Budget Based Selection
Request For Quotations	Least Cost Based Selection
Direct Selection	Quality Based Selection
Framework Agreement	Consultants Qualification Based Selection
Force Account	Direct Selection

The Procurement Manual provides a detailed description of each method of procurement mentioned above.

Project Procurement Thresholds

Table 33: Project procurement threshold

Method of Procurement	Thresholds for Method (US\$ equivalent)
Open International (Goods)	Equal to or more than 3 million
Open National (Goods)	< \$ 3 million and >\$100,000 as per the National Procurement Procedures (Para 5.3 to 5.5 of Section V of the Procurement Regulations and additional provisions for India
RFQ (Goods)	Less than \$100,000
Open International (Works)	Equal to or more than 40 million
Open National (Works)	Less than \$ 40 million and more than \$ 100,000 as per the National Procurement Procedures (Para 5.3 to 5.5 of Section V of the Procurement Regulations and additional provisions for India
RFQ (works)	Less than \$ 100,000
Open International (Non-consulting Services)	Equal to or more than 2 million
Open National (Non-consulting Services)	Less than 2 million and more than \$100,000 as per the National Procurement Procedures (Para 5.3 to 5.5 of Section V of the Procurement Regulations and additional provisions for India
Direct Contracting	Must meet the requirements as per para 6.8 to 6.10 of the Procurement Regulations, and with prior agreement in PP with the Bank

Method of Procurement	Thresholds for Method (US\$ equivalent)
Framework Agreements	All procurement that meets the requirements as per Para 6.57 to 6.59 of Procurement Regulations., and with prior agreement in PP with the Bank. All FAs are subject to prior review by Bank
Force Account	Must meet the requirements as per Para 6.54 and 6.55 of the Procurement Regulations, and with prior agreement in PP with the Bank

Prior Review Thresholds: The details of the Procurement review / oversight are defined in the Annex II of the Regulations for borrowers. The following would be subject to Prior review of the Bank regardless of the Method of selection:

Table 34: Prior review of the Bank

Type of procurement	Bank's Prior review threshold
Works	All procurements estimated to cost more than USD 10 million
Goods	All procurement estimated to cost more than \$2 Million
Consultant Firm	All procurements estimated to cost more than USD 1 Million
Consultant Individual	All procurement estimated to cost more than USD 300,000

**These thresholds are for the purposes of the initial procurement plan for the first 18 months. The thresholds may be increased or decreased based on re-assessment of risks assessment. All contracts not subject to prior review will be post-reviewed.*

Institutional Partnership through Direct Selection

It is envisaged to hire Govt. and semi government Institutions, including Research Institutes having unique and exceptional capacities, as consultants for different assignments with prior no objection of the World Bank, on a case-to-case basis. An MoU with a Strategic Partner shall be treated as a contract for all practical purposes (i.e. to be added to the Procurement Plan, to be reviewed by the Bank's etc.) irrespective of the nomenclature of the document as MoU. When a MoU is treated as a contract, it will be allowed only for small value assignments, i.e. up to US\$ 300,000. If a MoU is proposed to be used for an assignment having value greater than this threshold, it will be treated as non-standard procurement document and Bank prior clearance irrespective of the threshold is required before use. Justification for Direct Selection is mandatory by the Bank at respective specific approval stages.

Procurement Strategy for Procurement of Seed, Fertilizer and Biopesticides

In Phase II of the project, the procurement of climate resilient seeds, bio-fertilizers, and bio pesticides are required for demonstration purpose. These will be small value procurements at SDAO level. The estimated total cost towards seeds, bio-fertilizers, and bio pesticides to be used as consumables for demonstration purposes only for 55 Sub division covering 7196 villages for entire project period will not exceed USD 1 Million (appx 8.5 cr). These will be procured by individual's sub-divisions as consumables which will be used from time to time for farmer demonstration. Given the specific requirement that seeds /fertilizer are to be well suited to the local agro climatic conditions the same are proposed to be procured directly through Maharashtra State Seed Corporation (Mahabeej), Agricultural Universities, and/ or Research Institutes. These institutions are government entities are uniquely placed due to their own production capacities, ensuring authenticity and the availability of high-quality agricultural inputs which are well suited to the agro climatic conditions of the project area.

8.6 Procurement Cell and Management

Procurement Cell at PMU

Procurement under project will be managed by the Project Management Unit (PMU) headed by the Project Director (NDKSP phase II) who is overall in-charge of project implementation, the PMU shall be staffed with 1 Procurement Specialist, 1 Associate and 1 Assistant adequate staff for overall Procurement Management under the project. The PMU will be the Project Nodal Agency responsible for guiding, monitoring and supervising the overall procurement implementation under the project. It would be responsible for all contracting and procurements in the project. The State level PMU shall have the oversight and do the needful handholding and training as required for district and sub-district level procurement. It will also do the supervision of contracting and procurements at State/ Regional/district/ Sub division, community level, across all components as advised by technical staff, consultants and specialized agencies contracted for this purpose.

In addition to overall responsibility of procurement activities, the Procurement Cell will:

- a. Review the Procurement Plan for procurement to be conducted at the respective level and submit for Bank's clearance through STEP.
- b. Ensure that consistent and correct methods of procurement as specified in the Financing Agreement and as per procedures outlined in this Manual are consistently followed by all the procurement entities of the Project;
- c. Finalize technical specifications for goods/civil works and terms of references for consulting services respectively and preparation of bid documents;

- d) Evaluation of quotations/ bids for goods & works and proposals for consultancies followed by preparation of evaluation of reports;
- e) Develop and maintain Procurement Management Information System through banks STEP portal and project MIS of all procurements done in the Project;
- f) Maintain records of all procurement related complaints received and their disposal;
- g) Collect, collate all procurement related information for review by Bank and its auditors;
- H) Ensure that all procurement related records and documents required by the Bank Procurement Regulations are properly maintained, kept by the various procurement entities; with a system to retrieve the required document within a reasonable time.
- I) Seek Bank's No Objection for all Prior Review cases and wherever required as per the Bank's regulations or Procurement Plan.
- J) PMU will provide a Community Training Module to all district's Procurement Specialists. In the project period community procurement training will be conducted by District Procurement Specialists and other project staff in all project Areas to minimise errors.
- K) PMU Staff will give handholding support to the community at every stage of the Procurement Process.

Procurement Cells at Regional / District /Sub-Division Level

At the Regional/District/Sub-division Level the procurements under project will be managed by the PIUs led by the Regional Joint Directors of Agriculture / District Superintending Agricultural Officers/Project Director ATMA/Sub-divisional Agriculture Officer of respective Region / District, who are overall in-charge of implementation of project activities under the NDKSP phase II within the Region /District/Sub-division. The Procurement Cells under DPIUs will be created at District level supported by designated Supervising Officers at the Regional / District /Sub-division level and designated procurement/finance management support staff. (1 Procurement Specialist at district (PD ATMA) level and 1 project assistant, procurement point person at SDAO level)

Procurement capacity building:

For enhancing procurement capacity, agreed mitigation measures with the WB are up-skilling of procurement staff, availing training and on-going support from the WB. Key procurement staff will be sent for training at one of the Procurement Training Institutes. (Manage, Hyderabad/IIM Lucknow)

PMU Support and Training to DPIU's and community

The PMU will provide training of trainers to DPIU procurement persons and offer continuous handholding support. As part of capacity building, a series of procurement training sessions will be conducted on the World Bank regulation and SPD's , along with dedicated sessions on the Maha-Tender portal by PMU. Additionally, the PMU will help in the preparation of training

module for community procurement training and oversight. From time-to-time PMU will act as an observer in the community procurement training, ensuring effective oversight and guidance. Additionally, the PMU will organize workshops and clinical workshops once a year to discuss audit observations, facilitating better understanding and cross-learning among community members.

Community Capacity Building

The PMU is committed to strengthening and empowering the community through capacity-building initiatives. To support this, comprehensive training materials, along with a structured training roster, will be provided. The PMU trained procurement person in each district will facilitate training for community groups, including FPOs, FPCs, SHGs, and GKVS committees as per the training module developed and approved by PMU. This will be ensuring compliances, consistency and effective implementation at the grassroots level. The planning for the training will be based on the roster agreed with the PMU.

Record Keeping:

All records pertaining to procurement including the award of contract, bid notification, register pertaining to sale and receipt of bids, bid opening minutes, bid evaluation reports, tender register; and all correspondence pertaining to bid evaluation, communication sent to /with the WB in the process, bid securities, approval of invitation /evaluation of bids by the PMU would be retained by the concerned IAs.

Procurement Audit:

All contracts not covered under prior review by the WB will be subject to Post Procurement Audit Review by the WB and also by the PMU through auditors appointed for the purpose. Procurements covered under operating cost at PMU and DPIU's level will be included under audit by the Internal Auditor. Project will appoint a consultancy service firm for Post Procurement Audit Review. For the procurement audit, the project will plan to select 10% of samples for the contracts below INR 40 Lakh & all the contracts above INR 40 Lakh.

STEP

Contracts Above \$100,000 USD: These will be uploaded to the **STEP portal**. The STEP (Systematic Tracking of Exchanges in Procurement) portal is commonly used for tracking and managing procurement processes, often in international development projects.

Contracts Below \$100,000 USD: These will be tracked on a separate **Procurement MIS (Management Information System) portal**. An MIS portal typically allows for detailed tracking and management of procurement data, ensuring transparency and accountability.

View rights to World Bank: Sharing the view rights of the Procurement MIS with the World Bank allows the institution to monitor and oversee the procurement processes for contracts below \$100,000 USD. This can enhance transparency, ensure compliance with procurement standards, and facilitate communication between your organization and the World Bank. It's an important step, especially in projects funded or supported by the World Bank, to maintain accountability and integrity in the use of funds.

In Phase II, the total number of contracts is projected to exceed 16,000, with more than 2,500 contracts individually valued at over ₹50 lakh and remaining are below INR 50 Lakh. To handle this substantial volume effectively and ensure efficient administration, it is proposed that all contracts valued above USD 100,000 be uploaded to the STEP portal. This platform will facilitate streamlined management and allow for greater accountability in handling high-value contracts.

For contracts below USD 100,000, the monitoring and management will be carried out using the Project Management Information System (PMIS), a sophisticated platform developed by the Project Management Unit (PMU). The PMIS is designed to capture real-time data throughout the entire contract lifecycle, encompassing stages such as training, procurement planning, bid evaluation, and awarding contracts and complaint redressed as per the bank regulation. Each DPIU will be responsible for updating the data which will be accessible to PMU. PMU will be sharing the data on quarterly basis with the World Bank. In addition, an independent Procurement Auditor will be hired by PMU as per the ToR agreed with the Bank covering 15 % of the contracts or as agreed with the Bank. The report will be shared with World Bank. In view of the above arrangements and given the volume and geographical spread these activities are proposed not be included in STEP system. The working of the Procurement MIS is explained below.

To further reinforce transparency and strengthen oversight mechanisms, the PMIS will provide view-only access to the World Bank, enabling regular monitoring and ensuring alignment with project objectives and international standards. This dual-platform approach leverages the unique strengths of the STEP portal and the PMIS to address the diverse requirements of contract management in Phase II.

Procurements at community level not to be included in STEP

Like Phase I of PoCRA the Phase II will also have very large number of small value procurements at various locations i.e. at village level. There are 21 districts and more than 7000 villages in the project area where these procurements will take place. The community procurements under the project are being carried out by (a)community organizations i.e FPOs, FPC's, SHG's and GVSK for value upto INR 1CR (\$120,000). Project will be providing matching subsidies on cost sharing norms where these community organizations will contribute an equal amount of money directly

or through loan mobilized through a Bank. These community organizations are registered organizations which can execute activities like setting up a custom hiring center, post-harvest, and value addition activities like warehouses etc. In addition, there are Natural Resource Management activities like catchment and drainage treatment works which will be 100% project assisted activities. The subsidies will supplement the FPO own funding and will be used to procure civil works, storage facilities, farm implements and machinery. The State has mandated the use of e-procurement for activities estimated to cost Rs300,000 and above. Community organizations will be trained to carry out procurement directly following RFQ or DC method for contracts upto \$100,000. It may be noted that the state has price norms for equipment and schedule of rates for civil works, the FPO will be putting in their own funds and may procure the equipment using appropriate procedures i.e direct contracting or request for quotations. DPIU staff will provide support to the communities in carrying out e-procurement on behalf of the community. An internal audit or an independent post procurement review will regularly review these procurement activities (b) Activities at the individual beneficiary level shall be executed as per the state mechanism and since these are subsidies being reimbursed these are not considered as procurement. However, digitization of records and capturing of geo-spatial coordinates for every activity executed under the project leading to Direct Benefit Transfer of financial assistance under the project is itself a controlling mechanism developed under this project. PMU is already in the process of developing a digital platform which will integrate every activity for smooth execution of the project

Given the size and scale of the project the community level activities below \$100,000 will not be included in STEP. However, the PMU should require these communities with assistance of district staff to prepare procurement plans and provide the information about planned and completed procurements (similar information required in STEP) in an excel table on a quarterly basis which shall be updated in the **procurement MIS** by the DPIU's. PMU will remain fully accountable to ensure full compliance with the agreed procurement procedures (including eligibility and debarment) in conducting such small procurements. Each FPO/FPC /SHG and GVSK will have a procurement point person trained by DPIU. Third party firm will be hired to monitor procurement activities at this level (as part of financial or internal audit or **standalone procurement review** with TOR agreed with the Bank) and the findings shall be shared with the Bank regularly at least at the end of each fiscal year.

Use of Government e-marketplace (GeM)

The project will use the GeM for PMU and DPIU level procurements under the following conditions:

- a. upto INR 25,000 in catalogue mode (Direct Purchase) (viz. any available item could be selected by IA without further competition), provided selected Item/Supplier meeting the requisite quality, specification, and delivery period.
- b. upto INR 5,00,000 lakh from the Supplier having lowest price amongst at least three Suppliers meeting the requisite quality, specification and delivery period. The tools for online bidding and online reverse auction available on eGeM may be used by the Purchaser.
- c. Above INR 5,00,000 till INR 85,00,000 from the Supplier having lowest price and meeting the requisite quality, specification, and delivery period after **mandatorily obtaining** bids from at least three Suppliers, using online bidding or reverse auction tool provided on eGeM.”
- d. The required inclusion of relevant documents for eGem shall be defined in the Procurement manual which shall be fully complied with by all the IA’s.
- e. Threshold for RFQ contracts is \$100,000
- f. In above all cases of assessment (a to e) price reasonableness shall be recorded to ensure transparency, value for money, and compliance with applicable norms and guidelines.

"The e-Gem platform shall not be used as a substitute for the Request for Bids (RFB) – Open (International/National) method. All software shall be procured directly from the original developers/suppliers or their authorized dealers."

Books and journals shall be purchased directly from the publishers.

Procurement MIS (Management Information System)

Phase 1 of the Procurement MIS, successfully launched on 30th June 2022, provides comprehensive tracking of procurement activities by capturing data at the community level. View access has been shared with the World Bank, promoting transparency and efficiency in the process.

In Phase 2, the Project Management Unit (PMU) is developing a Procurement Approval Desk, which will be integrated into the FPO/FPC and NRM portal. This desk aims to streamline the entire procurement process by capturing data at each stage, from initiation to completion, ensuring greater efficiency and transparency.

Procurement Manual:

More details will be provided in the Procurement Plan, Procurement Manual for the Project.

Annexures

Annexure 1: Climate profile of Maharashtra

Climate profile of Maharashtra

Maharashtra state features a diverse climate, which is influenced by the Arabian Sea, Western Ghats, and the Deccan Plateau. It can be classified into several main types, including tropical monsoon, tropical semi-arid, and tropical wet and dry climate (IMD 2005). Figure 1 represents the Koppen-Geiger classification of climate over the state of Maharashtra.

The state of Maharashtra experiences distinct seasons, with a hot summer from March to May, a monsoon season from June to September and a cool winter from November to February. The primary period of rainfall in Maharashtra is during June to September with the southwest monsoon being the dominant monsoon phenomenon. Additionally, the state also receives some amount of its annual rainfall in the pre-monsoon (March to May) and post monsoon (October to December) months. Different regions of Maharashtra experience varying climatic conditions, with coastal regions experiencing a maritime climate and the central plateau region having a semi- arid climate. Overall Vulnerability Index Score of Maharashtra is 0.478 (3rd rank) and the other two states are Assam, Andhra Pradesh, (0.616 and 0.483 respectively), these are the top three climate vulnerable states in India. Maharashtra is prone to various climatic disasters such as droughts, floods, cyclones and heat waves.

Maharashtra has a diverse climate influenced by the Arabian Sea, Western Ghats, and Deccan Plateau. The state's climate ranges from tropical monsoon to semi-arid and wet and dry. Key seasons include a hot summer (March-May), monsoon (June-September), and cool winter (November-February). The state receives most of its rainfall during the southwest monsoon, with some additional rainfall in pre- and post-monsoon months. Maharashtra is highly vulnerable to climate change, ranking third in overall vulnerability in India, and is prone to disasters like droughts, floods, cyclones, and heatwaves.

Climatic variabilities and its impact on agriculture in Maharashtra

The monsoon is crucial as it serves as the primary water source for irrigation, ensuring crop productivity and maintaining soil moisture essential for seed germination and plant growth. The onset and withdrawal of the monsoon directly impact crop yield, agriculture productivity, and water management practices. Insufficient or delayed monsoon rainfall can lead to water scarcity for irrigation, affecting crop growth, while excessive rainfall from extreme precipitation events can cause waterlogging, crop damage, and hinder irrigation activities.

Mapping monsoon and temperature variability

To map the changing monsoon patterns in Maharashtra, a granular taluka-level analysis was conducted by downscaling the Pan-India findings of Prabhu and Chitale (2024). Utilising IMDAA reanalysis data at a spatial resolution of 12 km, changes during the southwest monsoon (June to

September) and post-monsoon (October to December) periods were mapped. The assessment quantified changes over the last decade (2012-2022) compared to the climatic baseline (1982-2011) for both the entire season and individual months. Furthermore, an examination of changes in dry and wet precipitation extremes was conducted using localised percentile thresholds.

Changing monsoon patterns in Maharashtra

The analysis revealed that over the last 40 years (1982-2022), 21 percent of talukas experienced both high number deficient as well as excessive rainfall years. If the total rainfall received in the year is below 20 percent of the long-period average (calculated based on 1982-2011), it is considered a deficient rainfall year, and if above 20 per cent, then an excessive rainfall year. This trend is witnessed because historically drought-prone Marathwada and Khandesh talukas transitioned to wetter conditions in the recent decades.

Decoding these trends further, in the last decade (2012-2022), it was observed that nearly 87 per cent of the talukas witnessed an over 10 per cent increase in southwest monsoon rainfall compared to the climatic baseline of 1982-2011. This trend suggests that the majority of the areas in Maharashtra are shifting towards wetter conditions. Figure 2 illustrates the taluka-wise changes in southwest monsoon rainfall over the last decade.

The increase in southwest monsoon rainfall was continuous throughout the months of June and July, which are considered important for the sowing phase of Kharif crops. However, in August, particularly in the eastern region districts like Nagpur, Gondia, Gadchiroli, Chandrapur, Yavatmal and Nanded 14 percent of talukas witnessed more than a 10 percent decrease in the last decade. Figure 3 represents the month-wise quantification of changes in southwest monsoon.

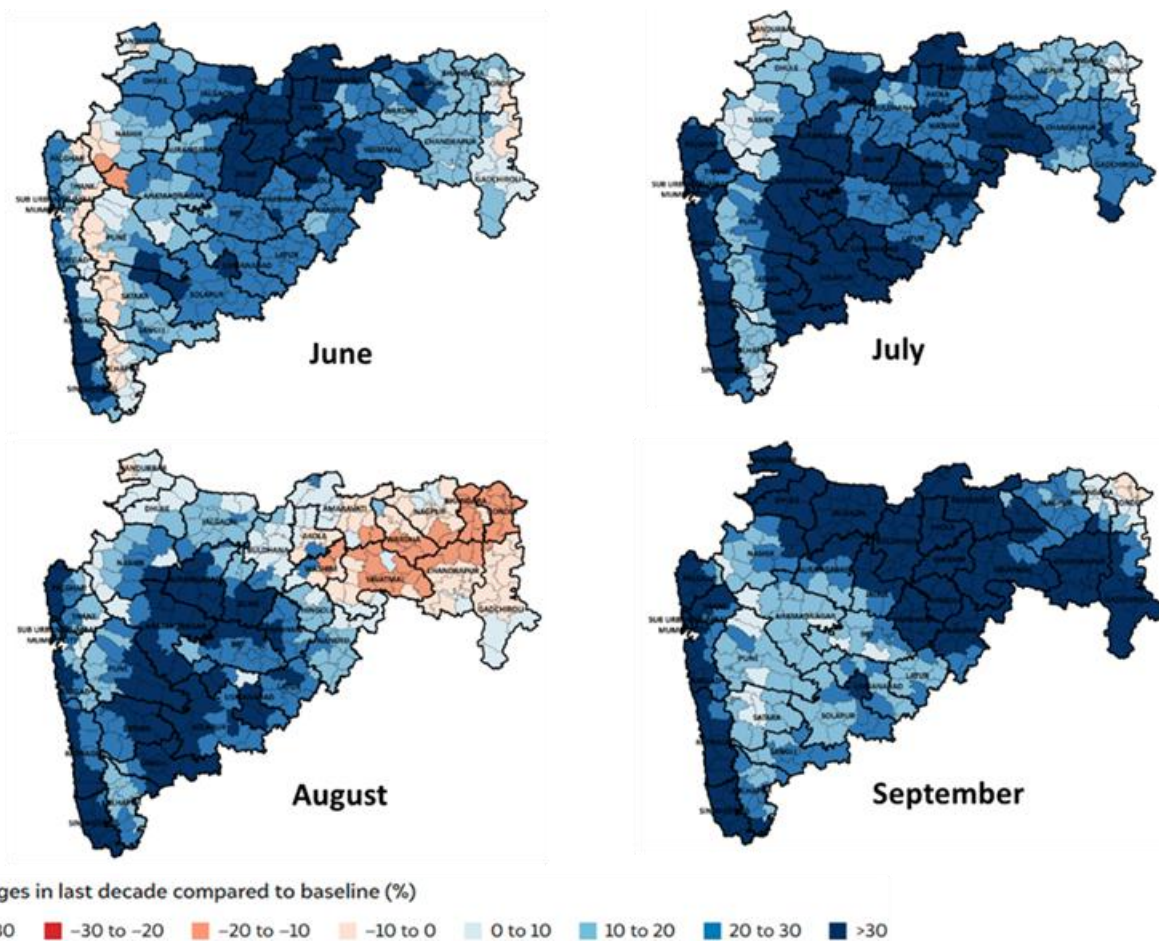


Figure 23: Month-wise changes in the southwest monsoon rainfall in the last decade (2012-2022) compared to the climatic baseline (1982-2011).

Additionally, a majority of talukas have seen a statistically significant increase in October-December rainfall in the last decade (2012-2022) compared to the climatic baseline (1982-2011), as seen in figure 4. 58 percent of Maharashtra talukas saw increased October to December rainfall by more than 10 percent, especially in October, which can be attributed to the delayed southwest monsoon withdrawal.

Analysis of the extreme rainfall events found that nearly 69 percent of talukas observed a rise in heavy rainfall days in the last decade (2012-2022), as can be seen in Figure 5. The highest increase was seen in the talukas of Konkan districts, which include Sindhudurg, Ratnagiri, Palghar, Raigad, Thane, and Mumbai, among others. The increase is in the range of 10 to 15 days per year in these talukas. The increase in extreme rainfall can be attributed to the increasing frequency of cyclones in the Arabian Sea by nearly 52 per cent in the last two decades. Moreover, the translation speed of the cyclones was found to have decreased by nearly 82 per cent, which means that the cyclone

moves slowly along the western coast before making a landfall, bringing in heavy rainfall and other associated events such as landslides, as seen in 2023 in Raigad.

Change in frequency of heavy rainfall events in the last decade (2012-2022) compared to climatic baseline (1982-2011)

b) One-day maximum rainfall received in the last decade (2012-2022)

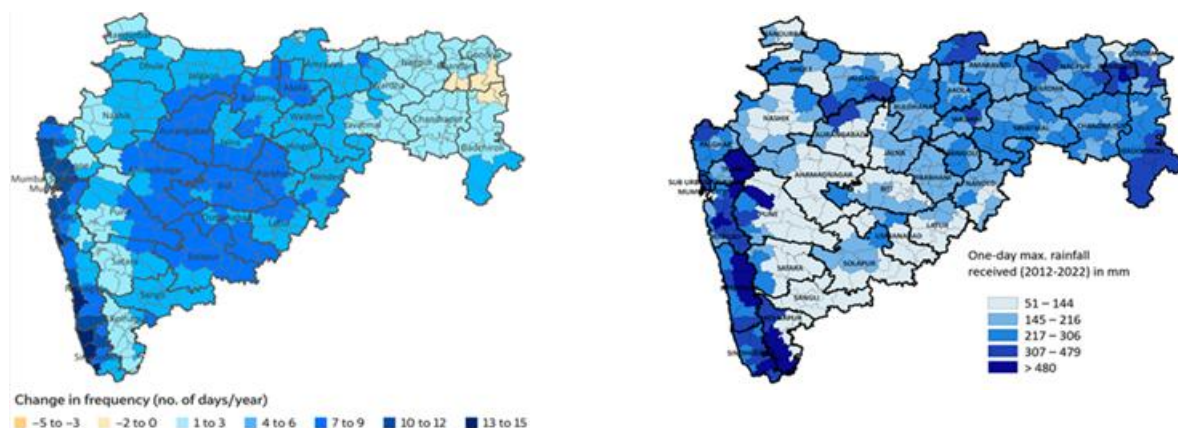


Figure 24: Heavy rainfall events over Maharashtra

Drought-prone talukas in districts such as Aurangabad, Jalgaon and Ahmednagar also saw an increase in the frequency of heavy rainfall events by 4 to 9 days per year. The analysis of heavy rainfall events suggests that the increase in total southwest monsoon rainfall as seen in **Figure 7**, is not well-distributed and is more due to an increase in heavy rainfall events. In Maharashtra, the southwest monsoon's onset and withdrawal are vital, particularly for Kharif crops. A timely onset marks the beginning of the sowing season, providing essential moisture for crop germination and early growth, including rice, millets, pulses and cotton. Delayed or erratic onset can lead to delayed sowing, reducing crop yields or even causing crop failures due to insufficient soil moisture. Similarly, timely withdrawal is necessary for proper harvesting and post-harvest activities. Prolonged monsoon duration can result in waterlogging, fungal diseases, and delayed harvesting, affecting crop quality and market availability.

The onset typically begins from the Coastal Maharashtra and North Maharashtra regions, which are closer to the Arabian Sea. According to the IMD, traditionally, the monsoon arrived in the Maharashtra region around the normal date of 10 June and withdrew from the North Maharashtra and Vidarbha regions around the normal date of 1 October. However, further decoding of these trends at regional level indicates the impact of break spells in the onset dates. The tables 1 and 2 represent the region-wise onset and withdrawal dates for the last 10 years. For example, in 2023, the onset was seen in Coastal Maharashtra on 11 June but the onset dates

over North Maharashtra and Marathwada was 24 June. This variation and the duration of monsoon season will significantly impact the agriculture activities including crop planting and harvesting, as well as water availability for irrigation and disrupt the weekly strategies outlined in the district-level cropping calendars.

Looking ahead, climatological projections for the monsoon over Maharashtra suggest a continuation of trends of increasing southwest monsoon rainfall as observed in the last decade, extending up to 2050 under both Representative Concentration Pathways (RCP) 4.5 and 8.5. As depicted in Figure 6, projections indicate that the highest rainfall will be witnessed in the Konkan region, following historically observed trends. Additionally, rainfall is expected to significantly increase in the central Maharashtra region, with parts of Pune and Ahmednagar districts, traditionally dry areas, becoming comparatively wetter than in the historical period. Conversely, the eastern part of Maharashtra is projected to receive slightly less rainfall, particularly in parts of

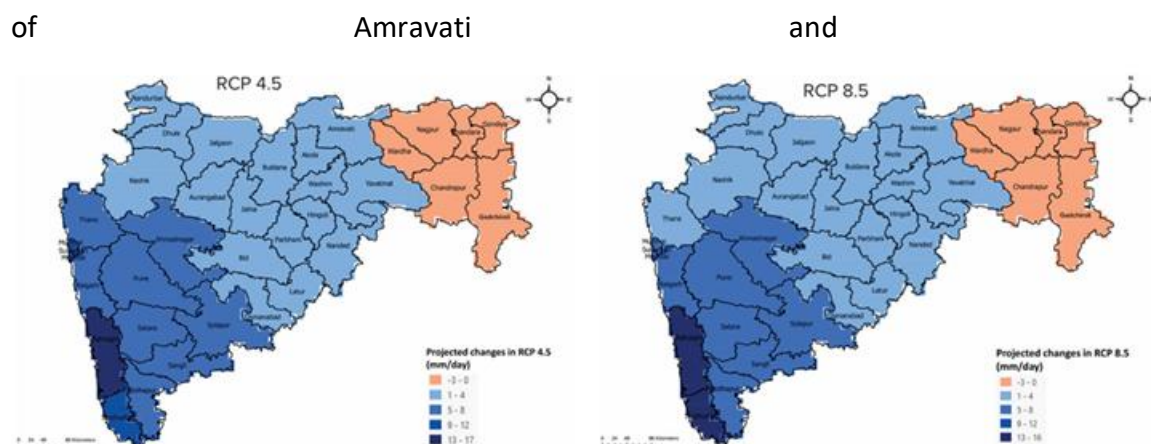


Figure 25: Projections of changes in the southwest monsoon rainfall in future (2023-2050) compared to the climatological baseline

In addition to changes in monsoon patterns, summer heat waves, typically occurring from March to May, consisting of rising daytime and night-time heat extremes, coincide with the pre-monsoon season and pose significant risks to Kharif crops in Maharashtra. Elevated temperatures during these months induce heat stress in plants, hindering their growth and development.

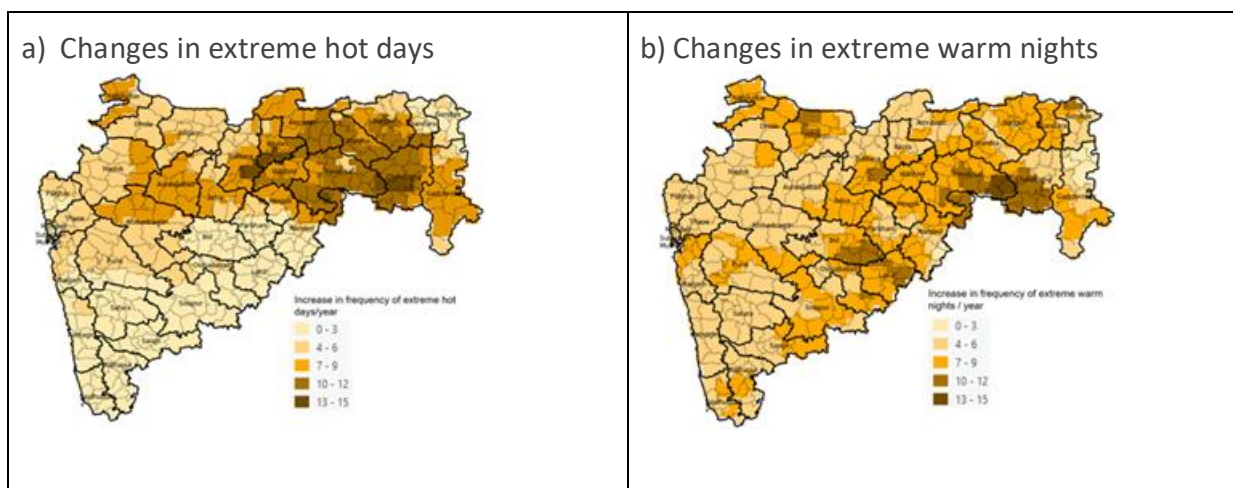


Figure 26: Changes in frequency of heat extremes in Maharashtra in the last decade (2012-2022) compared to climatic baseline (1982-2011)

To understand these compounding risks and the impacts of heat extremes, frequency of extreme hot days and warm nights in the last decade (2012-2022) compared to the climatic baseline (1982-2011) was quantified based on the 95th percentile threshold. As seen in Figure 7, a similar magnitude of increase in frequency can be seen in both day-time and night-time extremes over Maharashtra across all talukas. Moreover, coastal districts and cities face an additional challenge in the form of humidity, contributing an extra 3-4°C when the dry temperature exceeds 30°C and relative humidity surpasses 60 per cent. Figure 8 represents the trends in felt and dry heat over the coastal city of Thane in Maharashtra. It suggests that while both the dry and felt heat is increasing in the city, the difference between the both was nearly 3°C in recent years due to the added impact of relative humidity.

The National Action Plan on Climate Change (NAPCC), which was formulated by the Government of India in 2008, is geared towards the formulation of SAPCCs for all states. The NAPCC recognised that each state should have its own action plan to deal with climate change extremes and their impacts. The SAPCCs are intended to create a framework considering all the potential adverse effects of climate change and provide recommendations, suggestions, and guidelines for improved ways to protect communities and the environment, including strategies pertaining to adaptation and mitigation. The Ministry of Environment, Forest and Climate Change (MoEFCC) has created and distributed a common framework for the revision of all SAPCCs in 2019.

Maharashtra State Action Plan on Climate Change (MSAPCC)

Government of Maharashtra had prepared the State Action Plan on Climate Change in 2014 which was followed by formulation of Climate Change Policy of the State. The insights and recommendations of the action plan were referred to while designing the Maharashtra Project on Climate Resilient Agriculture (Phase I).

Due to the emergence of new climate models that can project climatological variables with a fair amount of greater precision, quantification of different consequences of climate change has become more accessible, which will help create sector-specific solutions. Considering this, the existing SAPCC, which was published in 2014, is revised with an updated methodology, datasets and quantification of impacts of climate change on different sectors.

Impacts of climate change on the cereal and pulses crops

The agrarian economy of Maharashtra predominantly practises rain-fed agricultural production with rain-fed crops such as sorghum, pearl millet, pulses, and oilseeds collectively covering about 60 per cent of the sown area. Major cash crops produced in the state, including sugarcane, rice, cotton, and maize account for about 27 per cent of the cropped area. The impact of changing climatic conditions is expected to affect the productivity of key crops. The following section provides a brief overview of the impact of climate change on major crops.

Table 35: Impact of climate change on major crops

Crop	Ideal Temperature (°C)	Rainfall Requirement (mm)	Climate Change Impact	Mitigation Strategies
Rice	20–35	1200–1500	Rising temperatures (above 35°C) cause spikelet sterility; erratic rainfall leads to drought/waterlogging ; a 1°C increase can reduce yields by 10%	Use heat/drought-tolerant varieties, AWD water management, adjust sowing/transplanting schedules, improve soil health, integrated pest management
Cotton	21–27 (can grow up to 43°C with moisture)	600–1000	High temperatures (>27°C) affect boll development; heavy rainfall leads to boll shedding; temperature increase may boost yields by	Use drought-tolerant varieties, improve moisture retention, adjust sowing schedules, pest management

			20-30% by 2040 but may damage buds	
Soybean	22–35	100 mm before sowing, 50–80 mm after	1°C rise reduces yield by 3–7%; excessive rain leads to root rot, delayed monsoon causes fungal infestations, untimely rain damages harvest	Adjust planting time, improve drainage, use disease-resistant varieties, adopt water-efficient farming practices
Sorghum	26–30 (can tolerate up to 45°C)	50% rainfall reduction can cause failure	Below 8°C harms flowering; extreme heat delays flower initiation, reducing yields by 6–18% in affected regions	Use heat-tolerant varieties, optimize irrigation, adopt moisture conservation techniques

Impact of climate change on soil health and water

Effects of climate change on soil are apparent in soil salinisation resulting from excessive groundwater irrigation and flooding, adversely impacting agricultural production and stability. This is predominantly observed in the sugarcane belts of western Maharashtra. With projections signalling an increase in rainfall intensities and extreme rainfall days, soil erosion and loss of critical soil nutrients will directly be responsible for low soil fertility and crop yields.

Agriculture consumes around 70% of the world's freshwater, making it the largest user of this vital resource. The agriculture sector in Maharashtra is the highest user of freshwater, withdrawing more than 80 per cent of the water (blue water), which is the water from surface water or groundwater which, and this is explicitly made available to the farms by diverting them through either irrigation systems or through groundwater abstraction structures such as wells and bore-wells. Moreover, almost 80 per cent of the agricultural land in the state is rainfed, and water use in rainfed agriculture comes from soil moisture, and the same is called the green water. Owing to the significant variabilities in rainfall pattern, such circumstances make a few regions severely water stressed. The state has an estimated annual availability of water resources of nearly 198 bn cubic metres (BCM), of which 164 BCM are surface water and 34 BCM groundwater, in addition to more than 21 lakh wells.

Climate change has reduced the number of rainy events, induced erratic rainfall, and increased average temperatures leading to groundwater depletion and causing water scarcity for irrigation. This will eventually affect agricultural production and cause water availability issues for allied activities and livelihood purposes. Therefore, the on farm or *in situ* water conservation is a critical area for maintaining soil moisture which is important for plant water uptake.

Impact of climate change on horticulture crops and its production

The horticulture sector in Maharashtra also faces increasing risk from climate change, which will potentially impact the productivity of key fruits and vegetables. Adverse impacts on some horticulture crops are given below:

Table 36: Impact of climate change on horticulture crops

Crop	Ideal Temperature (°C)	Rainfall Requirement (mm)	Climate Change Impact	Mitigation Strategies
Orange	Moderate temperatures, avoiding extremes	Unpredictable rainfall	Rising temperatures, heatwaves, and drought stress reduce yields and fruit quality	Use drought-resistant varieties, improve irrigation, adopt mulching to conserve moisture, and integrate agroforestry
Mango	10–31	890–1,015	Increased drought and vapor pressure deficit affect photosynthesis; rising humidity (60–90%) increases risk of powdery mildew, potentially causing 80–90% crop loss; potential shift in Alphonso production from Ratnagiri	Use drought-resistant varieties, improve irrigation and drainage, manage pest and disease risks, shift cultivation zones based on climate projections
Pomegranate	25–35 (tolerates -12 to 44)	500–1,000	Increased temperature and humidity (80–95%) favor anthracnose disease, reducing marketable yield by 10–80%	Disease-resistant varieties, proper spacing and pruning, balanced fertilization, moisture conservation techniques
Onion	27–32 (vegetative: 15–20, bulb development: 20–27)	500–1,000 (irrigation needed)	Soil-water stress at early stages causes up to 26% yield loss; temperatures above 40°C reduce bulb size; a 3.5°C increase above 38°C can reduce yield by 10–20%	Regular irrigation, maintain soil pH (6.0–7.0), soil moisture conservation, shade netting in extreme heat

Actions to cope with Climate Change

The proposed key actions for the agricultural sector focus on improving the research capacity in key areas pertaining to climate change adaptation of the agriculture sector. They also focus on risk mitigation and expanding capacities with a recognition that vulnerability to climate change is framed by social contexts, too. Some of the focus areas in the context of the project are:

- Improving the capacity of the stakeholders (farmers, Civil Society Organisations (CSOs), government institutions) to take up climate change adaptation measures,
- Mainstreaming climate-resilient agricultural practices,
- Increasing coverage of crop insurance programme to small and marginal farmers in the most vulnerable districts in the state,
- Providing hand-holding support and building capacities of the tribal communities to improve the
- Agri-allied based livelihoods opportunities and
- Supporting and strengthening the livelihood activities of tribal communities
- Resilience in the agriculture sector cannot exist without a vibrant allied sector. Allied activities help in withstanding climatic shocks and creating alternative streams of income. They also help in reducing farm distressing and forced migration.

Annexure 2: Note on PoCRA Phase 1

The project covered approximately 10% of the state's cropped area, focusing on 16 districts, including 670 vulnerable mini-watershed clusters and 5220 villages, with special attention to 932 salinity-affected villages in the Purna river basin.

Key Components:

Microplanning: Community-led, data-driven village plans for optimal natural resource use, appropriate cropping patterns, and improved market access.

Digital Agriculture Platform: A cutting-edge ICT system for climate-resilient technology delivery, including Direct Beneficiary Transfers (DBTs) via Aadhaar-enabled systems.

Agri-business Support: Establishment of value chains and support for Farmer Producer Organizations (FPOs).

Natural Resource Management (NRM): Comprehensive water conservation and soil management activities.

Achievements and Impact:

1.3 million farmland owners benefited, with increased crop yields, improved water productivity, and higher incomes.

The project gained high visibility nationally and internationally, influencing state and national agriculture policies.

The **Maha DBT portal** was established, ensuring transparency and efficiency in fund transfers.

Digital Approach:

A robust ICT system, including cloud, mobile, and GIS technologies, facilitated easy access to project benefits.

The DBT system promoted faceless contact and automatic RTI, ensuring transparency and efficiency.

Efficient Planning and Social Mobilization:

3959 Village Climate Resilient Agriculture Management Committees were formed, engaging **50,000 members** in decentralized decision-making.

The project empowered communities to identify beneficiaries, plan NRM, and resolve disputes.

Climate-Resilient Technologies:

Farmer Field Schools (FFS):

37,184 schools and **2 lakh sessions** conducted in **4800 villages**.

Focused on participatory learning and transfer of Climate Resilient Agriculture Technologies (CRATs).

Agro-Advisory Services:

Provided crop, weather, soil health, and irrigation advisories.

60% of farmers followed agro-advisory services delivered via social media.

Integrated Nutrient Management:

Promoted soil fertility and carbon sequestration through compost units and saline soil treatment.

98% of beneficiaries adopted soil treatment measures by the endline.

Integrated Pest Management:

40% of project farmers adopted eco-friendly pest management practices, compared to **34% in the control group**.

5. Micro-Irrigation Systems :431,328 beneficiaries adopted drip and sprinkler irrigation, reducing water consumption by 83% among project farmers

6.Protected Irrigation:11,120 farm ponds were constructed, increasing surface water storage by 30,375 TCM.

7. Protected Cultivation: Advanced technologies like shade nets and polyhouses were implemented on 1658.3 ha, with Rs. 651.44 Cr assistance.

8. Horticulture Plantation : 29,270 ha of horticulture and 613 ha of agroforestry plantations promoted carbon sequestration.

9. Integrated Farming System : Activities like sericulture, backyard poultry, and inland fisheries benefited 5391 small ruminant farmers, 2153 inland fishery farmers, and 234 backyard poultry farmers.

Increased Implementation of Climate-Resilient Technologies:Water Use Efficiency and Crop Productivity:

25% of project farmers adopted drip irrigation, up from **15% at baseline**.

31% of project farmers used sprinkler irrigation, up from **14% at baseline**.

Soybean productivity increased by **26%** among project farmers, compared to **14% in the control group**.

Cotton productivity increased by **84%** among project farmers, compared to **57% in the control group**.

Reduced Cultivation Costs:

The project reduced cultivation costs by **10%** compared to the control group, attributed to improved farming techniques and resource management.

Catchment Area Development:

4315 water conservation works were created, increasing surface water storage by **32,866 TCM**.

Micro-irrigation was implemented on **488,747 ha**, reducing costs and conserving soil.

Agribusiness Interventions:

FPOs were strengthened through capacity development, access to credit, and infrastructure provision.

Custom Hiring Centres (CHCs) promoted farm mechanization, reducing hiring costs and labor drudgery.

Farm incomes in project households were **46% higher** than in control households.

Learning and Way Forward:

- To ensure long-term sustainability, the project recommends:
- Strengthening institutional and individual capacity through training and exposure visits.
- Leveraging digital technology for farmer outreach and information dissemination.
- Promoting renewable energy initiatives like solar pumps.
- Enhancing market linkages, financial support, and environmental consciousness.
- Converging with other government schemes and strengthening partnerships for a multiplier effect.
- The NDKSP has significantly improved climate resilience and farmer livelihoods in Maharashtra, with a commitment to transparency, accountability, and continuous improvement to address future challenges posed by climate change.

Annexure 3: Vulnerability Assessment

As per the above-mentioned resolution, a Village Selection Committee was formed under the chairmanship of the Additional Chief Secretary (Agriculture) for the selection of villages for the second phase of the project. This committee was mandated to formulate an appropriate village selection criterion in a scientific manner. The committee decided to adopt a mini-watershed, cluster approach for project execution. The finalised parameters have been derived from the indicators mentioned in the ICAR-CRIDA report namely, *Risk and Vulnerability Assessment of Indian Agriculture to Climate Change, 2019* with relevant modifications. The Committee indicated that the parameters approved may be applied in the 21 districts to identify approximately 6800 villages. It was also decided that villages included in the first phase of the project shall not be part of the second phase. As per guiding principles of Village Selection Committee, the primary objective of vulnerability assessment is to compute a vulnerability index for clusters and select the most vulnerable.

Village Selection Criteria

The villages in the project districts have been grouped into clusters aligned with the mini watersheds identified with the help of GIS technology. This ensures that the interventions made under the project contribute to the comprehensive treatment of the mini watershed as per the 'ridge to valley' approach. These clusters have been prioritised on the basis of an 'Overall Sensitivity Index' as per the criteria finalised by the Village Selection Committee.

In the NDKSP Phase I, the framework from the Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report was utilised. However, the dynamic nature of vulnerability and its components is not fully addressed in both the Third and Fourth IPCC Assessment Reports.

It is now better understood that the risks due to climate change are a result of complex interactions among social and ecological systems and the hazards arising out of climate change. Various facets of these interactions have to be carefully differentiated to inform policy making for risk management. IPCC, 2014 has evolved an AR 5 framework which emphasises these aspects. The approach also recognises that the components of vulnerability and risk will also interact with the contextual factors of development pathways and the climate systems. Such a strategy also helps in identifying and managing risk and thus views vulnerability as a determinant. Based on this framework, ICAR-ICAR published the report titled '*Risk and Vulnerability Assessment of Indian Agriculture to Climate Change, 2019*'. As the project is focused on climate resilience, the climate change vulnerability approach adopted by ICAR-CRIDA has been considered for the selection of villages.

The definitions given in AR 5 (Risk and Vulnerability Assessment of Indian Agriculture to Climate Change, 2019) for risk and its components are given below:

Exposure is identified as "the presence of people, livelihoods, species or ecosystems, environmental functions, services, and resources, infrastructure, or economic, social, or cultural assets in places and settings that could be adversely affected".

Vulnerability is defined as “the propensity or predisposition to be adversely affected. Vulnerability encompasses a variety of concepts and elements including sensitivity or susceptibility to harm and lack of capacity to cope and adapt”.

Hazard is defined as “the potential occurrence of a natural or human-induced physical event or trend or physical impact that may cause loss of life, injury, or other health impacts, as well as damage and loss to property, infrastructure, livelihoods, service provision, ecosystems, and environmental resources. In this report, the term hazard usually refers to climate-related physical events or trends or their physical impacts”.

Risk refers to “the potential for consequences where something of value is at stake and where the outcome is uncertain, recognizing the diversity of values. Risk is often represented as probability of occurrence of hazardous events or trends multiplied by the impacts if these events or trends occur”.

Thus, we can classify risk as an interaction of vulnerability, exposure, and hazard as below:

$$\text{Risk} = (\text{Probability of Events or Trends}) \times \text{Consequences}$$

AR 5 of IPCC is summarised in the following figure.

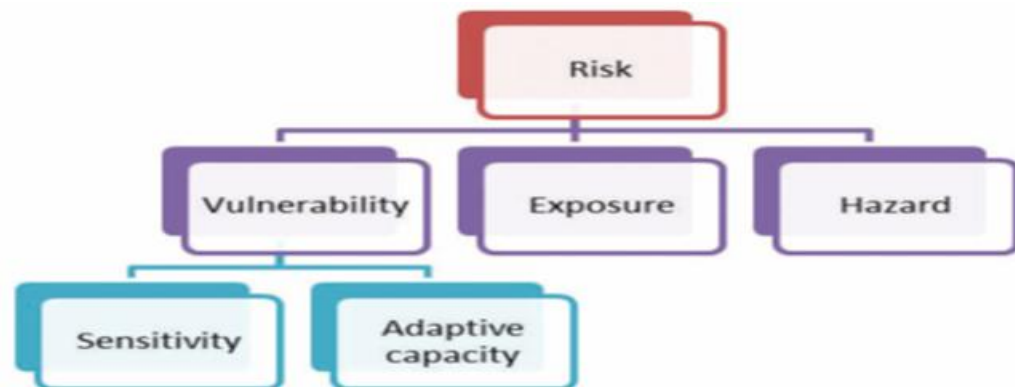


Figure 27: AR 5 of IPCC

The project has used existing village level as well as cluster level databases of all the indicators to compute the indices of hazard, exposure and vulnerability. The indicators were normalised to bring all the indicators to a common scale. Then the indices were aggregated into historical hazard, future hazard, exposure, vulnerability and finally of risk.

The following databases are used for compiling village wise/cluster wise information.

- Weather data from MAHAVEDH
- Groundwater prospects from GSDA
- Geo-spatial information of NBSS&LUP, 2022
- LULC of MRSAC 2015-16
- Livestock census 2020
- General Census 2011
- MAVIM and MSRLM, 2022
- Market data from APMC, 2023
- Warehouse data from MSWC, 2023

- SECC, 2011
- Drought declaration by Relief and Rehabilitation Department

The indicators related to future hazard are expressed in terms of changes from the baseline climate. In India, regional level modelling data is available at IMD, Pune for predictions which are not suitable for village selection. Hence, the Village Selection Committee advised that the Future Hazard index may not be practicable because of scale concerns and other indicators and their weightages are decided. The committee has finalised the indicators to assess the vulnerability of the clusters in the selected districts as mentioned in the Table 6 The indicators are categorised as Vulnerability, Exposure and Historical hazard with aggregate weightage of 60%, 30% and 10% respectively. Each category consists of indicators which are either directly or inversely related to overall vulnerability.

Table 37: Indicators and its Weightage

Climate Change Vulnerability Indicators for selection of clusters for project interventions				
	Indicator	Rationale	Relations hip with vulnerabi lity	Weightage assigned
A	Vulnerability (60%)			
1	Annual rainfall with temporal & spatial variation			
1.1	Annual rainfall	Increase in rainfall is favourable to agricultural productivity	Inverse	5%
1.2	No. of rainy days	Increase in number of rainy days implies a better distribution of rainfall	Inverse	4%
1.3	No. of dry spells	Higher the number of dry spells, less is productivity	Direct	4%
2	Ratio of Degraded land to Geographical area	Productivity levels would be low and highly risky, if crops are grown on degraded and wastelands	Direct	6%
3	Available water holding capacity of soil	Capacity of soils to hold larger amount of water can save crops during dry spells	Inverse	9%
4	Ground water availability	Groundwater availability reflects the scope to exploit ground water resources for irrigation	Inverse	11%
5	Livestock density	This is an indicator of diversification of agriculture and livelihoods and hence enhances the ability to cope with.	Inverse	9%
6	Literacy	Higher literacy enables people to adapt better and also enhances their ability to diversify livelihoods	Inverse	4

7	Gender gap	A lower gap indicates better gender equity	Direct	3
8	Self-help groups	A higher % include prevalence of farmers' organizations and thus help higher ability to adapt	Inverse	3
9	Net irrigated area	Irrigation is an important adaptation-enabler as it enables farmers to save crops during dry spells or droughts. It is also strongly related to technology adoption.	Inverse	20
10	Market Access including warehouses and transportation			
10.1	Market distance	Less distance from market shows better market access	Direct	3
10.2	Warehouse distance	Less distance from warehouse shows better market access	Direct	3
10.3	Farm machinery	Less houses with farm machinery shows more dependency on external factors	Inverse	2
11	Income	A higher per capita income implies better adaptive capacity.	Direct	4
12	Agri. labour	This indicates a relatively higher importance of agriculture in the livelihoods of population compared to other sectors	Direct	5
13	Farmer suicides	Agrarian distress due to crop failures, low prices, arrears of loans, non -profitability etc. indicates more vulnerability	Direct	5
B	Exposure (30%)			
1	Ratio of Net sown area to geographical area	A relatively higher area under cultivation implies higher relative importance of Agriculture and also that more area would be affected	Direct	40
2	population density	Higher density is an indication of more people at risk. It also implies high population pressure on land resources and since the livelihoods of rural population are heavily dependent on agriculture, it means higher exposure	Direct	15
3	Small and marginal farmers	Higher numbers imply more farmers being exposed. Smaller farm size limits marketable surplus and also the opportunity to diversify the cropping	Direct	20

		pattern and the low investment capacity of farmers make agriculture riskier.		
4	SC/ST Population	The SC/ST population is, in addition to being relatively poor, also less educated, poorly integrated with main-stream economy and heavily-dependent on natural resources for their livelihoods and more of them implies higher exposure.	Direct	15
5	Cross-bred cattle	Cross-bred cattle are more productive, require more investments in feed, fodder and management and are more sensitive to climate change. Higher the number, more is the exposure	Direct	10
C	Historical Hazard (10)			
	Proneness to extreme events (Drought & Flood)	Incidence of more frequent droughts implies more sensitivity to climate change	Direct	100

The combined vulnerability index for each of the clusters was calculated by aggregating individual indices, after multiplying them with the weightage assigned to the respective indicators. Based on their combined vulnerability index values, the clusters were then prioritized within each taluka.

Annexure 4: Soil and water conservation interventions

Once the village appreciates the context of water balance during the Micro-Planning exercise, it can plan for and address the supply side concerns through soil and water conservation interventions. The soil and water conservation interventions are mainly divided into Area treatment and drainage line treatment. Project will implement the following various measures to achieve water security at farm level. For carrying out these activities existing cost norms of the Department of Agriculture will be followed. These activities will be implemented through the Department of Agriculture with the engagement of GKVS.

Catchment Area Treatment

Area Treatment

Area Treatment refers to watershed management practices aimed at improving land productivity, reducing soil erosion, and conserving water resources within a specific geographical area. It involves various interventions that focus on soil and water conservation, enhancing groundwater recharge, and supporting sustainable agricultural practices.

The area treatment consists of following activities

CCT/Deep CCT

This activity will be carried out on barren land having slope up to 8%. Based on the contour survey and the site conditions, continuous contour trenches (CCT) & deep CCT will be constructed. The seeds or seedlings of trees of suitable local varieties will be planted on the downward slope side.

Compartment Bunding

Compartment bunding activity will be carried out on the private arable land of the farmer., having slope up to 4% and average annual rainfall up to 750 mm. This will help in increasing water holding capacity at field level and increase fertility of the soil by reducing the soil erosion.

Graded bunding

Graded bunding activity will be carried out in arable land having slope up to 4% in assured rainfall region. In this intervention, the purpose is not to store water, but to give water a gentle slope to the extent of 0.20%. This will ensure that the water collected in the area does not accumulate on the upper side of the Bund but flows slowly from the upper side of the field and is evacuated out of the field. This will arrest soil erosion and excess water will be drained out of the field.

Paddy Bunding/Terracing (Majagi)

Paddy Bunding activity will be carried out in rice growing areas of eastern Vidarbha and the Majgi scheme is implemented to improve the fallow area and bring the area under cultivation. The aim of both activities is to increase the production of rice by increasing the area under the crop. For rice cultivation, it is necessary to store some amount of water in all places on the farm. For this, changes are made to the natural slope of the land to convert it into a flat field. Project will implement both Paddy Bunding and Majagi Activity

Repairs of old paddy bunds

This activity will be carried out in paddy growing regions where the existing section of bund is less than 0.40 sqm. Paddy bunds are a very important part of paddy field. However, due to continuous heavy rains and other factors, it is seen that the bunds of paddy field have been

damaged due to various erosion. Due to partial or complete bursting and collapse of the bunds water is not stored or retained in full capacity, thus affecting the yield of rice crop.

Drainage line treatment

Drainage line treatment refers to the set of measures undertaken to manage water flow in natural drainage lines to prevent soil erosion, enhance groundwater recharge, and control runoff. It is a critical part of watershed development aimed at stabilizing drainage systems and ensuring sustainable water management.

It consists of following activities

Loose boulder structures

Loose Boulder Structures (LBS) are part of upper catchment treatment with the objectives of reducing the water flow rate, reducing soil erosion, and trapping silt for slowing down the rate of siltation in water bodies in the lower reaches of the watershed. LBSs enhance conversion of waste land into cultivable land, induce water recharge, and plantation activities around the LBS helps in increasing vegetative cover.

Gabion structure

The Gabion structure is constructed in a *Nala* having slope more than 3 % and width up to 10 m. Gabion structures are generally constructed in the upper catchment area of CNB, to avoid silting of CNB. This activity is carried out in the middle reaches of watersheds to reduce runoff and avoid soil erosion.

There are two types of Gabion structures:

Gabion structure using steel mesh

This is a low-cost gabion structure where stones are covered by using GI mesh.

Composite Gabion Structure

In this type of structure, a RCC wall is constructed, and stones are placed on both sides which are covered by using GI mesh.

Earthen Nala Bunds (ENB)

ENB is suitable for both flood control as well as gully control. Velocity of runoff is greater in the nala or gully due to heavy rainfall which erodes shores and increases the adjacent pan of nala. Earthen nala bunds are placed in gully or nala to restrict its widening. Water stored in ENBs percolates to wells and tube wells located in the lower part of the catchment. This structure will be carried out in a drain having slope up to 3% and bottom width should be generally 5 to 15 m.

Cement Nala Bunds

It is water storage structures constructed during various water conservation programmes implemented by central & state governments. With the major objectives to harvest surface runoff, recharging of groundwater, retain water carrying capacity of stream the construction of CNBs is done in the lower part of the watersheds. This structure will be carried out in a drain having slope up to 2% and bottom width should be generally 10 to 30 m.

Water harvesting structures

Groundwater serves as the primary source of water for both drinking and irrigation in the country, yet more than 50 percent of agricultural activities remain reliant on rainfed systems. Consequently, water conservation and decentralised rainwater harvesting are essential for enhancing productivity and promoting well-being (Narain, 2021). Water security can be enhanced through the various water harvesting structures such as Farm Pond, Open Dug Well etc.

Farm Pond

Farm ponds are structured to capture and store excess runoff carried by streams, drains, and this stored water is used for protective irrigation during dry spells. Farm ponds help to increase the availability of water through watershed-based water conservation. The project will promote construction of farm ponds on individual as well as community lands.

Generally, there are two main types of farm ponds

a) Farm pond with inlet outlet: - This farm pond is generally constructed where natural runoff accumulates from catchment areas. This type of farm pond helps in recharging of ground water as water percolates.

b) Farm pond without inlet outlet: - This farm pond is used by farmers to store water during ample availability periods and use it for protective irrigation, growing rabbi/summer/horticulture crops. In this type of farm pond farmers generally prefer laying polythene to reduce percolation losses.

Open Dug well

Open dug wells are a common source of irrigation in the project region. The main purpose of this structure is to store the groundwater. It is also used to increase groundwater recharge by conserving rainwater. Currently this activity is implemented through various government schemes such as Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS), Dr. Babasaheb Krishi Swavalamban Yojana, Birsu Munda Krishi Kranti Yojana etc.

Bodi Structure

Bodi refers to small water bodies created in the upper reaches of paddy fields, to ensure protected irrigation during water stress. This intervention will contribute to protective irrigation of paddy crops and help in increasing yield of Kharif and Rabi crops.

The project will promote construction of new Bodi's on individual farmers lands.

Recharge of groundwater

Recharge occurs when water seeps into the ground to replenish underground aquifers. Although some recharge happens incidentally—water flowing into the ground from rivers, unlined canals, or excess irrigation—intentional recharge can restore groundwater levels and store water for later use.

Recharge of open dug well

Dug Well recharge is a method of Recharging ground water level. A dug well recharge augments the groundwater to ensure long term sustainability. Generally, through this method we can divert rainfall runoff into open wells as a means of harvesting rainfall runoff and increasing recharge

Recharge Shaft / Recharge Trench with Recharge Shaft

Rainfall pattern, hydro-geomorphology and hydrogeology of the area play an important role in groundwater resource availability in the area. Hence proper understanding of the

hydrogeological conditions of the area is an important aspect of groundwater management which leads to near accurate estimates of the groundwater resource, its recharge and withdrawal for all other uses.

Considering the above importance of groundwater project will implement “recharge shaft with recharge trench” on feasible sites. This is the most efficient and cost-effective technique to recharge unconfined aquifers overlain by poorly permeable strata. These recharge structures are very useful for village ponds where a shallow clay layer impedes the infiltration of water to the aquifer. This helps in controlling runoff, recharging of groundwater, and strengthening of aquifers.

The project will collaborate with Groundwater Surveys and Development Agency (GSDA) for preparation of Cluster wise groundwater recharge plan along with estimates and provide technical backstopping including protocol development for supervision of ground water recharge work and related training of field staff. These activities will be implemented by the Department of Agriculture.

Rejuvenation of water harvesting structures

Following are some different methods used for the rejuvenation of water harvesting structures.

Cement Nala Bund (CNB) Rejuvenation

This involves restoring the functionality and effectiveness of existing CNBs, which are vital for water conservation and groundwater recharge. Rejuvenation typically includes:

Desilting: Removing accumulated silt and sand to restore the original storage capacity of the CNB.

Structural Repairs: Fixing cracks, leaks, or damaged sections of the cement structure to ensure water retention. These structures will be brought to their original potential through Desiltation and or repairing.

Rejuvenation will be done based on existing guidelines of the Department of Agriculture GoM.

Bodi Structure Rejuvenation

Bodi refers to small water bodies created in the upper reaches of paddy fields, to ensure protected irrigation during water stress. As the previously built *Bodis* are now damaged and silt has accumulated in it, it has become necessary to deepen and renew these old *Bodis*. *Bodi* structures need to be rejuvenated for containment and recharge surface runoff. Also, such an intervention will contribute to protective irrigation of paddy crops and help in increasing yield of Kharif and Rabi crops. Considering the importance of these structures, the project will promote Construction of new *Bodi* and its rejuvenation.

Rejuvenation of Water bodies

The Eastern Vidarbha region boasts numerous water bodies, including *Maji Malgulari Talav* and *Gav Talav (Village Ponds)*. In areas governed by the Panchayats (Extension of the Scheduled Areas) Act, 1996 (PESA), water bodies with irrigation capacities up to 100 hectares have been transferred to the control of Gram Panchayats from the Minor Irrigation Department. These water bodies serve as crucial resources for irrigation and other ancillary activities. However, they currently grapple with significant challenges such as siltation, leakage, and seepage, leading to a decline in their water storage capacity. Project aims to revitalize these water bodies, restoring them to their original potential.

Annexure 5. Participatory Micro-planning process

The project will implement a participatory micro level planning process (MLP) in every project village. The micro planning process will be supported by an integrated MLP app, which will be used to collect data and information. This data will be instrumental in developing the village adaptation plan. **The Gram Krishi Vikas Samiti** will play a pivotal role in steering the process of Micro Planning. As it guides the process, it will also be learning about the potential that the village holds to benefit from the project by creating people's synergies. The MLP will be organised as follows:

Training For Microplanning: Training sessions for master trainers, volunteers, people's representatives and field staff will be organised to facilitate the micro-planning process. YASHADA will provide the following training:

- Training sessions for master trainers and Agriculture Assistants will be held.
 - All sarpanch from the project area will receive orientation training on micro-planning and the development of the village adaptation plan.
 - Members of the Gram Krishi Vikas Samiti (GKVS)* will also be oriented on the MLP exercise with the help of Sarpanch, Agriculture Assistant etc.
- Master trainers will conduct orientation sessions for MLP staff, volunteers, and resource groups.

Pre-Preparation Phase: This phase consists of identification of villages, preparation of primary information of various facets of villages, preparation of resource material as modules & formats, village maps, etc. and orientation of field officials and training of trainers and facilitators. Use of various databases and information systems such as General Census, Agriculture Census, Livestock Census, SECC, etc. will help inform the background for the villages and reduce data collection drudgery. Extensive training will be provided to resource persons to access MLP apps and use the available information skilfully with the village community.

The Participatory Micro Planning Process: This phase shall consist of Socio-economic information and secondary information gathering, resource mapping, target group discussion, value chain mapping, planning based on needs of stakeholders, and an eventual approval from the Gram Sabha of the village adaptation plan inclusive of gender specific concern raised in Women Aam sabha.

Post-Micro Planning Process: The technical vetting of the Village Adaptation Plan shall be taken up by a committee set up for this purpose in the Sub-Divisional Agriculture Office. Such technically vetted VAPs will be presented to the "District Coordination Committee" under the chairmanship of District Collector for final approval.

Micro level planning process) Pre-preparation à Preparation à Post Preparation = Village Adaptation Plan (VAP) à Approval of Village Adaptation plans à Training and mobilisation

Village Adaptation Plan (VAP)

After MLP exercise, the Village Adaptation Plan (VAP) will be prepared in participatory mode. The contents of VAP are summarised below.

Situation Analysis:

- Climate Change – status of rainfall and temperature – changes / trends in rainy days, onset, intense rainfall events observed
- Seed related constraints
- Cropping pattern related constraints
- Soil and input management related constraints
- Water resources – structures and irrigation related constraints
- Livestock related constraints
- Labour/ Mechanisation related constraints
- Post – harvest related constraints

Water Budget of the village would be an integral part of the VAP: The water budget computation of the village provides real time information about the amount of water received through rainfall and its distribution based on the soil characteristics, population, livestock and crops as well as the balance at the end of monsoon. Also, the computation reveals total runoff generated, runoff impounded in the existing structures, and balance runoff.

Value chain:

The existing commodity value chains with its strengths and weaknesses are described based on the focus group discussions. This section also gives account of the constraints, suggestions for improvement of the value chain in order to enhance resilience of the value chain with enhancement in profitability.

Gender and social inclusion:

This section will cover the gender related concerns raised in the women Aam Sabhas and in the various discussions during the planning exercise. Concerns highlighted could be related to participation of women in agriculture, specific drudgery concerns, identification of machineries which could help reduce drudgery, knowledge dissemination among women relating to climate resilient agriculture, etc.

Proposed plan of activities:

This section would propose a summary for the village in tabular format indicating village level interventions and proposed expenditure for the same.

Village Adaptation Plan (VAP) as a Live Document

Village Adaptation Plans are useful documents to understand the constraints, and the potential and opportunities available to the villages. However, village adaptation plans in many programs often end up as dormant documents which are seldom used by the administrative units at various levels. While VAPs are born out of a participatory exercise, in order for the Plans to have a value in Planning and goal-setting the Plans should have the following additional qualities—they should be retrievable, aggregative, and flexible. This is possible by use of the input and output material collated during the VAP exercise

to identify the gaps and opportunities in easily processed formats on the MLP App. Thus, for example, the MLP app will collate and process the data made available to and from the MLP exercise to throw up the gaps and opportunities in the village and use them to identify the problems and solutions, as well as to measure them across time against the progress made in the project. The aim is to use the advances in digital systems in phase-1 of the project to ensure that the VAP becomes a live document. An additional advantage would be the fact that the yearly Kharif and Rabi planning sessions will become sites for updating and measurement of VAP vis-a-vis stated goals.

The project is also committed to carry out, nearing the mid-term, another village outreach activity of at least 3 days which will be both a stock-taking exercise and a renewal of intentions expressed by the Gram Sabha. This would be a site to discuss course-correction mechanisms and to also tap the thought leadership within the community. This additional “mini-MLP exercise” would bring further value and refinement to the VAPs.

Annexure 6: Climate-resilient technologies

The following climate-resilient technologies will be introduced and demonstrated to the farmers in project villages for field crops, through various extension approaches including FFS.

i. Promotion of Broad Bed Furrow technology

Broad Bed Furrow (BBF) technology, developed by CRIDA Hyderabad and included in ICAR's NICRA, is promising climate resilient technology. NDKSP 1 has demonstrated that BBF helps manage climate shocks like dry spells and heavy rainfall. It involves wide raised beds with furrows to drain excess rainwater, ensuring supplemental soil moisture during dry spells. BBF reduces runoff, prevents soil loss, and enhances in-situ moisture conservation. The system promotes efficient rainwater use through vertical and lateral movement in soil pores. It has proven effective in protecting soybean, cotton, pigeon pea, and maize from water stagnation and excess moisture damage.

ii. Raised Bed and Dibbling Technology

Raised bed preparation is recommended by the SAUs and Agriculture Research Institutes for vegetable and other crops. The NDKSP Phase I has demonstrated that the crops such as cotton, soybean, maize, chickpea, pearl millet could be grown successfully on the raised beds. The size of the beds range from 120 cm to 150 cm and the height of the beds ranges from 22.5 cm to 30 cm. The beds are suitable for all crops in both Kharif and Rabi seasons. The crops are sown on the beds by dibbling method. The traditional dibbling is done by human labour. The project will promote use of manual / bullock drawn dibbling machines to reduce drudgery. The beds help crops withstand soil moisture stress during dry spells by retaining moisture for longer. The dibbling method on the beds allows for proper seed placement, saving seeds and boosting germination. It helps the weed management and promotes microbial growth and organic biomass in stable beds.

iii. Zero tillage/ reduced tillage:

The importance and intervention approach for Zero Tillage is well described under Conservation agriculture from the perspective of soil health enhancement. The Zero Tillage initiative as a climate resilient technology from an agronomic perspective is equally important. Since the invention of mould-board plough, the farmers are experiencing a large number of tillage benefits, but at the cost of the environment, and the natural resource base on which their farming depended. The utility of ploughing was first questioned by a forward-looking agronomist in the 1930s, Edward H. Faulkner, who wrote a revolutionary book 'Plowman's Folly' (Faulkner 1943). Since the last 80 years, the debate and research on the necessity of tilling the soil versus no-tilling have concluded that the way forward for agriculture is to minimise and to stop the disturbance of farmland soil. Moreover, this approach of Zero tillage has proven to be the most Climate Resilient approach under both rainfed and irrigated conditions in many parts of the globe. Zero Tillage offers the benefit of reduced fuel consumption and labour expenses, reduction of total cost of cultivation, timely sowing is possible which offers yield advantage, reduced soil loss and erosion ultimately improvement in soil health. The details about regenerative and conservation agriculture, and No tillage protocol along with the project's support system is described in further chapter 'Enhancing Soil health'.

iv. Promotion of climate resilient varieties

Climate-resilient varieties are developed to withstand climate change impacts like drought, heat, salinity, and pests, ensuring stable yields under challenging conditions. They offer drought and heat tolerance, extended growing seasons, and resistance to pests and diseases, reducing the need for chemical inputs. These varieties also improve nutrient use efficiency, thriving in less fertile soils. The following are the key characteristics of climate-resilient varieties that contribute to their adaptability and effectiveness in diverse farming conditions

- Early maturity cycle
- Require less irrigation with higher yield than other variety
- Moisture stress tolerant & hence suitable for rainfed condition
- Suitable for rainfed condition in light to medium soil
- Escape terminal drought due to earliness
- Suitable for intercropping
- Tolerance to late heat stress
- Synchronised maturity
- Non- shattering pods
- Suitable for mechanical harvesting
- Moderately resistant to wilt
- Resistant to sterility mosaic
- Resistant to rust and leaf diseases
- Tolerant to Jassids

Climate-resilient varieties will be promoted through Farmers Field Schools (FFS) in project villages, educating farmers about new varieties for improved productivity and climate resilience. Additionally, the project offers subsidies to support seed production programs, ensuring the availability of locally adapted, climate-resilient seeds.

v. Intercropping

- In Maharashtra, particularly Marathwada and Vidarbha region intercropping system of soybean + pigeon pea (4:2 row ratio) is very profitable under rainfed conditions and is recommended particularly for rainfed cultivation. Early maturing (short duration) pigeon pea varieties are more suitable for intercropping. Soybean + sorghum or soybean + cotton intercropping is also recommended by SAUs. It is also advisable to farmers that continuous growing of soybean on the same piece of land should be avoided. Crop rotation tactics should be followed for sustainable soybean production. The recommended intercropping systems include
- Soybean based cropping system: Soybean + pigeon pea, soybean + corn, soybean + sorghum, soybean + sugarcane, soybean in mango/guava orchards, soybean in agroforestry.
- Cotton based cropping system: Cotton = pigeon pea, cotton+ soybean, cotton + black gram, cotton + green gram
- Cereal based cropping system: Pearl millet (Bajra) + Pigeon pea, Sorghum + Safflower, Wheat + Mustard

Intercropping allows farmers to grow multiple crops together, reducing dependence on a single crop and mitigating economic risks from price fluctuations or crop failures. It enhances pest management by using natural repellents and trap crops, reducing the need for pesticides. It also improves soil health by preventing erosion, enhancing aeration, and boosting fertility through nitrogen fixation by legumes. Intercropping increases biodiversity supports ecosystem services like pollination and reduces the risk of total crop failure. Additionally, it optimizes resource use, suppresses weeds, creates a favourable microclimate, and contributes to climate change mitigation by sequestering carbon.

vi. Integrated pest management

Integrated Pest Management (IPM) is an ecological approach, which aims at keeping pests below economic threshold level by employing all available alternate pest control methods and techniques such as cultural, mechanical and biological control with greater emphasis on use of bio-pesticides and pesticides of plant-origin like Neem formulation. The project will promote specific Integrated Pest Management (IPM) measures to enhance crop protection while minimizing chemical inputs. These include seed treatment, growing pest and disease-resistant varieties, and ensuring timely and synchronous sowing with optimum plant spacing. Trap and border crops will be used alongside post-sowing cultural practices, balanced fertilization, proper water management, and timely weed control. Pest monitoring will be strengthened using light, sticky, and pheromone traps, along with regular observation of pests and their natural enemies. Conservation of beneficial organisms such as parasites, predators, and pathogens will be encouraged, while bio-pesticides will be promoted for pest control. Farmers will be trained to assess the pest-defender ratio (2:1) before taking action and apply pesticides only as a last resort based on Economic Threshold Levels (ETL). Additionally, recycling and proper disposal of crop residues and weeds will be emphasized to maintain field hygiene and reduce pest incidence.

To enhance farmers' understanding, the project will provide training, exposure, and extension services on IPM through FFS and field-level interventions. NDKSP has partnered with NIPHM, Hyderabad, through an MoU to offer specialized training and capacity-building support. This collaboration ensures continuous technical guidance, strengthening IPM adoption among farmers and field staff.

vii) Integrated nutrient management

Integrated Nutrient Management (INM) optimizes nutrient use by combining organic and chemical inputs to enhance soil health, crop productivity, and environmental sustainability. It integrates soil, water, and crop management practices to balance productivity, economic viability, and ecological stewardship while minimizing environmental impact.

The project will promote INM through measures such as soil testing, organic matter use, crop rotation, crop Diversification intercropping, crop cover, biofertilizers, conservation agriculture, fertigation, and optimal use of chemical fertilizers.

The INM package will be disseminated to the project farmers through FFS and various other interventions at field level. The collaboration between the NDKSP and NIPHM, Hyderabad, under the MoU, will focus on providing training and support to stakeholders on Integrated Nutrient Management (INM) practices.

NIPHM will assist in enhancing the knowledge and implementation of INM through various training sessions, workshops, and continuous guidance at the field level.

viii) Integrated Weed Management

Integrated Weed Management (IWM) is a holistic approach that combines multiple strategies to control weeds effectively while minimizing environmental impact. The key components of IWM include cultural, mechanical, biological, chemical, and preventive methods. Cultural practices such as crop rotation, cover cropping, mulching, and optimal planting density help suppress weed growth naturally by enhancing crop competition. Mechanical methods, including hand weeding, tillage, and weed seed destruction, physically remove or disrupt weed development. Biological control leverages natural enemies like insects, fungi, and allelopathic crops to suppress weeds. Chemical control, when necessary, involves the strategic use of herbicides, ensuring rotation and precision application to prevent resistance. Preventive measures such as using clean seeds, quarantine practices, and timely weed management help reduce weed infestation in the long run. By integrating these diverse approaches, IWM ensures effective, sustainable, and eco-friendly weed control, reducing dependency on chemical herbicides and improving soil and crop health.

ix) Incorporation of Crop diversification

Crop diversification is a key strategy for enhancing agricultural resilience, improving soil health, and increasing farmers' income. It involves cultivating a variety of crops in a single growing season or in rotation over different seasons, reducing dependency on a single crop. Diversification can include intercropping, mixed cropping, relay cropping, and crop rotation, which help in efficient resource utilization, pest and disease management, and risk reduction due to climate variability.

By incorporating diverse crops, and multi-cropping system, farmers can improve soil fertility through nitrogen-fixing legumes, break pest and disease cycles, and enhance biodiversity as well as income resilience. Additionally, growing climate-resilient crops such as millets, pulses, and oilseeds alongside staple cereals can ensure better adaptability to erratic rainfall and temperature fluctuations. To encourage crop diversification and enhance climate resilience, the project will provide mini kits of oilseed crops and minor millets to farmers as well as project will promote production of new crops like chia, makhana, saffron and emerging technology like hydroponics on a lighthouse demonstration mode. This initiative aims to reduce dependency on single-crop farming, improve soil health, and increase farmers' income by introducing high-value, climate-resilient crops into traditional farming systems.

x) Retention and In-Situ Decomposition of Weed and Crop Biomass

Retaining weed and crop biomass in the soil and facilitating its in-situ decomposition is an effective practice for enhancing soil health, improving nutrient cycling, and promoting sustainable agriculture. Instead of removing or burning crop residues and weeds, their incorporation into the soil helps in maintaining soil organic matter, enhancing microbial activity, and improving moisture retention.

Weed and crop biomass decompose naturally in the soil, releasing essential nutrients like nitrogen, phosphorus, and potassium, which contribute to improved soil fertility. This practice also aids in carbon sequestration, reducing dependency on synthetic fertilizers, and suppressing weed growth by forming a

natural mulch layer. Additionally, the decomposition process encourages beneficial soil microbes, enhancing soil structure and overall productivity.

By adopting this approach, farmers can reduce soil erosion, retain soil moisture, and minimize greenhouse gas emissions from biomass burning. This method aligns with conservation agriculture principles and supports climate-resilient farming systems by improving the long-term sustainability of agricultural lands.

xi) Reducing methane emission from Paddy:

The project will expand and deepen technology interventions to sustain productivity of Paddy while keeping its GHG emissions competitive, including precision agriculture techniques such as Direct Seeded Rice, Systemic Intensification of Rice, Alternate wet and dry method and Zero Tillage for promoting low-methane paddy.

The details of this activity are explained in B3.

The following climate-resilient technologies will be introduced and demonstrated to the farmers in project villages for horticulture crops, through various extension approaches including FFS.

1. **Plantation on Ridges:** - Planting horticultural crops on raised ridges is an effective method to manage excess water during heavy rainfall, prevent waterlogging, and enhance root aeration. This technique, often used in high-density fruit plantations like Mango, Pomegranate, Citrus, Banana, and Guava in Maharashtra, ensures better water infiltration, reduces soil erosion, and supports healthier root systems. Inspired by the Indo-Israel technique, it promotes efficient resource use through proper spacing, sunlight exposure, and micro-irrigation practices, resulting in improved crop health and yield.
2. **Mulching:** - Mulching involves covering soil with organic materials (straw, leaves, compost) or synthetic materials (plastic sheeting) to conserve moisture, regulate soil temperature, suppress weeds, and enhance water use efficiency. In arid regions, it protects crops from temperature extremes, while organic mulches improve soil fertility and structure over time.
3. **Micro irrigation and fertigation:** - Drip and sprinkler irrigation deliver water efficiently, conserving resources and preventing soil erosion and waterlogging. Drip systems target plant roots directly, while sprinklers cover larger areas with even distribution. Combined with fertigation—applying fertilizers through irrigation—these methods ensure precise water and nutrient delivery, boosting crop health, yields, and resource efficiency, especially for fruit crops.
4. **Climate-Resilient Agriculture (CRA) Techniques:** CRA techniques improve horticultural adaptability to climate change through soil and water conservation measures like contour ploughing, terracing, and cover crops to reduce erosion and retain moisture. They also promote drought-tolerant and disease-resistant crop varieties, enhancing resilience and productivity amid extreme weather conditions.
5. **Canopy Management (Training, Pruning, Staking):** Canopy management optimizes sunlight, air circulation, and nutrient distribution, enhancing crop health and yield. Practices like pruning remove diseased branches to prevent pests, while training techniques like staking support plant structure, protect against weather damage, and ease harvesting. This is crucial for high-value crops like mango, guava, and grapes.
6. **Integrated Pest and Nutrient Management (IPNM):** IPM is a sustainable approach that combines biological, cultural, mechanical, and chemical methods to manage pests while

minimizing environmental risks. It includes monitoring pests, using natural predators, adopting cultural practices like crop rotation, employing mechanical controls, and applying pesticides only when necessary.

Integrated Nutrient Management (INM): - INM enhances soil fertility and crop nutrition by integrating organic, inorganic, and biological nutrient sources. It involves using compost, manure, and biofertilizers, applying chemical fertilizers based on soil tests, and practicing crop rotation and intercropping to maintain soil health and productivity.

7. **Bagging of Fruits:** Bagging protects fruits like mango, guava, and pomegranate from pests, diseases, and environmental damage, reducing pesticide use and preventing blemishes, sunburn, and wind damage. It enhances fruit quality, retaining natural colour and flavour, which boosts market value and ensures higher returns.
8. **Good Harvesting Practices:** - Proper harvesting ensures quality and minimizes post-harvest losses. Crops should be harvested at peak maturity using appropriate tools to prevent damage. Post-harvest management, including sorting, grading, and storage, is crucial for maintaining quality and maximizing returns, especially for fruits like tomatoes, bananas, and citrus.

Annexure 7: Project Results Framework

Table 38: PDO Indicators by PDO Outcomes

Baseline	Period 1	Closing Period
PDO Indicators		
Enhanced Farm Income: Increase in Farm Income at Household level		
Enhanced Farm Income: Increase in Farm Income at Household level (Percentage)		
Jul/2025	Jun/2028	Jun/2031
0	15	30
Enhanced Farm Income: Increase in Farm Income at Household level-Female headed households (Percentage)		
Jul/2025	Jun/2028	Jun/2031
0	20	35
Increase in Water Use Efficiency		
Increase in Water use efficiency (Percentage)		
Jul/2025	Jun/2028	Jun/2031
0	5	8
People Using digitally-enabled services		
People using digitally enabled services (enhanced services) (Number of People) CRI (Number)		
Jul/2025	Jun/2028	Jun/2031
0	1,000,000	2,000,000
ØPeople using digitally enabled services (enhanced services) - Female (Number of people) CRI (Number)		
Jul/2025	Jun/2028	Jun/2031
0	150,000	300,000
Beneficiaries with enhanced resilience to climate risks		
People benefitting from enhanced resilience of terrestrial and aquatic systems (Number of people) CRI (Number)		
Jul/2025	Jun/2028	Jun/2031
0	640,000	1,600,000
ØPeople benefitting from enhanced resilience of terrestrial and aquatic systems - Female (Number of people) CRI (Number)		
Jul/2025	Jun/2028	Jun/2031
0	96,000	240,000

Intermediate Indicators by Components

Baseline	Period 1	Closing Period
Resource Efficient Productivity Enhancement		
Enhanced Fertilizer use efficiency (Percentage)		
Jul/2025	Jun/2028	Jun/2031
0	3	7
Increase in cropping intensity (Percentage)		
Jul/2025	Jun/2028	Jun/2031
0	10	25
Promoting Precision Agriculture for GHG Emissions Management		
Agricultural area where climate-resilient agriculture practices are implemented (Hectare(Ha))		
Jul/2025	Jun/2028	Jun/2031
0	270,000	900,000
Net greenhouse gas emissions (GHG) reductions (Metric ton)		
Jul/2025		Jun/2031
0		-1,200,000
Improving and Building Income Resilience		
Project-supported farmer producer companies with growth in annual profits (Number)		
Jul/2025	Jun/2028	Jun/2031
0	0	250
Exclusive women Self Help Groups supported by the project (Number)		
Jul/2025	Jun/2028	Jun/2031
0	100	500
Project Management and Institutional Strengthening		
Knowledge products delivered (Number)		
Jul/2025	Jun/2028	Jun/2031
0	4	10
Participatory Village adaptation plans prepared (Number)		
Jul/2025	Jun/2028	Jun/2031
0	7,196	7,196

Annexure 8 - Current Support Structure for FPOs

The state has long been supporting the group formation and mobilisation of farming communities through various interventions and extension mechanisms such as ATMA, MACP-World Bank aided project etc. The first phase of the project included a value chain component that supported 1,799 farmers' groups, 1,204 self-help groups (SHGs), and 1,698 farmer producer companies (FPCs), with a total investment of Rs 501.70 crores as of 31st December 2023. Apart from this NDKSP has also extended capacity building, market led agricultural planning and other support to the FPOs in the project area. Table 28 shows the initiatives of the Government of Maharashtra to promote FPOs and to support the agribusiness activities.

Table 39: Current Programmes in the State to Support FPOs

Programme/Project	Salient feature
State of Maharashtra's Agribusiness and Rural Transformation project (SMART)	Support to development of inclusive and competitive agriculture value chains, focusing on small holder farmers and agri-entrepreneurs.
Formation and promotion of 10,000 Farmer Producer Organisations	(i) To promote holistic system for sustainable income-oriented farming, (ii) To enhance productivity through efficient, cost-effective and sustainable resource use, (iii) To provide support to new FPOs for management of agricultural activity SFAC, NABARD, NCDC, MCDC and NAFED are the implementing agencies for these initiatives
Chief Minister Agriculture and Food Processing Scheme	To encourage modern technology-based projects for quality enhancement of agriculture produce,
Prime Minister Formalisation of Micro Food Processing Enterprises Scheme	'One District One Product' intervention aimed at financial support to individual micro food processing units
Maharashtra Agribusiness Network Project (MAGNET)	Support to FPOs by improving a network of postharvest marketing and value chains while focusing on specific horticulture crops
Dr. Panjabrao Jaivik Kheti Mission	This scheme helps the farmers in production, certification, processing and marketing of organic farm products.

Annexure 9: Farmer Producer Organisations in Project districts

Table 40: Farmer Producer Organisations in Project districts

Sr. No.	District	No. of FPCs	No. of SHGs	No. of FIGs (ATMA Registered)	No. of revenue villages under PESA
1	Akola	188	12393	2751	0
2	Amaravati	299	19137	3028	72
3	Chatrapati Sambhaji Nagar	1060	16552	4705	0
4	Beed	573	18046	4909	0
5	Buldhana	426	22561	3215	0
6	Hingoli	229	9188	3200	0
7	Jalgaon	284	25656	3527	19
8	Jalna	1026	16194	3200	0
9	Latur	594	18990	4930	0
10	Nanded	484	17454	5120	36
11	Dharashiv	360	14503	11841	0
12	Parbhani	193	11378	2438	0
13	Wardha	124	14337	2227	0
14	Washim	189	10802	2449	0
15	Yavatmal	206	24280	3439	87
16	Nasik	896	24006	4219	294
17	Bhandara	110	13956	2754	0
18	Chandrapur	128	19958	4133	55
19	Gadchiroli	59	15064	1864	329
20	Gondia	99	16958	4033	0
21	Nagpur	156	18720	3280	0
	Total	7683	360133	81262	892



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